Polylogarithms and their Arithmetic Bridges

Volume 1: Polylogarithm Identities and Arithmetic Reductions

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About this volume

Nested sums and iterated integrals establish the common algebra. Order resonance, nonlinear feedback and inversion connect analytic continuation to uniform coefficient laws. Cyclotomic Fourier transforms organize special values; algebraic ladders and Gaussian/Eisenstein depth reductions turn experiments into exact identities.

The four volumes form one collective manuscript. They share definitions, notation, source evidence and a bibliography. Chapter, theorem and equation numbers are retained from the unified manuscript so references in research reports remain stable. Cross-volume references link to the companion PDF; keep the four PDF files together when downloading them.

Volume 1	Polylogarithm Identities and Arithmetic Reductions
Volume 2	Signed Kernels, Zero Geometry and Experimental Discovery
Volume 3	Hurwitz Jets, Stieltjes Calculus and Gamma Correlations
Volume 4	Gamma Grids, CM Periods and Herglotz Arithmetic

A rigorous analytic proof, a finite exact certificate and an independently repeated numerical diagnostic carry different evidence. Computation drives conjecture formation and exposes missing structure; successful numerical recognition does not itself prove an identity or arithmetic independence. The source inventory and editorial ledger record the accepted corrections and the scope of material still undergoing analytic reconciliation.

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Chapter 1

From one polylogarithm to many

An ordinary polylogarithm has two descriptions: a series and an iterated integral. Nesting the first and extending the alphabet of the second produce the same larger class of functions. Their two multiplication laws then provide exact relations. This chapter fixes conventions before those relations are used to organize special values.

1.1 Multiple polylogarithms: two faces

The classical polylogarithm $\mathrm{Li}_s(z) = \sum_{n\geq 1} z^n/n^s$ has, for the reader of Lewin, two complementary descriptions: a power series and an iterated integral $\mathrm{Li}_s(z) = \int_0^z \mathrm{Li}_{s-1}(t)\,dt/t$. Each survives the passage to many indices, and the resulting two descriptions of one and the same function are the source of essentially every nontrivial identity in this circle of ideas. We set up both here; the products they generate occupy the next section.

1.1.1 The nested sum

Definition 1.1. *For $s_1, \dots, s_k \in \mathbb{C}$ and $z_1, \dots, z_k \in \mathbb{C}$ the* multiple polylogarithm *is the strictly nested sum*

$$\mathrm{Li}_{s_1,\dots,s_k}(z_1,\dots,z_k) \;=\; \sum_{n_1>n_2>\cdots>n_k\geq 1} \prod_{i=1}^{k} \frac{z_i^{n_i}}{n_i^{s_i}}.$$

Its weight *is $w = s_1 + \cdots + s_k$ and its* depth *is k. The single-index case $k = 1$ recovers $\mathrm{Li}_s(z)$.*

Two features of this definition demand comment before anything is proved.

Remark 1.2 (Conventions differ; ours is fixed). The inequality is *strict*, $n_1 > n_2 > \cdots > n_k$. A sizable part of the literature (and several computer-algebra systems) instead uses a non-strict summation $n_1 \geq n_2 \geq \cdots \geq n_k$, or reverses the role of the indices so that the *innermost* summation variable carries the first parameter. These choices are genuinely different functions (they differ by lower-depth terms, exactly the diagonal contributions $n_i = n_{i+1}$), and the discrepancy is the single most common source of sign and indexing confusion when comparing sources. We adhere throughout to the strict, "outermost variable carries s_1" convention above. The arguments are likewise ordered with z_1 paired to the largest index n_1. When checking against `MultiplePolyLog`, confirm the convention numerically on a small case rather than assuming agreement.

Remark 1.3 (Multiple zeta values). Setting every $z_i = 1$ gives the *multiple zeta value*

$$\zeta(s_1,\dots,s_k) = \mathrm{Li}_{s_1,\dots,s_k}(1,\dots,1) = \sum_{n_1>\cdots>n_k\geq 1} \prod_i n_i^{-s_i},$$

the central special values of the theory (Zagier 1994; Hoffman 1992). The strict nesting is what makes $\zeta(s_1, \ldots, s_k)$ finite for $s_1 \geq 2$.

Convergence. The single-variable series $\mathrm{Li}_s(z)$ converges absolutely for $|z| < 1$ (any s), on $|z| = 1$ for $\Re s > 1$, and conditionally on $|z| = 1$, $z \neq 1$, for $\Re s > 0$. For the nested sum the corresponding statement is clean.

Proposition 1.4 (A sufficient absolute-convergence domain)**.** *For positive integer indices, the nested series converges absolutely when $|z_i| \leq 1$ for all i and $s_1 \geq 2$. It also converges absolutely for $|z_1| < 1$, $|z_i| \leq 1$ for $i > 1$, with $s_1 \geq 1$.*

Proof. The absolute value of the inner nested sum is at most $H^{k-1}_{n_1-1} = O((1 + \log n_1)^{k-1})$. The outer series is therefore bounded by $\sum |z_1|^{n_1}(1 + \log n_1)^{k-1}/n_1^{s_1}$, which converges in either stated case. □

These are sufficient conditions, not a complete classification of every boundary point. Oscillation can give conditional convergence: $\mathrm{Li}_1(-1)$ converges, but $\sum |(-1)^n/n|$ diverges. More generally, away from the polydisc an interior domain is given by $|z_1 \cdots z_j| < 1$ for each $1 \leq j \leq k$. At unity, for positive integer indices, admissibility is exactly $s_1 \geq 2$.

1.1.2 The iterated integral

The second face is Chen's iterated integral. Fix the two differential one-forms

$$\omega_0 = \frac{dt}{t}, \qquad \omega_1 = \frac{dt}{1-t},$$

and, for a word $u = a_1 a_2 \cdots a_w$ in the letters $\{0, 1\}$, write the iterated integral over the standard simplex

$$\int_0^z u \;=\; \int_{0<t_1<t_2<\cdots<t_w<z} \omega_{a_w}(t_1)\, \omega_{a_{w-1}}(t_2) \cdots \omega_{a_1}(t_w),$$

i.e. the leftmost letter a_1 is attached to the *outermost* (largest) variable t_w and one integrates from the top down. This is the orientation under which d/dz strips the outermost form. With this normalization the one-variable multiple polylogarithm is an iterated integral over a single explicit word.

Theorem 1.5 (Iterated-integral representation)**.** *For $s_1 \geq 2$ and the remaining $s_i \geq 1$,*

$$\mathrm{Li}_{s_1,\ldots,s_k}(z, 1, \ldots, 1) \;=\; \int_0^z \underbrace{0^{s_1-1}1}_{\text{block } 1}\, \underbrace{0^{s_2-1}1}_{\text{block } 2} \cdots \underbrace{0^{s_k-1}1}_{\text{block } k},$$

the iterated integral of the binary word

$$u(\mathbf{s}) = 0^{s_1-1}1\, 0^{s_2-1}1 \cdots 0^{s_k-1}1$$

of length w. Equivalently, with the alternative kernel $\widetilde{\omega}_1 = dt/(t-1) = -\omega_1$ one carries one sign per 1-letter, giving the textbook normalization

$$\mathrm{Li}_{s_1,\ldots,s_k}(z, 1, \ldots, 1) = (-1)^k \int_{0<t_1<\cdots<t_w<z} \prod_{j=1}^{w} \eta_{a_{w+1-j}}(t_j), \qquad \eta_0 = \tfrac{dt}{t},\ \eta_1 = \tfrac{dt}{t-1},$$

the factor $(-1)^k$ being exactly the number of 1-letters in $u(\mathbf{s})$, one per block.

The composition ↔ word dictionary. The map $\mathbf{s} = (s_1, \dots, s_k) \mapsto u(\mathbf{s})$ is a bijection between compositions (ordered tuples with each $s_i \geq 1$) and binary words that *end in a* 1. Reading u from left to right: a 1 *closes a block* and contributes a new summation index; the run of 0's immediately preceding that 1 sets the corresponding exponent, $s_i - 1$ zeros for exponent s_i. Thus depth $k =$ (number of 1's) and weight $w =$ (length of the word). Admissibility $s_1 \geq 2$ is the requirement that u *begin with a* 0: a leading 1 would attach ω_1 to the top variable and produce the logarithmic divergence at $t_w \to 1$ when $z = 1$. The endpoints play dual roles — ω_0 is singular at 0, ω_1 at 1 — so a convergent word neither starts with 1 nor ends with 0.

Origin. Expand the innermost form geometrically. The depth-one case is Lewin's $\mathrm{Li}_s(z) = \int_0^z 0^{s-1}1$, obtained by iterating $\mathrm{Li}_s(z) = \int_0^z \mathrm{Li}_{s-1}(t)\,\omega_0$ down to $\mathrm{Li}_1(z) = -\log(1-z) = \int_0^z \omega_1$. For general depth, $\int_0^z \omega_1$ on the innermost block introduces $\sum_{n_k \geq 1} t^{n_k}/n_k$, the subsequent ω_0's raise the power of n_k, and the next ω_1 couples a strictly larger index $n_{k-1} > n_k$; the strict inequality is forced by the nesting of the simplex $t_1 < \cdots < t_w$. Carrying this out reproduces the nested sum, and Chen's theory (1977) guarantees convergence and homotopy invariance of the integral. The general framework is due to Goncharov; the systematic exploitation for these functions is Goncharov's and Borwein–Bradley–Broadhurst–Lisoněk's. □

1.1.3 General alphabets and a fixed integral orientation

For a path from 0 to z avoiding its letters, use the outer-letter-first recursion

$$G(a_1, \dots, a_w; z) = \int_0^z \frac{G(a_2, \dots, a_w; t)}{t - a_1}\, dt, \qquad G(;z) = 1.$$

The all-zero word is regularized as $G(0^w; z) = \log^w z/w!$. For positive integer indices the dictionary in our decreasing-index convention is

$$\mathrm{Li}_{s_1,\dots,s_k}(z_1, \dots, z_k) = (-1)^k G\left(0^{s_1-1}, z_1^{-1}, 0^{s_2-1}, (z_1 z_2)^{-1}, \dots, 0^{s_k-1}, (z_1 \cdots z_k)^{-1}; 1\right).$$

Thus the letters are inverse *prefix* products. Reversing the integration order also reverses the letter word; mixing these conventions changes the function. The formula follows by expanding each nonzero kernel geometrically near the origin and integrating recursively, then continuing along a common path. Wolfram's generalized-polylogarithm interface must be calibrated against this orientation before numerical comparisons.

1.1.4 Two facts to carry forward

(1) **The differential strips the first letter.** In the integral orientation above,

$$z\frac{d}{dz}\,\mathrm{Li}_{s_1,\dots}(z, 1, \dots) = \begin{cases} \mathrm{Li}_{s_1-1,s_2,\dots}(z, 1, \dots), & s_1 \geq 2, \\ \dfrac{z}{1-z}\,(\text{remove the leading block}), & s_1 = 1, \end{cases}$$

i.e. differentiation removes one outer letter — a 0 lowers s_1, the leading 1 deletes the first index. This recursion (equivalently the integral equation defining the word) drives every analytic continuation and every series-rearrangement proof. It is also the recursion behind the appearance of `HarmonicPolyLog` in `DSolve`/`AsymptoticDSolveValue` output for Fuchsian systems with rational coefficients (Remiddi–Vermaseren 1999 for the $\{0,1\}$-alphabet case).

(2) **One function, two presentations.** The *same* $\mathrm{Li}_{s_1,\dots,s_k}$ is simultaneously the nested sum and the iterated integral. Multiplying two nested sums and re-sorting the indices yields the *stuffle* (harmonic, quasi-shuffle) product $*$; multiplying two iterated integrals over a common path and shuffling the letters yields the *shuffle* product Π. Both express a product of polylogarithms as a $\mathbb{Q}$-linear combination of polylogarithms, but through different combinatorics; equating them is the engine of the next section.

1.1.5 The smallest nontrivial example, in full

Take depth $k=2$, $\mathbf{s}=(2,1)$, the lightest admissible non-classical case. The nested sum is

$$\mathrm{Li}_{2,1}(z,1)=\sum_{m>n\geq 1}\frac{z^m}{m^2\,n}=\sum_{m=2}^{\infty}\frac{z^m}{m^2}\,H_{m-1},\qquad H_{m-1}=\sum_{n=1}^{m-1}\tfrac{1}{n},$$

exhibiting the characteristic harmonic-number coefficient. The associated word is

$$u(2,1)=\underbrace{0^{2-1}1}_{\text{block }(s_1=2)}\ \underbrace{0^{1-1}1}_{\text{block }(s_2=1)}=011,$$

a word of length $w=3$ (weight 3) with two 1's (depth 2), beginning in 0 (admissible) and ending in 1 (convergent). By Theorem 1.5,

$$\mathrm{Li}_{2,1}(z,1)=\int_0^z 011=\int_{0<t_1<t_2<t_3<z}\frac{dt_3}{t_3}\,\frac{dt_2}{1-t_2}\,\frac{dt_1}{1-t_1},$$

the leading 0 sitting on the outermost variable t_3 and the two 1's on t_2,t_1. (Equivalently $(-1)^2=+1$ in the $\eta_1=dt/(t-1)$ normalization, consistent with the unsigned $\omega_1=dt/(1-t)$ form.) A numerical check at $z=0.7$ gives both sides equal to $0.2384965\ldots$, and at $z=1$ the integral $\int_0^1 011$ recovers Euler's celebrated

$$\zeta(2,1)=\sum_{m>n\geq 1}\frac{1}{m^2 n}=\zeta(3),$$

the depth-two value whose collapse to a depth-one zeta was Euler's first multiple-zeta identity — and, in Wolfram 15, exactly the symbolic output of `MultipleZeta` on $\{2,1\}$.

1.2 Multiple zeta values and their relations

Specializing the multiple polylogarithm at $z_1=\cdots=z_k=1$ produces the central objects of this chapter, the *multiple zeta values* (MZVs):

$$\zeta(s_1,\dots,s_k)\;=\;\mathrm{Li}_{s_1,\dots,s_k}(1,\dots,1)\;=\sum_{n_1>n_2>\cdots>n_k\geq 1}\prod_{i=1}^{k}\frac{1}{n_i^{s_i}}. \tag{1.1}$$

The outermost summation index n_1 carries the exponent s_1, and the strict inequalities make the sum a genuine iterated, rather than ordinary, multiple series. The series (1.1) converges if and only if $s_1\geq 2$; such an index tuple is called *admissible*. (Any $s_i\geq 1$ for $i\geq 2$ is allowed; only the leading entry is constrained.) We call $w=s_1+\cdots+s_k$ the *weight* and k the *depth*. In Wolfram Language these are `MultipleZeta`, with $\zeta(2,1)$ written `MultipleZeta[{2,1}]`.

The depth-one values are the classical zeta values. Euler's formula is

$$\zeta(2n) = \frac{(-1)^{n+1} B_{2n} (2\pi)^{2n}}{2(2n)!} \in \pi^{2n}\mathbb{Q}.$$

The first genuinely new phenomenon is that distinct MZVs can collapse onto a single lower-depth value. The prototype is Euler's identity

$$\zeta(2,1) \;=\; \sum_{n>m\geq 1} \frac{1}{n^2\, m} \;=\; \zeta(3), \tag{1.2}$$

already known to Euler in the 1740s. The conjecture is that the $\mathbb{Q}$-vector space spanned by all MZVs is *weight-graded*: products and relations never mix weights, so one studies each weight w separately.

The sum theorem

Euler's $\zeta(2,1) = \zeta(3)$ is the depth-two case of the first structural theorem. Summing all admissible MZVs of a fixed weight w and fixed depth k recovers the single zeta value $\zeta(w)$.

Theorem 1.6 (Sum theorem; Euler for $k = 2$, Granville and Zagier in general, 1990s)**.** *For all* $w > k \geq 1$,

$$\sum_{\substack{s_1+\cdots+s_k=w \\ s_1\geq 2,\ s_i\geq 1}} \zeta(s_1,\ldots,s_k) \;=\; \zeta(w). \tag{1.3}$$

At $w = 3, k = 2$ the only admissible tuple is $(2,1)$, recovering (1.2). At $w = 4, k = 2$ the admissible tuples are $(3,1)$ and $(2,2)$, so

$$\zeta(3,1) + \zeta(2,2) = \zeta(4) = \tfrac{\pi^4}{90}. \tag{1.4}$$

This one equation does not separate $\zeta(3,1)$ from $\zeta(2,2)$; for that we need a *second* multiplication law, to which we now turn.

The two products

MZVs admit two structurally different multiplication rules, because $\zeta(s_1,\ldots,s_k)$ has two faithful representations: as a nested sum (1.1), and as an iterated integral. Each representation makes the product of two MZVs expand into a sum of MZVs, but the two expansions differ — and that discrepancy is the engine of the theory.

(a) The stuffle (harmonic / quasi-shuffle) product. Multiply two nested sums directly. For depth one,

$$\zeta(a)\,\zeta(b) = \Big(\sum_{m\geq 1} m^{-a}\Big)\Big(\sum_{n\geq 1} n^{-b}\Big) = \sum_{m>n} + \sum_{n>m} + \sum_{m=n},$$

splitting the lattice $\{(m,n)\}$ by the trichotomy $m > n$, $m < n$, $m = n$. The three regions are exactly two depth-two MZVs and one depth-one MZV:

$$\zeta(a)\,\zeta(b) = \zeta(a,b) + \zeta(b,a) + \zeta(a+b). \tag{1.5}$$

The general rule is the same combinatorial device — the *quasi-shuffle* of Hoffman — applied to the index words $(s_1,\ldots,s_k)$ and $(t_1,\ldots,t_\ell)$: interleave the two words in all order-preserving ways, and

additionally allow any aligned pair s_i, t_j to *collide* into a single slot $s_i + t_j$. (Setting $a = b = 2$ in (1.5) gives $\zeta(2)^2 = 2\,\zeta(2,2) + \zeta(4)$, used below.) A depth-two example of the general rule:

$$\zeta(a)\,\zeta(b,c) = \zeta(a,b,c) + \zeta(b,a,c) + \zeta(b,c,a) + \zeta(a+b,c) + \zeta(b,a+c),$$

the first three terms inserting a into the three gaps of (b,c), the last two colliding a with b and with c.

(b) The shuffle product. The integral representation reads each MZV as a Chen iterated integral over the simplex $1 > t_1 > \cdots > t_w > 0$ against the two one-forms $\omega_0 = \frac{dt}{t}$ and $\omega_1 = \frac{dt}{1-t}$. Encode an admissible index as a binary word: each s_i becomes $\underbrace{0\cdots0}_{s_i-1}1$, so that, e.g., $\zeta(2) \leftrightarrow 01$, $\zeta(3) \leftrightarrow 001$, $\zeta(2,1) \leftrightarrow 011$, $\zeta(3,1) \leftrightarrow 0011$, $\zeta(2,2) \leftrightarrow 0101$. Admissibility ($s_1 \geq 2$) is the condition that the word *begins with* 0; convergence at the inner end requires the word to *end with* 1. Multiplying two iterated integrals over the same simplex produces the integral over the product region, which decomposes into the $\binom{w_1+w_2}{w_1}$ order-preserving interleavings — the *shuffle product* Π — of the two letter words, *without* any collision term. Thus

$$\zeta(a)\,\zeta(b) = \sum_{\text{words } u \in (\text{word } a)\,\Pi\,(\text{word } b)} \zeta(u). \tag{1.6}$$

The double-shuffle relation: $\zeta(3,1)$ in closed form

The canonical demonstration computes $\zeta(2)^2$ by both products and equates the results. Stuffle, from (1.5) with $a = b = 2$:

$$\zeta(2)^2 = 2\,\zeta(2,2) + \zeta(4). \tag{1.7}$$

Shuffle: $\zeta(2)$ is the word 01, and we shuffle $01 \,\Pi\, 01$. Writing the two factors as 0_a1_a and 0_b1_b and listing the $\binom{4}{2} = 6$ interleavings that preserve $0_a \to 1_a$ and $0_b \to 1_b$:

$$0_a0_b1_a1_b,\;\; 0_a0_b1_b1_a,\;\; 0_b0_a1_a1_b,\;\; 0_b0_a1_b1_a,\;\; 0_a1_a0_b1_b,\;\; 0_b1_b0_a1_a.$$

Forgetting the colour labels, these are the words 0011 (four times) and 0101 (twice). Since $0011 \leftrightarrow \zeta(3,1)$ and $0101 \leftrightarrow \zeta(2,2)$,

$$\zeta(2)^2 = 4\,\zeta(3,1) + 2\,\zeta(2,2). \tag{1.8}$$

Now subtract (1.7) from (1.8): the $2\,\zeta(2,2)$ terms cancel and we obtain the *double-shuffle relation*

$$4\,\zeta(3,1) = \zeta(4) \qquad \Longrightarrow \qquad \zeta(3,1) = \tfrac{1}{4}\zeta(4) = \frac{\pi^4}{360}. \tag{1.9}$$

Feeding this back into the sum-theorem relation (1.4) also pins $\zeta(2,2) = \zeta(4) - \zeta(3,1) = \frac{3}{4}\zeta(4) = \pi^4/120$ — consistent with the all-2's family below.

Remark 1.7 (Regularized double shuffle). The naive double-shuffle relations above are produced only by admissible words. To capture the boundary case, one regularizes the divergent $\zeta(1)$ (the word 1). Comparing the two regularizations — shuffle versus stuffle — as polynomials in a formal variable T standing for the regularized $\zeta(1)$ yields the *extended* or *regularized double-shuffle relations* of Ihara, Kaneko, and Zagier (2006). These are conjectured to generate *all* $\mathbb{Q}$-linear relations among MZVs.

Other relation families

The double-shuffle relations sit inside a larger, heavily overlapping web. We list the principal families, one line each.

Duality. The integral substitution $t \mapsto 1-t$ swaps $\omega_0 \leftrightarrow \omega_1$ and reverses the order of integration, giving $\zeta(\mathbf{s}) = \zeta(\mathbf{s}^\dagger)$, where the *dual* word $\mathbf{s}^\dagger$ is obtained from the binary word by reading it backwards and exchanging $0 \leftrightarrow 1$. The simplest instance is $011 \mapsto 011$ read backwards is 110, complemented is 001: thus $\zeta(2,1) = \zeta(3)$ (1.2) is precisely a self/dual coincidence. Likewise $0011\,(\zeta(3,1))$ is *self*-dual—read backwards and complemented it is again 0011—so duality says nothing new there; the cleanest nontrivial higher instance is the depth-three identity $\zeta(2,1,1) = \zeta(4)$, from $0111 \mapsto 0001$.

Hoffman's relations. Relations obtained by comparing the regularized product with $\zeta(1)$ provide useful index identities. Their precise insertion and splitting terms matter; a schematic insertion of a single 1 in every position is not a valid general formula.

Derivation relations. An infinite family produced by certain Hopf-algebra derivations acting on the algebra of admissible words (Ihara–Kaneko–Zagier), equivalent over $\mathbb{Q}$ to large portions of the regularized double shuffle.

Ohno's relation. A single generating identity (Ohno 1999) that simultaneously generalizes the sum theorem (1.3) *and* duality, obtained by summing MZVs over all ways of distributing a fixed total increment among the indices of $\mathbf{s}$ and of its dual.

Cyclic sum formula. Hoffman–Ohno (and Ohno–Wakabayashi): a cyclic sum of MZVs of weight w equals a weighted sum of weight-w values, refining the sum theorem with a cyclic symmetry on the index word.

Closed-form families

Beyond individual reductions, several infinite families of MZVs evaluate to powers of π in closed form. Throughout, $\{s\}_n$ denotes the index s repeated n times.

All 2's. The generating function $\prod_{n\geq 1}(1 - x^2/n^2) = \sin(\pi x)/\pi x$ yields, after extracting the relevant power sums,

$$\zeta(\{2\}_n) = \zeta(\underbrace{2,2,\ldots,2}_{n}) = \frac{\pi^{2n}}{(2n+1)!}. \tag{1.10}$$

For $n=1$ this is $\zeta(2) = \pi^2/6$; for $n=2$, $\zeta(2,2) = \pi^4/120$, matching (1.9); for $n=3$, $\zeta(2,2,2) = \pi^6/5040$.

The $\{3,1\}$ family. A celebrated result of Borwein, Bradley, Broadhurst, and Lisoněk (1998), originally conjectured experimentally:

$$\zeta(\{3,1\}_n) = \zeta(\underbrace{3,1,3,1,\ldots,3,1}_{n \text{ blocks}}) = \frac{2\,\pi^{4n}}{(4n+2)!}. \tag{1.11}$$

For $n = 1$ this reproduces $\zeta(3,1) = 2\pi^4/6! = \pi^4/360$ of (1.9); for $n = 2$, $\zeta(3,1,3,1) = 2\pi^8/10!$. The proof is a striking generating-function argument identifying $\sum_n \zeta(\{3,1\}_n)x^{4n}$ with a ratio of cosh and cos.

The $\{2\}_a\,3\,\{2\}_b$ family. Zagier (2012) evaluated the two-parameter family $\zeta(\{2\}_a, 3, \{2\}_b)$ as an explicit $\mathbb{Q}$-linear combination of products $\pi^{2m}\zeta(2n+1)$, a result whose proof unexpectedly drew on the arithmetic of the modular-form-like generating series; it remains one of the deepest known closed-form families.

Dimensions and the weight grading

Let D_w be the $\mathbb{Q}$-vector space spanned by all MZVs of weight w (with $D_0 = \mathbb{Q}$, $D_1 = 0$ since $\zeta(1)$ diverges), and $d_w = \dim_{\mathbb{Q}} D_w$. Zagier (1994) conjectured the remarkably simple recursion

$$d_w = d_{w-2} + d_{w-3}, \qquad d_0 = 1,\ d_1 = 0,\ d_2 = 1, \tag{1.12}$$

giving the sequence

$$d_0, d_1, d_2, \cdots = 1, 0, 1, 1, 1, 2, 2, 3, 4, 5, 7, 9, \ldots.$$

The *upper* bound $d_w \le$ (the recursion value) is a theorem, proved independently by Goncharov and by Terasoma (around 2002), with the conceptual framework of mixed Tate motives completed by Deligne and Goncharov; it amounts to showing that the known relations (e.g. associator / motivic) suffice to cut the dimension down. Broadhurst and Kreimer (1997) refined the count by also grading by depth, conjecturing a two-variable generating function that brings in the dimensions of spaces of cusp forms (the first modular obstruction appearing at weight 12, depth 4). The matching *lower* bound — equivalently, that no further relations exist and the conjectured spanning sets are linearly independent — is wide open; even $\dim D_w \ge 2$ for some specific large w would already imply transcendence statements (such as the linear independence of π^6 and $\zeta(3)^2$ over $\mathbb{Q}$ at $w = 6$) that are presently out of reach.

A table of low-weight reductions

The following table records all admissible MZVs through weight 5 together with their reductions to the standard spanning sets $\{\zeta(2)\}$, $\{\zeta(3)\}$, $\{\zeta(4)\} \sim \{\pi^4\}$, $\{\zeta(5), \zeta(2)\zeta(3)\}$. Every entry has been verified symbolically.

w	spanning count	MZVs of weight w and their reductions
2	1	$\zeta(2) = \dfrac{\pi^2}{6}$
3	1	$\zeta(3)$; $\quad \zeta(2,1) = \zeta(3)$
4	1	$\zeta(4) = \dfrac{\pi^4}{90}$; $\quad \zeta(3,1) = \dfrac{\pi^4}{360}$, $\quad \zeta(2,2) = \dfrac{\pi^4}{120}$, $\quad \zeta(2,1,1) = \zeta(4)$
5	2	$\zeta(5)$, $\zeta(2)\zeta(3)$ span; $\quad \zeta(4,1) = 2\zeta(5) - \zeta(2)\zeta(3)$,
		$\zeta(3,2) = -\frac{11}{2}\zeta(5) + 3\,\zeta(2)\zeta(3)$, $\quad \zeta(2,3) = \frac{9}{2}\zeta(5) - 2\,\zeta(2)\zeta(3)$,
		$\zeta(3,1,1) = 2\zeta(5) - \zeta(2)\zeta(3)$, $\quad \zeta(2,1,1,1) = \zeta(5)$ (duality of $\zeta(5)$)
		$\zeta(2,2,1) = \zeta(3,2) = -\frac{11}{2}\zeta(5) + 3\zeta(2)\zeta(3)$,
		$\zeta(2,1,2) = \zeta(2,3) = \frac{9}{2}\zeta(5) - 2\zeta(2)\zeta(3)$ (duality)

The table gives spanning sets of sizes $1, 1, 1, 2$ at weights $2, 3, 4, 5$. The two-element weight-five set is not known to be independent. Every depth-two value in that row reduces to products of single zeta values. At weight eight, values such as $\zeta(6, 2)$ are candidates for a primitive direction beyond the polynomial algebra of single zetas.

1.3 Harmonic polylogarithms

The multiple polylogarithms of the previous sections are nested *sums*. For the practitioner who must actually manipulate weight-w transcendentals—most insistently the particle physicist computing Feynman integrals—it is the *iterated-integral* avatar that is decisive, because integration variables, differential equations, and analytic continuation all speak the language of integrals. The harmonic polylogarithms (HPLs) of Remiddi and Vermaseren [42] are precisely this avatar, specialised to the three integration kernels that arise when a rational integrand has singularities only at $t = 0, 1, -1$. They have become the de facto standard basis for one-scale problems, and Wolfram 15 exposes them directly as `HarmonicPolyLog`.

1.3.1 Definition: kernels and iterated integrals

Fix the alphabet $\{0, 1, -1\}$ and the three rational *kernels*

$$f_0(t) = \frac{1}{t}, \qquad f_1(t) = \frac{1}{1-t}, \qquad f_{-1}(t) = \frac{1}{1+t}.$$

A harmonic polylogarithm $H(a_1, \dots, a_w; z)$ is indexed by a word $\vec{a} = (a_1, \dots, a_w)$ over this alphabet; the length w is the *weight*. The definition is recursive in the leftmost letter. When the leading letter is nonzero,

$$H(a_1, a_2, \dots, a_w; z) = \int_0^z f_{a_1}(t)\, H(a_2, \dots, a_w; t)\, dt, \qquad a_1 \neq 0, \tag{1.13}$$

with the depth-zero seed $H(; z) = 1$ (the empty word). The recursion also applies when $a_1 = 0$, provided the remaining word is not all zeros. Such an integrand is integrable at the origin. Only the all-zero words need the logarithmic regularization above; zeros at the end of a general word can be removed by shuffle identities. For example $H(0, 1; z) = \mathrm{Li}_2(z)$ is a convergent integral with a leading zero. The leftmost letter always labels the outermost integration.

1.3.2 The weight-one and weight-two cases

At weight one (1.13)–the all-zero regularization give the three building blocks

$$H(0; z) = \ln z, \qquad H(1; z) = -\ln(1-z), \qquad H(-1; z) = \ln(1+z), \tag{1.14}$$

since $\int_0^z dt/(1-t) = -\ln(1-z)$ and $\int_0^z dt/(1+t) = \ln(1+z)$. At weight two, integrating these against another kernel reproduces the classical dilogarithm and its companions:

$$H(0, 1; z) = \int_0^z \frac{dt}{t}\big(-\ln(1-t)\big) = \mathrm{Li}_2(z), \qquad H(0, -1; z) = \int_0^z \frac{dt}{t} \ln(1+t) = -\mathrm{Li}_2(-z), \tag{1.15}$$

$$H(1, 0; z) = \int_0^z \frac{\ln t}{1-t}\, dt = -\mathrm{Li}_2(z) - \ln z\, \ln(1-z), \qquad H(1, 1; z) = \tfrac{1}{2} \ln^2(1-z), \tag{1.16}$$

the last by (1.13) with $H(1;t) = -\ln(1-t)$ and $\int_0^z (-\ln(1-t))/(1-t)\,dt = \frac{1}{2}\ln^2(1-z)$. The pair $H(0,1;z) = \mathrm{Li}_2(z)$ and $H(0,-1;z) = -\mathrm{Li}_2(-z)$ shows the general pattern: the words $H(\underbrace{0,\dots,0}_{s-1},1;z) = \mathrm{Li}_s(z)$ and $H(\underbrace{0,\dots,0}_{s-1},-1;z) = -\mathrm{Li}_s(-z)$ recover the entire ladder of classical polylogarithms, so the HPLs are a strict superset of the functions of Lewin's treatise. In Wolfram 15 these read, e.g., `HarmonicPolyLog[{0,1},x]` $= \mathrm{Li}_2(x)$ and `HarmonicPolyLog[{0,-1},x]` $= -\mathrm{Li}_2(-x)$.

1.3.3 The condensed m-notation

Writing out long strings of 0's is wasteful, and Remiddi and Vermaseren introduced a condensed index that absorbs them. To each maximal block "$\underbrace{0,\dots,0}_{|m|-1}\,\sigma$" consisting of $|m|-1$ zeros followed by a sign letter $\sigma \in \{1,-1\}$ one assigns a single integer

$$m = \sigma \cdot (\text{number of zeros} + 1),$$

so the sign of m records σ and the magnitude $|m|$ records how many kernels (one $f_{\pm 1}$ preceded by $|m|-1$ copies of f_0) the block contributes. The resulting tuple is written as a subscript,

$$H_{m_1,\dots,m_k}(z) := H(\vec{a};z),$$

and the *depth* of the HPL is k, the number of nonzero letters, while the weight is $w = \sum_i |m_i|$. Thus (1.15) becomes $H_2(z) = \mathrm{Li}_2(z)$ and $H_{-2}(z) = -\mathrm{Li}_2(-z)$, the polylogarithm ladder is $H_s(z) = \mathrm{Li}_s(z)$, $H_{-s}(z) = -\mathrm{Li}_s(-z)$, and a mixed word such as $H(0,1,-1;z)$ is $H_{2,-1}(z)$. The m-notation is the input most reduction tables are organised by; the explicit word notation $H(\vec{a};z)$ is the one that makes the integral representation (1.13) transparent. (Some authors reverse the global sign or the letter ordering; we follow the convention compatible with (1.14)–(1.15) and with `HarmonicPolyLog`, where indices read outermost-integration-first.)

1.3.4 The shuffle algebra

Theorem 1.8 (Shuffle product)**.** *For a fixed argument z, the product of two harmonic polylogarithms is a $\mathbb{Q}$-linear combination of harmonic polylogarithms of the same argument:*

$$H(\vec{a};z)\,H(\vec{b};z) = \sum_{\vec{c}\in \vec{a}\,\Pi\,\vec{b}} H(\vec{c};z),$$

the sum running over all words $\vec{c}$ in the shuffle $\vec{a}\,\Pi\,\vec{b}$, i.e. all interleavings of $\vec{a}$ and $\vec{b}$ that preserve the internal order of each.

This is the iterated-integral analogue of integration by parts: a product of two iterated integrals over $[0,z]$ is the integral over the union of the two simplices, which decomposes into the $\binom{|\vec{a}|+|\vec{b}|}{|\vec{a}|}$ ordered shuffles. Concretely, at weight $1+1$,

$$H(0;z)\,H(1;z) = H(0,1;z) + H(1,0;z), \tag{1.17}$$

which is the statement $\ln z \cdot (-\ln(1-z)) = \mathrm{Li}_2(z) + \big(-\mathrm{Li}_2(z) - \ln z\,\ln(1-z)\big) = -\ln z\,\ln(1-z)$, i.e. the integration-by-parts consistency that re-expresses the product of the two weight-one HPLs through the two weight-two ones. (The dilogarithm *reflection* formula is a different relation—the

$z \mapsto 1-z$ image of $H(0,1;z)$, derived in (1.19) below.) The shuffle algebra is graded by weight, and crucially it lets one strip leading zeros: since $H(0;z)=\ln z$, repeated use of (1.8) expresses any $H(0,\vec{a};z)$ through $\ln z$ and HPLs whose words begin with a nonzero letter, which is the regularisation alluded to after the all-zero regularization.

1.3.5 Values at $z=1$: Euler sums and colored MZVs

Setting $z=1$ turns admissible HPLs (those whose word does not begin with 1, so that the boundary integral converges) into constants. Because the kernels live on the level-two alphabet $\{0,\pm1\}$, these constants are exactly the level-2, or *alternating*, multiple zeta values—the Euler sums. Words over $\{0,1\}$ give the ordinary MZVs,

$$H(\underbrace{0,\dots,0}_{s_1-1},1,\underbrace{0,\dots,0}_{s_2-1},1,\dots;1)=\zeta(s_1,s_2,\dots), \tag{1.18}$$

so for instance $H(0,1;1)=\mathrm{Li}_2(1)=\zeta(2)$ and $H(0,0,1;1)=\zeta(3)$. Admitting the letter -1 produces the alternating Euler sums: with $\bar{s}$ marking a sign,

$$H(0,-1;1)=-\mathrm{Li}_2(-1)=\tfrac{\pi^2}{12}=-\zeta(\bar{2}),\qquad H(0,0,-1;1)=-\mathrm{Li}_3(-1)=\tfrac{3}{4}\zeta(3),$$

in the alternating-MZV notation $\zeta(\bar{s}_1,\dots)=\sum_{n_1>\cdots}\prod(\mathrm{sgn})^{n_i}/n_i^{s_i}$. Thus the $\mathbb{Q}$-algebra of HPL values at 1 is the algebra of alternating MZVs; its dimensions and the structure of its relations were mapped by Borwein–Bradley–Broadhurst–Lisoněk and, on the motivic side, Deligne and Goncharov. This is the practical reason HPLs are the right basis: the boundary data of a Feynman integral land automatically in a well-understood number-theoretic ring.

1.3.6 Argument transformations and the alphabet

Changing variables transports the singular set of the differential forms. Negation and inversion, and the map $(1-z)/(1+z)$, preserve the extended three-letter singular set $\{0,1,-1,\infty\}$, with appropriate endpoint and branch constants. Reflection $1-z$ preserves the two-letter subalphabet $\{0,1\}$ but sends the letter -1 to 2. Thus $H(-1;1-z)=\log(2-z)$ cannot in general be expressed using only the original three-letter alphabet at z. Squaring likewise introduces letters $\pm i$ when applied to a word containing -1.

The useful algorithm is to differentiate, partially fraction the transported kernels, integrate in the resulting alphabet, and fix the endpoint constant. For the two-letter dilogarithm this gives

$$\mathrm{Li}_2(z)+\mathrm{Li}_2(1-z)=\frac{\pi^2}{6}-\log z\log(1-z),\qquad 0<z<1. \tag{1.19}$$

Analytic continuation requires a common choice of branches. Duplication is the special closed identity

$$H(0,1;z^2)=2\big(H(0,1;z)-H(0,-1;z)\big).$$

These examples illustrate transport of alphabets, rather than universal closure under every fractional-linear or quadratic substitution.

1.3.7 Relation to Nielsen polylogarithms and to Li

The harmonic polylogarithms contain the Nielsen generalised polylogarithms $S_{n,p}(z)$ of Nielsen (1909), studied analytically by Kölbig (1986). Indeed $S_{n,p}(z)$ is, exactly the single HPL whose word

is n zeros followed by p ones,

$$S_{n,p}(z) = H(\underbrace{0,\dots,0}_{n},\underbrace{1,\dots,1}_{p};z), \tag{1.20}$$

of weight $n+p$ and depth p; the classical case is $S_{n,1}(z) = \mathrm{Li}_{n+1}(z)$, consistent with (1.15) for $n=1, p=1$. Thus the chain of inclusions reads: classical Li_s (depth-1 HPLs over $\{0,1\}$) $\subset$ Nielsen $S_{n,p}$ (two-letter words 0^n1^p) $\subset$ harmonic polylogarithms (arbitrary words over $\{0,1,-1\}$) $\subset$ multiple polylogarithms (the $z=1$ specialisations being MZVs and Euler sums). Each enlargement adds exactly the words the previous class could not name.

1.3.8 Finite truncations and Wolfram 15

Behind every iterated integral stands a nested *sum*, and behind every value at $z=1$ stands the partial sum of that series. These partial sums are the multiple (finite) harmonic numbers

$$H_N^{(s_1,\dots,s_k)} = \sum_{N\geq n_1>n_2>\cdots>n_k\geq 1} \prod_{i=1}^{k} \frac{1}{n_i^{s_i}},$$

generalising the ordinary $H_N = H_N^{(1)} = \sum_{n\leq N} 1/n$; their $N\to\infty$ limits are the admissible MZVs, and they are the natural objects in which the series expansions of ε-dependent Feynman integrals are stored. Wolfram 15 provides all of these as first-class functions: `HarmonicPolyLog` for $H(\vec{a};z)$ (with the index/sign conventions used above; recall `HarmonicPolyLog[{1},x]` $= -\ln(1-x)$), `MultiplePolyLog` and `MultipleZeta` for the limiting transcendentals, and `MultipleHarmonicNumber` for the finite sums $H_N^{(\vec{s})}$. With `DSolve` and `AsymptoticDSolveValue` now returning harmonic polylogarithms for Fuchsian systems with rational coefficients, the entire Remiddi–Vermaseren workflow—solve the differential equation, expand at the boundary into Euler sums, continue by the transformations the transformations just discussed—is available without leaving the kernel.

Remark 1.9. The choice of alphabet $\{0,1,-1\}$ is what fixes the singular set at $\{0,1,-1,\infty\}$ and hence ties the boundary values to *alternating* MZVs. Allowing the additional kernels $f_{1/2}(t) = 1/(t-\frac{1}{2})$ or roots of unity of higher order enlarges the class to the two-dimensional harmonic polylogarithms and to Goncharov's multiple polylogarithms with general singular points (the `GeneralizedPolyLog` of the previous section), at the price of boundary constants drawn from level-N colored MZVs rather than the level-2 Euler sums. For one-scale problems the three-letter alphabet is almost always enough, which is the empirical reason HPLs, not their more general cousins, became the working basis of the field.

1.4 Nielsen's generalized polylogarithms

Long before multiple zeta values acquired their modern algebraic name, Nielsen (1909) introduced a two-parameter family of higher polylogarithms that already pervades Lewin's *Polylogarithms and Associated Functions* (1981, Chapter 7). They are the simplest objects that go beyond the ordinary $\mathrm{Li}_s(z)$ while remaining genuinely "one-variable": a single argument z, two integer indices $n,p\geq 1$, and a closed integral representation. The reader who is comfortable with the dilogarithm reflection and Landen's transformation will find here their direct higher-weight descendants, and – once we translate into the language of the previous sections – their place as one distinguished corner of the multiple-polylogarithm world.

1.4.1 Definition and elementary reductions

Definition 1.10. *For integers $n, p \geq 1$ the* Nielsen generalized polylogarithm *is*

$$S_{n,p}(z) \;=\; \frac{(-1)^{n+p-1}}{(n-1)!\,p!} \int_0^1 \frac{(\ln t)^{n-1}\,\ln^p(1-zt)}{t}\,dt, \tag{1.21}$$

analytically continued from $|z| < 1$ to $z \in \mathbb{C} \setminus [1, \infty)$ along the cut of $\ln(1-zt)$. *Its* weight *is $n+p$ and its* depth *is p.*

The normalizing factorials are chosen exactly so that two boundary specializations collapse onto familiar functions. Setting $p = 1$ and integrating $\ln(1-zt)/t$ termwise against $(\ln t)^{n-1}$ recovers the classical Euler integral for the polylogarithm:

$$S_{n,1}(z) \;=\; \mathrm{Li}_{n+1}(z), \qquad n \geq 1. \tag{1.22}$$

Thus the entire first column of the Nielsen table is nothing but the ordinary polylogarithm shifted by one weight; in particular the lowest member is the dilogarithm,

$$S_{1,1}(z) \;=\; \mathrm{Li}_2(z). \tag{1.23}$$

At the opposite corner $n = 1$ the inner factor $\ln t$ disappears and (1.21) reduces to a pure power of $\ln(1-zt)$, giving the elementary $S_{0,p}(z) = \frac{(-1)^p}{p!}\ln^p(1-z)$ in the degenerate $n = 0$ row (we keep $n \geq 1$ in the sequel, but this boundary is the seed for the recursions below).

1.4.2 Series and the multiple-polylogarithm identification

Expanding $\ln^p(1-zt)$ as a power of the series $-\sum_{m\geq 1}(zt)^m/m$ and integrating against $(\ln t)^{n-1}/t$ turns (1.21) into a single power series in z whose coefficients are nested harmonic sums:

$$S_{n,p}(z) \;=\; \sum_{k\geq 1} \frac{z^k}{k^{\,n+1}}\, c_k^{(p)}, \qquad c_k^{(p)} = \sum_{k > m_1 > \cdots > m_{p-1} \geq 1} \frac{1}{m_1 \cdots m_{p-1}}, \tag{1.24}$$

so $c_k^{(1)} \equiv 1$ (reproducing (1.22)) and in general $c_k^{(p)}$ is a depth-$(p-1)$ harmonic sum, an instance of `MultipleHarmonicNumber`. Comparing (1.24) with the strictly nested sum of the article preamble identifies $S_{n,p}$ as a multiple polylogarithm with a single repeated argument:

$$\boxed{S_{n,p}(z) \;=\; \mathrm{Li}_{n+1,\underbrace{1,\ldots,1}_{p-1}}\!\left(z, \underbrace{1,\ldots,1}_{p-1}\right)} \tag{1.25}$$

of weight $n+p$ and depth p. Its iterated-integral word over the alphabet $\{0,1\}$ (with $0 \leftrightarrow dt/t$, $1 \leftrightarrow dt/(1-t)$) is

$$0^{\,n}\,1^{\,p} \;=\; \underbrace{0\cdots 0}_{n}\,\underbrace{1\cdots 1}_{p},$$

a single block of n zeros followed by a single block of p ones. This is the defining structural fact: among all multiple polylogarithms, the Nielsen family is exactly the "one initial zero-block, one trailing one-block" subclass. Everything peculiar about $S_{n,p}$ – the two-index labelling, the clean ladders, the duality below – is a shadow of that two-block word.

Remark 1.11 (Sign and normalization). With the positive kernel $dt/(1-t)$ the word 0^n1^p equals $S_{n,p}(z)$ with no extra sign. Replacing each 1-kernel by $dt/(t-1)$ gives the factor $(-1)^p$. No factor $(-1)^{p-1}$ belongs to either convention.

1.4.3 Integral and differential relations

Differentiating (1.24) (or (1.21)) in z lowers the leading index by one and threads the function back into the ladder:

$$\frac{d}{dz}\,S_{n,p}(z) \;=\; \frac{1}{z}\,S_{n-1,p}(z) \quad (n\geq 1), \qquad \frac{d}{dz}\,S_{1,p}(z) \;=\; \frac{1}{z}\,S_{0,p}(z) \;=\; \frac{(-1)^p}{p\,!}\,\frac{\ln^p(1-z)}{z}, \tag{1.26}$$

the second relation being the $n=1$ boundary, where the dt/t block is exhausted and the degenerate row $S_{0,p}(z)=\frac{(-1)^p}{p!}\ln^p(1-z)$ is the pure power of $\ln(1-z)$. The two clean ladder moves are the integral forms of these derivatives,

$$\text{zero-block ladder:}\quad S_{n,p}(z)=\int_0^z S_{n-1,p}(u)\,\frac{du}{u} \qquad (n\geq 1), \tag{1.27}$$

$$\text{one-block ladder:}\quad S_{0,p}(z)=\int_0^z S_{0,p-1}(u)\,\frac{du}{1-u} \qquad (p\geq 1), \tag{1.28}$$

with seed $S_{0,0}(z)=1$ (the empty word): (1.27) peels the outermost 0 from the word 0^n1^p, and (1.28) peels the outermost 1 from the pure one-block 1^p; in iterated-integral terms both are simply "integrate against the next letter of 0^n1^p." The practical content is that the whole Nielsen table is generated from the empty word by first applying the 1-kernel integral $\int_0^z(\cdot)\,du/(1-u)$ a total of p times to build $S_{0,p}$ (so $S_{0,1}(z)=-\ln(1-z)$, $S_{0,2}(z)=\frac{1}{2}\ln^2(1-z)$, and so on), then the 0-kernel integral $\int_0^z(\cdot)\,du/u$ a total of n times to prepend the zeros; this is exactly the Fuchsian recursion that `HarmonicPolyLog` encodes, and it is why `DSolve` now returns Nielsen polylogs for the relevant rational-coefficient ODEs.

1.4.4 Special values

The point $z=1$ and Hoffman's duality

Setting $z=1$ in (1.25) turns the multiple polylogarithm into a multiple zeta value with the same two-block word. The first column reproduces Euler,

$$S_{n,1}(1)=\zeta(n+1), \qquad S_{1,2}(1)=\zeta(3), \tag{1.29}$$

and in general

$$S_{n,p}(1) \;=\; \zeta\big(n+1,\{1\}^{p-1}\big), \tag{1.30}$$

the MZV whose argument string is $n+1$ followed by $p-1$ ones (admissible because the leading entry $n+1\geq 2$). Now the iterated-integral word 0^n1^p of $S_{n,p}$ has a manifest symmetry: reversing it and exchanging $0\leftrightarrow 1$ sends $0^n1^p\mapsto 0^p1^n$, the word of $S_{p,n}$. This reversal-complement is exactly Hoffman's MZV *duality* (the Kontsevich integral symmetry $\omega\mapsto\overline{\omega}^{\mathrm{rev}}$), so

$$\boxed{\zeta\big(n+1,\{1\}^{p-1}\big) \;=\; \zeta\big(p+1,\{1\}^{n-1}\big), \qquad \text{equivalently} \qquad S_{n,p}(1)=S_{p,n}(1).} \tag{1.31}$$

This is a striking statement: the value of a Nielsen polylogarithm at 1 is invariant under swapping its two indices, even though $S_{n,p}(z)$ and $S_{p,n}(z)$ are wholly different functions of z. For $p=1$ it gives $\zeta(n+1)=\zeta(2,\{1\}^{n-1})$; the first nontrivial case is $S_{1,2}(1)=S_{2,1}(1)$, i.e. $\zeta(2,1)=\zeta(3)$, Euler's reflection. We confirmed (1.30) and (1.31) symbolically against `MultipleZeta` (e.g. $S_{1,2}(1)-\zeta(2,1)$ and $S_{n,p}(1)-S_{p,n}(1)$ vanish to >70 digits for all $n,p\leq 4$).

The points $z=-1$ and $z=\frac{1}{2}$ (Kölbig 1986)

At $z=-1$ the column reduction (1.22) gives $S_{n,1}(-1)=\mathrm{Li}_{n+1}(-1)=-(1-2^{-n})\,\zeta(n+1)$, and the first off-column value is the elegant

$$S_{1,2}(-1) \;=\; \tfrac{1}{8}\,\zeta(3). \tag{1.32}$$

Higher entries $S_{n,p}(-1)$ reduce (Kölbig 1986) to the alternating-MZV basis at their weight; for example $S_{2,2}(-1)$ and $S_{1,3}(-1)$ already require $\mathrm{Li}_4(\frac{1}{2})$ alongside π^4, $\ln^4 2$, $\pi^2\ln^2 2$ and $\zeta(3)\ln 2$, the unmistakable level-2 weight-4 signature (we recovered both reductions by integer-relation search). At the binary point $z=\frac{1}{2}$ the cleanest closed form is

$$\boxed{S_{1,2}\big(\tfrac{1}{2}\big) \;=\; \tfrac{1}{8}\,\zeta(3) \;-\; \tfrac{1}{6}\,\ln^3 2.} \tag{1.33}$$

We obtained (1.33) by PSLQ over $\{\zeta(3),\pi^2\ln 2,\ln^3 2\}$ – the relation $24\,S_{1,2}(\frac{1}{2})-3\,\zeta(3)+4\ln^3 2=0$ – and verified it to 40 digits. (Compare Lewin's dilogarithm value $\mathrm{Li}_2(\frac{1}{2})=\frac{\pi^2}{12}-\frac{1}{2}\ln^2 2$: the same flavour, one weight up.)

1.4.5 Functional equations

The reader who knows Lewin's dilogarithm chapter expects three staples – reflection $z\leftrightarrow 1-z$, inversion $z\leftrightarrow 1/z$, and the Landen ladder – to survive at higher weight. They do, as specializations of the harmonic-polylogarithm argument transformations of the previous section, but they grow extra lower-weight "tail" terms.

Reflection $z\mapsto 1-z$

At weight 2 the only reflection is the dilogarithm law itself, here in Nielsen dress ($n=p=1$):

$$S_{1,1}(z)+S_{1,1}(1-z) \;=\; \frac{\pi^2}{6}-\ln z\,\ln(1-z), \tag{1.34}$$

which is exactly Euler's reflection for Li_2. At weight 3 the genuinely new instance is the $S_{1,2}$ reflection; carrying out the integer-relation search over the natural weight-3 basis yields the clean

$$\boxed{S_{1,2}(z) \;=\; \zeta(3) \;-\; \mathrm{Li}_3(1-z) \;+\; \ln(1-z)\,\mathrm{Li}_2(1-z) \;+\; \tfrac{1}{2}\,\ln z\,\ln^2(1-z).} \tag{1.35}$$

We verified (1.35) to 40 digits at several rational and one negative argument. Two consistency checks are worth noting. Putting $z=1$ collapses the right side to $\zeta(3)-\mathrm{Li}_3(0)+0+0=\zeta(3)$, reproducing (1.29); and adding (1.35) to its $z\mapsto 1-z$ image recovers the symmetric Euler relation $S_{1,2}(z)+S_{1,2}(1-z)=2\zeta(3)-\mathrm{Li}_3(z)-\mathrm{Li}_3(1-z)+\cdots$ with the logarithmic tails arranging themselves into the $\zeta(3)+\mathrm{Li}_3$ shuffle. Identity (1.35) is precisely the statement that the iterated integral of 011 from 0 to z and from 0 to $1-z$ differ by an explicit lower-weight boundary correction – the weight-3 avatar of the dilogarithm reflection.

A finite inversion formula

The earlier draft inferred failure of closure from a search in the basket $\{S_{1,2}(z),\zeta(3),\pi^2\log z,\log^3 z\}$. That basket omits polylogarithms and mixed log products needed by the actual transformation. Inversion does close in the Nielsen family after those products and endpoint constants are included. The following direct coefficient construction makes the distinction precise.

Proposition 1.12 (Nielsen inversion with explicit endpoint data)**.** *For integers $n, p \geq 1$, let $u = \log(-z)$ on $\mathbb{C} \setminus [0, \infty)$ and define*

$$\mathcal{N}_{n,p}(z) = \frac{u^{n+p}}{(n+p)!} + \sum_{r=1}^{p} \sum_{\ell=0}^{p-r} (-1)^{\ell} \binom{n+\ell-1}{\ell} \frac{u^{p-r-\ell}}{(p-r-\ell)!} S_{n+\ell,r}(z).$$

Then

$$S_{n,p}(1/z) = (-1)^n \mathcal{N}_{n,p}(z) + C_{n,p}(u),$$

where $C_{n,p}$ is a polynomial of degree at most $n-1$. Its coefficients are determined entirely by Nielsen values at -1: for $0 \leq j < n$,

$$C_{n,p}^{(j)}(0) = (-1)^j S_{n-j,p}(-1) - (-1)^n \sum_{r=1}^{p} (-1)^{p-r} \binom{n-j+p-r-1}{p-r} S_{n-j+p-r,r}(-1).$$

All logarithms and polylogarithms use the common principal continuation.

Proof. First take $z < 0$. Extend the notation by $S_{0,r}(z) = (-1)^r \log^r(1-z)/r!$ and $S_{0,0} = 1$. Since $\log(1 - 1/z) = \log(1-z) - u$, the binomial theorem gives

$$S_{0,p}(1/z) = \sum_{r=0}^{p} \frac{u^{p-r}}{(p-r)!} S_{0,r}(z).$$

The ladder $dS_{m,r}/du = S_{m-1,r}$ implies, by Pascal's identity, that $d\mathcal{N}_{n,p}/du = \mathcal{N}_{n-1,p}$, with $\mathcal{N}_{0,p} = S_{0,p}(1/z)$. On the other hand $dS_{n,p}(1/z)/du = -S_{n-1,p}(1/z)$. Consequently the difference has vanishing nth derivative in u and is a polynomial of degree at most $n-1$. Evaluate its first n derivatives at $u = 0$, namely $z = -1$, to obtain the displayed endpoint coefficients. Both sides are holomorphic on the stated slit plane, so the identity theorem supplies the continuation. □

At weight three this gives the particularly useful exact identity

$$S_{1,2}(1/z) = -S_{1,2}(z) + \mathrm{Li}_3(z) - \log(-z)\,\mathrm{Li}_2(z) - \frac{1}{6}\log^3(-z) + \zeta(3).$$

Closure of a function representation does not establish independence of its values. Other argument changes should likewise be derived by transporting the kernels in the appropriate alphabet, rather than by an unsuccessful search in an underspecified basket.

1.4.6 Placement inside the multiple-polylogarithm world

Equation (1.25) identifies $S_{n,p}$ as the "one trailing block" corner of the depth-p multiple polylogarithms: the words $0^n 1^p$ form a single two-parameter slice of the full $\{0,1\}^*$ alphabet, the slice closest to the ordinary polylogarithm (recovered at $p = 1$) and stable under the two operations – evaluation at 1 to MZVs, and the duality swap $n \leftrightarrow p$ – that organize the modern theory. This is exactly why the Nielsen polylogarithms are the natural bridge between Lewin's classical apparatus and the MZV algebra: every Lewin-style identity (reflection, Landen, the special values at $\pm 1, \frac{1}{2}$) is the $p \leq 2$ or $n \leq 2$ shadow of a multiple-polylogarithm identity, and conversely the duality $S_{n,p}(1) = S_{p,n}(1)$ is the first place a classical reader meets a genuinely MZV-theoretic symmetry with no single-variable analogue. In the Wolfram Language the family is a first-class function: `PolyLog[n,p,z]` is exactly $S_{n,p}(z)$, with `PolyLog[n,1,z]` agreeing with `PolyLog[n+1,z]` per (1.22), and it sits one specialization step from `MultiplePolyLog` via (1.25) and one from `MultipleZeta` via (1.30).

1.5 The common algebra and the motivic boundary

The nested series and outer-letter-first integral describe one function. Choosing the alphabet distinguishes general multiple polylogarithms, cyclotomic multiple polylogarithms, HPLs and the Nielsen subfamily. General Goncharov letters can be arbitrary complex numbers; restricting them to roots of unity gives only the cyclotomic class.

One must distinguish the argument patterns of the two products. Shuffle at a common endpoint gives $\mathrm{Li}_1(z)^2 = 2\,\mathrm{Li}_{1,1}(z,1)$, whereas stuffle gives $\mathrm{Li}_1(z)^2 = 2\,\mathrm{Li}_{1,1}(z,z)+\mathrm{Li}_2(z^2)$. Both follow directly from their respective representations and therefore give a nontrivial identity when compared.

Iterated integrals also carry a coproduct organized by subsequences of marked points. Its reduced components and the associated symbol organize functional equations by weight. They lose some period constants: vanishing symbol or reduced coproduct alone is not a proof of a zero function. Endpoint constants and branches must be fixed separately. For the dilogarithm the differential $d\,\mathrm{Li}_2(z) = -\log(1-z)d\log z$ makes this elementary; differentiating Euler reflection and evaluating an endpoint proves the identity exactly.

Motivic MZVs [43] provide an algebraic setting in which dimension and spanning theorems can be proved. The numerical period map transports identities and upper bounds. Its injectivity, and thus the corresponding lower bounds for numerical periods, is a separate conjectural issue. The same caution will apply to the Gaussian and Eisenstein depth experiments.

1.6 Resonant order jets and exact primitives

The iterated-integral description organizes weight and depth. A different structure appears when the order varies near an integer while the argument approaches one: a Gamma pole cancels a zeta pole, leaving logarithms. Keeping this cancellation intact produces exact local identities and a finite calculus of order derivatives. The resonance continuations [142] supply a uniform sector theorem and a reciprocal-Gamma normalization; the order-lowering and repeated-primitive formulas below develop that normalization further.

1.6.1 Pole pairing on a fixed logarithmic sheet

For $\Re a > 0$ set

$$F(s,t,a) = e^{-at}\Phi(e^{-t},s,a) = \sum_{n=0}^{\infty} \frac{e^{-(n+a)t}}{(n+a)^s} \qquad (\Re t > 0).$$

Powers of $n+a$ use the logarithm in the right half-plane. On $0 < |t| < 2\pi$, $|\arg t| < \pi$, continue from positive t using $t^{s-1} = \exp((s-1)\operatorname{Log} t)$. The classical Lerch expansion [115] is

$$F(s,t,a) = \Gamma(1-s)t^{s-1} + \sum_{j=0}^{\infty} \zeta(s-j,a)\frac{(-t)^j}{j!}. \tag{1.36}$$

At integer s, the colliding terms must be combined before specializing. In particular, $F(s,t,1) = \mathrm{Li}_s(e^{-t})$.

Write $H_k^{(j)} = \sum_{n=1}^{k} n^{-j}$, $H_k = H_k^{(1)}$, and

$$T = -\operatorname{Log} t + H_k - \gamma,$$

$$E_k(w) = \exp\left(\sum_{j\geq 2} \frac{\zeta(j) + (-1)^j H_k^{(j)}}{j} w^j\right) = \sum_{j\geq 0} b_{k,j} w^j, \qquad b_{k,0} = 1, \quad b_{k,1} = 0. \tag{1.37}$$

The Taylor-subtracted remainder is

$$\mathcal{R}_k(\epsilon,t,a)=\frac{k!}{(-t)^k}\left(F(k+1+\epsilon,t,a)-\sum_{j=0}^{k-1}\zeta(k+1+\epsilon-j,a)\frac{(-t)^j}{j!}\right).$$

Empty sums apply at $k=0$. Gamma recurrence in (1.36) gives

$$\begin{aligned}\mathcal{R}_k(\epsilon,t,a)&=\frac{1-e^{-T\epsilon}E_k(\epsilon)}{\epsilon}+G_a(\epsilon)+B_k(\epsilon,t,a),\\ G_a(\epsilon)&=\zeta(1+\epsilon,a)-\epsilon^{-1}=\sum_{r\ge0}\frac{(-1)^r\gamma_r(a)}{r!}\epsilon^r,\\ B_k(\epsilon,t,a)&=\sum_{\ell\ge1}\frac{k!(-t)^\ell}{(k+\ell)!}\zeta(1+\epsilon-\ell,a).\end{aligned}\tag{1.38}$$

All apparent singularities at $\epsilon=0$ are removable.

Theorem 1.13 (Uniform resonance on closed complex sectors)**.** *Fix $0<t_0<2\pi$, $0<\delta<\pi$, compact sets $A\subset\{a:\Re a>0\}$ and $U\subset\mathbb{C}$, and nonnegative integers M,J,D. For $a\in A$, $u\in U$, $0<|t|\le t_0$, $|\arg t|\le\pi-\delta$, as $\Re T\to+\infty$, uniformly in every integer $k\ge0$,*

$$\begin{aligned}\mathcal{R}_k(u/T,t,a)&=T\frac{1-e^{-u}}{u}+\gamma_0(a)\\ &\quad+\sum_{r=1}^{M}\frac{u^r}{T^r}\left(\frac{(-1)^r\gamma_r(a)}{r!}-e^{-u}b_{k,r+1}\right)+O(|T|^{-M-1}).\end{aligned}\tag{1.39}$$

The quotient at $u=0$ is one. The same remainder order holds after at most J derivatives in u and D derivatives in a. The separated analytic tail is $B_k(u/T,t,a)=O(|t|/(k+1))=O(e^{-\Re T})$, also with these fixed derivatives.

Proof. Choose $0<\eta<1/2$ and $t_0<R<2\pi$. The inequality $|\zeta(j)+(-1)^jH_k^{(j)}|\le2\zeta(j)$ uniformly bounds E_k on $|w|\le\eta$, independently of k. The regular part G_a is bounded on an enlarged compact set of shifts and a small order disk.

For completeness, a common majorant for the tail follows from Hermite's Hurwitz formula [93]:

$$\zeta(s,a)=\frac{a^{-s}}{2}+\frac{a^{1-s}}{s-1}+i\int_0^\infty\frac{(a+iy)^{-s}-(a-iy)^{-s}}{e^{2\pi y}-1}\,dy.$$

For $s=1+\epsilon-\ell$ and $|\epsilon|\le\eta$, the elementary terms have factorially summable bounds on the enlarged shift compact. For $y\ge1$, summing bounds with weights $R^\ell/\ell!$ leaves a fixed polynomial times $e^{-(2\pi-R)y}$. For $0\le y\le1$, take the difference of powers before absolute values, or integrate its derivative along the segment from $a-iy$ to $a+iy$. The resulting factor y cancels the denominator's simple zero. Hence

$$\sup_{a,|\epsilon|\le\eta}\sum_{\ell\ge1}\frac{t_0^{\ell-1}}{\ell!}|\zeta(1+\epsilon-\ell,a)|<\infty.$$

Since $k!/(k+\ell)!\le1/((k+1)\ell!)$, this proves the tail bound. Moreover $H_k-\gamma\le\log(k+1)$, so $|t|/(k+1)\le e^{-\Re T}$.

Substitute $\epsilon=u/T$ in (1.38). The first term is $T(1-e^{-u})/u-e^{-u}\sum_{j\ge2}b_{k,j}(u/T)^{j-1}$. Uniform Taylor remainders for E_k and G_a give (1.39). On the fixed sector $\Im T$ is bounded, so $|T|\sim\Re T$ and the analytic tail is smaller than every fixed inverse power. Prove the bounds on enlarged compact u,a sets and apply Cauchy's formula to obtain the stated fixed derivatives. □

This proves the fixed-sector extension requested as R1 in the preserved resonance report. It retains the finite-k coefficients $b_{k,j}$. It does not replace them by their large-k limits. A logarithmic-cover version follows under $\Re T \geq c|T| \to \infty$ for fixed $c > 0$; unrestricted winding, growing jet order and general growing transition windows require additional analysis.

1.6.2 A finite change to pure logarithmic jets

For integers $m \geq 0$ and $|\epsilon| < \rho < 1$ define

$$D_m(\epsilon) = \frac{\prod_{j=1}^{m}(1+\epsilon/j)}{\Gamma(1-\epsilon)}, \qquad C_{m,a}(\epsilon) = D_m(\epsilon)\zeta(1+\epsilon,a) - \epsilon^{-1}. \tag{1.40}$$

The second expression is holomorphic at zero. Put

$$\mathcal{F}_{m,a}(\epsilon,t) = (-1)^m m! D_m(\epsilon) \left(F(m+1+\epsilon,t,a) - \sum_{j=0}^{m-1} \zeta(m+1+\epsilon-j,a)\frac{(-t)^j}{j!} \right) - t^m C_{m,a}(\epsilon). \tag{1.41}$$

Both the lower Taylor subtraction and the counterterm are part of this normalization.

Theorem 1.14 (Exact pure-logarithm normalization)**.** *On the chosen slit disk,*

$$\mathcal{F}_{m,a}(\epsilon,t) = t^m \left(\frac{1-t^\epsilon}{\epsilon} + \mathcal{B}_{m,a}(\epsilon,t) \right), \tag{1.42}$$

$$\mathcal{B}_{m,a}(\epsilon,t) = \sum_{n\geq 1} \frac{(-1)^n m! D_m(\epsilon)}{(m+n)!} \zeta(1+\epsilon-n,a) t^n. \tag{1.43}$$

The tail is jointly holomorphic for $|\epsilon| < 1$, $|t| < 2\pi$, $\Re a > 0$, and vanishes at $t = 0$. For $r \geq 0$ let

$$J_{m,r,a}(t) = (-1)^r (r+1)\, \partial_\epsilon^r \mathcal{F}_{m,a}(\epsilon,t)\big|_{\epsilon=0}.$$

Then, with $L = -\operatorname{Log} t$,

$$J_{m,r,a}(t) = t^m \{ L^{r+1} + B_{m,r,a}(t) \}, \qquad B_{m,r,a}(0) = 0, \tag{1.44}$$

where $B_{m,r,a}$ is holomorphic on $|t| < 2\pi$. If $D_m(\epsilon) = \sum_{j\geq 0} d_{m,j}\epsilon^j$, its tail coefficients are

$$[t^n] B_{m,r,a}(t) = \frac{(-1)^{n+r}(r+1) r! m!}{(m+n)!} \sum_{j=0}^{r} \frac{d_{m,j}\zeta^{(r-j)}(1-n,a)}{(r-j)!}. \tag{1.45}$$

Proof. Gamma recurrence gives $(-1)^m m! D_m(\epsilon)\Gamma(-m-\epsilon) = -1/\epsilon$. Thus the singular Gamma term becomes $-t^{m+\epsilon}/\epsilon$, and the zeta term minus the counterterm is t^m/ϵ. The remaining terms are exactly (1.43). The preceding majorant, on arbitrary smaller compact disks, proves normal convergence and parameter differentiation. Finally, $(1-t^\epsilon)/\epsilon = \sum_{r\geq 0}(-1)^r L^{r+1}\epsilon^r/(r+1)!$. Taking the specified derivative and applying Leibniz's rule proves (1.44)–(1.45). □

This is a finite triangular change of ordinary order derivatives, not an asymptotic replacement of them. Indeed

$$\log D_m(\epsilon) = (H_m - \gamma)\epsilon + \sum_{j\geq 2} \frac{(-1)^{j+1} H_m^{(j)} - \zeta(j)}{j} \epsilon^j.$$

Only its first r coefficients and finitely many Stieltjes coefficients are needed for the rth jet. The coefficient of the highest raw order derivative is nonzero, so the finite transformation is invertible.

Corollary 1.15 (Exact monodromy)**.** *Let* M *be one counterclockwise continuation around zero. For* $d = r + 1$*,*

$$(\mathsf{M} - I)^{d+1} J_{m,r,a} = 0, \qquad (\mathsf{M} - I)^{d} J_{m,r,a} = (-2\pi i)^d d!\, t^m \neq 0.$$

Proof. The analytic tail is single-valued and L changes to $L - 2\pi i$. Apply the finite-difference formula to the monic polynomial L^d. □

The last nonzero term is holomorphic. Consequently the nilpotence index changes if one passes to germs modulo holomorphic functions. The result describes logarithmic monodromy of a convergent local expression.

1.6.3 An order-lowering ladder and all repeated primitives

Theorem 1.16 (Exact lowering and one anchored primitive)**.** *For* $m \geq 1$*,*

$$\partial_t \mathcal{F}_{m,a}(\epsilon, t) = (m + \epsilon)\mathcal{F}_{m-1,a}(\epsilon, t) - t^{m-1}. \tag{1.46}$$

With $J_{m,-1,a} = 0$*, this gives*

$$\partial_t J_{m,r,a} = m J_{m-1,r,a} - (r + 1) J_{m-1,r-1,a} - \mathbf{1}_{\{r=0\}} t^{m-1}. \tag{1.47}$$

For $m, r \geq 0$ *the anchored primitive is*

$$\int_0^t J_{m,r,a}(u)\, du = \sum_{j=0}^{r} \frac{(r+1)!}{(r-j+1)!(m+1)^{j+1}} J_{m+1,r-j,a}(t) + \frac{(r+1)!\, t^{m+1}}{(m+1)^{r+2}}. \tag{1.48}$$

Proof. Since $D_m = (1 + \epsilon/m)D_{m-1}$, the tail coefficients give $(m + t\partial_t)\mathcal{B}_m = (m + \epsilon)\mathcal{B}_{m-1}$. For $K = (1-t^\epsilon)/\epsilon$, $(m+t\partial_t)K = (m+\epsilon)K - 1$. Multiplying by t^{m-1} proves (1.46). Order differentiation gives (1.47). Differentiate the finite sum in (1.48); adjacent jets cancel, and the final polynomial cancels the remaining endpoint term. All terms vanish at zero, since $t^{m+1} \operatorname{Log}^q t \to 0$. This fixes the constant, including for the integrable $m = 0$ jets. □

The same mechanism gives every iterated primitive in one finite formula. For $q \geq 1$ define

$$I^q f(t) = \frac{1}{(q-1)!} \int_0^t (t-u)^{q-1} f(u)\, du, \qquad A_{m,q} = (m+1)_q, \qquad x_i = \frac{1}{m+i} \quad (1 \leq i \leq q).$$

Integrate along a fixed ray in the slit disk. Let $h_j = h_j(x_1, \ldots, x_q)$ be the complete homogeneous polynomial, so $\prod_{i=1}^{q}(1 - x_i z)^{-1} = \sum_{j \geq 0} h_j z^j$ and $h_0 = 1$.

Theorem 1.17 (Finite repeated-primitive identity)**.** *For all integers* $m, r \geq 0$ *and* $q \geq 1$*,*

$$\boxed{I^q J_{m,r,a}(t) = \frac{1}{A_{m,q}} \left\{ \sum_{j=0}^{r} \frac{(r+1)!}{(r-j+1)!} h_j J_{m+q,r-j,a}(t) + (r+1)! h_{r+1} t^{m+q} \right\}.} \tag{1.49}$$

Proof. Set $P_{m,q}(\epsilon) = (m + 1 + \epsilon)_q$. Termwise beta integration of the normally convergent tail and integration of the two core powers in (1.42) give

$$I^q \mathcal{F}_{m,a} = \frac{\mathcal{F}_{m+q,a}}{P_{m,q}(\epsilon)} + \frac{t^{m+q}}{\epsilon} \left(\frac{1}{A_{m,q}} - \frac{1}{P_{m,q}(\epsilon)} \right). \tag{1.50}$$

For the tail, this follows explicitly from $D_{m+q} = P_{m,q} D_m / A_{m,q}$ and $I^q t^{m+n} = (m+n)! t^{m+n+q}/(m+n+q)!$. The fractional power uses $I^q t^{m+\epsilon} = t^{m+q+\epsilon}/P_{m,q}(\epsilon)$; $|\epsilon| < 1$ ensures endpoint integrability. Finally,

$$\frac{1}{P_{m,q}(\epsilon)} = \frac{1}{A_{m,q}} \sum_{j \geq 0} (-1)^j h_j \epsilon^j.$$

The correction in (1.50) is removable at zero. Take $(-1)^r (r+1) \partial_\epsilon^r$ at zero and use Leibniz's rule. This gives (1.49) with its stated polynomial. □

For $q = 1$, $h_j = (m+1)^{-j}$ and the formula reduces to (1.48). For general q the first two coefficients are $h_1 = \sum_{i=1}^{q} (m+i)^{-1}$ and $h_2 = (h_1^2 + \sum_{i=1}^{q} (m+i)^{-2})/2$. Thus finite harmonic intervals control the entire integration hierarchy.

1.6.4 Explicit endpoints and experimental checks

At $a = 1$ the first zeroth-order jets are

$$\begin{aligned} J_{0,0,1}(t) &= -\log(1 - e^{-t}), \\ J_{1,0,1}(t) &= \zeta(2) - \mathrm{Li}_2(e^{-t}) - t, \\ J_{2,0,1}(t) &= 2\{\mathrm{Li}_3(e^{-t}) - \zeta(3) + t\zeta(2)\} - \tfrac{3}{2} t^2. \end{aligned}$$

The repeated-primitive formula recovers the exact identity

$$\int_0^t (t-u)\{-\log(1-e^{-u})\}\, du = \mathrm{Li}_3(e^{-t}) - \zeta(3) + t\zeta(2),$$

with both endpoint constants determined, rather than inferred numerically. The first nontrivial normalized order derivative is

$$\begin{aligned} J_{1,1,1}(t) &= 2\left\{ \partial_s \mathrm{Li}_s(e^{-t})\big|_{s=2} - \zeta'(2) + (1-\gamma)(\mathrm{Li}_2(e^{-t}) - \zeta(2)) \right\} \\ &\quad - t\{\gamma^2 + \zeta(2) + 2\gamma_1\} = t\left\{ L^2 + \frac{\gamma - 1 - \log(2\pi)}{2} t + O(t^2) \right\}. \end{aligned} \tag{1.51}$$

The lower logarithmic powers cancel exactly.

The finite integration pattern was exposed by comparing the normalized coefficient arrays at successive integer orders. The ratio $D_m/D_{m-1} = 1 + \epsilon/m$ then gave the all-order proof. Independent exact basis checks and direct complex defining-series comparisons are recorded separately from this analytic argument. They test signs, coefficients and endpoint terms; no interval-certified digits or Lean formalization are claimed. The inverse-fold, sharp-radius and Borel-reconstruction branches of the incoming reports retain separate proof audits. Their shared use of order jets does not make those further conclusions consequences of this section.

1.7 Quadratic feedback: finite coefficients and boundary inversion

Order jets describe how a polylogarithm changes with its order. Nonlinear feedback asks a complementary question: what remains explicit when the unknown also occurs in the argument? Consider the germ

$$uv(r) - \frac{v(r)^2}{2} = \mathrm{Li}_s(re^{v(r)}), \qquad v(0) = 0, \qquad u > 0, \qquad a_n(s,u) = [r^n] v(r). \tag{1.52}$$

The feedback reports [143] establish its interior inverse geometry and complete the boundary regimes. We first develop an exact finite coefficient formula, exposing a classical Bessel-polynomial structure. We then reconcile the global radius theorem. This order keeps finite identities, local expansions and global continuation distinct.

1.7.1 Two Lagrange inversions and a Bessel-polynomial formula

Let

$$C_{n,k}(s) = \sum_{\substack{m_1+\cdots+m_k=n \\ m_i\geq 1}} (m_1\cdots m_k)^{-s} = [t^n]\,\mathrm{Li}_s(t)^k.$$

The sum is finite, so it is entire in s. Define the classical Bessel polynomial [144]

$$y_d(z) = \sum_{j=0}^{d} \frac{(d+j)!}{(d-j)!\,j!}\left(\frac{z}{2}\right)^j, \qquad y_0 = 1, \quad y_1 = 1+z.$$

The positivity of its coefficients will be useful, but its role here is an exact change of coordinates, not an asymptotic approximation.

Theorem 1.18 (Finite Bessel coefficient identity)**.** *For $n \geq 1$, $s \in \mathbb{C}$ and $u > 0$,*

$$a_n(s,u) = \sum_{k=1}^{n} Q_{n,k}(u) C_{n,k}(s), \tag{1.53}$$

$$\begin{aligned} Q_{n,k}(u) &= \frac{n^{k-1}}{k!\,u^k} y_{k-1}\left(\frac{1}{nu}\right) \\ &= \sum_{j=0}^{k-1} \frac{(k-1+j)!\,n^{k-1-j}}{k!\,j!\,(k-1-j)!\,2^j u^{k+j}}. \end{aligned} \tag{1.54}$$

More generally, replace $\mathrm{Li}_s(t)$ *by any holomorphic germ* $A(t)$ *with* $A(0) = 0$ *and replace* $C_{n,k}$ *by* $[t^n]A(t)^k$*. The same formula holds for every complex* $u \neq 0$*.*

Proof. The implicit derivative of the defining equation in v at the origin is $u \neq 0$, so the germ exists and is unique. Set $f(t) = u - \sqrt{u^2 - 2A(t)}$, choosing the square root with value u. Then $v = f(t)$ and $t = re^{f(t)}$. Lagrange inversion gives

$$[r^n]v = \frac{1}{n}[t^{n-1}]f'(t)e^{nf(t)} = \frac{1}{n}[t^n]e^{nf(t)}.$$

For the second inversion write $g(z) = u - \sqrt{u^2 - 2z}$. Its inverse is $z = g(u - g/2)$, whence

$$[z^k]e^{ng(z)} = \frac{n}{k}[w^{k-1}]e^{nw}(u - w/2)^{-k}.$$

Expand both factors on the right and divide by the first factor n. The result is (1.54). Composing $z = A(t)$ requires only $1 \leq k \leq n$, because $A(0) = 0$. This proves both assertions. □

For fixed n, the weights have a short exact recurrence:

$$k(k+1)u^2 Q_{n,k+1} = k(2k-1)Q_{n,k} + n^2 Q_{n,k-1} \qquad (k \geq 2), \tag{1.55}$$

with $Q_{n,1} = u^{-1}$ and $Q_{n,2} = n/(2u^2) + 1/(2u^3)$. It follows either by comparing (1.54), or from $y_k = (2k-1)zy_{k-1} + y_{k-2}$. For example,

$$\begin{aligned} a_1 &= \frac{1}{u}, \\ a_2 &= \frac{2^{-s}}{u} + \frac{1}{u^2} + \frac{1}{2u^3}, \\ a_3 &= \frac{3^{-s}}{u} + 2^{-s}\left(\frac{3}{u^2} + \frac{1}{u^3}\right) + \frac{3}{2u^3} + \frac{3}{2u^4} + \frac{1}{2u^5}. \end{aligned}$$

For real s, a_n is a polynomial in u^{-1} with strictly positive coefficients at every degree from 1 through $2n-1$. The highest coefficient is $\mathrm{Cat}_{n-1}/2^{n-1}$, where $\mathrm{Cat}_d = (2d)!/(d!(d+1)!)$; for $n \geq 2$ the next is $n\,\mathrm{Cat}_{n-1}/2^{n-1}$. Only the composition into n ones contributes to these last two powers, so they are independent of s.

1.7.2 Finite order spectra and exact recurrences

Let $\mathcal{M}_n$ be the set of products of all ordered compositions of n, with arbitrary positive length. If $N_{n,k}(M)$ counts the compositions of length k and product M, put

$$W_{n,M}(u) = \sum_{k=1}^{n} Q_{n,k}(u) N_{n,k}(M).$$

Corollary 1.19 (A finite Dirichlet polynomial in the order)**.** *For $u > 0$,*

$$a_n(s,u) = \sum_{M \in \mathcal{M}_n} W_{n,M}(u) M^{-s}, \qquad W_{n,M}(u) > 0. \tag{1.56}$$

Consequently

$$\prod_{M \in \mathcal{M}_n} (\partial_s + \log M)\, a_n(s,u) = 0. \tag{1.57}$$

This is the monic constant-coefficient annihilator of least degree. For a real $h \neq 0$, define $P_{n,h}(X) = \prod_{M \in \mathcal{M}_n}(X - M^{-h}) = \sum_{j=0}^{d} p_j X^j$. Then, identically in s,

$$\sum_{j=0}^{d} p_j a_n(s + jh, u) = 0. \tag{1.58}$$

In particular $W_{n,1} = Q_{n,n}$ and $\lim_{s \to +\infty} a_n(s,u) = Q_{n,n}(u)$.

Proof. Group the finite sum (1.53) by the composition product. Every product in the support occurs with a positive weight. Applying a constant-coefficient differential operator with polynomial P gives $\sum_M W_{n,M} P(-\log M) M^{-s}$. Distinct real exponentials are linearly independent: differentiating at one point gives a Vandermonde matrix in their distinct exponents. Hence any annihilator must vanish at every $-\log M$, proving minimality. The shift formula follows by the same calculation with the distinct eigenvalues M^{-h}. The product is one only for the composition entirely of ones, proving the final statements. □

This finite spectrum concerns a fixed Taylor coefficient, not the complete function as n varies. All order derivatives are explicit: $\partial_s^j a_n = \sum_M W_{n,M}(-\log M)^j M^{-s}$. For instance $\mathcal{M}_3 = \{1, 2, 3\}$, so the unit-order samples satisfy

$$a_3(s+3,u) - \frac{11}{6} a_3(s+2,u) + a_3(s+1,u) - \frac{1}{6} a_3(s,u) = 0.$$

This exact relation survives every choice of positive feedback u.

1.7.3 The positive branch and its exact radius

The finite formula proves $a_n > 0$ for real s. Now suppose $s > 0$ and write, for $s > 2$,

$$A = \zeta(s-1), \qquad U_c(s) = \sqrt{2\zeta(s) + A^2}.$$

Theorem 1.20 (Complete quadratic-feedback classification)**.** *The radius R of (1.52) is the unique singular point on its circle of convergence, and the Taylor series converges absolutely there. Its value and coefficient equivalents are as follows.*

If $0 < s \le 2$, or if $s > 2$ and $u < U_c(s)$, let $t_* \in (0,1)$ be the unique solution of

$$u^2 = 2\,\mathrm{Li}_s(t_*) + \mathrm{Li}_{s-1}(t_*)^2, \qquad w_* = \mathrm{Li}_{s-1}(t_*).$$

Then

$$R = t_* e^{w_* - u}, \qquad a_n \sim \sqrt{\frac{w_*}{2\pi(1 + \mathrm{Li}_{s-2}(t_*))}}\, R^{-n} n^{-3/2}. \tag{1.59}$$

For $s > 2$ and $u \ge U_c(s)$ set $b = \sqrt{u^2 - 2\zeta(s)}$, $v_c = u - b$, $R_b = e^{-v_c}$. Here $R = R_b$. If $b > A$, then

$$a_n \sim \frac{b^{s-1}}{(b-A)^s} R_b^{-n} n^{-s}. \tag{1.60}$$

If $b = A$, the three critical laws are

$$a_n \sim R_b^{-n} \begin{cases} \dfrac{(A/\Gamma(1-s))^{1/(s-1)}}{-\Gamma(-1/(s-1))} n^{-1-1/(s-1)}, & 2 < s < 3, \\ \sqrt{A/\pi}\, n^{-3/2} (\log n)^{-1/2}, & s = 3, \\ \sqrt{\dfrac{A}{2\pi(1+\zeta(s-2))}}\, n^{-3/2}, & s > 3. \end{cases} \tag{1.61}$$

Proof. The parametrization from the first Lagrange inversion has

$$\frac{d}{dt} \log r(t) = \frac{1}{t} \left(1 - \frac{\mathrm{Li}_{s-1}(t)}{\sqrt{u^2 - 2\,\mathrm{Li}_s(t)}} \right).$$

The characteristic right side $2\,\mathrm{Li}_s(t) + \mathrm{Li}_{s-1}(t)^2$ increases strictly from zero, with endpoint infinity for $s \le 2$ and $U_c(s)^2$ otherwise. Any characteristic point precedes a zero of the radicand. Thus the positive branch increases to the stated fold or endpoint.

For a global convergence argument, iterate $\Phi(r, v) = v^2/(2u) + u^{-1} \sum_{m \ge 1} r^m e^{mv}/m^s$ from zero. The coefficients are nonnegative and stabilize degree by degree; for each positive r below the proposed radius, the real iterates are bounded by the parametrized solution. The Taylor series therefore converges there and agrees with that solution by local uniqueness. Monotone convergence at the endpoint shows $\sum_n a_n R^n = v_c$, or $u - w_*$ in the interior case.

At $|r| = R$, $r \ne R$, strict positivity in every degree implies $|v(r)| < v(R)$. Hence $|re^{v(r)}| < 1$, or $< t_*$ in the interior case. Moreover

$$|\Phi_v(r, v(r))| \le \frac{|v(r)| + \mathrm{Li}_{s-1}(|re^{v(r)}|)}{u} < 1.$$

At the positive endpoint the upper comparison is $(v_c + A)/u \le 1$, or $(u - w_* + w_*)/u = 1$. The implicit function theorem extends across every other point of the circle. It remains to prove that the positive endpoint is singular.

For the boundary cases put $x = -\operatorname{Log} t$, $d = -\operatorname{Log}(r/R_b)$ and

$$D_s(x) = \zeta(s) - \operatorname{Li}_s(e^{-x}), \qquad K_s(x) = Ax + \frac{x^2}{2} - D_s(x).$$

The selected square root gives the exact identities

$$(b+x)d - \frac{d^2}{2} = (b-A)x + K_s(x), \qquad v = v_c + d - x, \tag{1.62}$$
$$K_s(x) = \int_0^x (x-t)\{1 + \operatorname{Li}_{s-2}(e^{-t})\}\, dt.$$

The integral converges at zero for $s > 2$. The paired expansion (1.36) at $a = 1$ gives, for noninteger s,

$$K_s(x) = \Gamma(1-s)x^{s-1} + \frac{1+\zeta(s-2)}{2}x^2 + O(x^3).$$

At $s = 3$ it gives instead

$$K_3(x) = \frac{x^2}{2}\left(\log\frac{1}{x} + \frac{5}{2}\right) + \frac{x^3}{12} - \frac{x^4}{288} + O(x^6). \tag{1.63}$$

The $5/2$ includes the quadratic feedback contribution.

If $b > A$, put $\kappa = 1 - A/b > 0$. The first nonanalytic term of v after inversion is $\Gamma(1-s)d^{s-1}/(b\kappa^s)$. At integer $s = m \geq 3$, pairing first gives the term $(-1)^m d^{m-1}\log d/(b(m-1)!\kappa^m)$. Power or integer-logarithm transfer cancels the Gamma factor or the signed factorial and proves (1.60).

At $b = A$, the leading terms of d are $\Gamma(1-s)x^{s-1}/A$, $x^2\log(1/x)/(2A)$, or $(1+\zeta(s-2))x^2/(2A)$ in the three regimes. Inverting and using $v = v_c + d - x$ gives (1.61) by power–logarithm transfer [145]. In particular the middle inverse has leading term $2\sqrt{A}\, d^{1/2}(\log(1/d))^{-1/2}$. At an interior fold, expansion of the parametrization gives the square-root amplitude $\sqrt{2w_*/(1+\operatorname{Li}_{s-2}(t_*))}$, which proves (1.59).

For completeness these transfers have the required complex domains. For $d \sim cx^p$ on a closed slit sector, use the model inverse $(d/c)^{1/p}$; for $d \sim cx^2\log(1/x)$ use $(2d/(c\log(1/d)))^{1/2}$. On a fixed small relative circle around that model, the normalized convergent expansion tends uniformly to the corresponding power equation. Rouché's theorem gives one root. Differentiated convergent expansions make its derivative nonzero, so the local inverses glue on smaller slit sectors. The analytic coordinate $d = 1 - r/R_b + O((1-r/R_b)^2)$ supplies the local dent. A finite cover of the remaining circle, using the already proved implicit continuations, completes the transfer domain. The nonanalytic leading terms prove genuine singularity and the asserted radius. □

Corollary 1.21 (Physical evaluation threshold)**.** *For each $s > 0$, let $t_s \in (0,1)$ be the unique solution of $\log t_s + \operatorname{Li}_{s-1}(t_s) = 0$, and put $u_s^2 = 2\operatorname{Li}_s(t_s) + \operatorname{Li}_{s-1}(t_s)^2$. The Taylor series at $r = -e^{-u}$ converges exactly when $u \geq u_s$, and converges absolutely at equality.*

Proof. The left side increases strictly from negative infinity to a positive limit. In the interior regime $e^{-u}/R = e^{-w_*}/t_*$, giving exactly the stated threshold. For $s > 2$ it occurs below $U_c(s)$; at and above that transition $e^{-u}/R_b = e^{-b} < 1$. Outside the radius the series diverges, and the theorem gives absolute convergence on its circle. □

1.7.4 Evidence and the next inverse questions

The interior result and boundary completion retain the attribution in [143]. The latter resolves an explicitly posed project question; it is not a correction to a theorem that claimed those cases. The finite Bessel formula and order-spectrum identities above develop the exact Lagrange representation using classical polynomial machinery. They make no publication-priority claim.

Independent rational computations compare (1.53) with coefficient iteration of the original equation, including signed input kernels; separate checks test the weight and order-shift recurrences. The incoming full suites also reproduce the marginal coefficient ratio 0.739119713 to its leading estimate at $n = 2048$, versus 1.010075657 after the second logarithmic refinement. The finite discrepancy motivates the refined inverse charts but is not a bound or a proof of an equivalent.

Section 1.8 develops the marginal inverse chart and complete logarithmic correction calculus. The nonmarginal critical charts and joint crossover refinements retain separate integration scopes. The report's local complex folds do not settle global access from the Taylor germ. Also, when $b < A$ is real, $R_b = e^{b-u}$ is an endpoint reference smaller than the actual interior radius. The compound-Poisson coefficient identity uses that reference on both sides of criticality; it must not silently replace the radius established above.

1.8 Lambert inversion and exact logarithmic coefficients

The marginal law at $s = 3$ in Theorem 1.20 converges slowly. Its logarithmic corrections nevertheless have exact finite formulas. We first reconcile the order-three inverse chart and correction calculus from [143]. A Laplace representation then combines the correction calculus into a single Gamma moment at each order. The same representation gives an exact convolution identity, while the chart coefficients have a Catalan principal part.

1.8.1 An exact marginal chart and its rational coefficients

Throughout this section put

$$A = \zeta(2), \quad u = U_c(3), \quad v_c = u - A, \quad R = e^{-v_c}, \quad d = -\operatorname{Log}(r/R), \quad x = -\operatorname{Log}(re^{v}).$$

Thus $v = v_c + d - x$, and (1.62) becomes

$$(A + x)d - \frac{d^2}{2} = \frac{x^2}{2}\left(\log\frac{1}{x} + \frac{5}{2}\right) + x^3 C_3(x), \qquad C_3(x) = \sum_{j\ge 0} \frac{(-1)^{j+3}\zeta(-j)}{(j+3)!}x^j. \tag{1.64}$$

The tail is holomorphic for $|x| < 2\pi$, with $C_3(x) = 1/12 - x/288 + O(x^3)$. All branches below continue from small positive d. Let W_{-1} denote the lower real Lambert branch [146], and set

$$\Lambda = -\frac{1}{2}W_{-1}(-4Ade^{-5}), \qquad \tau = \sqrt{\frac{2Ad}{\Lambda}}. \tag{1.65}$$

Exactly, $\Lambda = \log(1/\tau) + 5/2$ and $\tau^2\Lambda = 2Ad$.

Theorem 1.22 (Convergent Lambert chart)**.** *There is a holomorphic germ $U(p, w)$, $U(0, 0) = 1$, determined by*

$$U^2(1 - w\log U) + 2pw^2U^3C_3(pwU) = 1 + \frac{pw}{A}U - \frac{p^2w}{4A^2}. \tag{1.66}$$

The selected inverse of (1.64) is exactly $x = \tau U(\tau\Lambda, \Lambda^{-1})$. Moreover, $U - 1$ is divisible by pw and

$$x = \tau + \tau^2\frac{\Lambda/A - 1/6}{2\Lambda - 1} + O(\tau^3\Lambda). \tag{1.67}$$

Proof. Substitute $x = \tau U$, $p = \tau\Lambda$, $w = \Lambda^{-1}$, $\tau = pw$, and $d = p^2 w/(2A)$ in (1.64). Division by Ad gives (1.66). The derivative in U at $(0, 0, 1)$ is 2, so the holomorphic implicit function theorem applies. At either $p = 0$ or $w = 0$, uniqueness gives $U = 1$. The coefficient of p is

$$[p]U = \frac{w/A - w^2/6}{2 - w};$$

the remaining terms are $O(p^2 w)$. This proves (1.67). The positive solution and then analytic continuation identify the inverse already selected in Theorem 1.20. □

Writing $U = 1 + \sum_{n\geq 1} u_n(w)p^n$, define

$$F(h, w) = (1 + h)^2\{1 - w\log(1 + h)\} - 1$$

and

$$\Psi(p, w, h) = \frac{pw}{A}(1 + h) - \frac{p^2 w}{4A^2} - 2pw^2(1 + h)^3 C_3(pw(1 + h)).$$

An auxiliary multiplier on Ψ and Lagrange inversion give the finite identity

$$u_n(w) = \sum_{k=1}^{n} \frac{1}{k}[p^n h^{k-1}] \left\{ \frac{h\Psi(p, w, h)}{F(h, w)} \right\}^k. \tag{1.68}$$

Only $k \leq n$ contributes because Ψ is divisible by p.

Proposition 1.23 (Exact Catalan principal parts)**.** *Each $u_n(w)$ is rational in w, with coefficients in $\mathbb{Q}(A)$. Its only finite pole is at $w = 2$, has exact order $2n - 1$, and satisfies*

$$\lim_{w\to 2}(2 - w)^{2n-1}u_n(w) = 2^{n-1}\,\mathrm{Cat}_{n-1}\left(\frac{2}{A} - \frac{2}{3}\right)^n. \tag{1.69}$$

Proof. In (1.68), the only denominators involving w arise from $F/h = (2-w)+(1-3w/2)h+O(h^2)$. In the summand with index k, extracting h^{k-1} produces at most $2k - 1$ factors of $(2 - w)^{-1}$. The largest pole can therefore come only from $k = n$, taking p once from every Ψ and every h from the denominator expansion. At $w = 2$, the linear coefficient of F/h is -2 and $[p]\Psi(p, 2, 0) = 2/A - 2/3$. The resulting factor is $2^{n-1}\binom{2n-2}{n-1}/n$, which proves (1.69). It is nonzero since $A = \zeta(2) < 2$. This is a statement about the rational coefficient functions; the chart is proved near $w = 0$ and no global physical singularity at $w = 2$ is inferred. □

1.8.2 A Laplace identity and an exact convolution

The Lambert core has a useful identity beyond its asymptotic expansion. For real $H > 1$, let $T(H) > 1$ be the solution of $T - \log T = H$, equivalently $T = -W_{-1}(-e^{-H})$. For $\nu > 0$ and $t > 0$ define

$$\rho_\nu(t) = \frac{\nu\, t^{t+\nu-1}}{\Gamma(t + \nu + 1)}. \tag{1.70}$$

Connections between the lower Lambert branch, Laplace transforms and probability laws are classical; see [147]. We give a direct proof in the normalization needed here, without a priority claim.

Theorem 1.24 (Gamma density identities)**.** *For $\nu > 0$ and $H > 1$,*

$$T(H)^{-\nu} = \int_0^\infty e^{-Ht} \rho_\nu(t)\, dt. \tag{1.71}$$

For every $\mu, \nu, t > 0$,

$$\int_0^t \rho_\mu(y)\rho_\nu(t-y)\, dy = \rho_{\mu+\nu}(t). \tag{1.72}$$

In particular, $e^{-t}\rho_\nu(t)$ integrates to one, and these densities form a convolution family in ν.

Proof. The map $z \mapsto z - \operatorname{Log} z$ is one-to-one on $\Re z > 1$: its derivative has positive real part, and integrating that derivative along a line segment proves injectivity. Its boundary has real part $1 - \frac{1}{2}\log(1+y^2) \le 1$ and imaginary part $y - \arctan y$, strictly increasing onto the real line. Also its modulus tends to infinity with $|z|$. The argument principle, or continuation from the positive axis in this boundary description, therefore gives a holomorphic inverse $T(H)$ for $\Re H > 1$, with $T(H) = H + \operatorname{Log} H + o(1)$ as $|\Im H| \to \infty$ on each fixed vertical line.

Initially take $\nu > 1$, so Bromwich inversion of $T(H)^{-\nu}$ is absolutely convergent. Set $H = z - \operatorname{Log} z$ in that integral. The image contour has $\Re z = c + \log|z|$, where $c > 1$ is the original Bromwich abscissa. It can be deformed to a fixed vertical line $\Re z = b > 1$. On the horizontal connectors the integrand is $O(|\Im z|^{-\nu})$ and their lengths are $O(\log|\Im z|)$; their contributions tend to zero. Inverse Laplace inversion for powers now gives

$$\begin{aligned}\frac{1}{2\pi i}\int e^{tz} z^{-t-\nu}(1-z^{-1})\, dz &= \frac{t^{t+\nu-1}}{\Gamma(t+\nu)} - \frac{t^{t+\nu}}{\Gamma(t+\nu+1)} \\ &= \rho_\nu(t).\end{aligned}$$

This proves (1.71) for $\nu > 1$ by Laplace uniqueness. Near zero, $\rho_\nu(t) \sim t^{\nu-1}/\Gamma(\nu)$; Stirling's formula gives $\rho_\nu(t) \sim \nu e^t/(\sqrt{2\pi}\, t^{3/2})$ at infinity. The integral is thus holomorphic in ν on $\Re\nu > 0$, locally uniformly, and analytic continuation proves the assertion for every positive ν.

Tonelli's theorem and (1.71) show that the Laplace transform of the convolution in (1.72) is $T(H)^{-\mu-\nu}$. Laplace uniqueness gives equality almost everywhere, and continuity for $t > 0$ gives equality everywhere. Finally let $H \downarrow 1$ and use monotone convergence and $T(H) \downarrow 1$. □

1.8.3 Convergent core polynomials and asymptotic coefficient polynomials

Put

$$L = \log(1/d), \qquad c = 5 - \log(4A), \qquad T = 2\Lambda.$$

Then $T - \log T = L + c$ and $\tau = 2\sqrt{Ad/T}$. The report's convergent logarithmic core expansion is

$$\tau = 2\sqrt{Ad}\, L^{-1/2} \sum_{j \ge 0} L^{-j} P_j(\log L + c), \tag{1.73}$$

where $P_0 = 1$ and, for $j \ge 1$,

$$P_j(Y) = -\frac{1}{2j}[h^{j-1}](1+h)^{-3/2}\{Y + \log(1+h)\}^j. \tag{1.74}$$

Indeed $T = L(1+V)$ gives $V = \eta + p\log(1+V)$ with $p = L^{-1}$ and $\eta = p(\log L + c)$. The implicit germ is holomorphic in (p, η) at zero. Lagrange inversion after $\eta = pY$ gives (1.74); grouping its normally convergent double series proves (1.73). In particular, $P_1 = -Y/2$ and $P_2 = 3Y^2/8 - Y/2$.

For comparison with the report's finite transfer formula, set

$$G(a)=\frac{1}{\Gamma(-a)},\qquad M_k(a)=\frac{(-1)^kG^{(k)}(a)}{k!\,G(a)}$$

and define

$$Q_N(Y)=\sum_{j=0}^{N}M_{N-j}(1/2)\prod_{\ell=0}^{N-j-1}\left(\frac{d}{dY}-j-\frac{1}{2}-\ell\right)P_j(Y). \tag{1.75}$$

An empty product is one. This expression is finite.

Theorem 1.25 (One Gamma moment for every logarithmic correction)**.** *The polynomials in* (1.75) *have the exact formulas*

$$Q_N(Y)=\frac{1}{\sqrt{\pi}\,N!}\Re\int_0^\infty e^{-y}y^{N-1/2}(\log y-Y+i\pi)^N\,dy \tag{1.76}$$

$$=\frac{1}{\sqrt{\pi}\,N!}\Re(\partial_z-Y+i\pi)^N\Gamma(z)\Big|_{z=N+1/2}. \tag{1.77}$$

Here Y is real; the resulting real polynomial extends uniquely to complex Y. Its leading coefficient is $(-1)^N\binom{2N}{N}/4^N$, and

$$Q'_{N+1}=Q'_N-(N+\tfrac{1}{2})Q_N. \tag{1.78}$$

For $a_n=[r^n]v(r)$ in the critical order-three model and every fixed integer $M\ge 0$,

$$R^na_n=\sqrt{\frac{A}{\pi}}\,n^{-3/2}(\log n)^{-1/2}\left\{\sum_{N=0}^{M}\frac{Q_N(\log\log n+c)}{(\log n)^N}+O\left(\frac{(\log\log n)^{M+1}}{(\log n)^{M+1}}\right)\right\}. \tag{1.79}$$

Unlike the core expansion, this coefficient expansion is asserted asymptotically to each fixed order, not as a convergent infinite sum.

Proof. First, power–logarithm transfer [145] applied to (1.73) gives (1.75). On a Hankel contour, $1-r/R=t/n$ changes the large logarithm to $\log n-\log t$. Taylor expansion contributes $(-1)^k(\log t)^k/k!$, whose integral against the power kernel is $G^{(k)}(1/2)$. Repeated differentiation of $L^{-j-1/2}P_j(\log L+c)$ gives exactly the product in (1.75). The transfer domain was established in Theorem 1.20. The analytic chart gives $x-\tau=O(\tau^2)$, and $d=1-r/R+O((1-r/R)^2)$; both changes affect only terms smaller by a positive power of n, up to logarithms. Finite differentiated remainders therefore give (1.79).

To combine the finite operations, use (1.71) with $\nu=1/2$ and $H=L+c$. It gives the exact local identity

$$\tau=2\sqrt{A}\int_0^\infty e^{-ct}\rho_{1/2}(t)d^{t+1/2}\,dt.$$

For logarithmic asymptotics it suffices to retain $0\le t\le\epsilon$ with a fixed $0<\epsilon<1/4$. The tail is $O(d^{1/2+\epsilon'})$ on each closed slit subsector, for any $0<\epsilon'<\epsilon$, as are its appropriately scaled derivatives. Uniform power transfer on this compact t-interval reduces the leading logarithmic sector of $-\tau$ to

$$\frac{\sqrt{A}}{\pi}n^{-3/2}\int_0^\epsilon e^{-(\log n+c)t}t^{t-1/2}\cos(\pi t)\,dt.$$

The simplification uses the exact reflection identity

$$\frac{1}{\Gamma(t+3/2)\Gamma(-t-1/2)} = -\frac{\cos(\pi t)}{\pi}.$$

Set $t = y/\log n$. Watson expansion of $\exp\{t(\log t - c)\}\cos(\pi t)$ gives precisely (1.76). Only finitely many Taylor terms are used for a fixed target order; no integral with the transferred density over the whole positive axis is asserted. Both calculations give polynomial asymptotic expansions of the same function, whose coefficients are unique, proving equality with (1.75).

Differentiation under the convergent Gamma integral proves (1.77). Its highest power of Y is $(-1)^N\Gamma(N+1/2)/(\sqrt{\pi}\,N!)$. Finally apply $\Gamma(z+1) = z\Gamma(z)$ in (1.77) and differentiate with respect to Y. The product rule gives (1.78). □

The centering which removes the digamma constant is especially simple. Let $H = Y - \psi(-1/2)$ and $\widehat{Q}_N(H) = Q_N(H + \psi(-1/2))$. Then

$$\begin{aligned}
\widehat{Q}_0(H) &= 1, \qquad \widehat{Q}_1(H) = -H/2,\\
\widehat{Q}_2(H) &= \frac{3}{8}H^2 - \frac{1}{2}H - \frac{3\pi^2}{16} - \frac{3}{2},\\
\widehat{Q}_3(H) &= -\frac{5}{16}H^3 + H^2 + \left(\frac{13}{4} + \frac{15\pi^2}{32}\right)H + 1 - \frac{\pi^2}{2} - \frac{35}{8}\zeta(3).
\end{aligned}$$

In (1.79), the centered argument is $H = \log\log n - \log A + \gamma + 3$. The first two corrections agree with the incoming report; the Gamma moment formula supplies every further coefficient directly.

Independent symbolic checks compare the two formulas through $N = 8$, verify their derivative recurrence, and reproduce the exact pole orders through $n = 5$ using implicit coefficient recursion at a rational diagnostic value of A. Two precisions of quadrature check the Laplace and convolution identities, separately from those exact checks. The conventional proofs above establish the all-order statements. The nonmarginal inverse charts and joint crossover branches retain separate integration scopes.

1.9 All late corrections of a factorial Lambert inverse

The preceding Lambert chart resolves a logarithmic marginal scale. A different Lambert structure arises when the forward tail itself has factorial growth. Here the small-index coefficients of that tail determine every inverse-index correction by a finite polynomial calculation. The leading hyperbolic amplitudes and their first correction were proved in the reversion continuation [150], resolving the conjectures of [149]. We give a uniform extension to every fixed order, motivated by its all-order question. The experimental calculation of the second correction guides the finite formula; the factorial estimates below justify the full infinite asymptotic statement. The continuation report [151] independently obtains the first correction and supplies the analytic input for continuing the inverse Borel transform beyond its initial disk. We combine these results below.

1.9.1 The inverse equation and finite coefficient extraction

Put $A = 4\pi^2$ and let $d(n)$ count positive divisors. Mellin inversion of the double Lambert sum gives

$$\mathcal{F}(t) = \sum_{n\ge1}\frac{d(n)}{n}e^{-nt}, \qquad \Gamma(w)\zeta(1+w)^2t^{-w}$$

as function and Mellin integrand. With the Stieltjes convention $\zeta(1+w) = w^{-1} + \sum_{j\ge 0}(-1)^j \gamma_j w^j/j!$, its triple pole at zero contributes

$$\frac{1}{2}\big(\log(1/t) + \gamma\big)^2 + C_*, \qquad C_* = \frac{\zeta(2)}{2} - \gamma^2 - 2\gamma_1.$$

The remaining residues define the formal tail

$$R(q) = \sum_{m\ge 1} r_m q^m, \quad c = r_1 = -\frac{1}{4}, \quad r_{2m} = \frac{4\Gamma(2m)\zeta(2m)^2}{2mA^{2m}}, \quad r_{2m+1} = 0 \quad (m \ge 1). \tag{1.80}$$

Equivalently $r_m = (-1)^m \zeta(1-m)^2/m!$, so all coefficients are rational; in particular $r_2 = 1/288$. The exact real completion is treated in [149]; the coefficient argument here uses only (1.80).

For a large target Y, set $X = \sqrt{2(Y - C_*)}$, $u = X^{-1}$, $q = e^{\gamma - X}$ and $t = qe^{-V}$. The corresponding formal equation is

$$V + \frac{u}{2}V^2 + uR(qe^{-V}) = 0, \qquad V(q,u) = \sum_{n\ge 1} C_n(u) q^n. \tag{1.81}$$

In the following theorem q and u are independent variables. Formal implicit inversion gives a unique solution over $\mathbb{Q}[u]$. Introduce the finite polynomial

$$H(w,k,u) = \sum_{v=0}^{k-1} \binom{k-1}{v} (k)_v (u/2)^v w^v, \qquad k \in \mathbb{Z}_{>0}, \tag{1.82}$$

where $(k)_v$ is the rising factorial. Auxiliary-variable Lagrange inversion yields the exact identity

$$C_n(u) = -\sum_{k=1}^{n} \frac{u^k}{k!} n^{k-1} [q^n] R(q)^k H(1/n, k, u). \tag{1.83}$$

Indeed, apply Lagrange inversion to $V = -zuR(qe^{-V})(1 + uV/2)^{-1}$. The coefficient of $z^k q^n$ is $(-u)^k [q^n] R^k / k$ times $[v^{k-1}] e^{-nv}(1 + uv/2)^{-k}$, which is the displayed term. Only $k \le n$ occurs, so setting $z = 1$ is a formal operation.

1.9.2 A finite polynomial for every correction

Let $S_r(L)$ be the Faulhaber polynomial equal to $\sum_{i=1}^{L} i^r$ for nonnegative integers L. Define the formal series

$$B(w,L) = \exp\left(\sum_{r\ge 1} \frac{L^r + S_r(L)}{r} w^r \right). \tag{1.84}$$

For integer $L \ge 0$ this is $(1 - Lw)^{-1} \prod_{i=1}^{L} (1 - iw)^{-1}$. When extracting a fixed power of w, we interpret (1.82) polynomially in k, using falling factorials for $\binom{k-1}{v}$. Thus both expressions make sense as finite coefficient operations even before an integer value is substituted.

Define the linear Poisson polynomial transform by

$$\mathcal{P}_z(L^r) = e^{-z} \sum_{\ell \ge 0} \frac{\ell^r z^\ell}{\ell!} = \sum_{a=0}^{r} \left\{ {r \atop a} \right\} z^a. \tag{1.85}$$

For $r = 0$ its value is 1, with $0^0 = 1$ in the sum. The Stirling-number formula makes the operation finite.

Fix $j \geq 0$. Range over nonnegative multiplicities $\mathbf{a} = (a_2, \dots, a_{j+1})$ satisfying

$$h = \sum a_m, \qquad s = \sum m a_m, \qquad d = s - h = \sum (m-1) a_m \leq j.$$

The empty tuple is allowed. For $\epsilon \in \{1, -1\}$ set $\theta = Au/4$ and $z_\epsilon = \epsilon A c u = -\epsilon\theta$. The correction polynomial is

$$T_j^\epsilon(u) = \sum_{\mathbf{a}:\ d \leq j} u^h (\epsilon A)^s \prod_{m=2}^{j+1} \frac{r_m^{a_m}}{a_m!}\, \mathcal{P}_{z_\epsilon}\Big([w^{j-d}] B(w, L+s) H(w, L+h+1, u) \Big). \tag{1.86}$$

The transform acts on L only. Every sum and coefficient extraction in (1.86) is finite; its only forward coefficients are $r_1, \dots, r_{j+1}$. In particular $T_0^\epsilon = 1$.

Theorem 1.26 (All fixed-order Lambert corrections)**.** *For each integer $J \geq 0$ and $U > 0$, uniformly for complex $|u| \leq U$,*

$$C_n(u) = -\frac{2u\Gamma(n)}{nA^n} \sum_{\epsilon = \pm 1} \epsilon^n e^{-\epsilon\theta} \sum_{j=0}^{J} \frac{T_j^\epsilon(u)}{n^j} + O_{J,U}\left(\frac{|u|\Gamma(n)}{A^n n^{J+2}} \right). \tag{1.87}$$

This is an additive expansion, including parameter values where a parity amplitude vanishes. It asserts an asymptotic expansion at every fixed order, with no assertion of convergence as $J \to \infty$.

1.9.3 Uniform control of the omitted configurations

We supply the estimates that allow the finite calculation to be summed over all factor counts in (1.83).

Lemma 1.27 (Factorial vertices)**.** *For positive integers $N \geq p$,*

$$\sum_{\substack{\ell_1 + \cdots + \ell_p = N \\ \ell_i \geq 1}} \prod_i (\ell_i - 1)! \leq 3^{p-1} (N-p)!. \tag{1.88}$$

Fix $J \geq 0$. If the configurations with $N - \max_i \ell_i \leq J$ are removed, then the remaining sum is at most $D_J^p (N-J-2)!$, for all sufficiently large N depending only on J, with a constant $D_J \geq 1$ independent of p and N.

Proof. The two-factor inequality $\sum_{a=0}^{M} a!(M-a)! \leq 3M!$ follows by dividing by $M!$: the endpoint terms sum to 2 and each interior reciprocal binomial coefficient is at most $1/M$. The cases $M = 0, 1$ are immediate. Induction proves (1.88).

If $p \geq J+2$, this bound already suffices since $(N-p)! \leq (N-J-2)!$. If $p = 1$ nothing remains. For $2 \leq p \leq J+1$, choose a maximal coordinate and put $d = N - \max_i \ell_i$. The remainder is at most

$$p 3^{p-2} \sum_{d=J+1}^{\lfloor N - N/p \rfloor} (d-p+1)!(N-d-1)!.$$

Write $M = N - p$ and $b = J + 2 - p \geq 1$. For sufficiently large N, both factorial indices in this sum lie between b and $M - b$. For each fixed b,

$$\sum_{t=b}^{M-b} t!(M-t)! \leq K_b (M-b)!. \tag{1.89}$$

To verify this uniformly, in the lower outer third the ratio of successive terms is $(t+1)/(M-t) \le 3/4$ for large M. Sum that geometric majorant starting at $b!(M-b)!$ and use symmetry for the upper third. The central terms are bounded by $b!(M-b)!(3/4)^{\lfloor M/3 \rfloor - b}$, times a fixed factor; their number is at most $M+1$. This is uniformly bounded after division by $(M-b)!$. The finitely many small M can be absorbed in K_b. Equation (1.89) now gives $(N-J-2)!$. Only finitely many $p \le J+1$ remain, so a common D_J suffices. □

Proof of Theorem 1.26. Choose $K \ge 1$ such that $A|r_1| \le K$ and $A^m|r_m| \le K(m-2)!$ for $m \ge 2$. Such a K exists by (1.80). For a composition $m_1 + \cdots + m_k = n$, put $\ell_i = m_i - 1$ and $N = n - k$. Choosing the p positive increments and applying (1.88) gives, for $k < n$,

$$A^n \left| [q^n] R^k \right| \le (4K)^k \Gamma(n-k). \tag{1.90}$$

Absolute values may also be placed inside the coefficient sum. Since $(k)_v \le (2k)^v$ for $v \le k-1$, $|H(1/n, k, u)| \le (1+U)^k$ for $k \le n$. After division by $\Gamma(n)/(nA^n)$, the terms with $k \le n/2$ are bounded by $D^k/k!$ for a constant $D = D_U$: indeed $n^k \Gamma(n-k)/\Gamma(n) \le 2^k$. The sum over $n/2 < k < n$ is at most

$$n \max(1, 2eD)^n \frac{\Gamma(\lceil n/2 \rceil)}{\Gamma(n)},$$

using $n^k/k! \le (2e)^k$. It is smaller than every inverse power of n. The $k = n$ term, where all indices are 1, has the same property directly from $n^n/(n!\Gamma(n))$. All these estimates retain an overall factor $|u|$, by bounding $|u|^k$ by $|u| \max(1, U)^{k-1}$.

For $k \le n/2$, retain exactly the configurations with one large increment and total other increment at most J. Lemma 1.27, including the choices of positive positions, bounds the omitted absolute coefficient sum by $A^{-n} D_J^k \Gamma(n-k-J-1)$, with an enlarged constant depending also on K. For sufficiently large n,

$$\frac{n^k \Gamma(n-k-J-1)}{\Gamma(n)} \le n^{-J-1} 3^{k+J+1}.$$

Together with the $k!$ denominator and the bound on H, this proves that the total omitted contribution is $O_{J,U}(|u|\Gamma(n)/(A^n n^{J+2}))$.

In a retained configuration let ℓ be the number of indices 1, and let a_m count the other small indices. Then $k = \ell + h + 1$ and the large index is $m = n - \ell - s$. The large increment is unique once n is sufficiently large, since $N > 2J$. The contribution to $[q^n]R^k$ is exactly

$$\frac{k!}{\ell! \prod a_m!} c^\ell \prod r_m^{a_m} \, r_{n-\ell-s}.$$

Here $n - \ell - s \ge n/2 + 1 - J$. Replacing $\zeta(n-\ell-s)^2$ by 1 in the large even coefficient has an exponentially small relative error, uniformly in these configurations. The odd large coefficients vanish exactly. Thus use

$$r_m = \frac{2\Gamma(m)}{mA^m} \sum_{\epsilon = \pm 1} \epsilon^m$$

at a total exponentially small cost under the same factorial majorant. The exact gamma factor is

$$\frac{n^{L+1}\Gamma(n-L)}{(n-L)\Gamma(n)} = B(1/n, L), \qquad L = \ell + s. \tag{1.91}$$

Consequently, for each small multiset the normalized action contribution is the finite sum of

$$\frac{z_\epsilon^\ell}{\ell!} u^h (\epsilon A)^s \prod \frac{r_m^{a_m}}{a_m!} \, n^{-d} B(1/n, \ell + s) H(1/n, \ell + h + 1, u).$$

Its common prefactor is $-2u\epsilon^n\Gamma(n)/(nA^n)$.

Expand the product BH through degree $J-d$ in $1/n$. Here is a uniform remainder estimate for this step. We have $L \le n/2 + J$, so $L/n \le 3/4$ for large n. On $0 \le w \le 1/n$, the product defining B is at most 4^{L+1}. The derivatives of its logarithm of order $a \ge 1$ are bounded by a constant times $(1+L)^{a+1}$, with constants depending on a. Its fixed-order Taylor remainder is therefore bounded by n^{-b-1} times a polynomial in L times 4^{L+1} when truncated at order b. For H, the terms beyond degree b are bounded by

$$\sum_{v=b+1}^{k-1} \binom{k-1}{v} (kU/n)^v \le (k/n)^{b+1}(1+U)^k.$$

The fixed retained coefficients of H are polynomials in k, u. These bounds give the same polynomial-times-exponential remainder for the product. It is summable against $|z_\epsilon|^\ell/\ell!$, uniformly on $|u| \le U$. Extending the resulting finite ℓ sums to infinity costs less than every inverse power of n, since only fixed-degree polynomials in ℓ remain. The coefficient of n^{-j} is now precisely the Poisson transform in (1.86), multiplied by e^{z_ϵ}. This proves (1.87). □

1.9.4 The explicit second correction

Let $b = uA^2/288$ and define

$$P_1 = \frac{1+u}{2}\theta^2 + b, \qquad D_1 = u+2, \tag{1.92}$$

$$P_2 = \frac{(1+u)^2}{8}\theta^4 + \frac{3u^2+11u+11+(u+1)b}{2}\theta^2 + (u+5)b + \frac{b^2}{2}, \tag{1.93}$$

$$D_2 = \frac{6u^2+15u+10}{6}\theta^2 + (2u+4)b + 2u + 3. \tag{1.94}$$

Then $T_1^\epsilon = P_1 - \epsilon\theta D_1$ and $T_2^\epsilon = P_2 - \epsilon\theta D_2$. For $E_n(\theta) = \cosh\theta$ at even n and $E_n(\theta) = -\sinh\theta$ at odd n, the expansion becomes

$$\frac{nA^nC_n(u)}{4\Gamma(n)} = -u\left[E_n + \frac{P_1E_n + D_1\theta E_n'}{n} + \frac{P_2E_n + D_2\theta E_n'}{n^2}\right] + O_U(|u|n^{-3}). \tag{1.95}$$

The primes differentiate E_n with respect to its argument.

For an explicit check of the calculation, write

$$b_1 = \frac{\ell(\ell+3)}{2}, \quad b_2 = \frac{\ell(2\ell^2+9\ell+1)}{12} + \frac{b_1^2}{2}, \quad h_1 = \frac{u\ell(\ell+1)}{2}, \quad h_2 = \frac{u^2\ell(\ell-1)(\ell+1)(\ell+2)}{8}.$$

The configurations with only linear small indices give $b_2 + b_1h_1 + h_2$ before the Poisson transform. One small index 2 gives $b\{(\ell+2)(\ell+5) + u(\ell+2)(\ell+1)\}/2$; two such indices give $b^2/2$. There is no small-index-3 term because $r_3 = 0$. Applying (1.85) to these expressions gives (1.93)–(1.94).

1.9.5 Uniformity at zero coupling

The extra vanishing of odd coefficients improves the remainder in Theorem 1.26. For odd $n > 1$, its error is in fact

$$O_{J,U}\left(\frac{|u|^2\Gamma(n)}{A^n n^{J+2}}\right), \qquad |u| \le U.$$

To see this, divide the coefficient formula and its finite asymptotic sum by u, using removable values at zero. Both are entire in u. The first vanishes at zero because $C_n(u)/u$ has constant term $-r_n = 0$. In the second, $T_0^\epsilon(0) = 1$ and $T_j^\epsilon(0) = 0$ for $j > 0$; the remaining signed sum vanishes for odd n. Apply the uniform remainder estimate on $|u| \le 2U$ and Schwarz's lemma to their difference.

Consequently every fixed-order expansion is relatively uniform on $0 < u \le U$, with no lower cutoff on u: divide by the leading $\cosh(Au/4)$ for even n, and by $\sinh(Au/4) \ge Au/4$ for odd n. This extends the first-correction endpoint uniformity of [151] to all fixed orders. It does not assert uniformity for u growing with n or across zeros of complex hyperbolic amplitudes.

1.9.6 Higher polylogarithmic Borel boundary models

Use the shifted Borel convention $\widehat{V}_u(\xi) = \sum_{n\ge1} C_n(u)\xi^{n-1}/\Gamma(n)$. For each fixed $J \ge 0$, Theorem 1.26 gives the exact decomposition in $|\xi| < A$

$$\widehat{V}_u(\xi) = -\frac{2u}{\xi} \sum_{\epsilon=\pm1} e^{-\epsilon\theta} \sum_{j=0}^{J} T_j^\epsilon(u)\, \mathrm{Li}_{j+1}(\epsilon\xi/A) + K_{J,u}(\xi). \tag{1.96}$$

The apparent division by zero is removable. The remainder and its first J complex derivatives extend continuously to the closed disk, locally uniformly in u: the Jth differentiated remainder series is bounded by a constant times $\sum n^{-2}$ there. Thus (1.95) supplies an explicit trilogarithmic model with a twice continuously differentiable remainder. It extends the logarithmic and dilogarithmic models of [150] to arbitrary finite boundary regularity.

Since $\mathrm{Li}_1(z) = -\log(1-z)$, the two logarithmic coefficients at $\xi = A$ and $\xi = -A$ are respectively $2ue^{-\theta}/A$ and $-2ue^{\theta}/A$. Both are nonzero for $u \ne 0$. The continuous $J = 0$ remainder therefore proves genuine singularities at both endpoints and exact Borel radius A; the ordinary series has radius zero. Finite-order boundary regularity alone gives no continuation through those points and does not determine the full local singularity on other sheets.

1.9.7 Continuation and sectorial inverse identities

The arithmetic forward tail provides more information than the boundary models alone. Its shifted Borel transform is

$$\widehat{R}(\xi) = -\frac{1}{4} - \frac{2}{\xi} \sum_{m\ge1} d(m) \log\left(1 - \frac{\xi^2}{A^2m^2}\right). \tag{1.97}$$

Expanding the logarithms for $|\xi| < A$ and using $\sum d(m)m^{-s} = \zeta(s)^2$ proves this identity. Along any bounded path avoiding $A\mathbb{Z}$, all sufficiently large m retain their Taylor branches and contribute $O(d(m)/m^2)$, uniformly on compact neighborhoods of the path. The remaining finitely many logarithms can be continued individually. Thus the sum continues normally on the path surface avoiding $A\mathbb{Z}$. Zero is regular on the initial sheet, but a loop around a branch point adds a multiple of $1/\xi$; later passage through zero is therefore excluded from the allowed paths.

Theorem 1.28 (Borel continuation and summed inverse equation)**.** *For every fixed $u \in \mathbb{C}$, $\widehat{V}_u$ continues along every path starting at zero and thereafter avoiding $A\mathbb{Z}$. The formal series $V(q,u)$ is 1-summable in each direction $\vartheta \notin \pi\mathbb{Z}$. If R_ϑ and V_ϑ denote the corresponding Borel–Laplace sums, then*

$$V_\vartheta(q,u) + \frac{u}{2} V_\vartheta(q,u)^2 + uR_\vartheta\big(qe^{-V_\vartheta(q,u)}\big) = 0. \tag{1.98}$$

On every smaller common sector and for sufficiently small q, this is the unique solution satisfying $V_\vartheta(q,u) = O(q)$.

Proof. We give the reduction of [151], making its external analytic input explicit. For $\Omega = A\mathbb{Z}$, let $\mathcal{R}_\Omega$ consist of constants plus formal series in x^{-1} whose shifted Borel transforms continue on the path surface just described. Sauzin's Theorems 3 and 5 [152] imply two closure properties: a convergent power series may be evaluated at elements with zero constant term, and the maps $x+\mathcal{R}_\Omega$ form a group under composition. Their hypotheses hold because Ω is closed, discrete, contains zero, and is stable under addition. These statements also imply $f \circ G \in \mathcal{R}_\Omega$ whenever $f \in \mathcal{R}_\Omega$ and $G \in x + \mathcal{R}_\Omega$, by subtracting G from $(x + f) \circ G$.

For $u \neq 0$, introduce the formal maps

$$h(t) = \frac{\sqrt{1 - 2u^2 R(t)} - 1}{u}, \qquad G(x) = xe^{-h(1/x)}, \qquad T = G^{\langle -1 \rangle}. \tag{1.99}$$

The square root has constant term one, so analytic substitution gives $h(1/x) \in \mathcal{R}_\Omega$ and $h(t) + uh(t)^2/2 + uR(t) = 0$. Moreover $h(t) = ut/4 + O(t^2)$. Multiplication by x is legitimate here: if $f(x) = \sum_{n\geq 1} f_n x^{-n}$, then $\mathcal{B}(xf - f_1) = \partial_\xi \widehat{f}$, which preserves continuation. Hence $G = x - u/4 + O(x^{-1})$ belongs to the stated group, as does T. Closure under composition now gives

$$V(q, u) = h\left(\frac{1}{T(1/q)}\right) \in \mathcal{R}_\Omega.$$

Indeed, $G(T(1/q)) = 1/q$ says $q = te^{h(t)}$ with $t = 1/T(1/q)$, so $t = qe^{-V}$ and the defining inverse equation holds. Formal uniqueness identifies this series with the earlier V.

For summability, restrict $\arg \xi$ to a closed arc separated from the real directions. Write $\rho = |\xi|/A$ and split (1.97) at $m = 2\rho$. On that arc $|1 - se^{2i \arg \xi}| \geq c(1 + s)$ for $s \geq 0$, for some $c > 0$. For $m \leq 2\rho$ the logarithms are bounded by $C + 2\log(1 + \rho)$; the elementary divisor estimate $\sum_{m\leq M} d(m) \leq M(1 + \log M)$ bounds this part after division by $|\xi|$. For $m > 2\rho$, the logarithmic Taylor bound and partial summation give $\sum_{m>2\rho} d(m)m^{-2} = O(\log(2 + \rho)/(1 + \rho))$. Together with the regular germ near zero, these estimates give

$$\widehat{R}(\xi) = O\big(1 + \log^2(2 + |\xi|)\big),$$

which is stronger than the exponential bound required for 1-summability. Sauzin's summable closure Theorems $3'$ and $5'$ apply to the same substitutions and inverse maps. Borel differentiation preserves an exponential bound on a smaller arc by Cauchy's estimate. Summation commutes with these operations, proving (1.98). Finally $R_\vartheta(t) = -t/4 + O(t^2)$ and $R'_\vartheta(t) = -1/4 + O(t)$ on smaller sectors. The map $w \mapsto -uw^2/2 - uR_\vartheta(qe^{-w})$ is a contraction on $|w| \leq M|q|$ for suitable M and small q, proving the stated uniqueness. At $u = 0$, $V = 0$ gives all assertions directly. □

The theorem supplies continuation around the lattice points, without asserting regularity at them. The nearest two are genuine singularities for $u \neq 0$, as already proved. Whether every other nonzero lattice point is singular, and the full singular germs on their various sheets, remain separate questions. The sectorial identity also does not identify arithmetic averaging of lateral inverses with nonlinear median summation.

1.9.8 Exact checks and numerical guidance

The accompanying correction compiler constructs (1.86) through $j = 4$ using rational arithmetic and Stirling polynomials. Independent finite-product expansions check the Faulhaber coefficients at thirteen integer values of L, and exact simplification checks both explicit corrections. The comparison uses all forty fresh finite-sum coefficient values from [150], through $n = 201$, including

complex coupling and a vanishing leading parity amplitude. For example, at $u = 0.1, n = 200$, the absolute error in $nA^nC_n/(4\Gamma(n))$ decreases from approximately $2.078 \cdot 10^{-3}$ at leading order to $5.799 \cdot 10^{-5}$, $1.720 \cdot 10^{-6}$, $5.779 \cdot 10^{-8}$ and $2.155 \cdot 10^{-9}$ through orders one to four. These finite computations helped identify and check the correction formula. They are numerical diagnostics, not interval certificates or the proof of the uniform theorem. An isolated replay of [151] additionally passed 5890 exact checks through q^{12} and evaluated seventy complete finite Lagrange sums through $n = 1025$. Its thirty-five independent 80-digit comparisons include fifteen comparisons with exact coefficient polynomials. The replay uses the required extended-range arithmetic; these finite checks support the formulas, while the continuation theorem depends on the analytic argument above.

1.10 Order filters, neutral curves and the digamma inverse

The preceding sections follow singularities through feedback and inversion. We now ask a complementary design question: which exponentially small data can an order derivative cancel before inversion is performed? For the Fermi family, exact completion turns this into interpolation at conjugate logarithmic nodes. The resulting filters lead to moving-order curves with zero lateral difference, whose physical inverses still have a nonzero exponential displacement. At order zero, the same completion gives a digamma core and a globally invertible positive integral.

This develops the order-filter continuation [153] of the resonant polylogarithm report [143]. Its Fermi limit $-e^x \to -\infty$ differs from the fixed-sheet Lerch limit toward one earlier in this chapter. The common tools are order differentiation, exact completion and controlled local inversion. The neutral inverse chart below resolves a further question left open in that continuation.

A terminating or factorially divergent power–logarithmic expansion does not determine a unique analytic function. Here the distinction can be made exact. For $x > 0$ set

$$F_s(x) = -\operatorname{Li}_s(-e^x), \qquad T_s(x) = F_s(-x) = \sum_{n\geq 1} \frac{(-1)^{n-1}e^{-nx}}{n^s}. \tag{1.100}$$

The subscript is the *polylogarithm order*; the usual normalized Fermi–Dirac index is $s-1$ [96]. There are two distinguished lateral realizations $L_s^{\pm}$ of the Sommerfeld expansion, satisfying

$$L_s^{\pm} = F_s + e^{\mp \mathrm{i}\pi s} T_s. \tag{1.101}$$

Consequently the jump $L_s^+ - L_s^-$ and the physical correction $(L_s^+ + L_s^-)/2 - F_s$ are different data. Neither should be silently substituted for the other. We use *arithmetic average*, not "median," for $(L_s^+ + L_s^-)/2$.

For a real polynomial Π, applying $\Pi(\partial_s)$ changes the coefficient of the nth exponential into the value of Π at $-\log n \mp \mathrm{i}\pi$. We ask how many initial actions can be removed by an order jet of prescribed degree, how this cancellation behaves under small order detuning, and what analytic inversion does to the remaining completion. Here an *action* n simply means the exponential e^{-nx}; this terminology does not add a summability hypothesis.

1.10.1 The Fermi family and exact lateral completion

The analytic realization used throughout

We first state the existing completion in the exact normalization needed below, and give its derivation to fix signs and domains.

Proposition 1.29 (Classical completion in the repository normalization)**.** *For $x > 0$ and $\operatorname{Re} s > 0$, the formal Sommerfeld series is*

$$\widehat{P}_s(x) = \frac{x^s}{\Gamma(s+1)} + 2\sum_{k\geq 1} \frac{\eta(2k)}{\Gamma(s-2k+1)} x^{s-2k}, \qquad \eta(z) = (1-2^{1-z})\zeta(z). \tag{1.102}$$

In the Borel convention $\mathcal{B}(\sum_{j\geq 1} a_j x^{-j})(\xi) = \sum_{j\geq 1} a_j \xi^{j-1}/(j-1)!$, the normalized minor is

$$b_s(\xi) = s\sum_{n\geq 1} \frac{(-1)^{n-1}}{n} \left\{(1+\xi/n)^{s-1} - (1-\xi/n)^{s-1}\right\}. \tag{1.103}$$

The upper and lower lateral realizations

$$L_s^{\pm}(x) = \frac{x^s}{\Gamma(s+1)} \left(1 + \int_0^{\infty e^{\pm \mathrm{i}0}} e^{-x\xi} b_s(\xi)\, \mathrm{d}\xi\right) \tag{1.104}$$

satisfy (1.101)*. Equivalently, with $a_\pm = \frac{1}{2} \pm \frac{\mathrm{i}x}{2\pi}$,*

$$L_s^{\pm}(x) = -\frac{(2\pi)^s e^{\mp \mathrm{i}\pi s/2}}{\Gamma(s)} \zeta(1-s, a_\pm). \tag{1.105}$$

The last formula has a removable singularity at $s = 0$ and is entire in s. It supplies the continuation of the lateral realization to every order. All finite order derivatives may be taken in the exact completion.

Proof. For $\operatorname{Re} s > 0$ the Fermi integral is

$$F_s(x) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{1+e^{t-x}}\, \mathrm{d}t.$$

Split at $t = x$ and put $u = |t - x|$. This gives

$$F_s(x) = \frac{x^s}{\Gamma(s+1)} + \frac{1}{\Gamma(s)} \int_0^\infty \frac{(x+u)^{s-1} - \mathbf{1}_{u<x}(x-u)^{s-1}}{e^u + 1}\, \mathrm{d}u. \tag{1.106}$$

The binomial expansion of (1.103) at zero has coefficient $2\eta(2k)s^{\underline{2k}}/(2k-1)!$ at ξ^{2k-1}. This proves the formal Borel identity. The lattice sum converges normally on compact continued sheets away from $\pm 1, \pm 2, \ldots$: the bracket in its nth term is $O(n^{-1})$ and its prefactor contributes another n^{-1}. The same estimates, with logarithmic factors, hold for finitely many order derivatives on compact order sets.

On a ray strictly above or below the positive axis the kernel has at most polynomial–logarithmic growth. One can verify this by splitting the sum at $n \asymp |\xi|$: the larger n use the preceding difference estimate, and the finitely many smaller n use the power growth of their binomials. Thus the tilted Laplace integrals converge. On the boundary their local powers are integrable when $\operatorname{Re} s > 0$.

In the nth integral substitute $\xi = nu/x$. Summing $\sum_{n\geq 1}(-1)^{n-1}e^{-nu} = 1/(e^u+1)$ turns the integral part of (1.104) into

$$\frac{1}{\Gamma(s)} \int_0^\infty \frac{(x+u)^{s-1} - (x-u)_{\mp}^{s-1}}{e^u+1}\, \mathrm{d}u.$$

The upper path gives argument $-\pi$ to $x-u$ when $u>x$. Subtracting (1.106) therefore gives

$$-\frac{e^{\mp \mathrm{i}\pi(s-1)}}{\Gamma(s)}\int_x^\infty \frac{(u-x)^{s-1}}{e^u+1}\,\mathrm{d}u = e^{\mp \mathrm{i}\pi s}T_s(x).$$

Endpoint domination by $v^{\epsilon-1}(1+|\log v|^r)$ on compact sets $\operatorname{Re} s\ge\epsilon>0$, and exponential domination at infinity, justify the differentiations and interchanges. Finite Taylor expansion of the Borel germ with a remainder gives the asymptotic interpretation of (1.102); no infinite divergent sum was integrated term by term.

Jonquière's identity [96, Eq. 25.12.13], at $z=-e^x$, gives the lower formula in (1.105); conjugating for real s and then using analytic continuation gives the upper one. For $a_\pm$ in the right half-plane, $\zeta(1-s,a_\pm)$ has only its simple pole at $s=0$, cancelled by the simple zero of $1/\Gamma(s)$. The small-argument series T_s is entire in s, normally on compact sets. Hence both the completion and the physical F_s continue entirely in the order, and finite order derivatives remain valid. □

Remark 1.30 (Continuation is not an unregularized boundary integral). For $\operatorname{Re} s\le 0$, (1.105) is the definition of the continued lateral realization used here. We do not assert that the boundary powers in (1.104) become ordinary locally integrable functions there. In Subsection 1.10.6 we give a separate, ordinary meromorphic Borel representation for the first jet at zero. Differentiating the *whole* normalized expression matters: $\log x$ is not treated as a constant coefficient in an unjustified ordinary Borel transform.

A polynomial dictionary for the exponential data

For real σ and $\Pi\in\mathbb{R}[u]$, write

$$\begin{aligned}
\mathcal{F}_{\sigma,\Pi}(x) &= \Pi(\partial_s)F_s(x)\big|_{s=\sigma}\,,\\
\mathcal{L}^{\pm}_{\sigma,\Pi}(x) &= \Pi(\partial_s)L^{\pm}_s(x)\big|_{s=\sigma}\,,\\
C_{\sigma,\Pi} &= \tfrac12(\mathcal{L}^{+}_{\sigma,\Pi}+\mathcal{L}^{-}_{\sigma,\Pi})-\mathcal{F}_{\sigma,\Pi},\\
J_{\sigma,\Pi} &= \mathcal{L}^{+}_{\sigma,\Pi}-\mathcal{L}^{-}_{\sigma,\Pi}.
\end{aligned}\tag{1.107}$$

Put $D=\mathrm{d}/\mathrm{d}u$. The operators

$$R_\sigma=\cos(\pi\sigma+\pi D),\qquad I_\sigma=\sin(\pi\sigma+\pi D)\tag{1.108}$$

are their Taylor series in the nilpotent operator D on any fixed-degree polynomial space; all sums terminate.

Theorem 1.31 (Exact order-filter dictionary)**.** *Let $u_n=-\log n$ and $q=e^{-x}$. Then*

$$\mathcal{L}^{\pm}_{\sigma,\Pi}-\mathcal{F}_{\sigma,\Pi}=e^{\mp \mathrm{i}\pi\sigma}\sum_{n\ge1}\frac{(-1)^{n-1}}{n^\sigma}\Pi(u_n\mp\mathrm{i}\pi)q^n,\tag{1.109}$$

$$C_{\sigma,\Pi}=\sum_{n\ge1}\frac{(-1)^{n-1}}{n^\sigma}(R_\sigma\Pi)(u_n)q^n,\tag{1.110}$$

$$J_{\sigma,\Pi}=-2\mathrm{i}\sum_{n\ge1}\frac{(-1)^{n-1}}{n^\sigma}(I_\sigma\Pi)(u_n)q^n.\tag{1.111}$$

The series converge absolutely for $|q|<1$, normally on compact order sets. For real u,

$$(R_\sigma\Pi)(u)+\mathrm{i}(I_\sigma\Pi)(u)=e^{\mathrm{i}\pi\sigma}\Pi(u+\mathrm{i}\pi).\tag{1.112}$$

Proof. Every derivative of $e^{\mp i\pi s}n^{-s}$ multiplies it by $-\log n \mp i\pi$. Apply the finite operator to (1.101). A polynomial in $\log n$ times $n^{-\sigma}$ has subexponential growth in n, which proves the stated normal convergence. Taylor's formula for a polynomial gives $\Pi(u + i\pi) = \exp(i\pi D)\Pi(u)$ and hence (1.112). Taking the arithmetic average and difference proves the two remaining formulas, with the sign in the jump fixed by upper minus lower. □

1.10.2 Designing order filters: separate and joint cancellation

What a degree can and cannot remove

For a nonzero power series in q with no constant term, let ν be its first nonzero exponent; let $\nu(0) = \infty$.

Theorem 1.32 (Sharp separate degree costs)**.** *Fix $\Pi \neq 0$ of degree d and real σ. If $\sin \pi\sigma \neq 0$, then $\nu(J_{\sigma,\Pi}) \leq d + 1$; if $\sin \pi\sigma = 0$ and $d \geq 1$, then $\nu(J_{\sigma,\Pi}) \leq d$. Similarly, if $\cos \pi\sigma \neq 0$, then $\nu(C_{\sigma,\Pi}) \leq d + 1$; if $\cos \pi\sigma = 0$ and $d \geq 1$, then $\nu(C_{\sigma,\Pi}) \leq d$. Every displayed bound is attained by a polynomial of the indicated degree. The exceptional constant polynomials have zero jump at integer order, and zero arithmetic completion at half-integer order.*

Proof. The constant term of I_σ as a polynomial in D is $\sin \pi\sigma$. When this is nonzero, I_σ is an invertible triangular operator on degree-at-most-d polynomials and preserves the degree. When it is zero,

$$I_\sigma = (\pi \cos \pi\sigma)D + O(D^3),$$

so it has kernel exactly the constants and reduces every positive degree by exactly one. The analogous assertions for R_σ follow from its constant and linear terms, $\cos \pi\sigma$ and $-\pi \sin \pi\sigma$.

A nonzero polynomial of degree k cannot vanish at all of the $k + 1$ distinct real numbers $u_1, \ldots, u_{k+1}$. The coefficients multiplying these evaluations in (1.110)–(1.111) never vanish. This proves the upper bounds.

To attain a bound for a degree-preserving operator, take its inverse image of $\prod_{n=1}^{d}(u + \log n)$. For a degree-lowering operator, its map from polynomials of degree at most d, modulo constants, to polynomials of degree at most $d - 1$ is an isomorphism. Take the inverse image of $\prod_{n=1}^{d-1}(u + \log n)$. The next node is not a root, so the bound is attained exactly. The assertions for constants follow directly from (1.108). □

Thus, for $d \geq 1$, the optimal first surviving actions are

Order center	Jump	Arithmetic completion
$\sigma \in \mathbb{Z}$	d	$d + 1$
$\sigma \in \mathbb{Z} + \frac{1}{2}$	$d + 1$	d
Neither of the above	$d + 1$	$d + 1$

The two columns are separately optimized; they need not be realized by the same polynomial. In particular, a degree statement about one datum cannot be reused for their simultaneous cancellation.

For example, at integer order the operators are, up to $(-1)^\sigma$, $\cos(\pi D)$ and $\sin(\pi D)$. A polynomial that delays the completion to action $d + 1$ is

$$\Pi_d^C(u) = \sec(\pi D) \prod_{n=1}^{d} (u + \log n). \tag{1.113}$$

The power series for sec terminates on the polynomial. A primitive followed by the finite operator $\pi D/\sin(\pi D)$ similarly constructs a jump extremizer. These formulas are exact algebra, not asymptotic infinite operator expansions.

Simultaneous cancellation is a principal ideal

Theorem 1.33 (Joint cancellation ideal and optimal real jet degree)**.** *Let E be any finite set of positive integers. For every real order center σ, the real polynomials Π for which both the jump and the arithmetic completion vanish at every action in E form exactly*

$$\left(\prod_{n\in E}\big((u+\log n)^2+\pi^2\big)\right)\mathbb{R}[u]. \tag{1.114}$$

Consequently, for $E=\{1,\dots,N\}$ the minimum nonzero degree is $2N$, and the unique monic minimizer is

$$Q_N^{\mathrm{joint}}(u)=\prod_{n=1}^{N}\big((u+\log n)^2+\pi^2\big). \tag{1.115}$$

At action $N+1$, at least one of its jump and completion coefficients is nonzero. More generally a degree-d nonzero jet cannot make both data zero at all of the first $\lfloor d/2\rfloor+1$ actions.

Proof. By (1.112), both real components at the node u_n vanish if and only if $\Pi(u_n+\mathrm{i}\pi)=0$. Since Π is real, this is equivalent to zeros at both $u_n+\mathrm{i}\pi$ and $u_n-\mathrm{i}\pi$, or to divisibility by $(u-u_n)^2+\pi^2$. These real quadratic factors are pairwise coprime: all their complex roots are distinct. Divisibility by each is therefore equivalent to divisibility by their product. This proves the ideal identity and the degree assertions.

For the monic minimizer at the next node, putting $\ell_j=\log(j/(N+1))$ gives

$$Q_N^{\mathrm{joint}}(-\log(N+1)+\mathrm{i}\pi)=\prod_{j=1}^{N}\ell_j(\ell_j+2\mathrm{i}\pi)\neq 0. \tag{1.116}$$

After rotation by $e^{\mathrm{i}\pi\sigma}$ this still has at least one nonzero real component. This proves the final assertion. We do not claim that both components must separately be nonzero in this formula. □

Corollary 1.34 (First-order detuning robustness)**.** *For a fixed real polynomial Π and a finite action set E, the following requirements are equivalent: the jump coefficients at every $n\in E$ vanish together with their first order-center derivatives at one real center σ_0; both the jump and completion coefficients vanish there; and the full lateral completion coefficients vanish at those actions for* every *order center. Their common solution ideal is (1.114). In particular robustness through first order in detuning costs exactly $2N$ for the first N actions.*

Proof. Up to a nonzero fixed scalar, the nth jump coefficient is $n^{-\sigma}(I_\sigma\Pi)(u_n)$. Its derivative is

$$n^{-\sigma}\{-\log n\,(I_\sigma\Pi)(u_n)+\pi(R_\sigma\Pi)(u_n)\}.$$

When the jump coefficient is zero, its derivative is zero exactly when the completion coefficient is zero. The joint theorem then forces $\Pi(u_n\pm\mathrm{i}\pi)=0$, independently of the order center. The converse is immediate from the exact dictionary. □

Corollary 1.35 (Finite reconstruction from exponential data)**.** *Fix N and real σ. A polynomial of degree less than $2N$ is uniquely determined by its jump and arithmetic-completion coefficients at the first N actions. Every assignment of those $2N$ real normalized data is attained by exactly one such polynomial.*

Proof. Divide out the nonzero scalar coefficients in (1.110)–(1.111) and undo the rotation in (1.112). This specifies Π at the N upper nodes and, by conjugacy, at the N lower nodes. Ordinary interpolation at $2N$ distinct complex points gives a unique degree-less-than-$2N$ polynomial. Its conjugate-reflection polynomial has the same values, so uniqueness makes its coefficients real. Equivalently, injectivity follows from Theorem 1.33 and equal finite dimensions give surjectivity. □

The first joint filter. For $N = 1$, the operator is $\partial_s^2 + \pi^2$. It removes the entire first action, not just its imaginary part. At integer m the next completion coefficient is $(-1)^{m+1}(\log 2)^2/2^m$ and the next jump coefficient is $-4\pi\mathrm{i}(-1)^m \log 2/2^m$. This is a simple exact test of the real-versus-imaginary distinction.

An explicit finite-tail estimate

For numerical evaluation the exact coefficient dictionary also supplies an elementary bound. Let $\Pi(u) = \sum_{j=0}^{d} p_j u^j$ and $K_\Pi = \sum_{j=0}^{d} |p_j|(1+\pi)^j$. For real $\sigma \geq 0$, $0 < q < 1$, and integer $M \geq 0$, each full lateral completion tail obeys

$$\left| \sum_{n>M} \frac{(-1)^{n-1}}{n^\sigma} \Pi(-\log n \pm \mathrm{i}\pi) q^n \right| \leq K_\Pi (M+1)^{d-\sigma} q^{M+1} \frac{E_d(q)}{(1-q)^{d+1}}, \tag{1.117}$$

where E_d is the polynomial defined by $\sum_{k\geq 0}(k+1)^d q^k = E_d(q)/(1-q)^{d+1}$, with $E_0 = 1$. Indeed $|\Pi(-\log n \pm \mathrm{i}\pi)| \leq K_\Pi n^d$ for $n \geq 1$, $n = M + 1 + k \leq (M+1)(k+1)$, and $n^{-\sigma} \leq (M+1)^{-\sigma}$. This proves the bound. The jump tail has at most twice the right side; the arithmetic-completion tail has at most the right side. This is a finite evaluation safeguard, not the basis of the cancellation theorem.

1.10.3 Arithmetic rigidity and moving-order cancellation

A transcendence input rules out accidental zeros

Lemma 1.36. *For every integer $n \geq 2$, $\log n/\pi$ is transcendental.*

Proof. Use the classical Gelfond–Schneider theorem in its complex form: if $\alpha \neq 0, 1$ and $\beta \notin \mathbb{Q}$ are algebraic, every fixed value of α^β is transcendental [156, Theorem 3.1]. If $a = \log n/\pi$ were algebraic, $\beta = -\mathrm{i}a$ would be nonreal algebraic, hence irrational. With $\log(-1) = \mathrm{i}\pi$, $(-1)^{-\mathrm{i}a} = \exp(\pi a) = n$, a contradiction. □

Theorem 1.37 (Parity and nonvanishing for every pure jet)**.** *Fix an integer order m and $r \geq 1$, and take $\Pi(u) = u^r$. At every action $n \geq 2$, both the jump coefficient and the arithmetic-completion coefficient are nonzero. Their first surviving actions are exactly*

$$(\nu(J_{m,u^r}), \nu(C_{m,u^r})) = \begin{cases} (1,2), & r \text{ odd}, \\ (2,1), & r \text{ even}. \end{cases} \tag{1.118}$$

For $r = 0$, the jump is identically zero and the completion starts at one. At half-integer centers the two columns of (1.118) are exchanged.

Proof. At $n = 1$, the relevant complex value is $(\mathrm{i}\pi)^r$; its real and imaginary parts alternate with the parity of r. At $n \geq 2$, divide $(-\log n + \mathrm{i}\pi)^r$ by π^r. Its real and imaginary parts are nonzero polynomials in $a = \log n/\pi$ with integer coefficients: the real polynomial has leading term $(-a)^r$, and the imaginary one has leading term $r(-a)^{r-1}$. Neither can vanish at the transcendental number in Lemma 1.36. At integer order the rotation is just $(-1)^m$; at half-integer order it is $\pm\mathrm{i}$, which exchanges the two components. The constant case follows from the completion formula. □

Corollary 1.38 (Rigidity for algebraic normalized coefficients)**.** *The nonvanishing at every action $n \geq 2$ extends from pure monomials to all nonconstant filters $\Pi(u) = c\,p(u/\pi)$, where $c \in \mathbb{R} \setminus \{0\}$ and p has real algebraic coefficients. At integer or half-integer centers no such filter can cancel even one of the two data at action two. In particular arbitrarily high degree in this coefficient class cannot reproduce the deep logarithmic-node filters of Theorem 1.32.*

Proof. The real and imaginary parts of $p(-a + \mathrm{i})$ are nonzero polynomials in a with real algebraic coefficients: their degrees are respectively $\deg p$ and $\deg p - 1$. A number transcendental over $\mathbb{Q}$ cannot be a root of a nonzero polynomial over the algebraic numbers. Apply Lemma 1.36 and then the integer or half-integer rotation. The common nonzero real scale c does not affect vanishing. □

This use of Gelfond–Schneider proves an all-order nonvanishing statement; a finite table of numerical coefficients would not prove it. Conversely, the freely chosen real coefficients in Theorem 1.33 deliberately permit cancellation. There is no contradiction: a pure monomial in the order derivative is a much smaller class of filters.

A normalized sharp jump filter

For $N \geq 1$ define

$$Q_N(u) = \prod_{j=1}^{N} (u + \log j). \tag{1.119}$$

There is a unique real polynomial Π_N with

$$\sin(\pi D)\Pi_N = Q_N, \qquad (\cos(\pi D)\Pi_N)(0) = 1. \tag{1.120}$$

It has degree $N + 1$. Existence and uniqueness follow from the isomorphism modulo constants used in Theorem 1.32; the second condition fixes that constant. One constructive formula is

$$\Pi_N = \frac{1}{\pi} \frac{\pi D}{\sin(\pi D)} \int_0^u Q_N(v)\,\mathrm{d}v + c_N,$$

where c_N enforces the normalization. All operators terminate. Writing $R_N = \cos(\pi D)\Pi_N$, the first two examples are

$$\Pi_1(u) = \frac{u^2}{2\pi} + 1 + \frac{\pi}{2}, \qquad R_1(u) = 1 + \frac{u^2}{2\pi}, \tag{1.121}$$

$$\Pi_2(u) = \frac{u^3}{3\pi} + \frac{\log 2}{2\pi}u^2 + \frac{\pi}{3}u + 1 + \frac{\pi \log 2}{2}, \qquad R_2(u) = 1 + \frac{u^3}{3\pi} + \frac{\log 2}{2\pi}u^2 - \frac{2\pi}{3}u. \tag{1.122}$$

The normalization is intentionally about the *completion*, not about $\Pi_N(0)$.

Theorem 1.39 (Convergent Stokes-neutral curve and its crossover)**.** *Fix an integer $m \geq 0$ and $N \geq 1$. For sufficiently small q, there is a unique holomorphic function $\epsilon_N(q)$ with $\epsilon_N(0) = 0$ for which*

$$J_{m+\epsilon_N(q),\Pi_N}(-\log q) = 0 \tag{1.123}$$

when $q > 0$. For complex q, this identity refers to the convergent q-series of the jump. The function has real Taylor coefficients and

$$\epsilon_N(q) = -\frac{\kappa_{m,N}}{\pi} q^N + O(q^{N+1}), \qquad \kappa_{m,N} = \frac{1}{(N+1)^m} \prod_{j=1}^{N} \log \frac{N+1}{j} > 0. \tag{1.124}$$

The associated completion does not *disappear:*

$$C_{m+\epsilon_N(q),\Pi_N}(-\log q) = (-1)^m q + O(q^2). \tag{1.125}$$

More generally, with $\epsilon = \lambda q^N$,

$$\frac{J_{m+\lambda q^N,\Pi_N}(-\log q)}{-2\mathrm{i}(-1)^m q^{N+1}} = \pi\lambda + \kappa_{m,N} + O(q), \tag{1.126}$$

uniformly for λ in any fixed compact subset of $\mathbb{C}$. At $m = 0$ these are analytically continued lateral realizations; no unregularized order-zero boundary integral is assumed.

Proof. Remove the factor $-2\mathrm{i}(-1)^m q$ from the jump and define

$$\Psi(\epsilon, q) = \sum_{n\geq 1} (-1)^{n-1} n^{-m-\epsilon} q^{n-1} \{\sin(\pi\epsilon) R_N(u_n) + \cos(\pi\epsilon) Q_N(u_n)\}. \tag{1.127}$$

The series is holomorphic for $|q| < 1$ and ϵ in a neighborhood of zero: on compact sets its coefficients have at most a fixed power of n times a fixed power of $\log n$. Its $n = 1$ term is precisely $\sin(\pi\epsilon)$, since $Q_N(0) = 0$ and $R_N(0) = 1$. Consequently

$$\Psi(0, 0) = 0, \qquad \partial_\epsilon \Psi(0, 0) = \pi.$$

The holomorphic implicit-function theorem gives the unique curve. Conjugation and uniqueness imply real Taylor coefficients.

At $\epsilon = 0$, the first N coefficients vanish. At $n = N + 1$,

$$(-1)^N (N+1)^{-m} Q_N(-\log(N+1)) = \kappa_{m,N}.$$

Thus $\Psi(0, q) = \kappa_{m,N} q^N + O(q^{N+1})$. Substitution in the implicit equation, whose ϵ derivative is $\pi + O(q)$, gives (1.124). Taylor expansion with $\epsilon = \lambda q^N$ proves (1.126); higher sine terms and mixed terms are of sufficiently high order, and holomorphy gives uniformity on compact λ sets.

The corresponding completion has normalized coefficient

$$\cos(\pi\epsilon) R_N(u_n) - \sin(\pi\epsilon) Q_N(u_n).$$

Its first term is $(-1)^m q \cos(\pi\epsilon)$. The rest is $O(q^2)$ uniformly for small ϵ, and the implicit curve tends to zero. This proves (1.125). □

Remark 1.40 (What the curve cancels). For a *fixed* order, the sharp filters remove individual initial action coefficients. On the moving-order curve, coefficients at different actions compensate after ϵ is substituted as a function of q. The result is an exact identity of analytic functions, not a statement that every fixed-order Borel singularity has disappeared. It also gives a precise example in which zero total lateral jump does not identify the physical function with the common lateral realization.

1.10.4 Transport along the exact neutral curve

The neutral curve cancels the entire lateral difference by allowing the order to move with the exponential parameter. Inversion must retain this dependence. The following result answers the inverse-transport question in [153] for every fixed filter size and nonnegative integer center; it also supplies an exact convergent coefficient formula.

Theorem 1.41 (A convergent neutral inverse chart and its physical displacement)**.** *Fix $N \geq 1$ and $m \in \mathbb{Z}_{\geq 0}$, and use the normalized filter Π_N and analytic function ε_N of Theorem 1.39, with this order center. Put*

$$\sigma(x) = m + \varepsilon_N(e^{-x}), \qquad P(x) = \Pi_N(\partial_s) F_s(x)\big|_{s=\sigma(x)}, \qquad L^{\pm}(x) = \Pi_N(\partial_s) L_s^{\pm}(x)\big|_{s=\sigma(x)}.$$

For all sufficiently large real x, the two lateral functions coincide: $L^+(x) = L^-(x) = L(x)$. The functions P and L are real analytic, strictly increasing and unbounded on a sufficiently late half-line. Let X denote the inverse of P there. Write

$$C_w = L(w) - P(w), \qquad H_w(t) = \begin{cases} (P(w+t) - P(w))/t, & t \neq 0, \\ P'(w), & t = 0. \end{cases}$$

For any fixed $0 < \rho < 1$, and all sufficiently large w, the series

$$X(P(w) + zC_w) - w = \sum_{k \geq 1} \frac{(zC_w)^k}{k} [t^{k-1}] H_w(t)^{-k} \tag{1.128}$$

converges for $|z| < \rho P'(w)/(2|C_w|)$, where the left side denotes the local holomorphic inverse extending the real one. This disk contains $z = 1$ for large w. In particular, the common lateral inverse at target $L(w)$ equals w, whereas the physical inverse satisfies

$$X(L(w)) - w \sim \frac{(-1)^m e^{-w}}{P'(w)} \sim \begin{cases} \dfrac{\pi w e^{-w}}{(\log w)^N}, & m = 0, \\ \dfrac{(-1)^m \pi (N+1)(m-1)!\, w^{1-m} e^{-w}}{(\log w)^{N+1}}, & m \geq 1. \end{cases} \tag{1.129}$$

The first two exact terms of (1.128) are $zC_w/P'(w) - z^2 P''(w) C_w^2/(2P'(w)^3)$.

Proof. The neutral-curve identity proves equality of the laterals and gives

$$C(x) := L(x) - P(x) = (-1)^m e^{-x} + O(e^{-2x}). \tag{1.130}$$

The underlying normally convergent power series in q and the analytic implicit function $\varepsilon_N(q)$ also give the differentiated estimate and its holomorphic version near the positive ray.

Set $d = N + 1$. The leading coefficient of Π_N is $1/(\pi d)$, since $\sin(\pi D)$ lowers degree by one with leading multiplier πd and its image is the monic Q_N. Uniformly for s in a small fixed disk about m, the Hurwitz realization (1.105) gives the finite Sommerfeld estimate

$$F_s(x) = \frac{x^s}{\Gamma(s+1)} + O(|x|^{m+\eta-2}), \qquad |s - m| \leq \eta < \tfrac{1}{2},$$

on a smaller sector about the positive ray. Here the exponential completion has been absorbed in the remainder. To obtain the estimate and any of its finitely many derivatives, take a slightly

larger compact order disk in the sectorial Hurwitz expansion [93, Eq. 25.11.43], then apply Cauchy's estimates in the order and in x. The odd x^{s-1} term cancels after the shift $1/2$ in the Hurwitz argument; equivalently the first correction in (1.102) has power x^{s-2}. This finite estimate, not termwise differentiation of a divergent infinite series, suffices.

At the fixed center m, applying $\Pi_N(\partial_s)$ to the leading term gives $x^m/m!$ times a polynomial in $\log x$ of degree d, with leading coefficient $1/(\pi d)$. The chain rule for the moving center is

$$P'(x) = \Pi_N(\partial_s)\partial_x F_s(x)|_{s=\sigma(x)} + \sigma'(x)\,\Pi_N(\partial_s)\partial_s F_s(x)|_{s=\sigma(x)}\,, \qquad \sigma'(x) = -e^{-x}\varepsilon_N'(e^{-x}). \tag{1.131}$$

Both $\sigma - m$ and σ' are $O(e^{-Nx})$; compact-order Cauchy estimates bound the terms they introduce by an exponential times a fixed power of x. They are therefore negligible relative to the leading derivative. Thus

$$P'(x) \sim \begin{cases} (\log x)^N/(\pi x), & m = 0, \\ x^{m-1}(\log x)^{N+1}/\big(\pi(N+1)(m-1)!\big), & m \geq 1. \end{cases} \tag{1.132}$$

The same sectorial estimates hold for derivatives and imply $P'(w+t)/P'(w) \to 1$ uniformly on every fixed complex disk in t. The leading expression for P tends to $+\infty$ also when $m = 0$. Together with (1.130), this proves eventual monotonicity, unboundedness and existence of both real inverses.

Uniformly for $|t| \leq \rho$, integration of the derivative estimate gives $H_w(t)/P'(w) \to 1$. For large w its modulus is greater than $1/2$. On $|t| = \rho$ the perturbation zC_w is therefore smaller than $P(w+t) - P(w)$ whenever $|z| < \rho P'(w)/(2|C_w|)$. Rouché's theorem gives exactly one zero, counted with multiplicity, of $P(w+t) - P(w) - zC_w$ in the disk. It is simple and varies holomorphically with z. Lagrange inversion yields (1.128) throughout this disk.

Finally $|C_w|/P'(w) \to 0$, so its radius in z tends to infinity. At $z = 1$ the solution t_w obeys $t_w H_w(t_w) = C_w$, hence $t_w \sim C_w/P'(w)$. Substitution of (1.130) and (1.132) proves the displayed displacement. Expanding $H_w(t) = P'(w) + P''(w)t/2 + \cdots$ gives the second coefficient. □

Thus an exactly vanishing lateral difference can coexist with a nonzero first exponential action in the physical inverse. The sign of that displacement alternates with m. The theorem concerns fixed m, N and an eventual inverse chart; it does not assert a growing-filter limit or an exact global complex inverse radius.

1.10.5 All order-zero jets in Stieltjes and integral coordinates

A complete finite formula in generalized Stieltjes constants

For $\operatorname{Re} a > 0$ use the Hurwitz–Stieltjes convention [93]

$$\zeta(1+z, a) = \frac{1}{z} + \sum_{n\geq 0} \frac{(-1)^n \gamma_n(a)}{n!} z^n. \tag{1.133}$$

In particular $\gamma_0(a) = -\psi(a)$. Define the complete exponential Bell polynomials by

$$\exp\left(\sum_{j\geq 1} b_j \frac{z^j}{j!}\right) = \sum_{r\geq 0} \mathcal{B}_r(b_1, \ldots, b_r)\frac{z^r}{r!}, \qquad \mathcal{B}_0 = 1. \tag{1.134}$$

The following theorem specializes classical Jonquière inversion to a resonance excluded from the positive-integer Gamma-kernel statement of [143].

Theorem 1.42 (Stieltjes coordinates for every zero-order lateral jet)**.** *Let $x > 0$, $a_\pm = \frac{1}{2} \pm \frac{\mathrm{i}x}{2\pi}$, and*

$$\begin{aligned} d_1^\pm &= \gamma + \log(2\pi) \mp \frac{\mathrm{i}\pi}{2}, \\ d_j^\pm &= (-1)^{j+1}(j-1)!\zeta(j), \qquad j \geq 2. \end{aligned} \tag{1.135}$$

For $r \geq 1$ put

$$Z_r^\pm(x) = \mathcal{B}_r(d^\pm) - r\sum_{n=0}^{r-1} \binom{r-1}{n} \gamma_n(a_\pm)\mathcal{B}_{r-1-n}(d^\pm), \qquad Z_0^\pm = 1. \tag{1.136}$$

Then $Z_r^\pm = \partial_s^r L_s^\pm|_{s=0}$ and

$$\partial_s^r F_s(x)|_{s=0} = Z_r^\pm(x) - \sum_{n\geq 1}(-1)^{n-1}(-\log n \mp \mathrm{i}\pi)^r e^{-nx}. \tag{1.137}$$

The formulas hold for every finite r and locally uniformly for $x > 0$.

Proof. The reciprocal-Gamma expansion [154, §5.7] gives

$$\frac{1}{\Gamma(s)} = s\exp\left(\gamma s + \sum_{j\geq 2}\frac{(-1)^{j+1}\zeta(j)}{j}s^j\right), \qquad |s| < 1.$$

In (1.133) substitute $z = -s$. Combining it with (1.105) yields the exact local generating identity

$$L_s^\pm(x) = \exp\left(\sum_{j\geq 1} d_j^\pm \frac{s^j}{j!}\right)\left(1 - \sum_{n\geq 0}\gamma_n(a_\pm)\frac{s^{n+1}}{n!}\right). \tag{1.138}$$

In particular the removable value at zero is 1, not zero. Multiplying Taylor series and taking $r!$ times the coefficient gives (1.136). Applying the exact completion formula at zero gives (1.137). The local Taylor identities and the normally convergent exponential series justify every finite coefficient extraction and the claimed local uniformity. □

For example,

$$Z_1^\pm = d_1^\pm - \gamma_0(a_\pm), \qquad Z_2^\pm = (d_1^\pm)^2 - \zeta(2) - 2d_1^\pm\gamma_0(a_\pm) - 2\gamma_1(a_\pm).$$

The order-zero second physical derivative is therefore an explicit combination of γ_1 at a conjugate pair of points, digamma values, and a convergent log-weighted exponential series. There is no finite-part ambiguity in these identities: the regularization is fixed by the holomorphic product in (1.105).

Independent endpoint-subtracted integral coordinates

A second exact representation is useful both for proof and for independent numerical evaluation. It avoids differentiating a numerically delicate polylogarithm continuation at an integer order.

Theorem 1.43 (Positive regularization and all zero-order moments)**.** *For any real x let $q = e^{-x}$ and $f_0 = (1+q)^{-1}$. Define*

$$M_j(x) = \frac{1}{1+q}\int_0^\infty \frac{(1-e^{-t})(\log t)^j}{t(1+e^{t-x})}\,\mathrm{d}t, \qquad j \geq 0. \tag{1.139}$$

All these integrals are absolutely convergent. With $g_1 = \gamma$ and $g_j = (-1)^{j+1}(j-1)!\zeta(j)$ for $j \geq 2$,

$$F_0(x) = f_0, \qquad \partial_s^r F_s(x)|_{s=0} = r\sum_{j=0}^{r-1}\binom{r-1}{j}\mathcal{B}_{r-1-j}(g)M_j(x) \quad (r \geq 1). \tag{1.140}$$

In particular $H(x) := \partial_s F_s(x)|_{s=0} = M_0(x) > 0$.

Proof. Let $f_x(t) = (1+e^{t-x})^{-1}$. The subtraction has the exact positive form

$$f_x(t) - f_0 e^{-t} = \frac{1-e^{-t}}{(1+e^{-x})(1+e^{t-x})}. \tag{1.141}$$

For $\operatorname{Re} s > 0$, subtract the elementary Gamma integral to get

$$F_s(x) = f_0 + \frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}(f_x(t) - f_0 e^{-t})\,\mathrm{d}t. \tag{1.142}$$

At zero the subtracted numerator is $O(t)$, so the remaining integral is holomorphic for $\operatorname{Re} s > -1$. At infinity it has exponential decay on compact real x sets. Multiplication by $1/\Gamma(s)$ continues (1.142) to a neighborhood of zero. Its jth integral derivative at zero is $M_j(x)$; the majorants $t^{-1+\epsilon}(1+|\log t|^j)$ after subtraction near zero, with $\epsilon > 0$, and exponential majorants at infinity justify this. The reciprocal-Gamma Bell expansion used above now gives (1.140). Positivity of the integrand for $j = 0$ proves the final assertion. □

1.10.6 The order-zero digamma core and its exact Borel kernel

An exact physical identity

Put

$$A(x) = \gamma + \log(2\pi) + \operatorname{Re}\psi\left(\tfrac{1}{2} + \frac{\mathrm{i}x}{2\pi}\right), \qquad x > 0, \tag{1.143}$$

$$B(z) = \sum_{n\geq 2}(-1)^n(\log n)z^n = (\log 2)z^2 - (\log 3)z^3 + (\log 4)z^4 - \cdots, \qquad |z| < 1. \tag{1.144}$$

The letter B here denotes a convergent analytic function, not the Borel transform $\mathcal{B}$.

Proposition 1.44 (Digamma core and both lateral first jets)**.** *For $x > 0$,*

$$H(x) = A(x) - B(e^{-x}), \tag{1.145}$$

$$Z_1^{\pm}(x) = A(x) \mp \mathrm{i}\pi\frac{e^{-x}}{1+e^{-x}}. \tag{1.146}$$

Thus the first zero-order jet has a jump beginning at action one, while its physical correction to the arithmetic average begins at action two.

Proof. By Theorem 1.42, $Z_1^+ = \gamma + \log(2\pi) - \mathrm{i}\pi/2 + \psi(a_+)$. The digamma reflection formula [154, §5.5] gives

$$\psi(1-z) - \psi(z) = \pi \cot(\pi z), \qquad \operatorname{Im} \psi\left(\tfrac{1}{2} + \frac{\mathrm{i}x}{2\pi}\right) = \frac{\pi}{2} \tanh(x/2).$$

Hence the imaginary part of Z_1^+ is $-\pi/(e^x+1)$, and its real part is A. The lower formula follows by conjugation. In (1.137) with $r = 1$, the imaginary tail cancels exactly, while the real tail is $B(e^{-x})$. □

Corollary 1.45 (Symmetry and the global positive core branch)**.** *The even function A satisfies*

$$H(x) + H(-x) = A(x), \qquad A(0) = \log(\pi/2), \qquad H(0) = \tfrac{1}{2}\log(\pi/2).$$

It is strictly increasing for $x > 0$ and maps that half-line onto $(\log(\pi/2), \infty)$.

Proof. For $x > 0$, $H(-x) = B(e^{-x})$, giving the symmetry from (1.145); continuity and evenness give the remaining real points. The classical value $\psi(1/2) = -\gamma - 2\log 2$ gives $A(0)$. For $v = x/(2\pi) > 0$, the convergent trigamma series gives

$$A'(x) = -\frac{1}{2\pi} \operatorname{Im} \sum_{k \geq 0} \frac{1}{(k + 1/2 + \mathrm{i}v)^2} > 0,$$

because each summand has strictly negative imaginary part. The range then follows from the logarithmic asymptotic proved below. This also shows that the exact positive core inverse used later is unique wherever its target exceeds $\log(\pi/2)$. □

The single meromorphic expression, holomorphic near large real points, is

$$A(z) = \gamma + \log(2\pi) + \frac{1}{2}\left\{\psi\left(\tfrac{1}{2} + \frac{\mathrm{i}z}{2\pi}\right) + \psi\left(\tfrac{1}{2} - \frac{\mathrm{i}z}{2\pi}\right)\right\}. \tag{1.147}$$

It is meromorphic in z, with possible poles on the imaginary axis, and agrees with (1.143) on the positive real axis. In particular it is holomorphic on every disk of a fixed radius around a sufficiently large positive real w. We use this expression, not the nonholomorphic operation Re, in complex inverse equations.

A meromorphic minor at the previously excluded order

Theorem 1.46 (Ordinary Borel reconstruction at order zero)**.** *The formal algebraic tail of $A(x) - (\log x + \gamma)$ is*

$$A(x) \sim \log x + \gamma - 2\sum_{k \geq 1} \eta(2k)(2k-1)! x^{-2k}. \tag{1.148}$$

Its ordinary Borel minor is the meromorphic function

$$b_0^{\mathrm{jet}}(\xi) = \frac{1}{\xi} - \frac{\pi}{\sin \pi\xi}, \tag{1.149}$$

with a removable zero at the origin and residue $(-1)^{n-1}$ at $\xi = n \geq 1$. For $x > 0$,

$$\log x + \gamma + \int_{\pm} e^{-x\xi} b_0^{\mathrm{jet}}(\xi)\, \mathrm{d}\xi = Z_1^{\pm}(x), \tag{1.150}$$

$$A(x) = \log x + \gamma + \mathrm{PV} \int_0^{\infty} e^{-x\xi} b_0^{\mathrm{jet}}(\xi)\, \mathrm{d}\xi. \tag{1.151}$$

The principal value is the arithmetic average of the lateral integrals, or the sum of symmetric local principal values at the positive integers.

Proof. The classical generator $\pi\xi/\sin\pi\xi = 1 + 2\sum_{k\ge1}\eta(2k)\xi^{2k}$ shows that

$$b_0^{\mathrm{jet}}(\xi) = -2\sum_{k\ge1}\eta(2k)\xi^{2k-1}.$$

The ordinary Borel convention gives exactly the coefficients in (1.148). The residue statement follows from $(\sin\pi\xi)'|_{\xi=n} = \pi(-1)^n$.

For an independent proof of the exact reconstruction, initially take $\operatorname{Re} x > 0$ and $\operatorname{Im} x < 0$. Rotate an upper tilted Laplace path to $\xi = \mathrm{i}t$. There are no poles inside the rotation sector. On its large arcs, the cosecant has exponential decay away from the real axis and the $1/\xi$ term is suppressed by $e^{-x\xi}$; the negative imaginary part of x ensures decay even at the final imaginary ray. The removable origin gives no small-circle contribution. Thus

$$\begin{aligned}\int_+ e^{-x\xi} b_0^{\mathrm{jet}}(\xi)\,\mathrm{d}\xi &= \int_0^\infty e^{-\mathrm{i}xt}\left(\frac{1}{t} - \frac{\pi}{\sinh\pi t}\right)\mathrm{d}t \\ &= \int_0^\infty \left\{\frac{e^{-\mathrm{i}xu/(2\pi)}}{u} - \frac{e^{-(1/2+\mathrm{i}x/(2\pi))u}}{1-e^{-u}}\right\}\mathrm{d}u.\end{aligned}$$

The standard digamma integral, combined with the Frullani difference, is [155, §5.9]

$$\psi(a) = \int_0^\infty \left(\frac{e^{-u}}{u} - \frac{e^{-au}}{1-e^{-u}}\right)\mathrm{d}u, \quad \operatorname{Re} a > 0,$$

where cancellation at zero is taken inside the integrand. Consequently the preceding integral is

$$\psi\left(\tfrac{1}{2} + \frac{\mathrm{i}x}{2\pi}\right) - \log x + \log(2\pi) - \frac{\mathrm{i}\pi}{2}. \tag{1.152}$$

Both sides continue to a neighborhood of the positive real axis. A fixed upper ray of angle less than $\pi/2$ has an absolutely convergent Laplace integral there, so (1.152) proves the upper formula for real positive x. Conjugation proves the lower formula. Its value is precisely (1.146).

On the real boundary the only singularities are simple poles. Upper minus lower indentation contributes minus $2\pi\mathrm{i}$ times their residues, and the weighted residue sum is absolutely convergent:

$$-2\pi\mathrm{i}\sum_{n\ge1}(-1)^{n-1}e^{-nx} = -2\pi\mathrm{i}\frac{e^{-x}}{1+e^{-x}}.$$

At infinity one may take boundaries midway between the integer poles; exponential decay controls the remaining contour and principal-value tails. Averaging proves (1.151). Finally, a finite Taylor expansion of the germ on a tilted ray, with a Laplace remainder, gives (1.148) to each finite order. □

The kernel (1.149) is the natural $m = 0$ extension of the subtracted Gamma kernel in [143, label `thm:Gamma-kernel`]. The extension has been proved here with a separate contour calculation, rather than by asserting that all hypotheses of the $m \ge 1$ theorem persist unchanged.

Corollary 1.47 (Derivative scales for the exact core)**.** *As $w \to +\infty$,*

$$A(w) = \log w + \gamma - \frac{\pi^2}{6w^2} + O(w^{-4}), \tag{1.153}$$

$$A'(w) = \frac{1}{w} + O(w^{-3}), \qquad A''(w) = -\frac{1}{w^2} + O(w^{-4}). \tag{1.154}$$

The estimates are holomorphically uniform in a fixed closed subsector about the positive real axis, away from the origin. In particular, $A'(w) > 0$ for all sufficiently large real w.

Proof. Use the first term of (1.148). Uniformity follows from the tilted Laplace proof, or the standard sectorial digamma expansion [155, Eq. 5.11.2], applied to the conjugate-pair formula (1.147). Cauchy estimates on disks of radius a fixed positive multiple of w, contained in such a subsector, justify differentiation of each finite asymptotic remainder and give (1.154). This uses holomorphic remainder control, not mere differentiation of a real-axis big-O statement. □

1.10.7 A global inverse and convergent charts at both endpoints

The real inverse exists globally

Theorem 1.48 (Global order-zero inverse)**.** *The function $H(x) = \partial_s F_s(x)|_{s=0}$ is a real-analytic strictly increasing bijection from $\mathbb{R}$ to $(0,\infty)$. Its inverse $X(y)$ is real analytic on $(0,\infty)$. Its two endpoint scales are*

$$H(x) \sim (\log 2)e^{2x} \quad (x \to -\infty), \qquad H(x) \sim \log x \quad (x \to +\infty). \tag{1.155}$$

Proof. Use the positive integral $H = M_0$ from Theorem 1.43. Both factors $(1+e^{-x})^{-1}$ and $(1+e^{t-x})^{-1}$ increase strictly with x for every $t > 0$. Differentiation on compact x intervals is justified by the same endpoint and exponential majorants, and the integrand derivative is strictly positive. Hence $H'(x) > 0$. Local holomorphic continuation of the integral in x in a strip of sufficiently small width gives real analyticity.

For $x < 0$, the original small-argument polylogarithm series and normal order differentiation give $H(x) = B(e^x)$, so $H(x) \sim (\log 2)e^{2x} \to 0$. For large positive x, (1.145) and (1.153) give $H(x) = \log x + \gamma + O(x^{-2}) \to \infty$. Continuity and strict increase prove the range assertion, and the analytic inverse-function theorem applies everywhere because $H' > 0$. □

The small-target logarithmic–square-root chart

Write $a = \log 2$ and

$$b(z) = \frac{B(z)}{az^2} = 1 - \frac{\log 3}{a}z + \frac{\log 4}{a}z^2 - \cdots, \qquad b(0) = 1. \tag{1.156}$$

The square-root parameter for a small target is $t = (y/a)^{1/2} > 0$.

Theorem 1.49 (All coefficients and an explicit convergent endpoint disk)**.** *For $0 < t < \sqrt{3}/40$, the global inverse has the convergent expansion*

$$X(at^2) = \log t + \sum_{N\ge 1} c_N t^N, \qquad c_N = \frac{1}{N}[z^N]b(z)^{-N/2}. \tag{1.157}$$

The analytic correction $X(at^2) - \log t$ extends holomorphically to $|t| < \sqrt{3}/40$. In particular,

$$X(y) = \frac{1}{2}\log\frac{y}{a} + \frac{\log 3}{2a}\left(\frac{y}{a}\right)^{1/2} + \left(\frac{(\log 3)^2}{2a^2} - 1\right)\frac{y}{a} + O(y^{3/2}). \tag{1.158}$$

The radius given here is a sufficient certificate, not an exact radius.

Proof. Let $r = 1/20$. Since $\log n \le n$ for $n \ge 1$,

$$\sup_{|z|\le r}|b(z) - 1| \le \frac{1}{a}\sum_{n\ge 3} nr^{n-2} = \frac{r(3-2r)}{a(1-r)^2} = \frac{58}{361a} < \frac{1}{4}.$$

For the strict last inequality, $\log 2 > 2/3$ follows, for example, from $\log 2 = 2\sum_{k\ge 0}(1/3)^{2k+1}/(2k+1)$; then $58/(361a) < 87/361 < 1/4$. There is a holomorphic square root of b on the disk, equal to one at zero, and $|\sqrt{b(z)}| > \sqrt{3}/2$ there. The function $f(z) = z\sqrt{b(z)}$ has exactly the simple zero 0, and on $|z| = r$ its modulus exceeds $\sqrt{3}/40$.

For $|t| < \sqrt{3}/40$, Rouché's theorem gives exactly one root, counting multiplicity, of $f(z) - t$ inside $|z| < r$. That root is simple, so it defines a holomorphic function $z(t)$ with $z(0) = 0$ and $z'(0) = 1$. Uniqueness glues the local roots throughout the disk. The quotient $z(t)/t$ extends holomorphically and has no zeros; its logarithm with value zero at $t = 0$ therefore exists on the disk. For positive small t, the root is real and positive and $H(\log z(t)) = B(z(t)) = at^2$. The global uniqueness theorem identifies $\log z(t)$ with $X(at^2)$.

For the coefficient formula, set $\Phi(z) = b(z)^{-1/2}$. The equation is $z = t\Phi(z)$ and $\log(z/t) = \log \Phi(z)$. Lagrange–Bürmann inversion gives, for $N \ge 1$,

$$\begin{aligned}[t^N]\log(z(t)/t) &= \frac{1}{N}[z^{N-1}](\log\Phi)'\Phi^N \\ &= \frac{1}{N^2}[z^{N-1}](\Phi^N)' = \frac{1}{N}[z^N]\Phi^N.\end{aligned}$$

This proves (1.157). Expanding the first two coefficients and using $\log 4 = 2\log 2$ proves (1.158). □

The large-target exact-core chart

For large $w > 0$, take the target $y = A(w)$, so w is the exact inverse of the algebraic digamma core rather than an approximation to that inverse. Put

$$q = e^{-w}, \qquad a_w = A'(w), \qquad b_w = A''(w), \qquad c_+ = -\mathrm{i}\pi, \quad c_- = \mathrm{i}\pi. \tag{1.159}$$

Let $G_\pm(y)$ denote the inverse branches near w of the lateral functions $Z_1^\pm(x)$ from (1.146).

Theorem 1.50 (Convergent inverse sectors and the exact second sector)**.** *There are constants $c > 0$ and $w_0 > 0$ such that for each $w > w_0$ the three inverse displacements are holomorphic for $|q| < c/w$ when q is treated as an independent variable. At the physical substitution $q = e^{-w}$, their convergent expansions give*

$$X(A(w)) = w + \frac{\log 2}{a_w}q^2 + O(w^3q^3), \tag{1.160}$$

$$G_\pm(A(w)) = w - \frac{c_\pm}{a_w}q + \left(-\frac{b_w c_\pm^2}{2a_w^3} - \frac{c_\pm^2}{a_w^2} + \frac{c_\pm}{a_w}\right)q^2 + O(w^3q^3). \tag{1.161}$$

The remainder constants can be chosen independently of large w. The physical branch in (1.160) is the global inverse of Theorem 1.48.

Proof. For independent q, the defining equations for the displacements are

$$A(w+\delta) - A(w) - B(qe^{-\delta}) = 0, \tag{1.162}$$

$$A(w+\delta) - A(w) + c_\pm\frac{qe^{-\delta}}{1+qe^{-\delta}} = 0. \tag{1.163}$$

Fix a small radius $\rho > 0$, for example $\rho = 1/4$. The uniform estimates in Corollary 1.47 imply

$$A(w+\delta) - A(w) = \frac{\delta}{w}(1 + O(w^{-1})) \qquad (|\delta| \le \rho).$$

For sufficiently large w, the core difference has exactly one zero in the disk and has boundary modulus at least $\rho/(2w)$; this follows by comparison with δ/w.

On the same disk, the lateral perturbation is bounded by $\pi|q|e^{\rho}/(1-|q|e^{\rho})$. Choose $c>0$ small enough that this is less than $\rho/(2w)$ whenever $|q|\le c/w$, for all sufficiently large w. The physical perturbation $B(qe^{-\delta})$ is $O(|q|^2)$ uniformly on the disk and satisfies the same strict inequality, after increasing w_0 if necessary. Rouché gives exactly one simple zero in each case. The implicit-function theorem and uniqueness give holomorphic roots on the parameter disk, bounded by ρ. Cauchy's estimate bounds their degree-n coefficients by $\rho(w/c)^n$. Hence on $|q|\le c/(2w)$ the remainder after degree two is at most

$$\rho\frac{(w|q|/c)^3}{1-w|q|/c}=O(w^3|q|^3).$$

The physical value $q=e^{-w}$ lies in this region for all large w. This proves actual sector convergence and a uniform remainder; it is not only a formal reversion computation.

For the coefficients, write $\delta=d_1q+d_2q^2+O(q^3)$ and use

$$A(w+\delta)-A(w)=a_w\delta+\tfrac12 b_w\delta^2+O(\delta^3),\qquad \frac{qe^{-\delta}}{1+qe^{-\delta}}=q-(d_1+1)q^2+O(q^3).$$

The first coefficient in the lateral equation gives $d_1=-c/a_w$. The second gives $a_wd_2+\frac12 b_wd_1^2-cd_1-c=0$, which is precisely (1.161). In the physical equation, $B(qe^{-\delta})=(\log 2)q^2+O(q^3)$, so $d_1=0$ and $d_2=\log 2/a_w$. For the physical real q, conjugation and uniqueness make the physical root real. Its defining equation is $H(w+\delta)=A(w)$, and the global monotonicity of H identifies it with $X(A(w))$. □

A larger inverse error than the physical completion

Corollary 1.51 (Digamma-core arithmetic-averaging defect)**.** *With the preceding exact-core normalization,*

$$\frac{G_+(A(w))+G_-(A(w))}{2}-X(A(w))=\left(\frac{\pi^2}{a_w^2}+\frac{\pi^2b_w}{2a_w^3}-\frac{\log 2}{a_w}\right)e^{-2w}+O(w^3e^{-3w}) \tag{1.164}$$

$$\sim\pi^2w^2e^{-2w}. \tag{1.165}$$

By contrast,

$$X(A(w))-w\sim(\log 2)we^{-2w},\qquad G_+(A(w))-G_-(A(w))\sim 2\pi\mathrm{i}we^{-w}. \tag{1.166}$$

In particular the inverse-average error divided by the physical core-inverse correction tends to infinity, with equivalent $\pi^2w/\log 2$.

Proof. Average (1.161), using $c_++c_-=0$ and $c_\pm^2=-\pi^2$, then subtract (1.160). This gives (1.164). Since $a_w=w^{-1}(1+O(w^{-2}))$ and $b_w=-w^{-2}(1+O(w^{-2}))$, the coefficient of e^{-2w} in that formula is $\pi^2w^2+O(w)$. The uniform remainder is smaller by a factor $O(we^{-w})$. The same estimates give the first equivalent in (1.166). Subtracting the two lateral expansions gives its second equivalent. Division proves the ratio. □

This is a model-specific quantitative consequence of the general nonlinear averaging mechanism already proved in [157] and restated in [143]. What is added here is the explicit core, the exact second-sector coefficient, a uniform convergent chart, and the separation by an unbounded factor from the physical correction. It is *not* a counterexample to a correctly formulated nonlinear median summation. Nor does it allege that either cited repository article confused arithmetic averaging with such a summation.

In the target variable. Let $v = e^{y-\gamma}$. Inverting (1.153) gives $w = v+\pi^2/(6v)+O(v^{-3})$. Therefore the physical inverse correction and the inverse-average error have the respective equivalents

$$(\log 2)e^{y-\gamma}\exp(-2e^{y-\gamma}), \qquad \pi^2 e^{2(y-\gamma)}\exp(-2e^{y-\gamma}). \tag{1.167}$$

These are doubly exponential scales in y. Using the exact w for finite computations remains preferable: an algebraic approximation to the core must not be confused with its much smaller exponential correction.

1.10.8 Experimental guidance and remaining questions

The delivered filter verifier was replayed on an isolated copy with mpmath 1.3.0 and SymPy 1.14.0 at seventy decimal digits. Its 165 exact checks include finite operator inversions, conjugate-node interpolation and the first nonlinear inverse coefficients. All 46 numerical checks passed: regularized Stieltjes extraction against positive-integral jets, tilted Borel integrals, noninteger completion, six neutral roots and the small- and large-target inverse charts. A separate replay already filed by the parallel intake is retained as independent run evidence.

For the new neutral inverse theorem, an independent implementation uses Cauchy extraction of order jets from the entire Hurwitz realization, with 64-to-96-point refinement, instead of tiny-step order differentiation. It solves the neutral equation and the physical inverse separately, retaining the moving-order term in (1.131). For $N = 1$, centers $m = 0, 1, 2$ and $w = 10, 20$, the displacement agrees with $C_w/P'(w)$ at the predicted leading scale. At $m = 0, w = 20$, their ratio is approximately 1.00000000065748. The logarithmic leading formula alone converges more slowly: the exact derivative retains the lower powers of $\log w$. These finite floating-point comparisons are diagnostics, not interval certificates or proofs of the infinite assertions. The analytic proof and exact coefficient identity are given above.

The higher-zero-jet Borel germs, conditioning of logarithmic-node interpolation, growing filter size, and the exact complex radius of the small-target inverse remain open directions. The neutral inverse theorem settles fixed m, N transport along the constructed curve, without extending it to those different uniform or global questions.

1.11 Joint coefficient laws and nonlinear composition

The boundary classification in Section 1.7 separates fixed-order coefficient regimes. We now allow the order and feedback to move together through the order-three boundary. Exact Lagrange inversion turns the coefficients into a positive compound-Poisson law; its Fourier integral produces a common sequence of Gaussian order kernels. This connects the marginal Lambert coefficients to a uniform moving-order expansion and to its positive-tail transition.

The joint-coefficient continuation [158] supplies the complete fixed-logarithmic-order theorem and the positive-tail matching below. Its two-derivative composition principle also motivates a further result: Theorem 1.63 isolates the first differing higher endpoint derivative by comparing two models. This supplies an exact hierarchy of spatially differentiated correction kernels beyond the common logarithmic expansion.

Throughout the coefficient discussion, $L = \log n$, $s = 3 + \varepsilon$ and $\lambda = \varepsilon L/2$ are real, with $|\lambda| \le \Lambda < \infty$ fixed. The feedback parameter lies in a fixed compact interval $[b_-, b_+] \subset (0, \infty)$. Constants may depend on these compact sets and the fixed truncation order, but not on the varying parameters or on a lattice location when uniformity in that location is asserted. The logarithm is

principal, with Fourier convention

$$\operatorname{Log}(-iy) = \log|y| - \frac{\pi i}{2}\operatorname{sgn}(y) \qquad (y \in \mathbb{R} \setminus \{0\}). \tag{1.168}$$

We write $\mathfrak{g}(\xi) = (2\pi)^{-1/2}e^{-\xi^2/2}$ for the normal density. Square brackets in $\mathcal{H}_a^{[j]}$ indicate order derivatives, and $\left\{ {j \atop k} \right\}$ denotes a Stirling number of the second kind. Every truncation order is fixed as $n \to \infty$.

1.11.1 The exact coefficient law and the resonant scale

Quadratic feedback as a compound-Poisson coefficient

For real $s > 2$ and $b > 0$, set

$$u = \sqrt{b^2 + 2\zeta(s)}, \qquad A = \zeta(s-1), \qquad v_c = u - b, \qquad R_b = e^{-v_c}. \tag{1.169}$$

Consider the unique germ

$$uv(r) - \frac{v(r)^2}{2} = \operatorname{Li}_s(re^{v(r)}), \qquad v(0) = 0, \qquad a_n = [r^n]v(r). \tag{1.170}$$

Define

$$f(z) = u - \sqrt{u^2 - 2\operatorname{Li}_s(z)} = \sum_{k\geq 1} c_k z^k, \tag{1.171}$$

using the square root equal to u at zero. Since $2\zeta(s) < u^2$, the Taylor expansion of the outer square root converges absolutely on the whole image of $|z| \leq 1$. In particular,

$$c_k \geq \frac{1}{uk^s} > 0, \qquad \sum_{k\geq 1} c_k = f(1) = v_c, \qquad f'(1) = \frac{A}{b}. \tag{1.172}$$

Proposition 1.52 (Exact coefficient–probability identity)**.** *Let X_n be the nonnegative integer-valued compound-Poisson variable with probability generating function*

$$\mathbb{E}z^{X_n} = \exp\{n(f(z) - f(1))\}. \tag{1.173}$$

Then

$$R_b^n a_n = \frac{1}{n}\mathbb{P}(X_n = n), \qquad \mathbb{E}X_n = \frac{nA}{b}. \tag{1.174}$$

The identity holds for every $b > 0$, including $b < A$.

Proof. Put $z = re^v$ in (1.170). Solving the quadratic equation on its germ gives $v = f(z)$, so $z = re^{f(z)}$. Lagrange inversion gives

$$a_n = \frac{1}{n}[z^{n-1}]f'(z)e^{nf(z)} = \frac{1}{n}[z^n]e^{nf(z)}.$$

For the second equality differentiate $e^{nf(z)}$ and compare its z^{n-1} coefficient. Positivity and summability in (1.172) make $\exp\{n\sum c_k(z^k - 1)\}$ a probability generating function: take a Poisson number of jumps of total intensity $nf(1)$, with jump probabilities $c_k/f(1)$. Multiplication by $e^{-nf(1)}$ proves the coefficient identity. Differentiation at $z = 1$ proves the mean formula. □

The endpoint reference R_b is the actual Taylor radius at and above $b = A$. Below that threshold the branch first encounters an interior fold and the radius is larger than R_b [143]. No radius claim is needed for (1.174).

Pairing the colliding Gamma and zeta terms

Write $s = 3 + \varepsilon$ and define

$$E_2(\varepsilon) = \frac{\Gamma(1-\varepsilon)}{(1+\varepsilon)(1+\varepsilon/2)}, \qquad Z(\varepsilon) = \zeta(1+\varepsilon) - \varepsilon^{-1}, \tag{1.175}$$

$$K_\varepsilon(\ell) = Z(\varepsilon) + \frac{1 - E_2(\varepsilon)e^{-\varepsilon\ell}}{\varepsilon}. \tag{1.176}$$

Both expressions are interpreted by removal of the singularity at $\varepsilon = 0$. The elementary values are

$$E_2(0) = 1, \quad E_2'(0) = \gamma - \frac{3}{2}, \quad Z(0) = \gamma, \quad K_0(\ell) = \ell + \frac{3}{2}. \tag{1.177}$$

Gamma recurrence gives $2\Gamma(-2-\varepsilon) = -E_2(\varepsilon)/\varepsilon$. Pairing that term with the zeta pole in the convergent Jonquière expansion yields the following uniform local statement.

Lemma 1.53 (The centered local exponent)**.** *There are $r > 0$ and $C < \infty$, uniform for $b \in [b_-, b_+]$ and real $|\varepsilon| \le 1/4$, such that, for $0 < |t| < r$ and $|\arg t| \le \pi/2$,*

$$f(e^{-t}) - f(1) + \frac{A}{b}t = \frac{t^2}{2b}\{K_\varepsilon(-\operatorname{Log} t) + \nu\} + \mathcal{E}(t), \qquad \nu = \frac{A^2}{b^2}, \tag{1.178}$$

where $|\mathcal{E}(t)| \le C|t|^{5/2}$.

Proof. The classical normally convergent expansion [96, 25.12.12] gives, after pairing,

$$D_s(t) := \zeta(s) - \operatorname{Li}_s(e^{-t}) = At - \frac{t^2}{2}K_\varepsilon(-\operatorname{Log} t) + t^3 J_\varepsilon(t), \tag{1.179}$$

where J_ε is jointly holomorphic for t near zero and ε near the indicated compact interval. Normal convergence can also be checked by the zeta functional equation: the coefficients of the analytic tail have geometric majorants on each smaller disk $|t| < r < 2\pi$. All poles near $\varepsilon = 0$ occur in the displayed paired term.

Using

$$\frac{1 - e^{-\varepsilon\ell}}{\varepsilon} = \ell \int_0^1 e^{-\varepsilon\ell v}\, dv$$

and the bounded regular part of $(E_2(\varepsilon) - 1)/\varepsilon$ gives

$$|K_\varepsilon(-\operatorname{Log} t)| \le C|t|^{-1/4}(1 + |\log|t||) \le C'|t|^{-1/2}.$$

Thus $D_s(t) = At + O(|t|^{3/2}) = O(|t|)$. Uniformly in b,

$$b - \sqrt{b^2 + 2D} = -\frac{D}{b} + \frac{D^2}{2b^3} + O(|D|^3).$$

Inserting (1.179) leaves the displayed terms of (1.178); the difference between D^2 and A^2t^2 is $O(|t|^{5/2})$. This proves the bound, including the imaginary boundary rays by continuation and continuity. □

The nonlinear feedback contributes precisely ν to the quadratic regular part. At criticality $b = A$, this is 1, changing the resonant constant from 3/2 to 5/2.

A scalar equation absorbing the slow logarithms

Define $B = B(n, \varepsilon, b) > 1$ by

$$\boxed{V := \frac{bB^2}{n} = K_\varepsilon(\log B) + \nu.} \tag{1.180}$$

This defines a scale from the paired quadratic exponent; it does not assume that the underlying distribution has a finite variance. Put

$$q = \frac{E_2(\varepsilon)B^{-\varepsilon}}{V}, \qquad \xi_m = \frac{m - nA/b}{B}, \qquad \xi = \xi_n. \tag{1.181}$$

Theorem 1.54 (Existence, uniqueness, and size of the scale)**.** *Uniformly for $|\lambda| \le \Lambda$ and $b \in [b_-, b_+]$,* (1.180) *has exactly one solution $B > 1$ for all sufficiently large n. Let*

$$\varphi(\lambda) = \frac{1 - e^{-\lambda}}{\lambda}, \quad \varphi(0) = 1, \qquad B_0^2 = \frac{nL}{2b}\varphi(\lambda). \tag{1.182}$$

Then

$$\frac{B}{B_0} = 1 + \frac{C_\lambda}{L\varphi(\lambda)} + O\left(\frac{(\log L)^2}{L^2}\right), \tag{1.183}$$

$$C_\lambda = \gamma + \nu + e^{-\lambda}\left\{\frac{1}{2}\log L + \frac{1}{2}\log\frac{\varphi(\lambda)}{2b} - \gamma + \frac{3}{2}\right\}, \tag{1.184}$$

$$q = \frac{2e^{-\lambda}}{L\varphi(\lambda)} + O\left(\frac{\log L}{L^2}\right). \tag{1.185}$$

In particular $q > 0$ and $q \asymp L^{-1}$ on the specified parameter sets.

Proof. Set $H = \log B$ and $F(H) = be^{2H}/n - K_\varepsilon(H) - \nu$. For small real ε, $K_\varepsilon(0) + \nu$ is uniformly positive by (1.177). Thus $F(0) < 0$ for large n. Moreover

$$F'(H) = e^{-\varepsilon H}\left\{\frac{2b}{n}e^{(2+\varepsilon)H} - E_2(\varepsilon)\right\}.$$

The braced expression increases strictly, because $2 + \varepsilon > 0$. It follows that F first decreases and then increases, or increases throughout. In both cases $F(H) \to +\infty$ as $H \to \infty$: the growth exponent 2 dominates the possible growth exponent $-\varepsilon$ of K_ε. Hence there is exactly one positive zero.

Write $H_0 = \log B_0 = \frac{1}{2}\{L + \log L + \log(\varphi(\lambda)/(2b))\}$. The representation (1.176), with $\varepsilon = 2\lambda/L$, gives uniformly, including $\lambda = 0$,

$$K_\varepsilon(H_0) + \nu = \frac{L}{2}\varphi(\lambda) + C_\lambda + O((\log L)^2/L).$$

For example the first term follows from $(1 - e^{-\lambda})/\varepsilon = L\varphi(\lambda)/2$; the next term uses $E_2'(0) = \gamma - 3/2$. First bracket the root by $B = cB_0$ for fixed constants $c < 1$ and $c > 1$; the leading terms have opposite signs. Then the derivative at the bracketed root is of order L. Taylor's theorem in $H - H_0$ improves the bracket to $H - H_0 = C_\lambda/(L\varphi(\lambda)) + O((\log L)^2/L^2)$. Exponentiation gives (1.183). Finally $E_2(\varepsilon)B^{-\varepsilon} = e^{-\lambda}(1 + O(\log L/L))$ and $V = L\varphi(\lambda)/2 + O(\log L)$, proving the q formula. □

1.11.2 Gaussian order derivatives and finite correction formulas

For $\operatorname{Re} a > -1$ and real ξ, define

$$\mathcal{H}_a(\xi) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-i\xi y - y^2/2}(-iy)^a\,dy, \qquad \mathcal{H}_a^{[j]}(\xi) = \partial_a^j \mathcal{H}_a(\xi). \tag{1.186}$$

The logarithm convention is (1.168). All fixed order derivatives are legitimate by domination at zero and infinity. For real a and ξ the two halves of the integral are conjugates.

Proposition 1.55 (Parabolic cylinder and Gamma forms)**.** *In the standard parabolic-cylinder normalization,*

$$\mathcal{H}_a(\xi) = \frac{e^{-\xi^2/4}}{\sqrt{2\pi}} D_a(-\xi), \tag{1.187}$$

$$\mathcal{H}_a(0) = \frac{2^{(a-1)/2}}{\Gamma((1-a)/2)}, \qquad \partial_\xi \mathcal{H}_a = \mathcal{H}_{a+1}. \tag{1.188}$$

In particular $\mathcal{H}_0 = \mathfrak{g}$ and $\mathcal{H}_{2k} = \mathfrak{g}^{(2k)}$. Each function $\mathcal{H}_a^{[j]}$ used below is an ordinary convergent integral, even when a later asymptotic series containing it does not converge.

Proof. The derivative identity follows immediately under the integral. Splitting into $y > 0$ and $y < 0$, then using the Gamma integral, gives

$$\mathcal{H}_a(0) = \frac{2^{(a-1)/2}}{\pi}\Gamma((a+1)/2)\cos(\pi a/2).$$

The reflection formula gives (1.188). Integration by parts in y, with vanishing boundary term $y(-iy)^a e^{-y^2/2 - i\xi y}$, gives

$$\mathcal{H}_a'' + \xi\mathcal{H}_a' + (a+1)\mathcal{H}_a = 0.$$

The function on the right of (1.187) satisfies the same equation by the parabolic-cylinder differential equation. The values of $D_a(0)$ and $D_a'(0)$ [160, 12.2.4–12.2.7] give the same value and first derivative at $\xi = 0$, namely $2^{(a-1)/2}/\Gamma((1-a)/2)$ and $2^{a/2}/\Gamma(-a/2)$. Uniqueness for the ordinary differential equation proves the identity. □

The correction polynomials in two small parameters

Define the finite functions

$$\mathcal{C}_0(\xi; q, \varepsilon) = \mathfrak{g}(\xi),$$

$$\mathcal{C}_j(\xi; q, \varepsilon) = \frac{1}{j!}\sum_{k=1}^{j} \left\{ {j \atop k} \right\} \left(-\frac{q}{2}\right)^k \varepsilon^{j-k} \mathcal{H}_{2k}^{[j]}(\xi), \qquad j \geq 1. \tag{1.189}$$

They are homogeneous of degree j in (q, ε). The first three are

$$\mathcal{C}_1 = -\frac{q}{2}\mathcal{H}_2^{[1]}, \tag{1.190}$$

$$\mathcal{C}_2 = -\frac{q\varepsilon}{4}\mathcal{H}_2^{[2]} + \frac{q^2}{8}\mathcal{H}_4^{[2]}, \tag{1.191}$$

$$\mathcal{C}_3 = -\frac{q\varepsilon^2}{12}\mathcal{H}_2^{[3]} + \frac{q^2\varepsilon}{8}\mathcal{H}_4^{[3]} - \frac{q^3}{48}\mathcal{H}_6^{[3]}. \tag{1.192}$$

The spatial argument ξ is suppressed on the right. These expressions involve order derivatives of the Gaussian kernel; in general they are not a Gaussian times an ordinary polynomial in ξ.

There is also a finite divided-difference form. For a holomorphic function of a, put

$$\mathfrak{D}_\varepsilon^k F(a) = \frac{1}{\varepsilon^k} \sum_{h=0}^{k} (-1)^{k-h} \binom{k}{h} F(a + h\varepsilon), \qquad \mathfrak{D}_0^k F = F^{(k)}. \tag{1.193}$$

The apparent poles are removable. The identity

$$\left(\frac{e^{\varepsilon z} - 1}{\varepsilon} \right)^k = k! \sum_{j \geq k} \left\{ {j \atop k} \right\} \frac{\varepsilon^{j-k} z^j}{j!} \tag{1.194}$$

explains the appearance of the Stirling numbers. We will prove that both the q-ordered finite expression

$$\sum_{k=0}^{N} \frac{(-q/2)^k}{k!} \, \mathfrak{D}_\varepsilon^k \mathcal{H}_a(\xi) \Big|_{a=2k} \tag{1.195}$$

and the total-degree expression $\sum_{j=0}^{N} \mathcal{C}_j(\xi; q, \varepsilon)$ approximate the same lattice law with error $O(L^{-N-1})$. They differ from one another only at the omitted orders. Formula (1.195) can be useful for computation away from $\varepsilon = 0$; near zero it is better evaluated using derivatives or the removable integral formula.

Finite Gamma moments at the center

For integers $j, k \geq 1$ set

$$M_j(k) = \operatorname{Re} \left[(\partial_z + \log 2 + i\pi)^j \Gamma(z) \right]_{z=k+1/2}. \tag{1.196}$$

This is also the convergent moment

$$M_j(k) = \operatorname{Re} \int_0^\infty e^{-t} t^{k-1/2} (\log t + \log 2 + i\pi)^j \, dt. \tag{1.197}$$

Corollary 1.56 (Central coefficient calculus)**.** *For every $j \geq 1$,*

$$\frac{\mathcal{C}_j(0; q, \varepsilon)}{\mathfrak{g}(0)} = \frac{1}{2^j j! \sqrt{\pi}} \sum_{k=1}^{j} \left\{ {j \atop k} \right\} q^k \varepsilon^{j-k} M_j(k). \tag{1.198}$$

In particular

$$\frac{\mathcal{C}_1(0; q, \varepsilon)}{\mathfrak{g}(0)} = \frac{q}{4}(2 - \gamma - \log 2). \tag{1.199}$$

Proof. Differentiate the Gamma–cosine expression in the proof of Proposition 1.55. At $a = 2k$ it gives

$$\mathcal{H}_{2k}^{[j]}(0) = \frac{(-1)^k 2^{k-j-1/2}}{\pi} M_j(k).$$

Substitution in (1.189) cancels the two factors $(-1)^k$ and proves (1.198). For $j = k = 1$, use $\Gamma(3/2) = \sqrt{\pi}/2$ and $\psi(3/2) = 2 - \gamma - 2\log 2$. □

These formulas can be converted into polygamma values, harmonic numbers, zeta values, and the Euler constant by repeated differentiation of Γ and the half-integer polygamma recurrences. The moment form keeps all signs and factors of two in a single finite expression.

1.11.3 The uniform coefficient transseries

We now prove the main result. The definition of B in (1.180) incorporates all logarithmic changes of the quadratic coefficient, including the slowly varying feedback contribution.

Theorem 1.57 (Every fixed logarithmic order, uniformly on the lattice)**.** *Fix $\Lambda < \infty$, $0 < b_- < b_+ < \infty$, and an integer $N \geq 0$. For $L = \log n$, $|\varepsilon L/2| \leq \Lambda$, and $b \in [b_-, b_+]$, let X_n, B, q, and ξ_m be defined by (1.173), (1.180), and (1.181). Then*

$$\sup_{m\in\mathbb{Z}} \left| B\mathbb{P}(X_n = m) - \sum_{j=0}^{N} \mathcal{C}_j(\xi_m; q, \varepsilon) \right| \leq C_N L^{-N-1} \tag{1.200}$$

for all sufficiently large n, uniformly in the stated parameters. Consequently the feedback coefficients satisfy

$$\boxed{nBR_b^n a_n = \mathfrak{g}(\xi) + \sum_{j=1}^{N} \frac{1}{j!} \sum_{k=1}^{j} \left\{ {j \atop k} \right\} \left(-\frac{q}{2}\right)^k \varepsilon^{j-k} \mathcal{H}_{2k}^{[j]}(\xi) + O(L^{-N-1}).} \tag{1.201}$$

The finite divided-difference expression (1.195) has the same error bound.

The theorem resolves the joint coefficient target Q1 in [143]. It is an absolute estimate. For bounded ξ it gives relative Gaussian asymptotics because $\mathfrak{g}$ has a positive minimum on each compact real interval. Relative conclusions for unbounded ξ will be proved separately.

Uniform control of the whole Fourier contour

Lemma 1.58 (Uniform Fourier domination)**.** *Let*

$$\chi_n(y) = \mathbb{E} \exp\left(iy \frac{X_n - nA/b}{B} \right). \tag{1.202}$$

There are positive constants c_1, c_2, uniform in the moving-order window and in b, such that

$$|\chi_n(y)| \leq e^{-c_1 y^2} \qquad (|y| \leq Bn^{-1/4}), \tag{1.203}$$

$$|\chi_n(y)| \leq e^{-c_2\sqrt{n}} \qquad (Bn^{-1/4} \leq |y| \leq \pi B). \tag{1.204}$$

Proof. By positivity of the intensities,

$$-\log|\chi_n(B\theta)| = n \sum_{k\geq 1} c_k (1 - \cos k\theta).$$

For $|\theta| \leq n^{-1/4}$ retain $1 \leq k \leq \lfloor n^{1/4} \rfloor$. Then $|k\theta| \leq 1$, so $1 - \cos k\theta \geq ck^2\theta^2$. The parameters keep u bounded above. Moreover $k^{-\varepsilon} \geq e^{-\Lambda/2}$ on this range, since $|\varepsilon| \log k \leq \Lambda/2$. Therefore

$$-\log|\chi_n(B\theta)| \geq cn\theta^2 \sum_{k\leq n^{1/4}} k^{-1-\varepsilon} \geq c' nL\theta^2.$$

Since $B^2 \asymp nL$, this proves (1.203). For $n^{-1/4} \leq |\theta| \leq \pi$, retain only $k = 1$. Here $c_1 = 1/u$ in the intensity notation and $1 - \cos\theta \geq 2\theta^2/\pi^2$, giving $-\log|\chi_n(B\theta)| \geq cn\theta^2 \geq c\sqrt{n}$. The centering phase has modulus one, proving the second assertion. □

This argument is insensitive to the cancellation of the Gamma and zeta poles. It uses the exact nonnegative coefficients and supplies the uniformity that a fixed-order local-limit theorem alone would not provide.

The exact central reduction

For $z \in \mathbb{C}$, define the entire divided exponential

$$h_\varepsilon(z) = \frac{e^{\varepsilon z} - 1}{\varepsilon}, \qquad h_0(z) = z. \tag{1.205}$$

Take $t = -iy/B$ in Lemma 1.53. The scale equation removes the constant quadratic term and gives

$$\chi_n(y) = \exp\left\{ -\frac{y^2}{2} + \frac{q}{2} y^2 h_\varepsilon(\mathrm{Log}(-iy)) + R_n(y) \right\}, \quad |R_n(y)| \le C n B^{-5/2} |y|^{5/2}, \tag{1.206}$$

whenever $0 < |y| < rB$. The limit at zero is understood continuously. The sign in this formula follows from $(-iy)^2 = -y^2$ and

$$K_\varepsilon(\log B - \mathrm{Log}(-iy)) - K_\varepsilon(\log B) = -E_2(\varepsilon) B^{-\varepsilon} h_\varepsilon(\mathrm{Log}(-iy)).$$

Since $nB^{-5/2} = O(n^{-1/4} L^{-5/4})$, the local nonlinear remainder is smaller than every fixed inverse power of L.

A finite expansion with an integrable remainder

Proof of Theorem 1.57. Choose $T_L = \sqrt{D_N \log L}$ with a sufficiently large constant D_N. By Lemma 1.58, the part of the scaled Fourier integral with $T_L < |y| \le \pi B$ has absolute size $O(L^{-N-1})$: use the Gaussian bound up to $Bn^{-1/4}$ and the bound $2\pi B e^{-c_2\sqrt{n}}$ beyond it.

On $|y| \le T_L$, put $z = \mathrm{Log}(-iy)$ and $w = (q/2) y^2 h_\varepsilon(z)$. The inequality

$$|h_\varepsilon(z)| \le |z| e^{|\varepsilon z|} \tag{1.207}$$

follows from $h_\varepsilon(z) = z \int_0^1 e^{v\varepsilon z}\, dv$. It proves $\sup_{|y| \le T_L} |w| = o(1)$. Near $y = 0$ one uses $y^2 |\log|y|| \, |y|^{-|\varepsilon|} \to 0$; no uniform bound on $|\varepsilon \log|y||$ is assumed. For $1 \le |y| \le T_L$ the bound is $O(L^{-1} \log L \log\log L)$. Equation (1.206) and the estimate on R_n therefore allow us to replace χ_n by $e^{-y^2/2} e^w$ in the central integral with error $O(nB^{-5/2})$.

Expand the exponential only to the finite order N:

$$e^w = \sum_{k=0}^{N} \frac{w^k}{k!} + O_N(|w|^{N+1}).$$

The error is integrable against $e^{-y^2/2}$ with bound $C_N q^{N+1}$. Indeed (1.207) bounds the integrand near zero by a constant times $|y|^{2N+2-(N+1)|\varepsilon|} (1 + |\log|y||)^{N+1}$; at infinity a Gaussian dominates the polynomial and logarithmic factors. For the fixed N, take n large enough that $(N+1)|\varepsilon| < 1/2$. All these integrals then have a common majorant.

For each $k \le N$, use

$$h_\varepsilon(z)^k = z^k \int_{[0,1]^k} e^{\varepsilon z (v_1 + \cdots + v_k)}\, dv_1 \cdots dv_k. \tag{1.208}$$

Taylor expansion in ε through degree $N - k$ gives

$$h_\varepsilon(z)^k = k! \sum_{j=k}^{N} \left\{ {j \atop k} \right\} \frac{\varepsilon^{j-k} z^j}{j!} + \mathcal{R}_{k,N}(z),$$

where

$$|\mathcal{R}_{k,N}(z)| \le C_N|\varepsilon|^{N+1-k}|z|^{N+1}e^{k|\varepsilon z|}.$$

After multiplication by $q^k|y|^{2k}e^{-y^2/2}$, its integral is $O(q^k|\varepsilon|^{N+1-k}) = O(L^{-N-1})$, by the same domination. Thus the total-degree polynomial in q,ε has an $L^1(dy)$ error of the asserted order. Its finite terms may be extended from $|y| \le T_L$ to the full real axis: their Gaussian tails are $O(L^{-N-1})$ after enlarging D_N if needed.

The resulting integrals are

$$\frac{1}{2\pi}\int_{\mathbb{R}} e^{-i\xi_m y - y^2/2}(-iy)^{2k}(\mathrm{Log}(-iy))^j\,dy = \mathcal{H}_{2k}^{[j]}(\xi_m).$$

Here the factor y^{2k} in w^k contributes $(-1)^k$, producing $(-q/2)^k$. Lattice Fourier inversion says

$$B\mathbb{P}(X_n = m) = \frac{1}{2\pi}\int_{-\pi B}^{\pi B} e^{-i\xi_m y}\chi_n(y)\,dy.$$

Every error bound above is an L^1 bound before insertion of the phase. Since $|e^{-i\xi_m y}| = 1$, it is uniform in all integers m. This proves (1.200). The exact coefficient identity gives (1.201).

If one retains $h_\varepsilon(z)^k$ instead of Taylor-expanding in ε, apply finite Fourier inversion to

$$(-iy)^{2k}h_\varepsilon(\mathrm{Log}(-iy))^k.$$

This gives precisely (1.195). The same q^{N+1} estimate proves its assertion. □

Remark 1.59 (The full formal exponential is not the Fourier law). The proof integrates a finite expansion in the central region and uses the exact characteristic function on its complement. It does not replace the full Fourier law by $\exp\{-y^2/2 + (q/2)y^2h_\varepsilon(\mathrm{Log}(-iy))\}$ on the whole real axis. For fixed positive q,ε the latter expression can grow at large $|y|$. Nor does the theorem assert convergence of the infinite sum over j. The finite integrals and all fixed-order remainders are the claimed objects.

The elementary normalization requested in Q1

Put $\xi_0 = n(1 - A/b)/B_0$, using (1.182). On compact ξ_0 sets, Theorem 1.54 and Theorem 1.57 give

$$\begin{aligned} nB_0R_b^n a_n = \mathfrak{g}(\xi_0) + \frac{C_\lambda\mathcal{H}_2(\xi_0) - e^{-\lambda}\mathcal{H}_2^{[1]}(\xi_0)}{L\varphi(\lambda)} \\ + O\left(\frac{(\log L)^2}{L^2}\right). \end{aligned} \tag{1.209}$$

To verify the scale-change term, write $B/B_0 = 1 + \delta$; then $(B_0/B)\mathfrak{g}(\xi_0B_0/B) = \mathfrak{g}(\xi_0) - \delta(\mathfrak{g} + \xi_0\mathfrak{g}')(\xi_0) + O(\delta^2)$, and $-(\mathfrak{g} + \xi\mathfrak{g}') = \mathcal{H}_2$.

In particular, if ξ_0 remains bounded, the leading statement is

$$n\sqrt{\frac{n\log n}{2b}\varphi(\lambda)}\,R_b^n a_n = \mathfrak{g}\left(\frac{n(1-\zeta(2+\varepsilon)/b)}{\sqrt{n\log n\,\varphi(\lambda)/(2b)}}\right) + o(1). \tag{1.210}$$

At the critical choice $b = A = \zeta(2+\varepsilon)$,

$$a_n \sim R_b^{-n}\sqrt{\frac{A}{\pi\varphi(\lambda)}}\,n^{-3/2}(\log n)^{-1/2}, \tag{1.211}$$

uniformly over the bounded moving-order window. This is a joint transition law; letting λ tend to infinity is outside its stated uniform parameter set.

1.11.4 Lambert resummation and nonlinear universality

A convergent chart for the scale itself

The coefficient expansion is asymptotic. The scalar scale B has a stronger analytic structure that explains how it absorbs the $\log\log n$ corrections.

Proposition 1.60 (Convergent two-variable scale chart)**.** *Let $h = L^{-1}$, $w = L^{-1}\log L$, and $c = \log(\varphi(\lambda)/(2b))$. There is a holomorphic germ $Y(h, w; \lambda, b)$, with $Y(0, 0; \lambda, b) = 1$, uniform on compact real (λ, b) sets as above, such that*

$$B^2 = B_0^2 Y(L^{-1}, L^{-1}\log L; \lambda, b). \tag{1.212}$$

It is determined by the removable holomorphic equation

$$\begin{aligned} Y = {} & \frac{1 - e^{-\lambda} E_2(2\lambda h) e^{-\lambda w - \lambda h(c + \log Y)}}{\lambda\varphi(\lambda)} \\ & + \frac{2h}{\varphi(\lambda)} \left\{ Z(2\lambda h) + \frac{\zeta(2 + 2\lambda h)^2}{b^2} \right\}. \end{aligned} \tag{1.213}$$

Its Taylor series in the two independent variables converges after physical substitution for sufficiently large L.

Proof. The quotient is holomorphic at $\lambda = 0$, since its numerator vanishes there. At $h = w = 0$, the right side equals $(1 - e^{-\lambda})/(\lambda\varphi(\lambda)) = 1$ and its Y derivative is zero. The implicit function theorem therefore applies with derivative one after moving Y to the left. A finite compactness cover and uniqueness give a common neighborhood. Substituting $B^2 = B_0^2 Y$ into (1.180) gives (1.213) exactly. The unique large positive solution from Theorem 1.54 identifies the branch. □

Exact marginal Lambert scale

At $\varepsilon = 0$, equation (1.180) is

$$\frac{bB^2}{n} = \log B + \frac{3}{2} + \nu, \qquad \nu = \frac{\zeta(2)^2}{b^2}.$$

Set $T = 2bB^2/n$. Then

$$T - \log T = L + 3 + 2\nu - \log(2b), \tag{1.214}$$

$$T = -W_{-1}\left(-\frac{2b}{n} e^{-3-2\nu}\right), \qquad B^2 = \frac{n}{2b} T, \qquad q = \frac{2}{T}. \tag{1.215}$$

The lower real branch is required because $T \to \infty$ [146]. This is a scale for the coefficient law. The coefficient scale and the earlier inverse-function Lambert core have different constants; the following exact matching explains them.

Corollary 1.61 (All marginal central coefficients as constants)**.** *Let $s = 3$, $b = A = \zeta(2)$, and let $T > 1$ solve*

$$T - \log T = L + 5 - \log(2A). \tag{1.216}$$

Define

$$D_j = \frac{1}{\sqrt{\pi}\, j!} \operatorname{Re}\left[(\partial_z + \log 2 + i\pi)^j \Gamma(z)\right]_{z=j+1/2}, \qquad D_0 = 1. \tag{1.217}$$

For each fixed N,

$$R_b^n a_n = \sqrt{\frac{A}{\pi}}\, n^{-3/2} T^{-1/2} \left\{ \sum_{j=0}^{N} \frac{D_j}{T^j} + O(T^{-N-1}) \right\}. \tag{1.218}$$

If $Q_j(Y)$ denotes the Gamma-moment polynomial in Section 1.8, then exactly

$$D_j = Q_j(-\log 2). \tag{1.219}$$

Thus the exact Lambert scale turns the previous logarithmic polynomials into a sequence of constants.

Proof. At $\varepsilon = 0$ only $k = j$ survives in (1.189). Since $q = 2/T$, the normalized central coefficient of T^{-j} is $(-1)^j \mathcal{H}_{2j}^{[j]}(0)/(j!\mathfrak{g}(0))$. Equation (1.198) gives precisely (1.217). The leading prefactor is $\mathfrak{g}(0)/(nB) = \sqrt{A/\pi}\, n^{-3/2} T^{-1/2}$. Finally the predecessor defines

$$Q_j(Y) = \frac{1}{\sqrt{\pi}\, j!} \operatorname{Re}\left[(\partial_z - Y + i\pi)^j \Gamma(z) \right]_{z=j+1/2},$$

so substitution gives (1.219), with no asymptotic comparison needed. □

For $C = \gamma + \log 2$, the first constants are

$$\begin{aligned} D_1 &= 1 - \frac{C}{2}, \\ D_2 &= 1 - 2C + \frac{3C^2}{8} - \frac{3\pi^2}{16}. \end{aligned} \tag{1.220}$$

Re-expanding (1.216) in L gives, in particular,

$$R_b^n a_n = \sqrt{\frac{A}{\pi}}\, n^{-3/2} L^{-1/2} \left\{ 1 - \frac{\log L - \log A + \gamma + 3}{2L} + O\left(\frac{(\log L)^2}{L^2} \right) \right\}. \tag{1.221}$$

This agrees with the integrated marginal result, including its constant $\gamma + 3$. At higher orders the same conclusion follows from the exact identity (1.219) and the convergent scale chart, rather than from a comparison of a few numerical values.

Positive analytic outer functions

The square root is not essential to the all-order logarithmic calculus. Let G vary in a family of holomorphic functions satisfying the following uniform hypotheses for all sufficiently small real ε:

(i) $G(0) = 0$, and the Taylor coefficients of G at zero are nonnegative real numbers;

(ii) there are $\eta, M > 0$ such that G is holomorphic on $|w| < \zeta(3+\varepsilon) + \eta$ and bounded by M there;

(iii) $G'(0) \geq \eta$.

Shrinking η makes it harmless to use the same symbol for the uniform lower bound and the disk margin. Compact families with fixed positive margins satisfy these conditions. Put

$$f(z) = G(\mathrm{Li}_{3+\varepsilon}(z)), \quad c = G'(\zeta(3+\varepsilon)), \quad \kappa = \frac{A^2 G''(\zeta(3+\varepsilon))}{c}, \quad \mu = cA. \tag{1.222}$$

Then c is bounded above and away from zero, and κ is nonnegative and bounded. Define X_n using (1.173) and set

$$\frac{B^2}{nc} = K_\varepsilon(\log B) + \kappa =: V, \qquad q = \frac{E_2(\varepsilon) B^{-\varepsilon}}{V}, \qquad \xi_m = \frac{m - n\mu}{B}. \tag{1.223}$$

Theorem 1.62 (Two-derivative composition principle)**.** *Under the preceding uniform hypotheses, Theorem 1.57 holds with (1.223) in place of (1.180)–(1.181), and with precisely the same functions $\mathcal{C}_j$. The leading scale is*

$$B^2 \sim \frac{ncL}{2}\varphi(\lambda). \tag{1.224}$$

For the germ $v(r) = f(re^{v(r)})$, write $a_n = [r^n]v(r)$ and $R = e^{-f(1)}$. Then $nBR^n a_n$ has the same expansion at $\xi = (n - n\mu)/B$.

After normalization by $f(1)$ and centering, all inverse-logarithmic orders depend on the outer function only through its first two derivatives at $\zeta(3+\varepsilon)$. Higher derivatives contribute to the remainder beyond every fixed inverse-logarithmic order.

Proof. The positive Taylor coefficients and the linear term of G imply $[z^k]f(z) \geq \eta k^{-3-\varepsilon}$. Uniform boundedness on the larger disk supplies uniform endpoint derivatives by Cauchy's estimates. Writing $D = \zeta(s) - \mathrm{Li}_s(e^{-t})$, Taylor expansion at the endpoint gives

$$G(\zeta(s) - D) - G(\zeta(s)) = -cD + \frac{1}{2}G''(\zeta(s))D^2 + O(|D|^3).$$

The proof of Lemma 1.53 therefore yields

$$f(e^{-t}) - f(1) + \mu t = \frac{ct^2}{2}\{K_\varepsilon(-\operatorname{Log} t) + \kappa\} + O(|t|^{5/2}).$$

The scale existence proof applies with b replaced by $1/c$ and ν replaced by κ. The positive-intensity proof of Lemma 1.58 also applies verbatim with the indicated uniform lower bound. The entire finite-expansion proof depends only on these two facts, so it gives the same $\mathcal{C}_j$ and remainder. Lagrange inversion proves the last coefficient identity exactly as in Proposition 1.52. Terms involving higher derivatives of G begin at $O(D^3)$ and are covered by the uniform local remainder. □

For the square-root family, choose the disk margin smaller than $b_-^2/2$. Its branch point is at $w = u^2/2 = \zeta(s) + b^2/2$, and $G'(0) = 1/u$ is uniformly positive. Thus the main quadratic model satisfies the hypotheses, with uniform boundedness on a slightly smaller common margin.

Outer function $G(w)$	c	κ
aw	a	0
$u - \sqrt{u^2 - 2w}$	b^{-1}	A^2/b^2
$a(e^{\beta w} - 1)/\beta$	$ae^{\beta\zeta(s)}$	βA^2
$-(a/\beta)\log(1 - \beta w)$	$a/(1 - \beta\zeta(s))$	$\beta A^2/(1 - \beta\zeta(s))$

Here $a > 0$, $\beta > 0$, and in the last row $\beta\zeta(s) < 1$ with a uniform margin. The limit $\beta = 0$ recovers the linear row. Compact parameter subsets satisfying the stated margins provide concrete examples of the theorem.

The principle does not say that these models have the same full transseries. Their coefficients at exponential scales in L can depend on higher endpoint derivatives and on other singularities.

1.11.5 The first differing endpoint derivative

The composition principle identifies all logarithmic orders but leaves the higher derivatives of the outer function invisible at that scale. A comparison of two models removes the common terms exactly. It gives a hierarchy of smaller coefficient laws, without subtracting an infinite formal expansion or choosing a summation for that expansion.

Theorem 1.63 (Transport of a matched endpoint jet)**.** *Fix an integer $r \geq 3$. Let G_1, G_2 satisfy the uniform positivity and analyticity hypotheses of Theorem 1.62, and suppose at $z_s = \zeta(3+\varepsilon)$ that*

$$G_1^{(\ell)}(z_s) = G_2^{(\ell)}(z_s) \quad (1 \leq \ell < r), \qquad \delta_r = G_1^{(r)}(z_s) - G_2^{(r)}(z_s).$$

Their endpoint values need not agree. Define $f_i = G_i \circ \mathrm{Li}_s$ and the laws $\mathbb{E}z^{X_{i,n}} = \exp\{n(f_i(z) - f_i(1))\}$. They have the common parameters c, κ, μ, B, q in (1.222)–(1.223). For each fixed $N \geq 0$, uniformly in the compact moving-order window and all integers m,

$$\begin{aligned} B\{\mathbb{P}(X_{1,n} = m) - \mathbb{P}(X_{2,n} = m)\} = \frac{(-1)^r A^r \delta_r n}{r! B^r} \sum_{j=0}^{N} \partial_\xi^r \mathcal{C}_j(\xi_m; q, \varepsilon) \\ + O\Big(\frac{n}{B^r} L^{-N-1}\Big). \end{aligned} \tag{1.225}$$

The exact finite correction functions are

$$\begin{aligned} \partial_\xi^r \mathcal{C}_0(\xi) &= \mathcal{H}_r(\xi), \\ \partial_\xi^r \mathcal{C}_j(\xi; q, \varepsilon) &= \frac{1}{j!} \sum_{k=1}^{j} \left\{ {j \atop k} \right\} \Big(-\frac{q}{2}\Big)^k \varepsilon^{j-k} \mathcal{H}_{2k+r}^{[j]}(\xi) \quad (j \geq 1). \end{aligned} \tag{1.226}$$

For $v_i(\rho) = f_i(\rho e^{v_i(\rho)})$, $a_{i,n} = [\rho^n] v_i$ and $R_i = e^{-f_i(1)}$, the left side at $m = n$ is exactly $nB(R_1^n a_{1,n} - R_2^n a_{2,n})$.

Proof. Set $a_n^* = n/B^r$, and write $\Psi_i(y) = \log \chi_{i,n}(y)$ for the explicitly centered Poisson exponent, not an unspecified branch of the logarithm of its exponential. With $t = -iy/B$ and $D = \zeta(s) - \mathrm{Li}_s(e^{-t})$, endpoint Taylor expansion gives

$$\Psi_1(y) - \Psi_2(y) = \frac{n(-1)^r \delta_r}{r!} D^r + O(n|D|^{r+1}). \tag{1.227}$$

The uniform local estimate proved above has $D = At + O(|t|^{3/2})$. Consequently, for $|t|$ sufficiently small,

$$\frac{\Psi_1 - \Psi_2}{a_n^*} = \frac{(-1)^r \delta_r A^r}{r!} (-iy)^r + O(B^{-1/2}|y|^{r+1/2}), \qquad |\Psi_1 - \Psi_2| \leq C a_n^* |y|^r.$$

For $0 \leq u \leq 1$, the convex combination $G_u = (1-u)G_2 + uG_1$ obeys the same uniform hypotheses and has exactly the same first two endpoint derivatives. Its centered characteristic function is $\chi_u = \exp\{(1-u)\Psi_2 + u\Psi_1\}$, and

$$\chi_1 - \chi_2 = (\Psi_1 - \Psi_2) \int_0^1 \chi_u \, du. \tag{1.228}$$

The positive-intensity Fourier bound, uniformly in u, therefore gives

$$\frac{|\chi_1 - \chi_2|}{a_n^*} \leq C|y|^r e^{-cy^2} \qquad (|y| \leq Bn^{-1/4}).$$

Beyond this range, the separate bounds on χ_i give total integral at most $2\pi B \, 2e^{-c\sqrt{n}}/a_n^*$, smaller than every fixed inverse power of L. Choose $T_L = \sqrt{D_{N,r} \log L}$ with a sufficiently large fixed constant. The remaining tail $T_L < |y| \leq Bn^{-1/4}$ is $O(L^{-N-1})$ after division by a_n^*. Thus the

comparison retains its smaller scale on the whole contour; subtracting the two unscaled remainder estimates would not suffice.

On $|y| \le T_L$, each χ_u has the same quadratic core (1.206), with local error $O(nB^{-5/2}|y|^{5/2})$, uniformly in u. Apply the finite exponential and divided-exponential expansions from the all-order proof, now with the extra integrable weight $|y|^r$. They give an L^1 error $O(L^{-N-1})$. The additional errors $O(B^{-1/2})$ and $O(nB^{-5/2})$ are smaller than every such order. The Gaussian controls all fixed polynomial weights, while near zero the powers remain integrable for sufficiently small $|\varepsilon|$. Each resulting Fourier integral contains $(-iy)^r$, so differentiation in ξ shifts $\mathcal{H}_{2k}^{[j]}$ to $\mathcal{H}_{2k+r}^{[j]}$. The phase $e^{-i\xi_m y}$ has modulus one, proving uniformity in m. Finally apply Lagrange inversion separately to each normalized model. □

This result isolates the first discrepancy after matching $r-1$ endpoint derivatives. Its scale is $n/B^r \asymp n^{1-r/2}L^{-r/2}$ in the scaled probability. It does not assert a complete transseries for a single model or a uniform expansion as r or N grows. The absolute assertion also permits $\delta_r = 0$.

Corollary 1.64 (The cubic discrepancy at the critical center)**.** *Under the theorem's assumptions with $r = 3$ and $\mu = 1$,*

$$R_1^n a_{1,n} - R_2^n a_{2,n} = -\frac{A^3\delta_3 q}{3B^4} + O(B^{-4}L^{-2}). \tag{1.229}$$

In particular this is an asymptotic equivalence if $|\delta_3|$ is bounded away from zero. The first nonzero central term has order $n^{-2}L^{-3}$, although the leading shifted Gaussian vanishes there.

Proof. The exact central formula gives $\mathcal{H}_3(0) = 0$ and $\mathcal{H}_5^{[1]}(0) = -4$. The latter follows by differentiating $2^{(a-1)/2}/\Gamma((1-a)/2)$ at $a = 5$; the derivative of reciprocal Gamma at -2 is 2. Hence $\partial_\xi^3 \mathcal{C}_1(0) = 2q$. Use $N = 1$ in (1.225) and divide by nB. Since $q \asymp L^{-1}$, the error has smaller order as asserted. □

A positive test family

Put $z_s = \zeta(s)$, $d_s = 1 + 2z_s + 3z_s^2$, and

$$G_\theta(w) = \frac{w + w^2 + w^3 + \theta\{(w - z_s)^3 + z_s^3\}}{Ad_s}, \qquad 0 \le \theta \le \frac{1}{6z_s}. \tag{1.230}$$

All three Taylor coefficients are strictly positive and $G_\theta(0) = 0$. At the endpoint, $c = 1/A$, $\mu = 1$ and $\kappa = A^2(2 + 6z_s)/d_s$ are independent of θ, whereas $G_\theta''' - G_0''' = 6\theta/(Ad_s)$. Thus increasing θ makes the normalized diagonal coefficient eventually smaller, with the explicit leading difference (1.229). Each coefficient must use its own R_θ; the endpoint values differ by $\theta z_s^3/(Ad_s)$.

Finite positive coefficient recurrences provide a direct experimental test without using Fourier asymptotics to compute the coefficients. If $f(z) = \sum_{k\ge1} f_k z^k$, compute

$$p_0 = e^{-nf(1)}, \qquad p_m = \frac{n}{m}\sum_{k=1}^{m} k f_k p_{m-k}, \qquad R^n a_n = \frac{p_n}{n}.$$

The preserved diagnostics use these recurrences for both models and compare their difference with the shifted-kernel formula. Such finite floating-point comparisons guide and check the identities; the uniform remainder is supplied by the contour argument above.

1.11.6 A growing positive window and its Lambert transition

The absolute error in Theorem 1.57 is uniform in the lattice location. Its usefulness is therefore not confined to bounded standardized displacements. We now use its explicit kernels to obtain a relative approximation in growing positive windows.

Throughout this section, put $L = \log n$ and retain the scale B, the parameters q, ε, and the centered coordinate $\xi = (m - nA/b)/B$ of Theorem 1.57. Write

$$\mathcal{P}_n(\xi) = B\mathbb{P}(X_n = m), \qquad g(\xi) = \mathfrak{g}(\xi) = \frac{\exp(-\xi^2/2)}{\sqrt{2\pi}}.$$

Only lattice coordinates are meant when $\mathcal{P}_n$ occurs. The uniform parameter hypotheses of that theorem give

$$\frac{c}{L} \le q \le \frac{C}{L}, \qquad |\varepsilon| \le \frac{C}{L}, \tag{1.231}$$

with positive constants independent of the parameters. In particular, the constants in the statements below are uniform for the scaled order in a fixed compact real set and for b in a compact subinterval of $(0, \infty)$.

Right tails of the explicit kernels

We use the notation

$$\mathcal{H}_{2k}^{[j]}(x) = \frac{\partial^j}{\partial \alpha^j} \mathcal{H}_\alpha(x)\Big|_{\alpha=2k}, \qquad \mathcal{H}_\alpha(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-iyx - y^2/2} (-iy)^\alpha \, dy,$$

where $\operatorname{Re} \alpha > -1$ and the principal powers are used.

Lemma 1.65 (Kernel tails and order derivatives)**.** *For fixed integers $k \ge 1$ and $j \ge 1$, as $x \to +\infty$,*

$$\mathcal{H}_{2k}^{[j]}(x) = O\big(x^{-2k-1}(1 + \log x)^{j-1}\big). \tag{1.232}$$

More precisely, if $a(\alpha) = 1/\Gamma(-\alpha)$, then

$$\mathcal{H}_{2k}^{[j]}(x) = x^{-2k-1} \sum_{r=1}^{j} \binom{j}{r} a^{(r)}(2k)(-\log x)^{j-r} + O\big(x^{-2k-11/4}\big). \tag{1.233}$$

In particular,

$$\mathcal{H}_2^{[1]}(x) = -2x^{-3}\left\{1 + 6x^{-2} + O(x^{-4})\right\}. \tag{1.234}$$

The constants can depend on k and j, which are fixed here.

Proof. We give a convolution proof so that differentiation with respect to the order does not rest on an unstated uniformity of a special-function asymptotic formula. Fix k, set $d = 2k + 1$, and let $|\alpha - 2k| \le 1/4$. Then

$$\mathcal{H}_\alpha(x) = \frac{1}{\Gamma(d - \alpha)} \int_0^\infty t^{d-\alpha-1} g^{(d)}(x - t) \, dt. \tag{1.235}$$

To verify the identity, first insert $e^{-\eta t}$ with $\eta > 0$ in the integral before differentiating d times in x. The Fourier multiplier of the resulting convolution is $(\eta - iy)^{\alpha-d}$. Differentiation multiplies it by $(-iy)^d$. As $\eta \downarrow 0$, this tends to $(-iy)^\alpha$. It is dominated, up to a constant uniform on the indicated order disc, by $|y|^{\operatorname{Re}\alpha} e^{-y^2/2}$: here $\operatorname{Re}(\alpha - d) < 0$ and $|\eta - iy| \ge |y|$. This is integrable.

Dominated convergence on the physical side is also valid, because the power at $t = 0$ has real part greater than -1 and every derivative of g has Gaussian decay at infinity. This proves (1.235), jointly holomorphically in α on a neighborhood of the closed disc.

Put $p = d - \alpha - 1$ and change variables to $u = t - x$. On $|u| \le x/2$, expand $(x+u)^p$ in powers of u/x through degree $d + 2M + 1$, for any fixed $M \ge 0$. The Taylor remainder is bounded by

$$C_{k,M} x^{\operatorname{Re} p - d - 2M - 2} |u|^{d+2M+2}$$

uniformly on the order disc. Integration against $|g^{(d)}(-u)|$ is harmless. Replacing the central integrals by integrals over the whole real line changes the answer by a quantity bounded by $e^{-x^2/16}$ times a fixed power of x. The part of the original integral where $|u| > x/2$ has the same type of bound. For the left part, the integrable factor $(x+u)^{\operatorname{Re} p}$ near $u = -x$ causes no difficulty; for the right part, polynomial growth is dominated by the Gaussian.

Integration by parts gives

$$\int_{\mathbb{R}} u^r g^{(d)}(-u)\, du = 0 \quad (r < d \text{ or } r - d \text{ odd}),$$

and, for $r = d + 2\ell$,

$$\int_{\mathbb{R}} u^{d+2\ell} g^{(d)}(-u)\, du = \frac{(d+2\ell)!}{2^\ell \ell!}.$$

It follows that, uniformly on the order disc,

$$\mathcal{H}_\alpha(x) = \sum_{\ell=0}^{M} \frac{x^{-\alpha-2\ell-1}}{2^\ell \ell!\, \Gamma(-\alpha-2\ell)} + O\big(x^{-\operatorname{Re}\alpha - 2M - 3}\big). \tag{1.236}$$

The exponentially small terms just described can be retained separately; they are already absorbed by this remainder.

The remainder in (1.236) is holomorphic in α. Cauchy's formula on the circle of radius $1/4$ about $2k$ bounds any fixed order derivative of that remainder by $O(x^{-2k-2M-11/4})$. With $M = 0$, differentiating the leading term gives (1.233): the $r = 0$ term vanishes because $a(2k) = 0$. This proves (1.232). For (1.234), use $M = 2$ and $k = j = 1$. The reciprocal Gamma function has a simple zero at each nonpositive integer, and

$$\frac{d}{d\alpha} \frac{1}{\Gamma(-\alpha-2\ell)}\bigg|_{\alpha=2} = -(2+2\ell)!.$$

The first two terms are therefore $-2x^{-3}$ and $-12x^{-5}$; the remaining displayed term and the differentiated remainder are $O(x^{-7})$, as asserted. □

The same tail structure follows from the classical connection formula for parabolic cylinder functions and their large-variable expansions [160]: $\mathcal{H}_\alpha(x) = e^{-x^2/4} D_\alpha(-x)/\sqrt{2\pi}$,

$$D_\alpha(-x) = \cos(\pi\alpha) D_\alpha(x) + \frac{\pi}{\Gamma(-\alpha)} V(-\alpha - 1/2, x).$$

Here the first term contributes Gaussian decay and the second contributes the power tail. The convolution proof above supplies the parameter uniformity and the order derivatives directly.

A relative Gaussian and power-tail approximation

Theorem 1.66 (Positive windows of polynomial size in $\log n$)**.** *Fix $M > 0$, and let $r_L \to \infty$ with $r_L \le L^M$. Uniformly over lattice points and admissible parameters for which $r_L \le \xi \le L^M$,*

$$\mathcal{P}_n(\xi) = \left\{ g(\xi) + q\xi^{-3} \right\} \{1 + o(1)\}. \tag{1.237}$$

Consequently, for each fixed $\delta \in (0, 2)$, whenever the stated intervals are nonempty,

$$\mathcal{P}_n(\xi) \sim g(\xi), \qquad r_L \le \xi \le \sqrt{(2-\delta)\log L}, \tag{1.238}$$

$$\mathcal{P}_n(\xi) \sim q\xi^{-3}, \qquad \sqrt{(2+\delta)\log L} \le \xi \le L^M. \tag{1.239}$$

No relative conclusion about negative growing displacements is included in this theorem.

Proof. Choose an integer $N > 3M$ in Theorem 1.57. Its term of total degree one is $-q\mathcal{H}_2^{[1]}(\xi)/2$. By (1.234), this equals $q\xi^{-3}\{1 + O(\xi^{-2})\}$. For every term with total degree $j \ge 2$, divide its absolute value by $q\xi^{-3}$. Equations (1.231) and (1.232) bound the quotient by a constant times

$$q^{k-1}|\varepsilon|^{j-k}\xi^{-2k+2}(1+\log\xi)^{j-1} \le C_N \left(\frac{1+\log\xi}{L} \right)^{j-1}.$$

Only finitely many terms occur. Their sum is $o(1)$ uniformly because $\log\xi \le M \log L$. The absolute remainder divided by $q\xi^{-3}$ is

$$O\big(\xi^3 L^{-N}\big) = O\big(L^{3M-N}\big) = o(1).$$

Finally $\xi^{-2} \le r_L^{-2} \to 0$. We have thus proved the stronger additive statement $\mathcal{P}_n(\xi) = g(\xi) + q\xi^{-3}\{1 + o(1)\}$, which implies (1.237) since both main terms are positive.

For (1.238), the ratio of the power term to the Gaussian is at most a constant times $L^{-\delta/2} r_L^{-3}$. For (1.239), the ratio of the Gaussian to the power term is at most a constant times $L^{-\delta/2}(\log L)^{3/2}$: the function $x^3 e^{-x^2/2}$ decreases for $x > \sqrt{3}$. Both bounds tend to zero, proving the two consequences. □

In the coefficient specialization of Theorem 1.57, $m = n$, and hence the same conclusions apply directly to $nBR_b^n a_n$. Coordinates $\xi \le L^M$ force $b \to A$ because $1 - A/b = \xi B/n = O(L^{M+1/2}/\sqrt{n})$. Thus these are growing positive windows adjacent to the feedback threshold. They are distinct from a fixed supercritical parameter regime, whose standardized displacement is much larger.

Corollary 1.67 (Relative matching with the supercritical coefficient law)**.** *In the diagonal specialization, uniformly for each fixed $M > 0$ and $\delta > 0$ in the nonempty range*

$$\sqrt{(2+\delta)\log L} \le \xi \le L^M,$$

one has

$$R_b^n a_n \sim \frac{1}{b\{n(1-A/b)\}^{3+\varepsilon}} = \frac{b^{s-1}}{(b-A)^s}\, n^{-s}. \tag{1.240}$$

Proof. On this window $E_2(\varepsilon) \to 1$ and $\xi^{-\varepsilon} \to 1$ uniformly, since $|\varepsilon \log\xi| = O(\log L/L)$. Thus the power term in (1.239) is relatively equivalent to $q\xi^{-3-\varepsilon}/E_2(\varepsilon)$. The exact identities

$$\frac{q}{E_2(\varepsilon)} = \frac{n}{bB^{2+\varepsilon}}, \qquad B\xi = n(1-A/b)$$

give (1.240) after division by nB. □

There is an exact explanation for the amplitude cancellation. The first divided-difference correction is $-q(\mathcal{H}_{2+\varepsilon} - \mathcal{H}_2)/(2\varepsilon)$. Its leading algebraic tail is $q\xi^{-3-\varepsilon}/E_2(\varepsilon)$ because $1/\Gamma(-2-\varepsilon) = -2\varepsilon/E_2(\varepsilon)$. The polynomial-log window makes this term relatively equivalent to the simpler $q\xi^{-3}$ while preserving the original coefficient amplitude in (1.240).

An exact Lambert location and a two-term profile

Theorem 1.68 (Lambert location of the Gaussian–power transition)**.** *For sufficiently small $q > 0$, let $x_*(q) > \sqrt{3}$ be the larger positive solution of $g(x_*) = qx_*^{-3}$. It has the exact formula*

$$x_*(q)^2 = -3W_{-1}\left(-\frac{(2\pi)^{1/3}}{3}q^{2/3}\right). \tag{1.241}$$

If $Q = \log(1/q)$, then

$$x_*(q)^2 = 2Q + 3\log(2Q) - \log(2\pi) + O\left(\frac{\log Q}{Q}\right). \tag{1.242}$$

For the parameters in (1.231), uniformly for τ in compact real intervals and positive lattice coordinates $\xi_n > 0$ satisfying

$$\xi_n^2 = x_*(q)^2 + 2\tau + o(1), \tag{1.243}$$

where the final error is uniform, one has

$$\frac{\mathcal{P}_n(\xi_n)}{qx_*(q)^{-3}} \longrightarrow 1 + e^{-\tau}. \tag{1.244}$$

In particular $x_*(q) \sim \sqrt{2\log\log n}$.

Proof. The function $x^3 g(x)$ increases up to $\sqrt{3}$ and decreases thereafter, so the stated larger root exists and is unique for small q. The defining equation is equivalent to

$$x_*^2 e^{-x_*^2/3} = (2\pi)^{1/3} q^{2/3},$$

which gives (1.241) with the branch W_{-1} [146]. Alternatively, taking logarithms gives the exact scalar equation

$$y = 2Q + 3\log y - \log(2\pi), \qquad y = x_*^2. \tag{1.245}$$

It first implies $y \sim 2Q$. Substituting $y_0 = 2Q + 3\log(2Q) - \log(2\pi)$ into its two sides leaves a residual $O((\log Q)/Q)$, while the derivative of $y - 3\log y$ is $1 - 3/y$, bounded away from zero at the larger root. The mean value theorem proves (1.242).

By (1.231), $Q = \log L + O(1)$. Therefore (1.243) lies in the window of Theorem 1.66 for any fixed $M > 0$ and a suitable $r_L \to \infty$. Moreover,

$$\frac{g(\xi_n)}{qx_*(q)^{-3}} = \frac{g(\xi_n)}{g(x_*(q))} = e^{-\tau + o(1)}, \qquad \frac{q\xi_n^{-3}}{qx_*(q)^{-3}} = \left(\frac{x_*(q)}{\xi_n}\right)^3 \longrightarrow 1.$$

The relative mixture theorem now gives (1.244). □

The lattice restriction is immaterial to the transition coordinate: rounding $nA/b + B\sqrt{x_*(q)^2 + 2\tau}$ to the nearest integer changes ξ_n^2 by $O(x_*(q)/B) = o(1)$. For diagonal coefficients, the feedback parameter can instead be varied on the locally unique branch near $b = A$, choosing the compact feedback interval to contain $\zeta(2)$ in its interior. Here is an explicit verification of the latter assertion. Hold n, s fixed, put $V = bB^2/n$, and differentiate the defining scale equation. It gives

$$\beta := \partial_b \log B = -\frac{b^{-1} + 2A^2/(b^3 V)}{2 - q} = O(1), \qquad \partial_b \log q = -b^{-1} - (2 + \varepsilon)\beta = O(1).$$

For $\xi(b) = n(1 - A/b)/B$, consequently,

$$\xi'(b) = \frac{nA}{b^2 B} - \xi(b)\beta \asymp \sqrt{n/L} > 0$$

through any window $\xi = O(L^M)$ near $b = A$. The derivative of $x_*(q(b))$ is only $O(1/x_*)$, by (1.245). Starting from $\xi(A) = 0$, continuity and these derivative bounds yield a unique local feedback choice for each bounded τ in (1.243).

Relation to established probability results

The Gaussian and power-tail competition is an established phenomenon. Lam, Blanchet, Burch, and Bazant obtain nonclassical corrections to Gaussian limits for regularly varying densities, express corrections using Dawson and parabolic cylinder functions, and explain their connection with heavy-tail large deviations [161]. Berger, Birkner, and Yuan treat large and local large deviations, including the infinite-variance normal domain with tail index -2, and identify the transition between collective Gaussian behavior and a single exceptionally large summand [162].

Accordingly, neither the existence of such a competition nor the use of parabolic cylinder functions is claimed as new here. The result established for the present model is the uniform extension through the varying order $s_n = 3 + O(1/\log n)$ and varying nonlinear feedback, with the explicit paired normalization and the finite all-order Stirling formulas of Theorem 1.57. The growing windows above follow from its uniform absolute remainder, and the exact Lambert coordinate (1.241) describes their Gaussian–power transition in that normalization. The fixed-law results just cited do not, without additional uniform estimates, supply this triangular-parameter assertion.

1.12 Reciprocal orders and removable inverse endpoints

The coefficient transition concerns order three. A different exceptional set appears when directly inverting a polylogarithm at its finite endpoint: the reciprocal orders $s = 1 + 1/m$. We develop their convergent moving-order chart after recalling the local ramification classification. That classification and the cut-modulus obstruction also occur in the earlier reversion continuation [150]; the joint report [158] gives the proof route used here and the uniform chart. The direct-inverse predecessor [159] states its global endpoint-circle theorem only for integer orders. The obstruction below limits its proposed real-order extension and does not retract that theorem.

The nonintegral charts in the predecessor imply an exceptional phenomenon that is worth separating from the integer-order resonance theory. Write $s = 1 + \alpha > 1$ and

$$D_\alpha(t) = \zeta(1 + \alpha) - \mathrm{Li}_{1+\alpha}(e^{-t}).$$

For nonintegral α, the classical convergent expansion [96] gives

$$D_\alpha(t) = b(\alpha)t^\alpha + tA_\alpha(t), \qquad b(\alpha) = -\Gamma(-\alpha), \qquad A_\alpha(t) = \sum_{k\geq 1} \frac{(-1)^{k+1}\zeta(1 + \alpha - k)}{k!} t^{k-1}.$$

Initially $0 < |t| < 2\pi$, $-\pi < \arg t < \pi$, and $t^\alpha = \exp(\alpha \log t)$ uses the principal logarithm. Further continuation is on the logarithmic covering. The function $A_\alpha(t)$ is jointly holomorphic for t near zero and α in any sufficiently small neighborhood avoiding the nonnegative integers. Let $t_\alpha(y)$ denote the inverse which is positive for positive small y.

Theorem 1.69 (Minimal local ramification)**.** *For real $s > 1$ the minimal degree of a finite ramification $y = w^N$ making the endpoint inverse $t_{s-1}(y)$ holomorphic is as follows.*

1. *If $0 < \alpha = s - 1 < 1$ and $\alpha = p/q$ is in lowest terms, the minimal degree is p.*
2. *If $\alpha > 1$ is nonintegral and $\alpha = p/q$ is in lowest terms, the minimal degree is q.*
3. *If α is irrational, or if s is a positive integer at least two, no finite ramification suffices.*

In particular, the endpoint inverse is holomorphic without ramification if and only if $s = 1 + 1/m$ for an integer $m \geq 2$. The same classification holds for the endpoint germ $\mathrm{Li}_s^{\langle -1 \rangle}(\zeta(s) - y) = e^{-t_{s-1}(y)}$.

Proof. Suppose first that $0 < \alpha = p/q < 1$. The substitution $t = \xi^q$ makes the forward germ holomorphic:

$$D_\alpha(\xi^q) = b\xi^p + \xi^q A_\alpha(\xi^q).$$

The factor multiplying ξ^p is nonzero at zero, so an analytic change of coordinate reduces this germ to η^p. Its local inverse is therefore holomorphic in $y^{1/p}$, and its t coordinate starts with $b^{-q/p} y^{q/p}$. Since p and q are coprime, this leading term forces every possible ramification degree to be divisible by p. Thus the minimal degree is exactly p.

If $\alpha = p/q > 1$ is nonintegral, the same substitution gives a forward germ of order q, because $A_\alpha(0) = \zeta(\alpha) > 0$. It follows that the inverse t is holomorphic in $y^{1/q}$. To see minimality, let $T(y)$ be the analytic inverse of $tA_\alpha(t)$. Expansion of the inverse equation gives

$$t_\alpha(y) = T(y) - b\zeta(\alpha)^{-\alpha-1} y^\alpha + o(y^\alpha),$$

after retaining the analytic terms of T through the relevant degree. The displayed fractional coefficient is nonzero, so q divides every possible ramification degree.

For irrational $\alpha < 1$, the leading exponent $1/\alpha$ is irrational. For irrational $\alpha > 1$, the preceding first nonanalytic exponent is α. Neither can occur in a convergent Puiseux germ. At $s = 2$, $D_1(t) = t(1 - \log t) + O(t^2)$, whence $t_1(y) \sim y/\log(1/y)$, which is not Puiseux. At integer $s = m + 1 \geq 3$, the singular term of the forward deficit is $(-1)^m t^m \log t/m!$, and its linear coefficient is $a = \zeta(m) > 0$. Therefore the inverse has a nonzero logarithmic term $(-1)^{m+1} y^m \log y/(m! a^{m+1})$ in addition to its analytic part. Finite ramification multiplies the logarithm by a nonzero integer and does not remove it. Finally $t \mapsto e^{-t}$ is biholomorphic near zero, so it preserves these ramification degrees. □

1.12.1 Explicit coefficients at the removable endpoints

Fix an integer $m \geq 2$, put $\alpha = 1/m$ and $b_m = -\Gamma(-1/m) > 0$, and define

$$G_m(\xi) = b_m + \sum_{k \geq 1} \frac{(-1)^{k+1} \zeta(1 + 1/m - k)}{k!} \xi^{mk-1}.$$

Then $y = \xi G_m(\xi)$, $t = \xi^m$, and Lagrange inversion gives the exact finite extraction formulas

$$[y^n] t_{1/m}(y) = \frac{m}{n} [\xi^{n-m}] G_m(\xi)^{-n},$$

$$[y^n] e^{-t_{1/m}(y)} = -\frac{m}{n} [\xi^{n-m}] e^{-\xi^m} G_m(\xi)^{-n} \qquad (n \geq 1).$$

Coefficients with a negative extraction exponent vanish. In particular

$$t_{1/m}(y) = b_m^{-m} y^m - m\zeta(1/m) b_m^{-2m} y^{2m-1} + O(y^{3m-2}).$$

This last remainder follows because the next possible degree in $t(y)$ is $3m-2$ (the term with $k=2$ first contributes at $3m-1$). For the ordinary inverse endpoint germ, every derivative of order $1,\dots,m-1$ vanishes, and its mth derivative with respect to $w=\zeta(s)-y$ equals $(-1)^{m+1}m!b_m^{-m}$.

For example, at $s=3/2$ one has $b_2=2\sqrt{\pi}$ and

$$t_{1/2}(y)=\frac{y^2}{4\pi}-\frac{\zeta(1/2)}{8\pi^2}y^3+\frac{5\zeta(1/2)^2}{64\pi^3}y^4+O(y^5).$$

Thus the square-root singularity of the forward polylogarithm has become a quadratic stationary point of its holomorphic endpoint inverse.

1.12.2 A uniform chart crossing a removable endpoint

The apparent disappearance of branching is not uniform under order perturbation. Nevertheless a short exact equation gives a convergent chart that is uniform at the reciprocal order.

Theorem 1.70 (Reciprocal-order crossover chart)**.** *Let $m\geq 2$ and let K be a compact real interval. Put*

$$\alpha=\frac{1}{m}+\epsilon,\quad T=\log(1/y),\quad \tau=\epsilon T,\quad h=T^{-1},\quad r=y^{m-1},\quad q=y^m.$$

There is a function $U(h,r,q;\tau)$ holomorphic in its first three variables near zero, with a complex neighborhood of K for the last variable and neighborhoods uniform on K, such that the physical inverse satisfies

$$t_\alpha(y)=y^mU(T^{-1},y^{m-1},y^m;\epsilon T).$$

Its value at the origin is

$$U(0,0,0;\tau)=b_m^{-m}e^{m^2\tau}.$$

The Taylor series in the three independent small variables converges after physical substitution for sufficiently small positive y.

Proof. Substitute $t=y^mU$ into $D_\alpha(t)=y$ and divide by y. The exact equation is

$$b(1/m+h\tau)e^{-m\tau}U^{1/m+h\tau}+rUA_{1/m+h\tau}(qU)=1.$$

For $h=r=q=0$ its indicated solution is $U_0(\tau)=b_m^{-m}e^{m^2\tau}$. Choose the logarithm $\log U_0(\tau)=-m\log b_m+m^2\tau$ and write $U=U_0V$. In this coordinate the equation is holomorphic near $V=1$, and its V derivative at $h=r=q=0$, $V=1$ is $1/m$. The holomorphic implicit function theorem and a finite compactness cover of K give the claimed uniform chart. Local uniqueness glues the charts. The physical positive branch is selected by continuity from positive y and the strictly increasing real forward deficit. □

For each fixed $j\geq 1$, differentiation in α at fixed deficit y gives

$$\partial_\alpha^j t_\alpha(y)\big|_{\alpha=1/m}=b_m^{-m}(-m^2)^jy^m(\log y)^j+O\big(y^m(1+|\log y|)^{j-1}\big).$$

One may prove this either from the uniform chart or by differentiating $b(\alpha)^{-1/\alpha}y^{1/\alpha}$ and the jointly holomorphic two-generator remainder. Derivatives of the latter acquire at most fixed powers of $\log y$ and retain a positive additional power of y. Thus order differentiation exposes an infinite logarithmic monodromy which is absent from the base germ. Here the independent variable is the deficit y; differentiation at fixed w also differentiates $\zeta(s)-w$ and is a different operation.

1.12.3 Why the integer endpoint-circle proof cannot extend verbatim

For $0 < \alpha = s - 1 < 1/2$ and $u \downarrow 0$, the upper-bank value is

$$\mathrm{Li}_{1+\alpha}(e^u + i0) = \zeta(1+\alpha) + \Gamma(-\alpha)e^{-i\pi\alpha}u^\alpha + O(u).$$

Since $\Gamma(-\alpha) < 0$ and $\cos(\pi\alpha) > 0$, this yields

$$|\,\mathrm{Li}_s(e^u + i0)| = \zeta(s) + \Gamma(1-s)\cos(\pi(s-1))u^{s-1} + o(u^{s-1}) < \zeta(s)$$

for every sufficiently small positive u. Consequently the cut-modulus assertion in the integer-order theorem fails for every $1 < s < 3/2$. The predecessor states that assertion only for integers, so this is a negative answer to a proposed extension, not a counterexample to a printed theorem.

The cut entering the endpoint circle does not itself determine the Taylor radius of an analytically continued inverse: the cut is a boundary of the chosen forward sheet, not necessarily an inverse singularity. Likewise endpoint removability does not rule out another singularity of the same modulus. Determining the global radius at the reciprocal orders remains a separate problem.

1.12.4 Evidence and remaining analytic questions

The delivered joint report's three programs have a preserved isolated replay: exact original-equation versus Lagrange coefficients through degree eight, six finite Stirling identities, 36 central Gamma identities, five marginal coefficient identities, nine noninteger Fourier comparisons, 36 coefficient cases through $n = 4096$, nine endpoint diagnostics and six cut-bank diagnostics. The supplied article, PDF, code and outputs remain unchanged. The separate experiment for Theorem 1.63 uses the positive cubic family (1.230); its coefficient recurrence and order-kernel evaluation are independent routes. All floating-point results are non-interval diagnostics. The infinite claims rest on the written analytic proofs, and no proof-assistant formalization is asserted.

The joint theorem settles compact real $(s-3)\log n/2$ windows at every fixed logarithmic order. The positive relative theorem reaches every fixed power of $\log n$ in the standardized coordinate. Growing negative relative windows, unbounded moving-order parameters, and macroscopic positive-tail matching remain open. The matched-derivative theorem identifies the first discrepancy between suitable models; a complete single-model expansion of all smaller sectors is a separate question. Likewise the convergent reciprocal-order chart is local and does not determine the global Taylor radius after endpoint removability.

Chapter 2

Cyclotomic values and the polygamma bridge

Roots of unity turn additive Fourier coordinates into arithmetic character coordinates. The same finite transform carries Hurwitz zeta, positive-order polygamma and Clausen values. It explains which half of a polylogarithm is elementary at a given weight and which character values remain to be studied.

2.1 The two-way exact DFT

We use the Clausen convention

$$\mathrm{Cl}_{2m}(\theta) = \Im\,\mathrm{Li}_{2m}(e^{i\theta}), \qquad \mathrm{Cl}_{2m+1}(\theta) = \Re\,\mathrm{Li}_{2m+1}(e^{i\theta}).$$

The sine series at odd weight is a different function; its rational-angle values reduce to powers of π through Bernoulli polynomials.

Law 2.1 (forward direction; classical)**.** *For every $n \geq 2$, every $q \geq 1$, and $\gcd(p,q) = 1$,*

$$\mathrm{Li}_n\big(e^{2\pi i p/q}\big) = \frac{(-1)^n}{q^n\,(n-1)!}\sum_{k=1}^{q-1} e^{2\pi i k p/q}\,\psi^{(n-1)}\Big(\frac{k}{q}\Big) \;+\; \frac{\zeta(n)}{q^n}.$$

This is identity 10.08.03.0007.01 of the `functions.wolfram.com` *PolyLog* reference [2]; it is provable by rearranging the absolutely convergent Hurwitz-zeta series, and it is the workhorse of the project's cyclotomic reducer (`cycloExact`): a finite polygamma sum, exact and fast, resolving Li_n at *every* root of unity with no PSLQ, no squarefree-radical basis, and no nested-radical limitation.

Result 2.2 (inverse direction). Inverting the discrete Fourier transform over $\mathbb{Z}/q$ gives, with $\zeta_q = e^{2\pi i/q}$,

$$\psi^{(n-1)}\Big(\frac{k}{q}\Big) = (-1)^n\,(n-1)!\;q^{\,n-1}\sum_{p=0}^{q-1}\zeta_q^{-pk}\;\mathrm{Li}_n\big(\zeta_q^{\,p}\big).$$

[historical experiment: verified to 40 digits at weights $2, 3$ across $q \leq 21$; re-verified by mpmath at 70 dps for $n-1 = 1, 2, 3$ over the full store grid]

The bridge is therefore an *exact finite Fourier transform, in both directions.* Three non-obvious consequences are recorded.

1. **DFT-conjugacy of the relation lattices.** The distribution relation $\sum_j \mathrm{Li}_n(z\,\zeta_m^j) = m^{1-n}\,\mathrm{Li}_n(z^m)$ Fourier-transforms into the polygamma multiplication theorem, and polylogarithm inversion/reflection corresponds to the symmetry $k \leftrightarrow q-k$ of the ψ-grid. This is *why* the two relation lattices — computed independently by the project's two reducers — carry identical dimension counts $\varphi(q)/2$ per denominator grid.

2. **The parity split, and Catalan as the $q=4$ odd component.** The imaginary parts are the odd Fourier sector under $k \mapsto q-k$. The usual even-weight Clausen values belong to this sector, while odd-weight Clausen values are real parts in the even sector. In particular

$$G = \mathrm{Im}\,\mathrm{Li}_2(i) = \frac{\psi^{(1)}(1/4) - \psi^{(1)}(3/4)}{16},$$

   which identifies the well-known "arguments 1/4 and 3/4" pairing (Math.SE 893486) as the $q=4$ odd component of the parity split.

3. **Certificate transport.** Cyclotomic-polylogarithm proof certificates can in principle be transported to and from ψ-side lattice certificates through the exact $\mathbb{Q}(\zeta_q)$ DFT matrix. The ψ-side elementary atoms ($\pi^2\csc^2$ values) are far lighter than the sine-product algebraic objects that constrain the Γ-certificate sweeps on large grids, so the transport is a proposed escape route from that bottleneck (not yet implemented).

2.2 Clausen-type laws for polygamma values at rational arguments

MathWorld's *Polygamma Function* page lists fifteen closed forms in its special-cases block, eqs. (22)–(36) ($\psi^{(1)}$, $\psi^{(2)}$, $\psi^{(3)}$ at p/q with $q \in \{2,3,4,6\}$) in Catalan/Clausen/β atoms, several of them traditionally presented as isolated curiosities. All of them — and their extension to *every* rational argument — are instances of three sine/cosine-weighted Clausen-DFT laws, obtained by splitting the inverse DFT of Result 2.2 into its elementary (reflection) half and its Clausen half.

Law 2.3 (trigamma; Lewin 1991)**.** *For $0 < p < q$, $\gcd(p,q) = 1$,*

$$\psi^{(1)}\Big(\frac{p}{q}\Big) = \frac{\pi^2}{2\sin^2(\pi p/q)} + 2q \sum_{j=1}^{\lfloor (q-1)/2 \rfloor} \sin\Big(\frac{2\pi jp}{q}\Big)\,\mathrm{Cl}_2\Big(\frac{2\pi j}{q}\Big).$$

[historical experiment: verified at $q = 3,4,5,8,12$ to 10^{-46} or better, then over the full store grid to 10^{-50} in-kernel and at 70 dps in mpmath]

This law is classical — MathWorld's *Trigamma Function* eq. (14) cites Lewin (1991) [1] — and was independently re-derived here from the DFT bridge before the attribution was spotted; the project's records explicitly re-frame the finding as *placement*, not novelty. MathWorld's scattered coefficients are its small-q shadows: the $3\sqrt{3}$ of $q=3$ is $2q\sin(2\pi/3)$, the $8G$ of $q=4$ is $2q\ \mathrm{Cl}_2(\pi/2)$.

Law 2.4 (tetragamma)**.**

$$\psi^{(2)}\Big(\frac{p}{q}\Big) = -\pi^3\cot\Big(\frac{\pi p}{q}\Big)\csc^2\Big(\frac{\pi p}{q}\Big) - 2q^2\left(\zeta(3) + 2\sum_{j=1}^{\lfloor (q-1)/2 \rfloor} \cos\Big(\frac{2\pi jp}{q}\Big)\,\mathrm{Cl}_3\Big(\frac{2\pi j}{q}\Big) + \varepsilon_q\right),$$

where $\varepsilon_q = (-1)^p\,\mathrm{Cl}_3(\pi)$ for even q, $\varepsilon_q = 0$ for odd q, and $\mathrm{Cl}_3(\pi) = -\frac{3}{4}\zeta(3)$. [historical experiment: verified over the full store grid to 10^{-50} in-kernel and at 70 dps in mpmath]

Law 2.5 (pentagamma)**.**

$$\psi^{(3)}\Big(\frac{p}{q}\Big) = \pi^4\Big(2\cot^2\Big(\frac{\pi p}{q}\Big)\csc^2\Big(\frac{\pi p}{q}\Big) + \csc^4\Big(\frac{\pi p}{q}\Big)\Big) + 12\,q^3\sum_{j=1}^{\lfloor (q-1)/2\rfloor}\sin\Big(\frac{2\pi jp}{q}\Big)\ \mathrm{Cl}_4\Big(\frac{2\pi j}{q}\Big),$$

equivalently, in difference form, $\psi^{(3)}(p/q) - \psi^{(3)}(1-p/q) = 24\,q^3\sum_j \sin(2\pi jp/q)\ \mathrm{Cl}_4(2\pi j/q)$. *[historical experiment: same verification battery]*

The $\psi^{(3)}$ law is the Lewin law one weight up, by the same derivation; no literature citation has been located, and the project records it as "presumably in Lewin's orbit." MathWorld's $162\sqrt{3}$ and $768\beta(4)$ coefficients are $12q^3\sin(2\pi j/q)$-instances of the one-sided law ($24q^3$ in the difference form).

Result 2.6 (new instances beyond the MathWorld table). Among the instances at $q \in \{5, 8, 12\}$:

$$\psi^{(1)}\big(\tfrac{1}{5}\big) = \frac{\pi^2}{2\sin^2(\pi/5)} + 10\Big[\sin\tfrac{2\pi}{5}\ \mathrm{Cl}_2\big(\tfrac{2\pi}{5}\big) + \sin\tfrac{4\pi}{5}\ \mathrm{Cl}_2\big(\tfrac{4\pi}{5}\big)\Big],$$

$$\psi^{(1)}\big(\tfrac{1}{12}\big) = 40\,G + \frac{4\pi^2}{(\sqrt{3}-1)^2} + 30\sqrt{3}\ \mathrm{Cl}_2\big(\tfrac{2\pi}{3}\big),$$

the latter after duplication/triplication canonicalization collapses the raw five-atom DFT form onto the canonical atom set. Also, MathWorld's eq. (27) simplifies: from $\mathrm{Cl}_3(2\pi/3) = -\frac{4}{9}\zeta(3)$,

$$\psi^{(2)}\big(\tfrac{1}{3}\big) = -\frac{4\pi^3}{3\sqrt{3}} - 26\,\zeta(3).$$

Result 2.7 (the polygamma–Clausen store). The corpus now carries `identities/polygamma-clausen-values.wl`: 54 entries $\psi^{(m)}(p/q)$ for $m \in \{1, 2, 3\}$, $q \in \{3, 4, 5, 6, 8, 12\}$, all p coprime to q (18 arguments $\times$ 3 orders), in canonicalized Clausen atoms. The residual atom basis across the store is

$$\{G,\ \beta(4),\ \zeta(3)\}\ \cup\ \mathrm{Cl}_2\big\{\tfrac{2\pi}{3}, \tfrac{2\pi}{5}, \tfrac{4\pi}{5}, \tfrac{\pi}{4}\big\}$$
$$\cup\ \mathrm{Cl}_3\big\{\tfrac{2\pi}{5}, \tfrac{\pi}{4}, \tfrac{\pi}{6}\big\}\ \cup\ \mathrm{Cl}_4\big\{\tfrac{2\pi}{3}, \tfrac{2\pi}{5}, \tfrac{4\pi}{5}, \tfrac{\pi}{4}\big\}.$$

Every entry is verified in-kernel to $|\mathrm{lhs} - \mathrm{rhs}| < 10^{-50}$ at 60 digits (including a disk round-trip) and independently by an mpmath script at 70 dps (tolerance 10^{-55}) that re-derives the raw DFT bridge, the three laws, all 17 Clausen canonicalization relations, and the L-function bridges of Section 2.3 from mpmath primitives, with positive and negative PSLQ controls.

2.3 The boundary of the odd-weight Clausen collapse

A folklore rule says that odd-weight Clausen values at rational angles collapse to rational multiples of zeta values ($\mathrm{Cl}_3(2\pi/3) = -\frac{4}{9}\zeta(3)$ being the standard example). The store emission *sharpened the rule's scope*:

Proposition 2.8 (Crystallographic-angle reductions)**.** *At denominators* $q \in \{1, 2, 3, 4, 6\}$*, the odd Clausen value* $\mathrm{Cl}_3(2\pi j/q) = \Re\,\mathrm{Li}_3(e^{2\pi ij/q})$ *reduces to a rational multiple of* $\zeta(3)$*. At* $q = 5, 8, 12$ *the*

character decomposition instead gives

$$\mathrm{Cl}_3(2\pi/5) = -\frac{6}{25}\zeta(3) + \frac{\sqrt{5}}{4}L(3,\chi_5),$$
$$\mathrm{Cl}_3(\pi/4) = -\frac{3}{256}\zeta(3) + \frac{1}{\sqrt{2}}L(3,\chi_8),$$
$$\mathrm{Cl}_3(\pi/6) = \frac{1}{24}\zeta(3) + \frac{\sqrt{3}}{2}L(3,\chi_{12}).$$

Here χ_D is the primitive even quadratic character of conductor D.

These are exact coordinate identities from the residue-class Fourier sum. No theorem asserting that the three L-values are independent of $\zeta(3)$ or π^3 is supplied. The source searches at height 10^{10} are numerical evidence in a stated basis, not an "if and only if" classification of all possible reductions.

2.4 The fundamental constants, by conductor

The residue classes of a cyclotomic series can be organized either by additive Fourier modes or by Dirichlet characters. Sine, or imaginary, components use odd characters, while cosine, or real, components use even characters. With the Clausen convention used here, even-index Clausen functions are sine components and odd-index ones are cosine components. Distribution relations first remove non-coprime residues; the remaining finite character transform has $\varphi(n)/2$ formal odd coordinates. Its coefficients are generally algebraic, so this coordinate count is not a proof of the dimension of the numerical $\mathbb{Q}$-span of the periods.

For the levels $8, 12, 24$, convenient generators are

$$V = \mathrm{Cl}_2(\pi/3), \quad G = \mathrm{Cl}_2(\pi/2), \quad W_8 = \mathrm{Cl}_2(\pi/4), \quad W_{24} = \mathrm{Cl}_2(\pi/12).$$

There are no primitive odd characters of conductor 12; its odd characters are induced from conductors 3 and 4. This explains the distribution reductions below. Conductors 3 and 4 each have one primitive odd character, while conductor 5 has two conjugate primitive odd characters. The source's printed sequence of primitive-character counts started with two incorrect zeros; it is not used here.

Their canonical L-value (trigamma) forms are

$$G = \frac{\psi^{(1)}(\frac{1}{4}) - \pi^2}{8}, \qquad \sqrt{3}\,V = \frac{3\psi^{(1)}(\frac{1}{3}) - 2\pi^2}{6} \quad \Big(\text{i.e. } \psi^{(1)}(\tfrac{1}{3}) = \tfrac{2\pi^2}{3} + 2\sqrt{3}\,V\Big). \tag{2.1}$$

2.5 Reduction tables and the settled level-24 question

Every $\mathrm{Cl}_2(2\pi a/n)$ at $n \in \{8, 12, 24\}$ reduces to the basis above (PSLQ, ≥ 110 digits, Champernowne canary; all confirmed in Wolfram):

$$\begin{aligned}
\text{level } 12: &\quad \mathrm{Cl}_2(\tfrac{\pi}{6}) = \tfrac{2}{3}G + \tfrac{1}{4}V, \quad \mathrm{Cl}_2(\tfrac{5\pi}{6}) = \tfrac{2}{3}G - \tfrac{1}{4}V, \quad \mathrm{Cl}_2(\tfrac{2\pi}{3}) = \tfrac{2}{3}V;\\
\text{level } 8: &\quad \mathrm{Cl}_2(\tfrac{3\pi}{4}) = W_8 - \tfrac{1}{2}G;\\
\text{level } 24: &\quad \mathrm{Cl}_2(\tfrac{5\pi}{12}) = \tfrac{16W_8+24W_{24}-4G-3V}{24}, \quad \mathrm{Cl}_2(\tfrac{7\pi}{12}) = \tfrac{4W_8+6W_{24}-3G}{6},\\
&\quad \mathrm{Cl}_2(\tfrac{11\pi}{12}) = \tfrac{24W_{24}-8G-3V}{24}.
\end{aligned}$$

The displayed formulas prove that the level-24 values are generated by G, V, W_8, W_{24}. They give an upper bound of four on their rational span. The source did not prove that W_{24} lies outside the span of the other three, so the minimal number of numerical generators remains open. This distinction is central when a symbolic conductor count is compared with a PSLQ experiment.

2.6 A new closed form

Since $\mathrm{Cl}_2(\pi/12) = W_{24}$ is itself fundamental, the cleanest carrier is the dilogarithm at the primitive 24th root $e^{\mathrm{i}\pi/12}$:

$$\mathrm{Li}_2(e^{\mathrm{i}\pi/12}) = \frac{73\pi^2}{576} + \mathrm{i}\, W_{24}, \tag{2.2}$$

the real part being $\frac{\pi^2}{6} - \frac{\pi}{2}\cdot\frac{\pi}{12} + \frac{1}{4}(\frac{\pi}{12})^2$. Pulling this back through $z = 1/(1-e^{\mathrm{i}\theta})$ gives Reshetnikov-style closed forms at the corresponding $\mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{6})$ points; the cleanest stays the root-of-unity form (2.2).

2.7 A weight parity swap

The same character parity that makes Cl_2 (weight 2, odd characters) carry deep constants reverses one weight up. For primitive odd χ, the positive odd integer values are algebraic multiples of π^s; since $\operatorname{Im}\mathrm{Li}_3(e^{\mathrm{i}\theta}) = \sum \sin(k\theta)/k^3$ pairs with odd characters at $s = 3$,

$$\operatorname{Im}\mathrm{Li}_3(e^{2\pi\mathrm{i}a/n}) \in \mathbb{Q}\cdot\pi^3 \quad \text{for all } a, n, \tag{2.3}$$

e.g. $\operatorname{Im}\mathrm{Li}_3(e^{2\pi\mathrm{i}/3}) = \frac{2\pi^3}{81}$, $\operatorname{Im}\mathrm{Li}_3(\mathrm{i}) = \frac{\pi^3}{32}$, $\operatorname{Im}\mathrm{Li}_3(e^{\mathrm{i}\pi/3}) = \frac{5\pi^3}{162}$, $\operatorname{Im}\mathrm{Li}_3(e^{2\pi\mathrm{i}/5}) = \frac{4\pi^3}{125}$. Thus at weight 3 the *imaginary* part at roots of unity is elementary, and the deep constants ($\zeta(3)$ and the even-character $L(3,\chi)$) live in the *real* part—the exact reverse of weight 2. (On the line $\operatorname{Re} z = 1$ the picture is different again: there $\operatorname{Im}\mathrm{Li}_3(1+\mathrm{i})$ is the central-binomial Gaussian tower constant, because $1+\mathrm{i}$ is not a root of unity; see `polylog-vertical-line`.)

Chapter 3

Algebraic arguments, integrals and ladders

Away from roots of unity, functional equations still move values onto structured sets of arguments. Vertical lines and half-angle substitutions give tractable examples; unit relations in number fields lead to longer ladders. The organizing principle is to understand the map of arguments before choosing the constants in an integer-relation search.

3.1 The problem and the key geometric fact

MSE 1410276 records

$$\operatorname{Re}\operatorname{Li}_2(1\pm \mathrm{i}\sqrt{3}) = \frac{\pi^2}{24} - \tfrac{1}{2}\ln^2 2 - \tfrac{1}{4}\operatorname{Li}_2(\tfrac{1}{4}), \tag{3.1}$$

with $\operatorname{Im}\operatorname{Li}_2(1+\mathrm{i}\sqrt{3}) = \frac{\pi\ln 2}{2} + \frac{5\sqrt{3}}{72}\big[\psi_1(\frac{1}{3}) - \psi_1(\frac{2}{3})\big]$ (a/1410829, a/1410705). The point $1+\mathrm{i}\sqrt{3} = 2e^{\mathrm{i}\pi/3}$, but the structurally decisive fact is that for any $z = 1+\mathrm{i}y$ the reflection point $1-z = -\mathrm{i}y$ is *purely imaginary*. This is exactly what makes the dilogarithm functional equations collapse.

3.2 The weight-2 vertical-line theorem

Theorem 3.1. *For $y > 0$,*

$$\operatorname{Re}\operatorname{Li}_2(1+\mathrm{i}y) = \frac{\pi^2}{6} - \tfrac{1}{2}\ln(1+y^2)\ln y - \frac{\pi}{2}\arctan y - \tfrac{1}{4}\operatorname{Li}_2(-y^2), \tag{3.2}$$

$$\operatorname{Im}\operatorname{Li}_2(1+\mathrm{i}y) = \frac{\pi}{4}\ln(1+y^2) - \arctan y\,\ln y + \operatorname{Ti}_2(y), \qquad \operatorname{Ti}_2(y) = \operatorname{Im}\operatorname{Li}_2(\mathrm{i}y). \tag{3.3}$$

Proof. The reflection law $\operatorname{Li}_2(z)+\operatorname{Li}_2(1-z) = \zeta(2) - \ln z\ln(1-z)$ at $z = 1+\mathrm{i}y$, $1-z = -\mathrm{i}y$, gives $\operatorname{Li}_2(1+\mathrm{i}y) = \zeta(2) - \ln(1+\mathrm{i}y)\ln(-\mathrm{i}y) - \operatorname{Li}_2(-\mathrm{i}y)$. The duplication law $\operatorname{Li}_2(\mathrm{i}y)+\operatorname{Li}_2(-\mathrm{i}y) = \frac{1}{2}\operatorname{Li}_2(-y^2)$ together with $\overline{\operatorname{Li}_2(w)} = \operatorname{Li}_2(\bar{w})$ yields $\operatorname{Re}\operatorname{Li}_2(-\mathrm{i}y) = \frac{1}{2}\big(\operatorname{Li}_2(\mathrm{i}y) + \operatorname{Li}_2(-\mathrm{i}y)\big) = \frac{1}{4}\operatorname{Li}_2(-y^2)$ and $\operatorname{Im}\operatorname{Li}_2(-\mathrm{i}y) = -\operatorname{Ti}_2(y)$. Expanding $\ln(1+\mathrm{i}y) = \frac{1}{2}\ln(1+y^2) + \mathrm{i}\arctan y$ and $\ln(-\mathrm{i}y) = \ln y - \frac{\mathrm{i}\pi}{2}$ and taking real and imaginary parts gives (3.2)–(3.3). □

On the line $\operatorname{Re} z = 1$ the dilogarithm thus splits into a *real* part carried by the even real dilogarithm $\operatorname{Li}_2(-y^2)$ and an *imaginary* part carried by the odd inverse-tangent integral $\operatorname{Ti}_2(y)$. Equation (3.1) is the case $y = \sqrt{3}$ (using $\operatorname{Li}_2(-3) = \operatorname{Li}_2(\frac{1}{4}) - \zeta(2) + 2\ln 2\ln\frac{2}{3}$), and the trigamma form of the imaginary part is just $\operatorname{Ti}_2(\sqrt{3}) = \frac{\pi}{6}\ln 3 + \frac{5\sqrt{3}}{72}[\psi_1(\frac{1}{3}) - \psi_1(\frac{2}{3})]$ substituted into (3.3).

3.3 The Clausen form at $y = \tan\theta$

Since $1 + \mathrm{i}\tan\theta = e^{\mathrm{i}\theta}/\cos\theta$, Theorem 3.1 combines with Ramanujan's $\mathrm{Ti}_2(\tan\theta) = \theta\ln\tan\theta + \frac{1}{2}\big[\mathrm{Cl}_2(2\theta) + \mathrm{Cl}_2(\pi - 2\theta)\big]$ so that the $\theta\ln\tan\theta$ terms cancel, leaving a pure-Clausen imaginary part:

$$\operatorname{Im}\mathrm{Li}_2(1 + \mathrm{i}\tan\theta) = -\frac{\pi}{2}\ln\cos\theta + \tfrac{1}{2}\big[\mathrm{Cl}_2(2\theta) + \mathrm{Cl}_2(\pi - 2\theta)\big]. \tag{3.4}$$

Corollary 3.2. $\operatorname{Re}\mathrm{Li}_2(1+\mathrm{i}) = \dfrac{\pi^2}{16}$ *and* $\operatorname{Im}\mathrm{Li}_2(1+\mathrm{i}) = \dfrac{\pi\ln 2}{4} + G$ *(with G Catalan's constant), from $y = 1$ in Theorem 3.1 (using $\mathrm{Li}_2(-1) = -\pi^2/12$, $\mathrm{Ti}_2(1) = G$).*

3.4 Weight 3: a half-lifted symmetry

At weight 3 the real/imaginary split survives only in part. The *real* part still closes at special y:

$$\operatorname{Re}\mathrm{Li}_3(1+\mathrm{i}) = \frac{35}{64}\zeta(3) + \frac{\pi^2\ln 2}{32}, \qquad \operatorname{Re}\mathrm{Li}_3(1+\mathrm{i}\sqrt{3}) = \frac{35\zeta(3) + 18\,\mathrm{Li}_3(\frac{1}{4}) + 6\pi^2\ln 2 - 24\ln^3 2}{144}, \tag{3.5}$$

the second being the exact weight-3 analogue of the $\mathrm{Li}_2(\frac{1}{4})$ in (3.1). For the general real part, the displayed calculation introduces a further integral: differentiating, $\partial_y \operatorname{Re}\mathrm{Li}_3(1+\mathrm{i}y) = -\big(\operatorname{Im}\mathrm{Li}_2(1+\mathrm{i}y) - y\,\operatorname{Re}\mathrm{Li}_2(1+\mathrm{i}y)\big)/(1+y^2)$, and integrating the Ti_2 contribution produces $\int_0^y \arctan^2(t)/t\,dt$, a weight-three inverse-tangent-square integral. The displayed special cases at $y = 1, \sqrt{3}$ simplify, without giving an elementary-value classification for every rational angle. The *imaginary* part at $1+\mathrm{i}$ is represented here by the central-binomial Gaussian tower constant

$$\operatorname{Im}\mathrm{Li}_3(1+\mathrm{i}) = \mathcal{F}_3 - \frac{\pi^3}{192}, \qquad \mathcal{F}_3 = \sqrt{2}\,{}_4F_3\big(\{\tfrac{1}{2}\}^4; \{\tfrac{3}{2}\}^3; \tfrac{1}{2}\big), \tag{3.6}$$

the $n = 3$ entry of the $\operatorname{Im}\mathrm{Li}_n(1+\mathrm{i})$ tower (companion report `gaussian-1plusi-tower`). So on $\operatorname{Re} z = 1$ the imaginary part is precisely the deep object that the central-binomial ladder measures, while the real part remains tractable—the two halves of the polylogarithm part ways at weight 3.

Remark 3.3 (An unsupported nonreduction claim). Earlier prose asserted that this imaginary part was not elementary, without specifying a comparison algebra or supplying an obstruction. That assertion is an open reducibility question, rather than a consequence of (3.6). The displayed hypergeometric representation remains valid; any proposed reduction or nonreduction must name its algebra of constants and establish a separate proof.

3.5 The integral and its provenance

The object (Math.StackExchange question 564816) is

$$I = \int_0^{\pi/2} \arctan^2\!\left(\frac{6\sin x}{3+\cos 2x}\right) dx = 1.33097039684256666612532 7\ldots \tag{3.7}$$

Cleo gave the dilogarithmic form

$$I = \frac{\pi}{2}\Big(2\,\mathrm{Li}_2\big((1-\sqrt{2})(2-\sqrt{5})\big) - 2\,\mathrm{Li}_2\big((1-\sqrt{2})(\sqrt{5}-2)\big) + \mathrm{Li}_2(3-\sqrt{8}) - \mathrm{Li}_2(\sqrt{8}-3) + \mathrm{Li}_2(9-\sqrt{80}) - \mathrm{Li}_2(\sqrt{80}-9)\Big), \tag{3.8}$$

while Sangchul Lee gave the Legendre-chi form

$$I = \pi\Big(\chi_2(3-2\sqrt{2}) + \chi_2(9-4\sqrt{5}) + 2\chi_2\big((\sqrt{2}-1)(\sqrt{5}-2)\big)\Big), \tag{3.9}$$

where $\chi_2(x) = \frac{1}{2}\big(\mathrm{Li}_2(x) - \mathrm{Li}_2(-x)\big) = \sum_{k\ge 0} \frac{x^{2k+1}}{(2k+1)^2}$. These are identical: $\mathrm{Li}_2(x) - \mathrm{Li}_2(-x) = 2\chi_2(x)$ turns each Li_2 pair in (3.8) into a χ_2, and $3-\sqrt{8} = (\sqrt{2}-1)^2$, $9-\sqrt{80} = (\sqrt{5}-2)^2$, $(1-\sqrt{2})(2-\sqrt{5}) = (\sqrt{2}-1)(\sqrt{5}-2)$, so both equal

$$I = \pi\Big[\chi_2\big((\sqrt{2}-1)^2\big) + \chi_2\big((\sqrt{5}-2)^2\big) + 2\,\chi_2\big((\sqrt{2}-1)(\sqrt{5}-2)\big)\Big]. \tag{3.10}$$

The arguments are a silver $\otimes$ golden pair: $\sqrt{2}-1 = \tan\frac{\pi}{8} = \delta_{\mathrm{S}}^{-1}$ (reciprocal silver ratio $\delta_{\mathrm{S}} = 1+\sqrt{2}$) and $\sqrt{5}-2 = \phi^{-3}$ (golden $\phi = \frac{1+\sqrt{5}}{2}$, $\phi^3 = \sqrt{5}+2$), so $I = \pi[\chi_2(\delta_{\mathrm{S}}^{-2}) + \chi_2(\phi^{-6}) + 2\chi_2(\delta_{\mathrm{S}}^{-1}\phi^{-3})]$.

3.6 The master formula (Reshetnikov–Lee) and a differentiation proof

The closed forms (3.8)–(3.9) are instances of a two-parameter master formula. Its seed is Reshetnikov's single-arctangent ladder $\int_0^{\pi/2} \arctan(t\sin x)\,dx = 2\chi_2(u_t)$; inspired by it, Lee (a/562492, 2013) conjectured and proved

Theorem 3.4 (Lee 2013)**.** *For real p, q,*

$$J(p,q) := \int_0^{\pi/2} \arctan(p\sin x)\,\arctan(q\sin x)\,dx = \pi\,\chi_2\big(u_p u_q\big), \qquad \text{where} \quad u_p = \frac{\sqrt{1+p^2}-1}{p} = \tan\big(\tfrac{1}{2}\arctan p\big). \tag{3.11}$$

Lee gives two proofs (a triple integral and a Fourier-orthogonality argument). We record a short third proof by differentiation, which is the most direct route and is machine-verifiable.

Proof. Both sides vanish on the axes, and $\partial_p J(p,0) = \partial_p[\pi\chi_2(u_p u_0)] = 0$ (since $u_0 = 0$), symmetrically in q; hence each side equals the double integral of its mixed partial over $[0,p]\times[0,q]$, and it suffices to match mixed partials. Differentiating under the integral and using the partial fraction $\frac{s^2}{(1+p^2s^2)(1+q^2s^2)} = \frac{1}{p^2-q^2}\big(\frac{1}{1+q^2s^2} - \frac{1}{1+p^2s^2}\big)$ together with $\int_0^{\pi/2} \frac{dx}{1+c\sin^2 x} = \frac{\pi}{2\sqrt{1+c}}$,

$$\partial_p\partial_q J = \frac{\pi}{2(p^2-q^2)}\Big(\frac{1}{\sqrt{1+q^2}} - \frac{1}{\sqrt{1+p^2}}\Big). \tag{3.12}$$

On the right, $\chi_2'(w) = \operatorname{arctanh}(w)/w$ gives $\partial_p\partial_q[\pi\chi_2(u_p u_q)] = \pi\,u_p' u_q'/(1-u_p^2u_q^2)$. The substitution $p = \frac{2u_p}{1-u_p^2}$ yields $\sqrt{1+p^2} = \frac{1+u_p^2}{1-u_p^2}$ and $u_p' = \frac{(1-u_p^2)^2}{2(1+u_p^2)}$, whence both sides reduce to the same rational function $\frac{\pi(1-u_p^2)^2(1-u_q^2)^2}{4(1+u_p^2)(1+u_q^2)(1-u_p^2u_q^2)}$. (Wolfram confirms this reconciliation symbolically: `FullSimplify` of the difference returns 0.) □

3.7 Cleo's integral as the perfect-square instance

The argument in (3.7) collapses to a sum of arctangents. With $\cos 2x = 1 - 2\sin^2 x$,

$$\frac{6\sin x}{3+\cos 2x} = \frac{3\sin x}{2-\sin^2 x} = \frac{(1+\frac{1}{2})\sin x}{1-(1\cdot\frac{1}{2})\sin^2 x},$$

the tangent-addition shape with roots $1, \frac{1}{2}$ of $t^2 - \frac{3}{2}t + \frac{1}{2}$, so for $0 \le x \le \frac{\pi}{2}$

$$\arctan\left(\frac{6\sin x}{3+\cos 2x}\right) = \arctan(\sin x) + \arctan(\tfrac{1}{2}\sin x), \tag{3.13}$$

so that $I = \int_0^{\pi/2} \big(\arctan(\sin x) + \arctan(\frac{1}{2}\sin x)\big)^2 dx$. Applying Theorem 3.4 to the expanded square, with $u_1 = \sqrt{2}-1$, $u_{1/2} = \sqrt{5}-2$, reproduces (3.10). The coefficient pattern $1, 2, 1$ is a perfect square: writing $a = \sqrt{2}-1$, $b = \sqrt{5}-2$,

$$\frac{I}{\pi} = \sum_{k\ge 0} \frac{\big((\sqrt{2}-1)^{2k+1} + (\sqrt{5}-2)^{2k+1}\big)^2}{(2k+1)^2}. \tag{3.14}$$

The coefficients $a^{2k+1} + b^{2k+1}$ are the sin-power Fourier coefficients of $\arctan(\sin x) + \arctan(\frac{1}{2}\sin x)$; (3.14) is the Parseval form underlying Lee's Fourier proof.

3.8 Closed values and a bounded experimental survey

Three classical values supply useful exact specializations. With $\phi = (1+\sqrt{5})/2$,

$$\begin{aligned} \chi_2(\phi^{-1}) &= \frac{\pi^2}{12} - \frac{3}{4}\log^2\phi, \\ \chi_2(\phi^{-3}) &= \frac{\pi^2}{24} - \frac{3}{4}\log^2\phi, \\ \chi_2(\sqrt{2}-1) &= \frac{\pi^2}{16} - \frac{1}{4}\log^2(1+\sqrt{2}). \end{aligned} \tag{3.15}$$

The source surveyed inverse metallic means m_n^{-k} with $1 \le n, k \le 6$, selected rational points, and $\tan(\pi/k)$ for $k = 5, 8, 10, 12$ and their squares. Searches at 130–140 digits found no further accepted reduction in their selected logarithmic and π^2 baskets. This does not classify all elementary values. Nor would independence of the individual terms alone prove that their sum cannot simplify.

3.9 New closed-form integrals from elementary χ_2

Conversely, (3.15) feeds the master formula to produce integrals that *do* close. The cleanest cross integral uses $u_2 = \phi^{-1}$, $u_{2/\sqrt{5}} = \phi^{-2}$, so $u_2 u_{2/\sqrt{5}} = \phi^{-3}$:

$$\boxed{\int_0^{\pi/2} \arctan(2\sin x)\,\arctan\big(\tfrac{2}{\sqrt{5}}\sin x\big)\,dx = \frac{\pi^3}{24} - \frac{3\pi}{4}\ln^2\phi} \qquad (= 0.7463164407\ldots). \tag{3.16}$$

The diagonal $\int_0^{\pi/2} \arctan^2(\lambda \sin x)\, dx = \pi\, \chi_2(u_\lambda^2)$ closes at three clean λ, since $u^2_{\sqrt{\phi}} = \phi^{-3}$, $u^2_{\sqrt{2+2\sqrt{2}}} = \sqrt{2}-1$, and $u^2_{2\phi^{3/2}} = \phi^{-1}$:

$$\int_0^{\pi/2} \arctan^2(\sqrt{\phi}\, \sin x)\, dx = \frac{\pi^3}{24} - \frac{3\pi}{4} \ln^2 \phi, \quad \int_0^{\pi/2} \arctan^2\big(2\phi^{3/2} \sin x\big)\, dx = \frac{\pi^3}{12} - \frac{3\pi}{4} \ln^2 \phi. \tag{3.17}$$

The $(1,1)$ diagonal is $\int_0^{\pi/2} \arctan^2(\sin x)\, dx = \pi\, \chi_2(3-2\sqrt{2})$. No elementary reduction or nonreduction theorem for this value is established here.

3.10 A sibling trichotomy

The master formula has two siblings under the companion half-angle substitution $v_p = \frac{1-\sqrt{1-p^2}}{p} = \tanh(\frac{1}{2} \operatorname{arctanh} p)$ (for $|p| < 1$). Replacing one or both arctangents by arctanh produces:

$$\int_0^{\pi/2} \begin{cases} \arctan(p \sin x)\ \arctan(q \sin x) \\ \operatorname{arctanh}(p \sin x)\ \operatorname{arctanh}(q \sin x) \\ \arctan(p \sin x)\ \operatorname{arctanh}(q \sin x) \end{cases} dx = \begin{cases} \pi\, \chi_2(u_p u_q), \\ \pi\, \chi_2(v_p v_q), \\ \pi\ \mathrm{Ti}_2(u_p v_q), \end{cases} \tag{3.18}$$

where $\mathrm{Ti}_2(z) = \Im\, \mathrm{Li}_2(\mathrm{i}z) = \int_0^z \frac{\arctan t}{t}\, dt$ is the inverse-tangent integral: like pairs give the Legendre chi, the mixed pair gives the inverse-tangent integral. The linear shadows are $\int_0^{\pi/2} \arctan(\lambda \sin x)\, dx = 2\chi_2(u_\lambda)$ (Reshetnikov) and, for $-1 < \lambda < 1$ with $\theta = \arcsin \lambda$,

$$\int_0^{\pi/2} \operatorname{arctanh}(\lambda \sin x)\, dx = \theta\, \ln \left|\tan \tfrac{\theta}{2}\right| + 2\, \mathrm{Cl}_2(\theta) - \tfrac{1}{2}\, \mathrm{Cl}_2(2\theta), \tag{3.19}$$

with the value at zero defined by continuity. Indeed, differentiation of the integral with respect to θ and the substitution $t = \cos x$ give $\cos\theta \int_0^1 (\cos^2\theta + \sin^2\theta\, t^2)^{-1} dt = \theta / \sin\theta$. The derivative of the right side is the same, using $\mathrm{Cl}_2'(t) = -\log|2\sin(t/2)|$, and both sides tend to zero at $\theta = 0$. This proves the real branch on both sides of zero. Monotone convergence and oddness extend the formula to $\lambda = \pm 1$, with values $\pm 2\, \mathrm{Cl}_2(\pi/2)$. The latter shows that the arctanh branch is *not* a clean $2\chi_2(v_\lambda)$ mirror—that coincidence is special to arctan. By $x \mapsto \frac{\pi}{2} - x$ the cos versions of all of these equal the sin versions. The weighted analogue $\int_0^{\pi/2} \arctan(p \sin x) \arctan(q \sin x)/\sin x\, dx$ has a different parameter structure. Theorem 3.26 evaluates it by two real trilogarithmic kernels at the hyperbolic sum and difference. Proposition 3.28 proves that no function of $u_p u_q$ alone can represent this family near the origin, replacing the earlier bounded-search observation with an exact obstruction.

3.11 An alternate form, and minimality

Landen-mirror form. Applying the χ_2 Landen law

$$\chi_2(x) + \chi_2\left(\frac{1-x}{1+x}\right) = \frac{\pi^2}{8} - \frac{1}{2} \ln x\, \ln \frac{1-x}{1+x} \qquad (0 < x < 1) \tag{3.20}$$

to the three arguments of (3.10) sends $(\sqrt{2}-1)^2 \mapsto \frac{1}{\sqrt{2}}$, $(\sqrt{2}-1)(\sqrt{5}-2) \mapsto \sqrt{5}-\sqrt{2}$, $(\sqrt{5}-2)^2 \mapsto \frac{2}{\sqrt{5}}$, giving a second three-term representation

$$\frac{I}{\pi} = \mathcal{E} - \Big[\chi_2(\tfrac{1}{\sqrt{2}}) + 2\, \chi_2(\sqrt{5}-\sqrt{2}) + \chi_2(\tfrac{2}{\sqrt{5}})\Big], \tag{3.21}$$

with the elementary constant $\mathcal{E} = \frac{\pi^2}{2} - \frac{1}{2}\ln 2\,\ln(1+\sqrt{2}) + 3\ln 2\,\ln\phi - \frac{3}{2}\ln 5\,\ln\phi + \big(\ln(1+\sqrt{2}) + 3\ln\phi\big)\ln(\sqrt{5}-\sqrt{2})$. This representation trades the small original arguments for larger ones and explicit logarithms. It follows directly by applying the Landen law three times.

Search for shorter forms. The source tested 209 algebraic arguments in $\mathbb{Q}(\sqrt{2},\sqrt{5},\sqrt{10})$ at at least 120 digits, with residual threshold 10^{-105}. It found no accepted one-term or two-term representation in the tested baskets. The three-term formula is an exact evaluation, while its minimality and non-elementarity remain unproved.

3.12 Unit relations and the golden family

With $\phi = \frac{1+\sqrt{5}}{2}$, $\rho = \phi^{-1}$, the defining relation $1 - \rho^2 = \rho$ is the seed of the golden ladders. Reshetnikov's conjectures live among the values $\mathrm{Li}_n(\phi^{-2k})$. The source used PSLQ to recover coefficient vectors and tested the resulting relations numerically (`mpmath`, 130–420 digits, Champernowne canary), with Wolfram comparisons where the cancellation depth allowed. Exact functional-equation certificates, where supplied, are identified separately. This description concerns the evidence presented here; it does not assert that the corresponding identities lack proofs in the literature.

Remark 3.5. The source reports cancellation of $\sim 10^8$-sized terms with more than 100 matching decimal digits. Its historical Wolfram evaluation under-resolved some cancellations, so independently controlled precision in mpmath supplied a comparator. Agreement between numerical engines is a useful check, but does not establish exact equality.

3.13 Reshetnikov's formulas and their numerical evidence

The source recovered the following tetralogarithm relation (MSE 1373123, MO 261408) numerically. The exact proof below now establishes it. With $\alpha = \ln 2, \beta = \ln 3$, its form is

$$\begin{aligned} 720\,\mathrm{Li}_4(\tfrac{1}{2}) - 2160\,\mathrm{Li}_4(\tfrac{1}{3}) + 2160\,\mathrm{Li}_4(\tfrac{2}{3}) \\ + 270\,\mathrm{Li}_4(\tfrac{1}{4}) + 540\,\mathrm{Li}_4(\tfrac{3}{4}) + 135\,\mathrm{Li}_4(\tfrac{1}{9}) \\ = 19\pi^4 + 30(\pi^2-\beta^2)(10\alpha^2 - 12\alpha\beta + 3\beta^2) - 30\alpha^2(19\alpha^2 - 24\alpha\beta + 8\beta^2), \end{aligned} \tag{3.22}$$

The source recovered this relation in its weight-four basket; Theorem 3.7 proves it without a numerical-recognition premise. Uniqueness of all relations among the individual values remains a separate question. For the three Li_5 golden formulas below, the source reports residuals of approximately 10^{-257}. Their connection with the ladders of Abouzahra and Lewin [36] is described below:

$$\begin{aligned} 4455\,\mathrm{Li}_5(\phi^{-2}) - 1215\,\mathrm{Li}_5(\phi^{-4}) - 360\,\mathrm{Li}_5(\phi^{-6}) + 15\,\mathrm{Li}_5(\phi^{-12}) &= 3015\zeta(5) - 702\ln\phi^5 + 180\pi^2\ln\phi^3 - 38\pi^4\ln\phi, \\ 45000\,\mathrm{Li}_5(\phi^{-2}) - 16875\,\mathrm{Li}_5(\phi^{-4}) - 144\,\mathrm{Li}_5(\phi^{-10}) + 9\,\mathrm{Li}_5(\phi^{-20}) &= 28944\zeta(5) - 6000\ln\phi^5 + 1600\pi^2\ln\phi^3 - 356\pi^4\ln\phi, \\ 15660\,\mathrm{Li}_5(\phi^{-2}) - 19440\,\mathrm{Li}_5(\phi^{-4}) + 7680\,\mathrm{Li}_5(\phi^{-6}) & \\ +2430\,\mathrm{Li}_5(\phi^{-8}) - 15\,\mathrm{Li}_5(\phi^{-24}) &= 5025\zeta(5) + 2088\ln\phi^5 - 240\pi^2\ln\phi^3 - 28\pi^4\ln\phi. \end{aligned}$$

3.14 Reconstructing Tito's weight-6 and weight-7 formulas

Tito's "old answer" lists the golden family at weights 2–5 (numerically checked in the source), then states the weight-6 and weight-7 ladders only schematically, the integer coefficients "too

tedious to write." PSLQ recovers the following candidate coefficient vectors on the six-argument set $\{\phi^{-2},\phi^{-4},\phi^{-6},\phi^{-8},\phi^{-12},\phi^{-24}\}$:

$$\begin{aligned}
&98820\,\mathrm{Li}_6(\phi^{-2}) - 104490\,\mathrm{Li}_6(\phi^{-4}) + 33120\,\mathrm{Li}_6(\phi^{-6}) + 7290\,\mathrm{Li}_6(\phi^{-8}) - 50\,\mathrm{Li}_6(\phi^{-12}) - 15\,\mathrm{Li}_6(\phi^{-24}) \\
&\quad = -13032\ln\phi^6 + 3240\pi^2\ln\phi^4 - 424\pi^4\ln\phi^2 + \mathbf{32155}\,\zeta(6), \qquad (3.23)\\
&74707920\,\mathrm{Li}_7(\phi^{-2}) - 39497220\,\mathrm{Li}_7(\phi^{-4}) + 8346240\,\mathrm{Li}_7(\phi^{-6}) + 1377810\,\mathrm{Li}_7(\phi^{-8}) - 6300\,\mathrm{Li}_7(\phi^{-12}) - 945\,\mathrm{Li}_7(\phi^{-24}) \\
&\quad = 2814912\ln\phi^7 - 979776\pi^2\ln\phi^5 + 213696\pi^4\ln\phi^3 - 51448\pi^6\ln\phi + \mathbf{44689995}\,\zeta(7). \qquad (3.24)
\end{aligned}$$

The $\zeta(6)$ and $\zeta(7)$ coefficients reproduce Tito's stated 32155 and 44689995 exactly. The source also reports direct-substitution residuals 10^{-117} and 10^{-114} for (3.23)–(3.24), with primitive coefficient vectors. These are consistent numerical checks; an exact derivation of these displayed reconstructions is not supplied in this section.

3.15 Explicit candidate formulas at weights 8 and 9

Tito remarks that the index-24 ladder "can be extended even to $n = 9$" but gives no formula. The source recovers the following candidates. Its searches on the six-argument set of (3.23)–(3.24) detected numerical reductions at weights 2–7 and none at 8 or 9 within the tested bounds. The accepted higher-weight candidates use the *eight*-argument union of the index-12, 20, 24 sets, $\mathcal{A} = \{\phi^{-2},\phi^{-4},\phi^{-6},\phi^{-8},\phi^{-10},\phi^{-12},\phi^{-20},\phi^{-24}\}$:

$$\begin{aligned}
&-490795200\,\mathrm{Li}_8(\phi^{-2}) + 636131475\,\mathrm{Li}_8(\phi^{-4}) - 388684800\,\mathrm{Li}_8(\phi^{-6}) + 82668600\,\mathrm{Li}_8(\phi^{-8}) \\
&\quad + 9507456\,\mathrm{Li}_8(\phi^{-10}) + 5250000\,\mathrm{Li}_8(\phi^{-12}) - 74277\,\mathrm{Li}_8(\phi^{-20}) - 18900\,\mathrm{Li}_8(\phi^{-24}) \\
&\qquad = -23641920\ln\phi^8 + 7956480\pi^2\ln\phi^6 - 1434720\pi^4\ln\phi^4 + 184432\pi^6\ln\phi^2 - 148565886\,\zeta(8), \qquad (3.25)\\
&220857840000\,\mathrm{Li}_9(\phi^{-2}) - 143129581875\,\mathrm{Li}_9(\phi^{-4}) + 58302720000\,\mathrm{Li}_9(\phi^{-6}) - 9300217500\,\mathrm{Li}_9(\phi^{-8}) \\
&\quad - 855671040\,\mathrm{Li}_9(\phi^{-10}) - 393750000\,\mathrm{Li}_9(\phi^{-12}) + 3342465\,\mathrm{Li}_9(\phi^{-20}) + 708750\,\mathrm{Li}_9(\phi^{-24}) \\
&\qquad = -2364192000\ln\phi^9 + 1022976000\pi^2\ln\phi^7 - 258249600\pi^4\ln\phi^5 \\
&\qquad\quad + 55329600\pi^6\ln\phi^3 - 14149132\pi^8\ln\phi + 125596001790\,\zeta(9). \qquad (3.26)
\end{aligned}$$

The source reports a residual printed as 0 at 150 working digits for (3.25), and approximately 10^{-190} at 200 working digits for (3.26); both coefficient vectors are primitive. The weight-9 vector was recovered in three searches with different padding arguments, all of which received coefficient 0. These observations support the formulas numerically. A residual printed as zero at finite precision does not establish exact equality.

Remark 3.6. These experiments support golden formulas through weight 9, using four or five arguments per index at weight 5, six arguments at weights 6–7, and eight arguments at weights 8–9. Higher-order golden extrapolations and their numerical construction appear in Abouzahra–Lewin [36] and Bailey–Broadhurst [38]. The present numerical evidence does not prove a maximal ladder weight. Completeness of a multiplicative seed list, additional higher-weight compatibility conditions, and numerical period independence are separate questions.

3.16 Reducing the tetralogarithm relation to a functional identity

Duplication reduces the proved tetralogarithm relation (3.22) to the following equivalent cross-base relation. The complete certificate proof that follows establishes both; no independence claim about the six individual values is needed.

Two duplications $\mathrm{Li}_4(\frac14) = 8\big(\mathrm{Li}_4(\frac12) + \mathrm{Li}_4(-\frac12)\big)$ and $\mathrm{Li}_4(\frac19) = 8\big(\mathrm{Li}_4(\frac13) + \mathrm{Li}_4(-\frac13)\big)$ reduce the left side to the reflected arguments, giving the following *cross-base Kummer specialization*:

$$\begin{aligned}
\mathrm{Li}_4(\tfrac34) + 4\,\mathrm{Li}_4(\tfrac23) = {}& -\tfrac{16}{3}\,\mathrm{Li}_4(\tfrac12) - 4\,\mathrm{Li}_4(-\tfrac12) + 2\,\mathrm{Li}_4(\tfrac13) - 2\,\mathrm{Li}_4(-\tfrac13) + \tfrac{19}{540}\pi^4 + \tfrac59\pi^2\ln^2 2 + \tfrac16\pi^2\ln^2 3 \\
&- \tfrac23\pi^2\ln 2\ln 3 - \tfrac{19}{18}\ln^4 2 - \tfrac16\ln^4 3 + \tfrac43\ln^3 2\ln 3 - \ln^2 2\ln^2 3 + \tfrac23\ln 2\ln^3 3. \qquad (3.27)
\end{aligned}$$

Substituting the two duplication identities into (3.27) cancels the tetralogarithms in (3.22). The next proof supplies rational-function cancellation, real branch conventions, and endpoint descent directly, closing the earlier unproved-specialization gap.

3.17 An exact proof of the rational tetralogarithm conjecture

The six-term relation (3.22) was proposed in 2015 and 2017 [100, 101]. Its numerical discovery and the duplication reduction suggest a rational-function certificate. The continuation [102] supplies the forty-two rows below, together with the analytic descent and endpoint calculations. These jointly prove the relation; tensor cancellation alone would not fix the constants or branches. The proof uses elementary differential identities for classical polylogarithms and finite rational arithmetic, and also yields a one-parameter functional identity. No worldwide priority or period independence claim is needed.

Theorem 3.7 (The six rational arguments)**.** *Put $A = \log 2$ and $B = \log 3$. Then*

$$\begin{aligned} &720\,\mathrm{Li}_4(\tfrac{1}{2}) - 2160\,\mathrm{Li}_4(\tfrac{1}{3}) + 2160\,\mathrm{Li}_4(\tfrac{2}{3}) + 270\,\mathrm{Li}_4(\tfrac{1}{4}) \\ &\qquad + 540\,\mathrm{Li}_4(\tfrac{3}{4}) + 135\,\mathrm{Li}_4(\tfrac{1}{9}) \\ &= 19\pi^4 + 30(\pi^2 - B^2)(10A^2 - 12AB + 3B^2) - 30A^2(19A^2 - 24AB + 8B^2). \end{aligned} \tag{3.28}$$

In particular, the intermediate relation (3.27) above is also proved, by the two elementary duplications already recorded there.

3.17.1 A real generalized Rogers function and its exact descent

Write $\ell_n(x) = \mathrm{Re}\,\mathrm{Li}_n(x)$ on the principal branch. For $x \in \mathbb{R} \setminus \{0, 1\}$ set $L = \log|x|$, $M = \log|1 - x|$, so $\ell_1(x) = -M$. Define

$$\begin{aligned} R_2(x) &= \ell_2(x) - \tfrac{1}{2}L\ell_1(x), \\ R_3(x) &= \ell_3(x) - L\ell_2(x) + \tfrac{1}{3}L^2\ell_1(x), \\ R_4(x) &= \ell_4(x) - L\ell_3(x) + \tfrac{1}{2}L^2\ell_2(x) - \tfrac{1}{8}L^3\ell_1(x). \end{aligned}$$

These functions extend continuously by $R_n(0) = 0$ and $R_n(1) = \zeta(n)$. Direct telescoping differentiation gives

$$dR_n(x) = \frac{(-1)^n(n-1)}{n!}L^{n-2}(L\,dM - M\,dL), \qquad n = 2, 3, 4. \tag{3.29}$$

At a crossing of $x = 1$, all logarithmically singular terms have positive powers of L, which tend to zero; this proves the stated continuity and allows identities obtained on regular subintervals to be joined.

Let $V = \mathbb{Q}(t)^\times \otimes_{\mathbb{Z}} \mathbb{Q}$, written additively, and put

$$\beta_n[f] = f^{\otimes(n-2)} \otimes (f \wedge (1 - f)).$$

The sign of a rational function is torsion and vanishes in V. Rational prime factors do not vanish and must be retained.

Lemma 3.8 (Descent with several independent logarithms)**.** *Let $f_i(t)$ be nonconstant rational functions with no zero or pole on an interval I, and let $v_i \in \mathbb{Q}$ satisfy $\sum_i v_i\beta_4[f_i] = 0$. For every linear functional $\eta : V \to \mathbb{R}$ and each $n = 2, 3, 4$,*

$$\sum_i v_i\eta(f_i)^{4-n}R_n(f_i(t))$$

is constant on I, with the continuous interpretation at $f_i(t) = 1$.

Proof. Contract $4 - n$ of the first two tensor slots with η. The result is $\sum_i v_i \eta(f_i)^{4-n} \beta_n[f_i] = 0$. Evaluate the remaining ordinary tensor slots by $g \mapsto \log|g(t)|$ and the wedge by $g \wedge h \mapsto \log|g|\, d\log|h| - \log|h|\, d\log|g|$. Equation (3.29) gives a zero derivative on every regular subinterval. Continuity joins the constants across the finitely many crossings of $f_i = 1$. □

The point of the lemma is that $\eta(f_i)$ need not be a rational multiple of a single logarithm. At the final specialization $t = 1/2$, the needed values are independent linear forms in $A = \log 2$ and $B = \log 3$. Thus the one-logarithm golden descent does not have to be applied outside its stated hypotheses.

3.17.2 The complete finite certificate

For the following forty-two rows define

$$f_i(t) = \varepsilon_i 2^{d_i} t^{a_i} (1-t)^{b_i} (1+t)^{c_i}. \tag{3.30}$$

All $a_i \geq 0$, and all factors except the displayed sign are positive for $0 < t < 1$. The coefficients below are primitive integers.

i	v_i	ε_i	d_i	a_i	b_i	c_i	i	v_i	ε_i	d_i	a_i	b_i	c_i
1	-1	+	-1	0	0	1	22	3	+	-1	1	0	-1
2	-11	+	0	0	0	1	23	-4	+	0	1	0	-1
3	-6	-	0	0	0	1	24	6	-	0	1	0	-1
4	-3	+	1	0	0	1	25	1	+	1	1	0	-1
5	-4	+	0	0	1	-1	26	-3	+	-1	1	0	0
6	4	-	0	0	1	-1	27	-3	-	-1	1	0	0
7	-1	+	-1	0	1	0	28	6	+	0	1	0	0
8	22	+	0	0	1	0	29	-6	-	0	1	0	0
9	21	-	0	0	1	0	30	3	+	1	1	0	0
10	-3	+	1	0	1	0	31	3	-	1	1	0	0
11	4	+	0	0	1	1	32	1	+	-1	1	0	1
12	-1	+	0	0	2	1	33	1	-	-1	1	1	0
13	-1	-	0	0	3	0	34	3	+	0	1	1	0
14	-1	+	-1	1	-2	0	35	3	-	0	1	1	0
15	-6	-	0	1	-2	0	36	-4	-	0	2	-1	-1
16	-6	-	0	1	-1	-1	37	-3	-	0	2	-1	0
17	3	-	-1	1	-1	0	38	-1	+	1	2	-1	0
18	-21	+	0	1	-1	0	39	-3	+	0	2	0	-1
19	-7	-	0	1	-1	0	40	-1	+	1	2	0	-1
20	1	-	1	1	-1	0	41	1	+	0	3	-3	0
21	-1	-	-1	1	0	-2	42	1	-	0	3	-1	-2

Lemma 3.9 (Exact tensor cancellation)**.** *The displayed rational functions satisfy*

$$\sum_{i=1}^{42} v_i \beta_4[f_i] = 0. \tag{3.31}$$

Proof. The complete factor list needed for all f_i and $1 - f_i$, modulo sign, is

$$2, \quad t, \quad t-1, \quad t+1, \quad t-2, \quad t+2, \quad 2t-1, \quad 2t+1, \quad t^2-t-1, \quad t^2-t+1.$$

Use their exponent vectors as coordinates in V. For example, the constant 2 occurs in

$$1-\frac{1+t}{2}=\frac{1-t}{2},\qquad 1-\frac{t}{2(1-t)^2}=\frac{(2t-1)(t-2)}{2(1-t)^2}.$$

The first two slots of β_4 are symmetric. For each of the ten unordered pairs of coordinates of $2, t, 1-t, 1+t$, multiply the corresponding exponents and the two-by-two alternating minors of the vectors of $f_i, 1-f_i$, and sum with coefficient v_i. Every resulting coordinate is zero. There are 230 coordinates touched by the finite sum.

The accompanying `tetralogarithm_certificate.json` gives exactly the forty-two displayed rows. The self-contained verifier independently factors every rational function, checks the polynomial factorizations by multiplication, retains the prime-2 coordinate, and performs the indicated sums over rational integers. It does not search for a relation or use numerical polylogarithms in this check. This is a finite exact verification of (3.31). □

3.17.3 The functional identity proved by the certificate

For $0<t<1$, put

$$A=\log 2,\qquad X=\log t,\qquad Y=\log(1-t),\qquad Z=\log(1+t),$$

and define

$$\begin{aligned}P_4=&\frac{A^4}{4}-\frac{A^3(Y+Z)}{6}+\frac{A^2(Y^2+Z^2)}{4}+\frac{5A(Y^3+Z^3)}{6}-XY(3Y+Z)(Y+Z)\\&+\frac{17Y^4+8Y^3Z+9Y^2Z^2+2YZ^3+2Z^4}{6},\end{aligned}\tag{3.32}$$

$$P_2=-6A^2-4A(Y+Z)+\frac{5}{2}Y^2+8YZ-8Z^2.\tag{3.33}$$

Theorem 3.10 (A rational-function identity)**.** *With the forty-two rows of the table,*

$$\sum_{i=1}^{42}v_i\,\mathrm{Re}\,\mathrm{Li}_4(f_i(t))-4\,\mathrm{Li}_4(\tfrac12)=P_4+\zeta(2)P_2-\frac{71}{4}\zeta(4),\qquad 0<t<1.\tag{3.34}$$

The principal real parts and continuous values are used when an argument is greater than one or equals one.

Proof. Fix $t\in(0,1)$ and write $L_i=\log|f_i(t)|$. Apply Lemma 3.8 with the functional sending a prime or irreducible factor $2, s, 1-s, 1+s$ respectively to A, X, Y, Z, and extend it linearly to the factor space, assigning arbitrary values to its other basis elements. Unique factorization shows that such a functional exists, and its value on every $f_i(s)$ is L_i. This definition remains valid when one of the additional factors of $1-f_i(t)$ vanishes; no logarithm of zero is used to define the functional. The table functions have limits

$$e_i=\lim_{s\to0^+}f_i(s)=\begin{cases}0, & a_i>0,\\ \varepsilon_i 2^{d_i}, & a_i=0.\end{cases}$$

Thus for $n=2,3,4$,

$$\sum_i v_iL_i^{4-n}R_n(f_i(t))=\sum_{a_i=0}v_iL_i^{4-n}R_n(e_i).$$

The elementary triangular identity

$$\ell_4(x) = R_4(x) + LR_3(x) + \tfrac{1}{2}L^2R_2(x) + \tfrac{1}{24}L^3\ell_1(x)$$

then supplies an exact ordinary-polylogarithm identity. More explicitly, put $M_i = \log|e_i| = d_iA$ for $a_i = 0$, and set $Q_i = -M_i^3/8 + L_iM_i^2/3 - L_i^2M_i/4$. The result is

$$\sum_i v_i\ell_4(f_i(t)) = \sum_{a_i=0} v_i\big\{\ell_4(e_i) + (L_i - M_i)\ell_3(e_i) + \tfrac{1}{2}(L_i - M_i)^2\ell_2(e_i) + Q_i\ell_1(e_i)\big\} + \frac{1}{24}\sum_i v_iL_i^3\ell_1(f_i(t)). \tag{3.35}$$

When $e_i = 1$, $Q_i = 0$ and the corresponding last term in braces is zero; no divergent value is assigned to $\ell_1(1)$. Likewise $L_i^3\ell_1(f_i(t))$ is assigned its limiting value zero at $f_i(t) = 1$.

Only $e_i \in \{1, -1, 1/2, 2\}$ occur. The required lower values are

e	$\ell_1(e)$	$\ell_2(e)$	$\ell_3(e)$
-1	$-A$	$-\zeta(2)/2$	$-3\zeta(3)/4$
$1/2$	A	$\zeta(2)/2 - A^2/2$	$7\zeta(3)/8 - \zeta(2)A/2 + A^3/6$
2	0	$3\zeta(2)/2$	$7\zeta(3)/8 + 3\zeta(2)A/2$

and $\ell_2(1) = \zeta(2)$, $\ell_3(1) = \zeta(3)$. For completeness, the lower reflection and Landen identities are

$$\begin{aligned}
\operatorname{Li}_2(x) + \operatorname{Li}_2(1-x) &= \zeta(2) - \log x\log(1-x),\\
\operatorname{Li}_2\left(-\frac{x}{1-x}\right) &= -\operatorname{Li}_2(x) - \frac{1}{2}\log^2(1-x),\\
\operatorname{Li}_3(x) + \operatorname{Li}_3(1-x) + \operatorname{Li}_3\left(-\frac{x}{1-x}\right) &= \zeta(3) + \zeta(2)\log(1-x)\\
&\quad - \frac{1}{2}\log x\log^2(1-x) + \frac{1}{6}\log^3(1-x),
\end{aligned}$$

for $0 < x < 1$. Differentiating the first two identities gives zero differences, with constants fixed at $x \to 0^+$. Using them in the derivative of the third gives zero as well, and the same endpoint fixes its constant. Substitution $x = 1/2$ proves the displayed half values. The values at 2 follow by real inversion. Thus these endpoint reductions require no integer-relation recognition.

The endpoint weight-four sum is exactly

$$\sum_{a_i=0} v_i\ell_4(e_i) = 4\operatorname{Li}_4(\tfrac{1}{2}) + \frac{A^4}{4} - 6\zeta(2)A^2 - \frac{71}{4}\zeta(4).$$

Substitute $L_i = d_iA + a_iX + b_iY + c_iZ$ and the factorizations of $1 - f_i$ in (3.35). All logarithms of the six additional polynomial factors cancel, as do the $\zeta(3)$ terms. The remaining polynomial is precisely $P_4 + \zeta(2)P_2 - 71\zeta(4)/4$, which proves the theorem. The verifier expands both the master formula and the displayed P_4, P_2 independently and checks their exact polynomial difference is zero. □

3.17.4 The rational specialization, including all inversion terms

For $r > 1$, the real inversion formulas at weight four are

$$\begin{aligned}
\ell_4(r) + \operatorname{Li}_4(r^{-1}) &= -\frac{\log^4 r}{24} + \zeta(2)\log^2 r + 2\zeta(4),\\
\operatorname{Li}_4(-r) + \operatorname{Li}_4(-r^{-1}) &= -\frac{\log^4 r}{24} - \frac{1}{2}\zeta(2)\log^2 r - \frac{7}{4}\zeta(4).
\end{aligned}$$

They follow by repeated differentiation from the real logarithm and dilogarithm inversion laws, fixing the constants at $r = 1$. Negative arguments inside the unit interval are removed by

$$\mathrm{Li}_4(-r) = \tfrac{1}{8}\,\mathrm{Li}_4(r^2) - \mathrm{Li}_4(r), \qquad 0 \le r \le 1.$$

At $t = 1/2$ one has $X = Y = -A$, $Z = B - A$ and $f_i(1/2) = \varepsilon_i 2^{d_i - a_i - b_i - c_i} 3^{c_i}$. Applying the three displayed rules to the left side of (3.35), after subtracting its endpoint weight-four sum and multiplying by 180, gives exactly the six tetralogarithms in (3.28). The remaining polynomial from those inversions is

$$\begin{aligned} I = {} & 120A^4 - 510A^3B + 765A^2B^2 - 510AB^3 + 150B^4 \\ & + \zeta(2)(90A^2 + 2880AB - 1980B^2) - 1710\zeta(4). \end{aligned}$$

The non-weight-four part of the specialized descent master formula is

$$\begin{aligned} D = {} & -450A^4 + 210A^3B + 225A^2B^2 - 150AB^3 + 60B^4 \\ & + \zeta(2)(1890A^2 + 720AB - 1440B^2). \end{aligned}$$

Consequently the six-term sum equals $D - I$, namely

$$\begin{aligned} & -570A^4 + 720A^3B - 540A^2B^2 + 360AB^3 - 90B^4 \\ & \quad + \zeta(2)(1800A^2 - 2160AB + 540B^2) + 1710\zeta(4). \end{aligned}$$

Using $\zeta(2) = \pi^2/6$, $\zeta(4) = \pi^4/90$ gives exactly (3.28), proving Theorem 3.7.

3.17.5 A further proved rational specialization

The functional identity also produces explicit identities beyond the six-term conjecture. The specialization $t = 1/3$ is convenient because all logarithms reduce to $A = \log 2$ and $B = \log 3$. The following corollary is a new entry in this report, with no claim of priority among classical tetralogarithm identities.

Corollary 3.11 (Twenty-five positive rational arguments)**.** *For the argument–coefficient pairs (r, c_r) in the table,*

$$\begin{aligned} \sum_r c_r\,\mathrm{Li}_4(r) = {} & \frac{109}{3}A^4 - 116A^3B + 54A^2B^2 + 36AB^3 - 21B^4 \\ & + \zeta(2)(316A^2 - 408AB + 132B^2) + 53\zeta(4). \end{aligned} \tag{3.36}$$

All arguments lie in $(0, 1)$, and the integer coefficient list is primitive: its greatest common divisor is one.

r	c_r	r	c_r
$1/1024$	1	$9/64$	-6
$9/1024$	-1	$1/6$	16
$1/81$	1	$2/9$	8
$1/64$	-4	$1/4$	-107
$1/36$	-6	$8/27$	8
$1/32$	-8	$1/3$	80
$4/81$	3	$3/8$	64
$1/16$	9	$4/9$	24
$1/12$	-24	$1/2$	-200
$64/729$	-1	$16/27$	-8
$3/32$	8	$3/4$	112
$1/9$	-14	$8/9$	32
$1/8$	64		

Proof. Set $t = 1/3$ in Theorem 3.10. The logarithms are $X = -B$, $Y = A - B$, and $Z = 2A - B$. Apply the real inversion formulas to arguments whose absolute value exceeds one, then the duplication formula $\mathrm{Li}_4(-r) = \mathrm{Li}_4(r^2)/8 - \mathrm{Li}_4(r)$ to negative arguments in $[-1, 0]$. Multiplication by eight clears the resulting coefficient denominators. Collecting equal positive arguments gives exactly the table, and collecting all inversion terms gives the displayed polynomial. The same certificate verifier performs these reductions over $\mathbb{Q}$ and checks that the ordinary-polynomial residual is zero. As an independent audit only, a direct evaluation of the reduced identity at ninety working decimal digits has absolute residual below 8×10^{-90}. □

The argument list and polynomial are supplied separately in `tetralogarithm_one_third.json`, and are also included in the shared certificate data. The replay checks the original conjecture, the functional identity, and this corollary in a single run.

3.17.6 Verification, provenance, and further questions

The exact verifier finishes with a zero tensor residual, a zero ordinary-polynomial residual, and the six specified rational arguments with coefficients $(720, -2160, 2160, 270, 540, 135)$. Separately, it evaluates the functional identity at $t = 1/5, 37/100, 1/2, 4/5$ using 90 working decimal digits. The largest numerical discrepancy is approximately 7.9×10^{-90}. These calculations check the branch conventions; the proof is the finite tensor cancellation and the analytic descent.

The original questions are https://math.stackexchange.com/questions/1373123/ and https://mathoverflow.net/questions/261408/. The rational weight-four question and the separate golden weight-five ladders have different proof obligations: the certificate here settles the rational weight-four formula directly. The method is compatible with the higher-Bloch-condition approach to functional equations, but no unproved period conjecture or transcendence assertion is used.

Three useful next problems are to decompose the forty-two rows into a small set of named classical Kummer specializations; to specialize the proved functional identity at other rational or quadratic arguments and systematically reduce the resulting baskets; and to search for similarly

complete certificates for the unresolved golden weights six and seven. The Gaussian S_6 conjecture remains a distinct problem and is not promoted by this result. No minimality or arithmetic independence of the six rational tetralogarithms is asserted.

The complete delivered certificate and verifier are preserved under `reports/uniform-transition-continuation/continuations/tetralogarithm-euler-polynomial-zeros/code/`. The fresh isolated replay is recorded under `verification/seventh-replay/tetra/`. It checks all forty-two rows and 230 touched tensor coordinates, the exact lower-weight polynomial descent, both rational specializations, and separately labeled numerical branch diagnostics. The other claims in that package have independent proofs and retain their editorial-ledger audit status.

3.18 Canonical index-12/20/24 ladders and a ζ-elimination chain

Reshetnikov's Li_5 conjectures are the weight-5 entries of three first-degree golden ladders (Abouzahra–Lewin). For positive integer n, use the following canonical (monic, a^{n-1}-normalized) notation. Put $\rho = \phi^{-1}$, $\ell = \ln \rho$, and

$$\mathrm{tw}(n) = D_1 \frac{\ell^n}{n!} + D_2 \zeta(2) \frac{\ell^{n-2}}{(n-2)!} + D_3 \zeta(4) \frac{\ell^{n-4}}{(n-4)!},$$

where a summand with a negative factorial index is omitted. The previously implicit L_{12} is reconstructed from the first displayed weight-5 formula using the same normalization:

$$\begin{aligned}
L_{12}(n) &= \frac{\mathrm{Li}_n(\rho^{12})}{12^{n-1}} - \frac{3}{2}\frac{\mathrm{Li}_n(\rho^6)}{6^{n-1}} - \frac{\mathrm{Li}_n(\rho^4)}{4^{n-1}} + \frac{11}{48}\frac{\mathrm{Li}_n(\rho^2)}{2^{n-1}} + \mathrm{tw}(n),\\
L_{20}(n) &= \frac{\mathrm{Li}_n(\rho^{20})}{20^{n-1}} - \frac{\mathrm{Li}_n(\rho^{10})}{10^{n-1}} - 3\frac{\mathrm{Li}_n(\rho^4)}{4^{n-1}} + \frac{1}{2}\frac{\mathrm{Li}_n(\rho^2)}{2^{n-1}} + \mathrm{tw}(n),\\
L_{24}(n) &= \frac{\mathrm{Li}_n(\rho^{24})}{24^{n-1}} - \frac{4}{3}\frac{\mathrm{Li}_n(\rho^{12})}{12^{n-1}} - 2\frac{\mathrm{Li}_n(\rho^8)}{8^{n-1}}\\
&\quad + \frac{7}{3}\frac{\mathrm{Li}_n(\rho^4)}{4^{n-1}} - \frac{205}{576}\frac{\mathrm{Li}_n(\rho^2)}{2^{n-1}} + \mathrm{tw}(n).
\end{aligned}$$

For L_{12}, L_{20}, L_{24} respectively, the triples (D_1, D_2, D_3) are

$$(-13/48, 1/48, -19/1728), \quad (-1/2, 1/25, -89/4000), \quad (179/576, -5/192, 629/41472).$$

The displayed weight-5 formulas are equivalent, by exact rational recombination and normalization, to

$$\begin{aligned}
L_{12}(5) &= \frac{67}{6912}\zeta(5), \qquad L_{20}(5) = \frac{201}{10000}\zeta(5),\\
L_{24}(5) &= -\frac{1541}{110592}\zeta(5).
\end{aligned}$$

For the last formula, first add 64/3 times the first displayed weight-5 formula to the third, eliminating $\mathrm{Li}_5(\rho^6)$, then divide by $-15 \cdot 24^4$. This supplies the missing notation; it does not provide a new proof of those evaluations. The source's low-order vanishing statements, with the factorial convention above, concern $1 \le n \le 4$. The rational combinations $M_{20} = L_{20} - \frac{1296}{625} L_{12}$ and $M_{24} = L_{24} + \frac{23}{16} L_{12}$ cancel the displayed $\zeta(5)$ coefficients exactly. The source's subsequent numerical extrapolation reaches weight 7 and forms $T = M_{24} + \frac{409375}{373248} M_{20}$ to cancel its $\zeta(7)$ term and reach weight 9, proposing

$$T(9) = \frac{66452911}{41278242816000}\zeta(9) + \frac{3537283}{2063912140800}\zeta(8)\ell - \frac{11527}{2866544640}\zeta(6)\frac{\ell^3}{3!}.$$

The source reports direct substitution and an independent PSLQ check at 460 digits for this expression. Those checks record numerical evidence for the higher-order extrapolation, rather than a functional-equation certificate. This presentation follows the Abouzahra–Lewin ladder construction [36]; the single eight-argument formulas (3.25)–(3.26) are written separately.

3.19 The minimal Pisot ladder base

Beyond the golden ratio, the reciprocal of the minimal Pisot number $\omega = 0.7548776\ldots$, the real root of $x^3 + x^2 - 1$, carries its own ladders. Abouzahra, Lewin, and Xiao Hongnian studied this field in their published 1987 sequel [37]. Its seeds include the pure-unit relations $1-\omega = \omega^5$, $1-\omega^2 = \omega^3$, $1-\omega^3 = \omega^2$, and $1-\omega^5 = \omega$. The source records the following low-weight formulas, with $\lambda = \ln\omega$:

$$\tfrac{1}{2}\operatorname{Li}_2(\omega^2) - \operatorname{Li}_2(\omega^5) - \lambda^2 = 0, \qquad -\operatorname{Li}_3(\omega) + \tfrac{6}{4}\operatorname{Li}_3(\omega^2) + \tfrac{3}{9}\operatorname{Li}_3(\omega^3) - \operatorname{Li}_3(\omega^5) - \tfrac{6\lambda^3}{3!} = 0,$$
$$-71\operatorname{Li}_4(\omega) - \tfrac{120}{8}\operatorname{Li}_4(\omega^2) + \tfrac{1053}{27}\operatorname{Li}_4(\omega^3) - 17\operatorname{Li}_4(\omega^5) + \tfrac{5488}{343}\operatorname{Li}_4(\omega^7) - \operatorname{Li}_4(\omega^{14}) + 51\zeta(4) + \tfrac{2010\lambda^4}{4!} - \tfrac{84\zeta(2)\lambda^2}{2!} = 0.$$

The weight-four formula uses the extra legs ω^7, ω^{14}. The later Bloch-framework investigation identifies higher ladders for this field; the earlier assertion that the tower stops at weight four is discarded. A fixed argument set can stop yielding detected relations while larger sets continue to produce them.

3.20 Hunting more ladder bases, and a methodological caveat

Pure-unit seeds $1 - \rho^a = \rho^b$ offer a useful way to discover ladders. They do not exhaust general multiplicative relations, and a bounded search over polynomial coefficients does not cover all monic polynomials of a given degree. Root disguises are easy to miss: the root of $x^4 + x^2 - 1$ becomes a golden reciprocal after squaring. The following two cubic and quartic relations illustrate why negative searches require this care.

Two bases the earlier survey had dismissed as "weight-2 only" do, in fact, carry weight-3 ladders. With ρ the root of $x^3 + x - 1$ (the supergolden reciprocal) and of $x^4 + x - 1$ (the reciprocal of the second-smallest Pisot number $x^4 - x^3 - 1$),

$$-60\operatorname{Li}_3(\rho) + 54\operatorname{Li}_3(\rho^2) - 32\operatorname{Li}_3(\rho^3) - 3\operatorname{Li}_3(\rho^6) = 24\ln^3\rho - 4\pi^2\ln\rho - 36\zeta(3),$$
$$-12\operatorname{Li}_3(\rho) + 12\operatorname{Li}_3(\rho^3) - 12\operatorname{Li}_3(\rho^4) - 3\operatorname{Li}_3(\rho^6) = 22\ln^3\rho - 2\pi^2\ln\rho - 12\zeta(3),$$

each verified in two engines (and by direct substitution to 400+ digits). The lesson is methodological: **a PSLQ "no relation" is not a proof of non-existence.** The first relation above was *missed* by a pure-tower PSLQ search at 300 digits, yet passed direct substitution at 400 digits. Its exact proof below establishes that this was a false negative. Recomputing a frozen candidate at precision well above its discovery precision is a useful diagnostic; the analytic proofs below establish the two identities. Whether these two bases climb past weight 3 remains open in the present investigation; the question is delicate precisely because PSLQ negatives cannot settle it.

3.21 Proof certificates for the cubic and quartic trilogarithm ladders

The two displayed formulas record two trilogarithm combinations at the positive roots of $r^3 + r = 1$ and $r^4 + r = 1$, with substantial experimental evidence. Their exact functional-equation certificates now explain the coefficients. We give such derivations. The quartic identity belongs to a uniform family; the cubic identity has a short certificate using two specializations of Kummer's nine-term

relation and four Rogers pentagons. These are exact consequences of classical functional equations, not claims that the underlying equations are new.

3.21.1 A uniform ladder from a complementary pair of powers

For $0 < x < 1$ the real Landen relation for the trilogarithm is

$$\operatorname{Li}_3(x) + \operatorname{Li}_3(1-x) + \operatorname{Li}_3(1-x^{-1}) = \zeta(3) + \frac{\pi^2}{6}\log x - \frac{1}{2}\log^2 x \log(1-x) + \frac{1}{6}\log^3 x. \tag{3.37}$$

For a modern account and its arithmetic analogues see Nakamura and Shiraishi, *Landen's trilogarithm functional equation and ℓ-adic Galois multiple polylogarithms* [63]. For completeness, a real-variable proof is especially short. Differentiation of the left side, followed by

$$\operatorname{Li}_2(x) + \operatorname{Li}_2(1-x) = \frac{\pi^2}{6} - \log x \log(1-x), \qquad \operatorname{Li}_2(1-x^{-1}) = -\operatorname{Li}_2(1-x) - \frac{1}{2}\log^2 x,$$

gives

$$\frac{\pi^2/6 - \log x \log(1-x)}{x} + \frac{\log^2 x}{2x(1-x)}.$$

This is the derivative of the right side. The difference tends to zero as $x \to 1^-$, proving (3.37). The two dilogarithm relations used here follow themselves by one differentiation and an endpoint value. All logarithms in the calculation are real, so there is no branch ambiguity.

Theorem 3.12 (Complementary-power trilogarithm ladder)**.** *Let a, b be positive integers with $a \le b$, and let $0 < \rho < 1$ be the unique positive solution of $\rho^a + \rho^b = 1$. Set $\lambda = \log \rho$ and $d = b - a$. Then*

$$\begin{aligned} &\operatorname{Li}_3(\rho^a) + \operatorname{Li}_3(\rho^b) - \operatorname{Li}_3(\rho^d) + \frac{1}{4}\operatorname{Li}_3(\rho^{2d}) \\ &\qquad = \zeta(3) + \frac{a\pi^2}{6}\lambda + \frac{a^2(a-3b)}{6}\lambda^3. \end{aligned} \tag{3.38}$$

When $a = b$, the powers ρ^0 are interpreted as 1.

Proof. The polynomial $t^a + t^b$ is strictly increasing from 0 to 2 on $[0, 1]$, which proves existence and uniqueness. Substitute $x = \rho^a$ in (3.37); its other arguments are $1 - x = \rho^b$ and $1 - x^{-1} = -\rho^{b-a}$. The absolutely convergent series identity

$$\operatorname{Li}_3(z) + \operatorname{Li}_3(-z) = \frac{1}{4}\operatorname{Li}_3(z^2), \qquad 0 \le z \le 1,$$

eliminates the negative argument. The logarithmic terms become exactly the right side of (3.38). □

In particular, if $r_m^m + r_m = 1$ and $\lambda_m = \log r_m$, then

$$\operatorname{Li}_3(r_m) + \operatorname{Li}_3(r_m^m) - \operatorname{Li}_3(r_m^{m-1}) + \frac{1}{4}\operatorname{Li}_3(r_m^{2m-2}) = \zeta(3) + \frac{\pi^2}{6}\lambda_m - \frac{3m-1}{6}\lambda_m^3. \tag{3.39}$$

This supplies an infinite family of finite exact reductions, although different pairs (a, b) can describe the same base after a power substitution. It does not assert that these exhaust all ladders for any one number field.

Corollary 3.13 (The quartic identity)**.** *If $r^4 + r = 1$ and $\lambda = \log r$, then*

$$-12\operatorname{Li}_3(r) + 12\operatorname{Li}_3(r^3) - 12\operatorname{Li}_3(r^4) - 3\operatorname{Li}_3(r^6) = 22\lambda^3 - 2\pi^2\lambda - 12\zeta(3). \tag{3.40}$$

Proof. Take $(a,b)=(1,4)$ in Theorem 3.12 and multiply by -12. □

For $m=2$, (3.39) reduces to the familiar golden value

$$\mathrm{Li}_3(r_2^2)=\frac{4}{5}\zeta(3)+\frac{2\pi^2}{15}\log r_2-\frac{2}{3}\log^3 r_2.$$

For $m=3$ it gives a useful additional relation on the arguments r,r^2,r^3,r^4; the manuscript's cubic identity instead uses r,r^2,r^3,r^6. The change of the last argument requires more than the one-variable Landen relation.

3.21.2 An elementary dilogarithm certificate in the cubic field

Throughout this subsection let $r^3+r=1$, $0<r<1$, and put $\lambda=\log r$, $A=\log(1+r)$. Direct algebra in $\mathbb{Q}[r]/(r^3+r-1)$ gives

$$1-r=r^3, \qquad 1-r^2=r^3(1+r), \qquad 1-r^3=r, \tag{3.41}$$

$$1-r^4=r^2(1+r), \qquad 1+r^3=r^2(1+r)^2, \qquad 1-r^6=r^3(1+r)^2. \tag{3.42}$$

Define the Rogers dilogarithm on $(0,1)$ by

$$R(x)=\mathrm{Li}_2(x)+\frac{1}{2}\log x\log(1-x).$$

It satisfies $R(x)+R(1-x)=\zeta(2)$ and the pentagon

$$R(x)+R(y)=R(xy)+R\left(\frac{x(1-y)}{1-xy}\right)+R\left(\frac{y(1-x)}{1-xy}\right), \qquad 0<x,y<1. \tag{3.43}$$

The latter is a real-domain form of the classical five-term relation. It may also be verified by differentiating in x with fixed y and using the limit at $x=0$.

Lemma 3.14 (Cubic dilogarithm identity)**.**

$$\mathrm{Li}_2(r^6)=6\,\mathrm{Li}_2(r^2)-2\,\mathrm{Li}_2(r)-4\,\mathrm{Li}_2(r^3). \tag{3.44}$$

Proof. Set $s=r/(1+r)$ and $t=r^4/(1+r)$, so every argument below lies in $(0,1)$. Write $R_k=R(r^k)$ and $R_s=R(s)$, $R_t=R(t)$. Applying (3.43) at $(x,y)=(r,r)$ gives

$$2R_1=R_2+2R_s.$$

At $(x,y)=(r,r^3)$ its two remaining arguments are $1-s$ and t. Since $R_1+R_3=\zeta(2)$, reflection gives

$$R_t=R_s-R_4.$$

At $(x,y)=(r^2,r^4)$ its two remaining arguments are s and t, and therefore

$$R_6=R_2+2R_4-2R_s.$$

Finally, at $(x,y)=(r,r^2)$ these arguments are $1-r^2$ and r^4. Using reflection twice gives

$$R_4=2R_2-2R_3.$$

Eliminating R_s,R_t,R_4 now yields $R_6=6R_2-2R_1-4R_3$. The difference between $R(r^k)$ and $\mathrm{Li}_2(r^k)$ is $(k\lambda/2)\log(1-r^k)$. In the indicated combination these terms cancel: by (3.42), the required check is

$$3\lambda(3\lambda+2A)-6\lambda(3\lambda+A)+3\lambda^2+6\lambda^2=0.$$

This proves the claimed ordinary dilogarithm identity. □

3.21.3 Two Kummer substitutions prove the cubic trilogarithm

The single-valued trilogarithm used in the manuscript is

$$P_3(z) = \operatorname{Re}\operatorname{Li}_3(z) - \log|z|\operatorname{Re}\operatorname{Li}_2(z) - \frac{1}{3}\log^2|z|\log|1-z|.$$

The value at $z=1$ is defined by continuity. It obeys

$$P_3(z^{-1}) = P_3(z), \qquad P_3(z) + P_3(-z) = \frac{1}{4}P_3(z^2),$$

and the single-valued Landen relation $P_3(x) + P_3(1-x) + P_3(1-x^{-1}) = \zeta(3)$ for real $0 < x < 1$. The latter follows from (3.37) and the two dilogarithm identities in its proof.

We use Kummer's classical equation in the form stated and justified by Zagier, *Special values and functional equations of polylogarithms*, §7, Example 3, equation (7) [62]:

$$\begin{aligned} &2P_3(x) + 2P_3(y) + 2P_3\left(\frac{x(1-y)}{x-1}\right) + 2P_3\left(\frac{y(1-x)}{y-1}\right) \\ &\quad + 2P_3\left(\frac{1-x}{1-y}\right) + 2P_3\left(\frac{x(1-y)}{y(1-x)}\right) - P_3(xy) - P_3(x/y) \\ &\quad - P_3\left(\frac{x(1-y)^2}{y(1-x)^2}\right) = 2\zeta(3). \end{aligned} \tag{3.45}$$

We only use specializations with nonzero denominators and finite arguments. Since P_3 is single-valued, some real arguments exceeding 1 cause no branch correction in this equation; after inversion and duplication, all remaining arguments are $r^k \in (0,1)$.

Theorem 3.15 (The cubic identity)**.** *Let $r^3 + r = 1$ with $0 < r < 1$, and let $\lambda = \log r$. Then*

$$\begin{aligned} &-60\operatorname{Li}_3(r) + 54\operatorname{Li}_3(r^2) - 32\operatorname{Li}_3(r^3) - 3\operatorname{Li}_3(r^6) \\ &\qquad = 24\lambda^3 - 4\pi^2\lambda - 36\zeta(3). \end{aligned} \tag{3.46}$$

Proof. Write $P_k = P_3(r^k)$. Landen at r and duplication give

$$\mathsf{A} := P_1 - P_2 + P_3 + \tfrac{1}{4}P_4 = \zeta(3).$$

The nine arguments of (3.45) at $(x,y) = (r, r^3)$ and $(-r, r^2)$, in their displayed order, are

	1	2	3	4	5	6	7	8	9
(r, r^3)	r	r^3	$-r^{-1}$	$-r^5$	r^2	r^{-4}	r^4	r^{-2}	r^{-6}
$(-r, r^2)$	$-r$	r^2	r^4	$-r^{-1}$	r^{-3}	$-r^2$	$-r^3$	$-r^{-1}$	$-r^5$

respectively. Using inversion and $P_3(-r^k) = \frac{1}{4}P_{2|k|} - P_{|k|}$, these two equations reduce to

$$\begin{aligned} \mathsf{B} &:= \tfrac{3}{2}P_2 + 2P_3 + P_4 - 2P_5 - P_6 + \tfrac{1}{2}P_{10} = 2\zeta(3), \\ \mathsf{C} &:= -3P_1 + \tfrac{3}{4}P_2 + 3P_3 + \tfrac{5}{2}P_4 + P_5 - \tfrac{1}{4}P_6 - \tfrac{1}{4}P_{10} = 2\zeta(3). \end{aligned}$$

The exact rational row operation $4\mathsf{C} + 2\mathsf{B} - 48\mathsf{A}$ therefore proves

$$-60P_1 + 54P_2 - 32P_3 - 3P_6 = -36\zeta(3). \tag{3.47}$$

To recover ordinary trilogarithms, put $c_1 = -60, c_2 = 54, c_3 = -32, c_6 = -3$ and

$$D = \sum_{k\in\{1,2,3,6\}} c_k \operatorname{Li}_3(r^k), \quad S = \sum_{k\in\{1,2,3,6\}} c_k k \operatorname{Li}_2(r^k), \quad T = \sum_{k\in\{1,2,3,6\}} c_k k^2 \log(1-r^k).$$

Expansion of the definition gives $\sum c_k P_k = D - \lambda S - \lambda^2 T/3$. Lemma 3.14, together with $\operatorname{Li}_2(r) + \operatorname{Li}_2(r^3) = \pi^2/6 - 3\lambda^2$, gives

$$S = 72\lambda^2 - 4\pi^2.$$

The factorizations (3.42) give

$$T = -60(3\lambda) + 216(3\lambda + A) - 288\lambda - 108(3\lambda + 2A) = -144\lambda.$$

Consequently (3.47) becomes $D - 24\lambda^3 + 4\pi^2\lambda = -36\zeta(3)$, which is (3.46). □

3.21.4 What the certificate establishes and what remains open

The proofs settle the two displayed cubic and quartic trilogarithm identities without period-independence assumptions, numerical recognition, or unproved higher Bloch-group assertions. The universal family (3.38) also gives a principled generator of additional exact identities. It does not justify pseudointegration to weight four or above: differentiating with respect to an algebraic base does not preserve its isolated defining relation.

Natural next problems are to derive comparable local certificates for the weight-five through weight-nine golden formulas, find the smallest useful functional-equation closure for a chosen base, and extend the exact argument/substitution checker to arbitrary algebraic number fields. Any claim of minimality of the surviving period basket remains distinct from the existence of the displayed reductions.

3.22 Rigidity of complementary powers

The earlier complementary-power ladder starts from a relation $r^a + r^b = 1$, where $0 < r < 1$ and a, b are positive integers. That construction raises a concrete completeness question: can one base have many different pairs of complementary powers? We settle this question for pure powers, including arbitrary common divisors in the exponents. The decisive arithmetic input is a classical irreducibility theorem for trinomials. Our conclusions concern this precisely specified class of seeds; they do not classify general multiplicative relations among $1 - r^k$, or bound the maximum weight of polylogarithm ladders.

For $0 < r < 1$, define

$$\mathcal{C}(r) = \{(a, b) \in \mathbb{Z}^2_{>0} : a \le b,\ r^a + r^b = 1\}.$$

Let ω be the unique root in $(0, 1)$ of $X^3 + X^2 - 1$. Thus ω is the reciprocal of the plastic number.

Theorem 3.16 (Complete classification of collisions)**.** *For every real $0 < r < 1$, the set $\mathcal{C}(r)$ has at most two elements. It has two elements if and only if $r^d = \omega$ for some positive integer d, in which case*

$$\mathcal{C}(r) = \{(d, 5d), (2d, 3d)\}.$$

If $\mathcal{C}(r)$ contains (a, a), then $\mathcal{C}(r) = \{(a, a)\}$ and $r = 2^{-1/a}$.

The theorem supplies an exact answer to the seed-classification question posed after the earlier cubic and quartic ladder certificates. It is proved below as a consequence of the Ljunggren–Tverberg theorem, with all additional arguments included. We make no claim that this consequence is globally new in the literature. The related theorem of Aarts, Fokkink, and Kruijtzer classifies the two morphic numbers, by requiring both $p+1$ and $p-1$ to be powers of p; their proof also uses trinomial irreducibility [74]. Our formulation permits arbitrary positive exponent pairs and explicitly handles power substitutions.

3.22.1 The irreducibility input and the entire cyclotomic factor

For $1 \le a < b$, put $d = \gcd(a, b)$, $A = a/d$, $B = b/d$, and

$$F_{a,b}(X) = X^b + X^a - 1.$$

The following precise specialization is the arithmetic input.

Theorem 3.17 (Ljunggren–Tverberg, specialized)**.** *The polynomial $F_{a,b}$ has exactly one irreducible factor over $\mathbb{Q}$ whose roots are not roots of unity. Its other factor is*

$$C_{a,b}(X) = \begin{cases} X^{2d} - X^d + 1, & (A, B) \equiv (1, 5) \text{ or } (5, 1) \pmod 6, \\ 1, & \text{otherwise.} \end{cases}$$

In particular $H_{a,b} = F_{a,b}/C_{a,b}$ is irreducible, monic, and has constant term -1. Moreover $H_{a,b}(X) = H_{A,B}(X^d)$.

This is the trinomial result of Ljunggren and Tverberg [75, 76]; a modern explicit formulation, including the assertion that $H_{A,B}(X^d)$ remains irreducible, is Boyd–Martin–Thom, Lemma 4.1(a),(b) [77]. Only the trinomial theorem is used. Its separate quadrinomial analogue has a more delicate history and plays no role here.

For clarity, the displayed cyclotomic factor can be obtained directly, without relying on a table of sign cases. Suppose $|z| = 1$ and $z^a + z^b = 1$. The two unit vectors must be the two numbers $e^{i\pi/3}$ and $e^{-i\pi/3}$, in either order. If $w = z^d$, then w^A and w^B are these primitive sixth roots. A Bezout identity for A, B implies $w^6 = 1$, and w^A of order six then implies that w itself has order six. Consequently A, B have opposite residues $1, 5$ modulo six. Conversely those residues make every root of $X^{2d} - X^d + 1$ a root of $F_{a,b}$. These roots are simple: a common zero of $F_{a,b}$ and its derivative on the unit circle would give $a|z^a| = b|z^b|$, hence $a = b$, a contradiction. Thus the formula for $C_{a,b}$ accounts for the *entire* cyclotomic factor. The remaining assertion, irreducibility of $H_{a,b}$, is the cited theorem.

3.22.2 A conjugate-rotation invariant

Lemma 3.18 (The common divisor is intrinsic)**.** *Suppose $r^a + r^b = r^c + r^e = 1$, where $0 < r < 1$, $1 \le a < b$, and $1 \le c < e$. Then*

$$\gcd(a, b) = \gcd(c, e).$$

Proof. Set $d = \gcd(a, b)$. Since $0 < r < 1$, it is not a root of unity, so its minimal polynomial is $H_{a,b}$. Theorem 3.17 shows that this polynomial is a polynomial in X^d. If ζ is a primitive dth root of unity, $r\zeta$ is therefore a conjugate of r. The second equation also holds at this conjugate:

$$r^c\zeta^c + r^e\zeta^e = 1.$$

Taking absolute values gives equality in the triangle inequality, because $r^c + r^e = 1$. Both summands must have the argument of their positive real sum. Thus $\zeta^c = \zeta^e = 1$, proving $d \mid \gcd(c, e)$. Interchanging the pairs proves the reverse divisibility. □

This step prevents a gap that would arise from normalizing the two pairs independently. The cyclotomic factor can have degree $2d$, so a degree comparison before proving that the common divisors agree would not suffice.

3.22.3 The primitive coefficient comparison

Proof of Theorem 3.16. First suppose an equal pair occurs. Then $r^a = 1/2$, so r cannot be an algebraic integer: otherwise the rational number $r^a = 1/2$ would be an algebraic integer. Any unequal pair would make r a root of the monic integer polynomial $F_{c,e}$, a contradiction. There is only one equal pair because the powers of r strictly decrease.

Now suppose two distinct unequal pairs occur. By Lemma 3.18 they have a common gcd d. Replace r by r^d and divide all four exponents by d. Both pairs are now primitive. Their trinomials have the same irreducible noncyclotomic factor, and their cyclotomic factors belong to $\{1, \Phi_6\}$, where $\Phi_6 = X^2 - X + 1$. If the two cyclotomic factors agree, the trinomials agree, contrary to distinctness. After interchanging them if necessary, there must therefore be a polynomial identity

$$(X^2 - X + 1)(X^n + X^m - 1) = X^{n+2} + X^k - 1, \quad 1 \le m < n, \quad 1 \le k < n + 2. \tag{3.48}$$

The coefficient of X^{n+1} on the left is -1 unless $m = n - 1$, in which case it is zero. The right-hand coefficient is zero or one. Hence $m = n - 1$, and the identity reduces to

$$X^{n+2} + X^{n-1} - X^2 + X - 1 = X^{n+2} + X^k - 1.$$

For $n = 2$ the remaining polynomial is $2X - X^2$, which is not a monomial. For $n > 3$ its coefficient at X^2 is -1. Thus $n = 3$, which gives $m = 2$ and $k = 1$. The common factor is $X^3 + X^2 - 1$, and the two normalized pairs are $(1, 5)$ and $(2, 3)$. Conversely,

$$X^5 + X - 1 = (X^2 - X + 1)(X^3 + X^2 - 1)$$

shows that these two pairs do occur at ω. Finally, a third pair would have to be the same exceptional companion of either existing pair, so no third pair occurs. □

3.22.4 All signed pure-power unit equations

The functional equations for polylogarithms use signs and inverses as well as positive powers. The preceding classification automatically controls all such complementary arguments.

Corollary 3.19 (Sharp signed-unit count)**.** *Let* $\Gamma_r = \{\pm r^k : k \in \mathbb{Z}\}$ *and*

$$\mathcal{U}(r) = \{x \in \Gamma_r : 1 - x \in \Gamma_r\}.$$

Then $|\mathcal{U}(r)|$ *is one of* $0, 3, 6, 12$*. For every unequal pair* $(a, b) \in \mathcal{C}(r)$*, its contribution is the six-element set*

$$\left\{r^a, r^b, r^{-a}, r^{-b}, -r^{b-a}, -r^{a-b}\right\}. \tag{3.49}$$

These sets are disjoint for different unordered pairs. An equal pair contributes precisely $\{1/2, 2, -1\}$*. The case* $|\mathcal{U}(r)| = 12$ *occurs exactly when* $r^d = \omega$ *for some positive integer* d*.*

Proof. The transformations $x \mapsto 1-x$ and $x \mapsto 1/x$ preserve $\mathcal{U}(r)$. For example, $1 - 1/x = -(1-x)/x \in \Gamma_r$ whenever $x, 1-x \in \Gamma_r$. Their orbit consists of

$$x, \quad 1-x, \quad x^{-1}, \quad (1-x)^{-1}, \quad x/(x-1), \quad (x-1)/x.$$

Every real orbit has a representative in $(0,1)$: use $1/x$ if $x > 1$ and $1/(1-x)$ if $x < 0$. Such a representative and its complement are r^a, r^b with $a, b > 0$. The two members of a nondegenerate real orbit that lie in $(0,1)$ are exactly x and $1-x$, so its unordered positive pair is unique. Substituting $x = r^a$, $1 - x = r^b$ gives (3.49). All six entries are distinct unless $a = b$, when they reduce to the stated three values. The cardinalities now follow from Theorem 3.16. □

All four possibilities occur: $r = 1/3$, $1/2$, the reciprocal golden ratio, and ω give respectively $0, 3, 6, 12$. The conclusion classifies *pure signed-power complements*. It does not say that there are at most twelve useful polylogarithm arguments: products of units $1 - r^k$ can introduce many more arguments.

Corollary 3.20 (Rational exponents)**.** *The same collision classification holds for positive rational exponents, with the exceptional common scale d allowed to be a positive rational number. In particular there are at most two unordered positive rational pairs, and*

$$\#\{x \in \{\pm r^q : q \in \mathbb{Q}\} : 1 - x \in \{\pm r^q : q \in \mathbb{Q}\}\} \le 12.$$

Proof. Given two pairs, choose a common denominator N and apply Theorem 3.16 to $r^{1/N}$. A third rational pair could be included in the same denominator and would contradict the integer theorem. The orbit proof is unchanged. □

3.22.5 A provably complete bounded-degree search

Corollary 3.21 (Degree and an exhaustive exponent bound)**.** *If $r^a + r^b = 1$ with $1 \le a < b$, then*

$$[\mathbb{Q}(r) : \mathbb{Q}] = \begin{cases} b - 2d, & (a/d, b/d) \equiv (1,5) \text{ or } (5,1) \pmod 6, \\ b, & \text{otherwise}, \end{cases} \qquad d = \gcd(a,b).$$

Consequently all such bases of degree at most D are found by enumerating

$$1 \le a < b \le \left\lfloor \frac{5D}{3} \right\rfloor$$

and retaining those whose displayed degree is at most D. For primitive pairs it suffices to enumerate $b \le D + 2$. The only repeated bases in either enumeration are the classified plastic pairs. Equal pairs add the separate bases $2^{-1/a}$, with $a \le D$.

Proof. The degree formula is Theorem 3.17. In the exceptional case $b/d \ge 5$, so $b - 2d \ge 3d$. If the degree is at most D, then $d \le D/3$ and $b = (b-2d) + 2d \le 5D/3$. The nonexceptional case is immediate. For primitive pairs the possible degree loss is only two. The equal-pair polynomial $2X^a - 1$ has degree a and is irreducible: its reciprocal is, up to a nonzero scalar, $X^a - 2$, which is Eisenstein at 2. □

This bound is sharp for infinitely many D: the pair $(d, 5d)$ has degree $3d$ and largest exponent $5d$. Unlike a search over bounded polynomial coefficients, the algorithm has an explicit completeness theorem for its stated class. No numerical equality test is required. A base is stored by the exact

irreducible polynomial $H_{a,b}$ together with its unique root in $(0,1)$, and duplicate detection is exact polynomial comparison.

For illustration, the complete unequal-pair list through degree four is

degree	complement pair(s)	minimal polynomial
2	$(1,2)$	X^2+X-1
3	$(1,3)$	X^3+X-1
3	$(1,5),(2,3)$	X^3+X^2-1
4	$(1,4)$	X^4+X-1
4	$(2,4)$	X^4+X^2-1
4	$(3,4)$	X^4+X^3-1

The degree-four entry with pair $(2,4)$ is exactly a square root of the reciprocal golden ratio. Its presence illustrates why normalization of exponents is necessary when counting bases rather than primitive seed types.

The exact enumeration in the accompanying certificate gives:

D	1	2	3	4	5	6	7	8	9	10	11	12
unequal-pair bases	0	1	3	6	10	15	20	27	37	46	56	67
primitive bases	0	1	3	5	9	11	16	20	28	32	42	46
all bases	1	3	6	10	15	21	27	35	46	56	67	79

Each row counts bases of degree at most D. A primitive base has an unequal complementary pair of gcd one; the intrinsic-gcd lemma makes this definition independent of the choice of pair. The last row also includes the D equal-pair bases $2^{-1/a}$, $1 \le a \le D$.

3.22.6 An exact counting formula

The classification also counts bases without constructing or factoring their minimal polynomials. For $n \ge 1$ define

$$e(n) = \#\{a : 1 \le a < n,\ \gcd(a,n) = 1,\ (a,n) \equiv (1,5) \text{ or } (5,1) \pmod 6\}.$$

In particular $e(n) = 0$ unless $n \equiv 1$ or $5 \pmod 6$, and $e(1) = 0$.

Corollary 3.22 (Counting all bases by algebraic degree)**.** *Let $N(D)$ be the number of real $0 < r < 1$ with $[\mathbb{Q}(r) : \mathbb{Q}] \le D$ and $\mathcal{C}(r) \ne \varnothing$. For every positive integer D,*

$$N(D) = \frac{D(D+1)}{2} - \left\lfloor \frac{D}{3} \right\rfloor + \sum_{d=1}^{\lfloor D/3 \rfloor} \left\{ e\left(\left\lfloor \frac{D}{d} \right\rfloor + 1 \right) + e\left(\left\lfloor \frac{D}{d} \right\rfloor + 2 \right) \right\}. \tag{3.50}$$

In particular,

$$N(D) = \frac{1}{2}D^2 + O(D \log D), \qquad D \longrightarrow \infty.$$

Proof. There are $D(D-1)/2$ unequal pairs with largest exponent at most D, and their associated bases all have degree at most D. A pair with largest exponent greater than D can still have degree

at most D only in the exceptional cyclotomic case. Write that pair (dA, dB), where $\gcd(A, B) = 1$ and the residues are opposite $1, 5$ modulo six. The two inequalities are

$$dB > D, \qquad d(B-2) \le D.$$

Equivalently, $B \in \{\lfloor D/d \rfloor + 1, \lfloor D/d \rfloor + 2\}$. Since the exceptional normalized degree $B-2$ is at least three, only $d \le D/3$ can occur. This gives the sum in (3.50). Every base counted twice is $\omega^{1/d}$, of degree $3d$; there are exactly $\lfloor D/3 \rfloor$ such duplicates. Finally the D equal-pair bases are distinct from all unequal-pair bases and from each other. This proves the exact formula.

For the asymptotic statement, use $0 \le e(n) \le n$ to bound the sum by

$$\sum_{d \le D/3} \left(2 \left\lfloor \frac{D}{d} \right\rfloor + 3 \right) \le 2D \sum_{d \le D/3} \frac{1}{d} + D.$$

The harmonic sum is $O(\log D)$, proving the claim. □

For an alternative exact implementation, when $n > 1$ is coprime to six, Mobius inversion gives

$$e(n) = \sum_{d \mid n} \mu(d) \left\lfloor \frac{n/d + 1}{6} \right\rfloor. \tag{3.51}$$

Indeed, after writing $a = dj$, the condition $a \equiv -n \pmod 6$ becomes $j \equiv -(n/d) \pmod 6$ because every unit modulo six is its own inverse. For t coprime to six, the number of integers $1 \le j < t$ in the residue class $-t$ is $\lfloor (t+1)/6 \rfloor$. Summing with $\mu(d)$ imposes $\gcd(a, n) = 1$.

3.23 Exact proofs and log-free ladders at the plastic base

The exceptional base in Theorem 3.16 has two complementary pairs. This additional structure produces short proofs of two previously numerically supported formulas in the manuscript, as well as log-free identities for $\zeta(2)$ and $\zeta(3)$.

Throughout this section, $0 < q < 1$, $q^2 + q^3 = 1$, and $\lambda = \log q$. The factorization in the preceding proof gives

$$1 - q = q^5, \qquad 1 - q^2 = q^3, \qquad 1 - q^3 = q^2, \qquad 1 - q^5 = q. \tag{3.52}$$

All logarithms used in the final identities are real.

3.23.1 The dilogarithm formula: one Rogers pentagon

Write

$$R(x) = \mathrm{Li}_2(x) + \frac{1}{2} \log x \log(1-x), \qquad 0 < x < 1.$$

The Rogers reflection and pentagon equations are

$$R(x) + R(1-x) = \zeta(2), \tag{3.53}$$

$$R(x) + R(y) = R(xy) + R\left(\frac{x(1-y)}{1-xy} \right) + R\left(\frac{y(1-x)}{1-xy} \right). \tag{3.54}$$

They hold for $0 < x, y < 1$ and follow from the classical dilogarithm functional equations; alternatively one can differentiate in x and evaluate at $x = 0$. This real-domain formulation avoids branch choices [62].

Theorem 3.23 (The manuscript's plastic dilogarithm formula)**.**

$$\frac{1}{2}\operatorname{Li}_2(q^2) - \operatorname{Li}_2(q^5) = \lambda^2. \tag{3.55}$$

Proof. Set $R_k = R(q^k)$. At $(x, y) = (q, q^2)$ the three pentagon arguments are

$$xy = q^3, \qquad \frac{x(1-y)}{1-xy} = q^2, \qquad \frac{y(1-x)}{1-xy} = q^5.$$

Hence $R_1 + R_2 = R_3 + R_2 + R_5$, or $R_1 = R_3 + R_5$. Reflection gives $R_1 + R_5 = R_2 + R_3 = \zeta(2)$, so $R_2 = 2R_5$. Expanding the logarithmic corrections by (3.52) gives

$$\operatorname{Li}_2(q^2) + 3\lambda^2 = 2\operatorname{Li}_2(q^5) + 5\lambda^2,$$

as required. □

3.23.2 The trilogarithm formula: one Kummer specialization

For real nonzero $z \neq 1$, define the single-valued trilogarithm

$$P(z) = \operatorname{Re}\operatorname{Li}_3(z) - \log|z|\operatorname{Re}\operatorname{Li}_2(z) - \frac{1}{3}\log^2|z|\log|1-z|,$$

with the continuous value $P(1) = \zeta(3)$. The inversion and duplication equations are

$$P(z^{-1}) = P(z), \qquad P(z) + P(-z) = \frac{1}{4}P(z^2).$$

Kummer's equation is (3.45), with $P = P_3$. We use the form recorded by Zagier, Section 7, Example 3, equation (7) [62].

The single-valued Landen equation is

$$P(x) + P(1-x) + P(1-x^{-1}) = \zeta(3), \qquad 0 < x < 1.$$

Its ordinary real counterpart is

$$\operatorname{Li}_3(x) + \operatorname{Li}_3(1-x) + \operatorname{Li}_3(1-x^{-1}) = \zeta(3) + \zeta(2)\log x - \frac{1}{2}\log^2 x \log(1-x) + \frac{1}{6}\log^3 x. \tag{3.56}$$

One obtains it by differentiation using dilogarithm reflection and Landen, with the integration constant determined at $x = 1$. See also Nakamura–Shiraishi [63].

Theorem 3.24 (The manuscript's plastic trilogarithm formula)**.**

$$-\operatorname{Li}_3(q) + \frac{3}{2}\operatorname{Li}_3(q^2) + \frac{1}{3}\operatorname{Li}_3(q^3) - \operatorname{Li}_3(q^5) = \lambda^3. \tag{3.57}$$

Proof. Put $P_k = P(q^k)$. At $(x, y) = (q, q^2)$, the nine arguments in (3.45), in displayed order, are

$$q, \quad q^2, \quad -q^{-1}, \quad -q^4, \quad q^2, \quad q^{-3}, \quad q^3, \quad q^{-1}, \quad q^{-5}.$$

Using inversion and duplication, the equation becomes

$$\mathsf{K} = -P_1 + \frac{9}{2}P_2 + P_3 - 2P_4 - P_5 + \frac{1}{2}P_8 = 2\zeta(3).$$

Landen at $x = q$, with $1 - q = q^5$, similarly gives

$$\mathsf{A} = P_1 + P_5 - P_4 + \frac{1}{4}P_8 = \zeta(3).$$

Consequently the exact row operation $(\mathsf{K} - 2\mathsf{A})/3$ proves

$$-P_1 + \frac{3}{2}P_2 + \frac{1}{3}P_3 - P_5 = 0. \tag{3.58}$$

It remains to convert this to ordinary trilogarithms. Denote the left side of (3.57) by D, and put

$$S = -\operatorname{Li}_2(q) + 3\operatorname{Li}_2(q^2) + \operatorname{Li}_2(q^3) - 5\operatorname{Li}_2(q^5).$$

The two ordinary reflection equations and Theorem 3.23 give

$$\begin{aligned} \operatorname{Li}_2(q) + \operatorname{Li}_2(q^5) &= \zeta(2) - 5\lambda^2, \\ \operatorname{Li}_2(q^2) + \operatorname{Li}_2(q^3) &= \zeta(2) - 6\lambda^2, \\ \tfrac{1}{2}\operatorname{Li}_2(q^2) - \operatorname{Li}_2(q^5) &= \lambda^2. \end{aligned}$$

Taking minus the first row, plus the second, plus four times the third yields $S = 3\lambda^2$. The remaining logarithmic coefficient in the definition of P is

$$T = \{-1 \cdot 1^2 \cdot 5 + (3/2) \cdot 2^2 \cdot 3 + (1/3) \cdot 3^2 \cdot 2 - 1 \cdot 5^2 \cdot 1\}\lambda = -6\lambda.$$

Thus the left side of (3.58) is

$$D - \lambda S - \frac{\lambda^2}{3}T = D - 3\lambda^3 + 2\lambda^3 = D - \lambda^3.$$

This proves (3.57). □

The proof promotes the two displayed manuscript formulas from numerical support to exact consequences of classical functional equations. It does not rely on pseudointegration at an isolated algebraic base.

3.23.3 Two log-free identities

Theorem 3.25 (Log-free plastic ladders)**.** *At the same base q, the following identities hold:*

$$6\operatorname{Li}_2(q) - 5\operatorname{Li}_2(q^2) - 5\operatorname{Li}_2(q^3) + 6\operatorname{Li}_2(q^5) = \zeta(2), \tag{3.59}$$

$$12\operatorname{Li}_3(q) - 5\operatorname{Li}_3(q^2) - 4\operatorname{Li}_3(q^3) - 8\operatorname{Li}_3(q^4) + 8\operatorname{Li}_3(q^5) + 2\operatorname{Li}_3(q^8) = 4\zeta(3). \tag{3.60}$$

Proof. Six times the first ordinary reflection equation minus five times the second proves (3.59). For the trilogarithm, write $L_k = \operatorname{Li}_3(q^k)$. Equation (3.56) at $x = q$ and $x = q^2$, with duplication to eliminate negative arguments, gives

$$\begin{aligned} A &= L_1 + L_5 - L_4 + \frac{1}{4}L_8 = \zeta(3) + \zeta(2)\lambda - \frac{7}{3}\lambda^3, \\ B &= \frac{5}{4}L_2 + L_3 - L_1 = \zeta(3) + 2\zeta(2)\lambda - \frac{14}{3}\lambda^3. \end{aligned}$$

The logarithmic tails are proportional. Therefore $4(2A - B) = 4\zeta(3)$, exactly the claimed six-argument identity. □

These formulas are explicit additional identities for the package. Their derivations and the classical origins of the functional equations are given in full; no global priority claim is made. They also give exponentially convergent algebraic series, for example

$$\zeta(3) = \sum_{n=1}^{\infty} \frac{3q^n - \frac{5}{4}q^{2n} - q^{3n} - 2q^{4n} + 2q^{5n} + \frac{1}{2}q^{8n}}{n^3}. \tag{3.61}$$

Absolute convergence justifies expansion of the polylogarithms. If this series is truncated after N terms, its absolute tail is at most

$$\frac{39}{4}\,\frac{q^{N+1}}{(N+1)^3(1-q)},$$

obtained by summing the absolute coefficients and using $q^{kn} \le q^n$ for $k \ge 1$. The constant $39/4$ is deliberately simple; the sharper sum of the six individual geometric tails can also be used.

3.24 The modified polylogarithm

Zagier's one-valued modified polylogarithm (his eq. (33)) is

$$P_m(x) = \mathrm{Re}_m\Big(\sum_{j=0}^{m-1} \frac{2^j B_j}{j!}\,(\log|x|)^j\ \mathrm{Li}_{m-j}(x)\Big), \qquad \mathrm{Re}_m = \begin{cases} \mathrm{Re}, & m \text{ odd}, \\ \mathrm{Im}, & m \text{ even}, \end{cases} \tag{3.62}$$

with B_j the Bernoulli numbers and $\mathrm{Li}_1(x) = -\log(1-x)$. It is continuous on the Riemann sphere away from $\{0, 1, \infty\}$, satisfies $P_m(\bar{x}) = (-1)^{m-1}P_m(x)$, and $P_m(1) = \zeta(m)$ for odd m, 0 for even m; $P_2(\mathrm{i})$ is Catalan's constant. We verified our implementation against Zagier's worked examples $P_5(-\frac{1}{8}) - 126P_5(\frac{1}{2}) - 162P_5(-\frac{1}{2}) = \frac{403}{16}\zeta(5)$ (his eq. (825)) and the long thirteen-term relation (825)/(833), to 60 digits.

3.25 Ladders are Bloch-group elements

The Bloch-group viewpoint records the obstruction to simplifying a dilogarithm combination. At weight two the boundary sends $[x] \mapsto x \wedge (1-x)$; a vanishing boundary makes a combination a candidate for the Bloch group, modulo the five-term relation. Higher weights require further relations and compatibility conditions: a weight-one unit relation alone does not prove that its pseudointegrations are higher Bloch-group elements. Modified polylogarithms remove explicit logarithmic tails and make the following ladder combinations transparent:

$$\text{golden}: \quad 15\,P_3(\phi^{-2}) = 12\,\zeta(3), \tag{3.63}$$

$$\text{supergolden}: \quad -60\,P_3(\rho) + 54\,P_3(\rho^2) - 32\,P_3(\rho^3) - 3\,P_3(\rho^6) = -36\,\zeta(3), \tag{3.64}$$

both exact. Compare (3.63) with the "raw" form $15\,\mathrm{Li}_3(\phi^{-2}) = 10\ln^3\phi - 2\pi^2\ln\phi + 12\zeta(3)$: the modified trilogarithm removes the explicit logarithmic tail. At odd weight, comparable modified combinations require the appropriate lower-weight identities. At even weight the definition takes an imaginary part, and every term is real on $0 < x < 1$, so $P_{2j}(x) = 0$. The ordinary even-weight golden formulas retain their content; under this definition they cannot become nonzero even-zeta multiples. This distinction concerns the chosen modification, not a universal bound on regulator directions.

3.26 The bases as number fields

base ρ	minimal polynomial of $1/\rho$	field F	signature	disc
golden ϕ^{-1}	$x^2 - x - 1$	$\mathbb{Q}(\sqrt{5})$	$(2,0)$ totally real	5
minimal Pisot ω	$x^3 - x - 1$	$\mathbb{Q}(\theta)$	$(1,1)$	-23
supergolden recip.	$x^3 - x^2 - 1$	$\mathbb{Q}(\sigma)$	$(1,1)$	-31

The middle row is Zagier's own running example (§4–5, §9C): the field $\mathbb{Q}(\theta)$, $\theta^3 - \theta - 1 = 0$, whose ladder he calls "particularly spectacular," and whose defining unit relations $1 - \theta = -\theta^{-4}$, $1 + \theta = \theta^3, \ldots$ are precisely the seeds we rediscovered. Our supergolden base gives a *different* complex cubic field, discriminant -31.

3.27 Regulator parity and the limits of ladder counting

For $m \geq 2$, Borel's rank theorem gives rank $r_1 + r_2$ for $K_{2m-1}(F)$ when m is odd, and r_2 when m is even. The modified polylogarithm uses its real part at odd weight and imaginary part at even weight. In particular P_{2m} vanishes on real arguments, even when the number field has nonzero regulator directions at its complex embeddings. This explains why a real-argument search can miss arithmetic content visible at a complex embedding.

Cyclotomic unit searches compute $\operatorname{Norm} \Phi_k(\rho) = \operatorname{Res}(p, \Phi_k)$ and select the unit cases. For a fixed finite seed set, imposing successive regulator constraints is a finite linear-algebra problem. A dimension count can give a lower bound for a kernel or exhaust that chosen construction. It does not prove there are no additional seeds, no dependencies among constraints, and no other functional identities at higher weight.

The later project source identifies minimal-Pisot ladders through weight six. The published literature already includes the 1987 Abouzahra–Lewin–Xiao paper [37], which distinguishes analytic low-weight results from numerical higher ladders, and Gangl's proved weight-six and weight-seven relations for single-valued modified polylogarithms [39]. The modified and ordinary polylogarithms must be distinguished when comparing these statements, particularly at real arguments and even weight. Cohen–Lewin–Zagier's published numerical sixteenth-order Lehmer ladder [40] supplies another substantial example. These results supersede earlier bounded negative searches. The project source's stronger assertion that the supergolden field has no weight-five real ladder is not justified by exhausting its selected seed set, and is omitted. A maximal-weight theorem needs completeness for the chosen class of higher-weight relations; completeness of multiplicative seeds alone does not establish it.

3.28 Attributed dilogarithm and inverse-binomial examples

The following examples are attributed to Hakimoglu-Brown [41]. Their displayed numerical checks in the source are historical.

All to ≥ 38 digits (`tools/scratch/dilog-inverse-binomial/`); a representative selection:

- **Sun's open conjecture** (eq. (24)), proved in the paper via the $\mathbb{Q}(\sqrt{3})$ evaluation (25):

$$\sum_{n\geq 0} \frac{\binom{2n}{n}}{(2n+1)8^n} \Big(\sum_{0\leq k<n} \frac{(-1)^k}{2k+1} - \frac{(-1)^n}{2n+1} \Big) = -\frac{\sqrt{2}}{16}\pi^2.$$

- **Theorem 2** ($\mathbb{Q}(\sqrt{3})$): $8\,\mathrm{Li}_2(2-\sqrt{3}) - \mathrm{Li}_2(4\sqrt{3}-7) = \frac{5\pi^2}{12} + \ln(2-\sqrt{3})\ln(2+\sqrt{3})$, and $3\,\mathrm{Li}_2(7-4\sqrt{3}) = 6\,\mathrm{Li}_2(2-\sqrt{3}) + 6\,\mathrm{Li}_2(\sqrt{3}-2)$.
- **Lima's identity** ($\mathbb{Q}(\sqrt{2})$, eq. (26)): $\mathrm{Li}_2(\frac{1}{\sqrt{2}}) - \mathrm{Li}_2(1-\sqrt{2}) = \frac{\pi^2}{6} - \frac{\ln^2 2}{8} + \frac{1}{2}\ln 2\,\ln(\sqrt{2}-1)$.
- **Zagier's four-term** (eq. (27)): $3L(-\frac{1}{6}) - L(\frac{1}{8}) + L(\frac{1}{9}) + L(\frac{1}{28}) = -\frac{\pi^2}{12}$, with the (unnormalized) Rogers $L(x) = \mathrm{Li}_2(x) + \frac{1}{2}\ln|x|\ln|1-x|$; and its compressed three-term form.
- the rational-argument $\frac{\pi^2}{18} + 2\ln 2\ln 3 - 2\ln^2 2 - \frac{2}{3}\ln^2 3 = \mathrm{Li}_2(\frac{1}{4}) + \frac{1}{3}\mathrm{Li}_2(\frac{1}{9})$, the inverse-tangent/$\mathrm{Cl}_2(\frac{\pi}{3})$ relation (30), and $\mathrm{Li}_2(\frac{4}{3}) + \mathrm{Li}_2(4) = \frac{1}{2}(\pi^2 - \ln^2 3)$ (real parts).
- the **engine**: ${}_4F_3(\{1,1,1,\frac{3}{2}\};\{\frac{5}{2},\frac{5}{2},\frac{5}{2}\};1) = \frac{27}{2}\big(7\zeta(3) + (3-2G)\pi - 12\big)$ (D'Aurizio) and the cube-binomial ${}_4F_3$ of eq. (1).
- the paper's **claimed-original elliptic ${}_4F_3(1)$ evaluations** (eqs. 32, 33, 35):

$$\begin{aligned}
\mathrm{Cl}_2(\tfrac{\pi}{3}) &= \tfrac{\sqrt{3}}{2}\,{}_4F_3\big(\{1,\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}\};\{\tfrac{3}{2},\tfrac{5}{6},\tfrac{7}{6}\};1\big),\\
{}_4F_3\big(\{1,\tfrac{1}{2},\tfrac{2}{3},\tfrac{1}{3}\};\{\tfrac{3}{2},\tfrac{5}{4},\tfrac{3}{4}\};1\big) &= \tfrac{3\sqrt{3}}{4}\big(2\ln(4+3\sqrt{2}) + \ln(\tfrac{7}{2}-2\sqrt{3})\big),\\
\ln^2\!\Big(\tfrac{5+\sqrt{13}}{2\sqrt{3}}\Big) - \tfrac{\pi^2}{12} &= \tfrac{1}{180}\,{}_4F_3\big(\{1,1,1,\tfrac{3}{2}\};\{\tfrac{7}{6},\tfrac{11}{6},2\};-\tfrac{1}{324}\big).
\end{aligned}$$

All check out. One caution worth recording: equation (27) uses the *unnormalized* Rogers L (so the right side is $-\frac{\pi^2}{12}$, not a pure rational); with the $\frac{6}{\pi^2}$-normalized L the identity reads $3L(-\frac{1}{6}) - L(\frac{1}{8}) + L(\frac{1}{9}) + L(\frac{1}{28}) = -\frac{1}{2}$.

Connection. The paper evaluates inverse-binomial series with $\binom{2n}{n}^2$ and $\binom{3n}{n}$ via beta integrals; this is the reciprocal-binomial sibling of the central-binomial ladder $\sum\binom{2n}{n}4^{-n}(2n+1)^{-p}$ studied in `harvested-central-binomial-arcsin.wl`. The verified identities are stored (with attribution) in `src/PolyLog/identities/harvested-dilogarithm-ladders-2506.10206.wl`.

3.29 Weighted arctangent families in classical trilogarithms

The bilinear integral underlying Section 3.7 reduces to the product of its half-angle parameters. Inserting the weight $1/\sin x$ changes that structure, but it still leaves an exact evaluation in classical dilogarithms and trilogarithms. Hyperbolic sums and differences of the parameters replace their product. This provides a new arithmetic route at real algebraic parameters, alongside the unit-circle route of the inverse-hyperbolic companion below. The incoming continuation [120] supplies these two families; we give their complete proofs and strengthen its coefficient obstruction to arbitrary one-variable functions.

$$K(p,q) = \int_0^{\pi/2} \frac{\arctan(p\sin x)\arctan(q\sin x)}{\sin x}\,dx, \qquad p,q\in\mathbb{R}. \tag{3.65}$$

The integral in (3.65) is ordinary and convergent. Its integrand is $pq\,x + O(x^3)$ at zero. The principal real arctangent is used throughout. The polylogarithms below use convergent real series or absolutely convergent unit-circle boundary series, with the classical conventions of DLMF §25.12 [96].

3.29.1 The hyperbolic addition formula

Write

$$\chi_j(z) = \frac{\mathrm{Li}_j(z) - \mathrm{Li}_j(-z)}{2} = \sum_{k=0}^{\infty} \frac{z^{2k+1}}{(2k+1)^j} \qquad (|z| < 1)$$

for the Legendre chi function. The values used in the next theorem are real and lie in its ordinary convergent series domain, including the endpoints $\chi_2(1) = \pi^2/8$ and $\chi_3(1) = 7\zeta(3)/8$.

Theorem 3.26 (A real two-term weighted evaluation)**.** *Define the even function*

$$\mathfrak{F}(s) = \frac{\pi^2|s|}{8} - \frac{7}{4}\zeta(3) + |s|\chi_2(e^{-|s|}) + 2\chi_3(e^{-|s|}), \qquad s \in \mathbb{R}. \tag{3.66}$$

Its value at zero is $\mathfrak{F}(0) = 0$. The apparent lack of smoothness from the absolute values is removable: $\mathfrak{F}$ is real analytic on $\mathbb{R}$, and it is equivalently characterized by

$$\mathfrak{F}''(s) = \frac{s}{2\sinh s}, \qquad \mathfrak{F}''(0) = \frac{1}{2}, \qquad \mathfrak{F}(0) = \mathfrak{F}'(0) = 0. \tag{3.67}$$

For every $p, q \in \mathbb{R}$, put $\alpha = \operatorname{arcsinh} p$ and $\beta = \operatorname{arcsinh} q$. Then

$$\boxed{K(p,q) = \mathfrak{F}(\alpha+\beta) - \mathfrak{F}(\alpha-\beta).} \tag{3.68}$$

In particular the formula includes both diagonals $p = \pm q$, either zero parameter, and arbitrary signs without branch changes.

Proof. First, for $s > 0$ use $d\chi_j(e^{-s})/ds = -\chi_{j-1}(e^{-s})$ to obtain

$$\mathfrak{F}'(s) = \frac{\pi^2}{8} - \chi_2(e^{-s}) - s\chi_1(e^{-s}), \qquad \mathfrak{F}''(s) = s\chi_0(e^{-s}) = \frac{s}{2\sinh s}.$$

The limits at zero follow from the endpoint values of χ_2 and χ_3 and from $s\chi_1(e^{-s}) = \frac{s}{2}\log\coth(s/2) \to 0$. The unique twice-normalized primitive of the real analytic even function $s/(2\sinh s)$ is real analytic and even. Hence (3.66) agrees with this primitive on the whole real line; this proves all assertions in (3.67).

Differentiate (3.65) once in each parameter. On compact real parameter sets the differentiated integrand is bounded by an integrable function, and therefore

$$\begin{aligned} K_{pq}(p,q) &= \int_0^{\pi/2} \frac{\sin x\, dx}{(1+p^2\sin^2 x)(1+q^2\sin^2 x)} \\ &= \frac{p\operatorname{arcsinh} p/\sqrt{1+p^2} - q\operatorname{arcsinh} q/\sqrt{1+q^2}}{p^2-q^2} \qquad (p^2 \neq q^2). \end{aligned}$$

For the second equality use partial fractions and

$$\int_0^{\pi/2} \frac{\sin x}{1+p^2\sin^2 x}\, dx = \frac{\operatorname{arcsinh} p}{p\sqrt{1+p^2}},$$

whose zero-parameter value is 1 and which follows by $v = \cos x$.

Let $\widetilde{K}(\alpha,\beta)=K(\sinh\alpha,\sinh\beta)$. Multiplication by $\cosh\alpha\cosh\beta$ and the addition formulas for sinh now give

$$\begin{aligned}\widetilde{K}_{\alpha\beta}&=\frac{\alpha\sinh\alpha\cosh\beta-\beta\sinh\beta\cosh\alpha}{\sinh^2\alpha-\sinh^2\beta}\\&=\frac{1}{2}\left\{\frac{\alpha+\beta}{\sinh(\alpha+\beta)}+\frac{\alpha-\beta}{\sinh(\alpha-\beta)}\right\}.\end{aligned}\tag{3.69}$$

Both quotients on the last line have their removable value 1 at zero. Thus the equality holds at $\alpha=\pm\beta$ by continuity. By (3.67), this is the mixed derivative of $\mathfrak{F}(\alpha+\beta)-\mathfrak{F}(\alpha-\beta)$. Both this expression and $\widetilde{K}$ vanish on $\alpha=0$ and $\beta=0$. Integration of their equal mixed derivatives over the oriented rectangle from $(0,0)$ to (α,β) proves (3.68) for every real α,β. □

For algebraic real p,q, the two arguments $e^{-|\alpha+\beta|}$ and $e^{-|\alpha-\beta|}$ are algebraic. Indeed $e^{\alpha}=p+\sqrt{1+p^2}$ and $e^{\beta}=q+\sqrt{1+q^2}$ are positive algebraic numbers, and taking the absolute value in either exponent selects an algebraic number or its reciprocal. Thus (3.68) expresses every such integral by classical polylogarithms at real algebraic arguments and logarithms of positive algebraic numbers. It asserts no independence or minimality of the resulting constants.

The diagonal specializes particularly simply:

$$K(p,p)=\mathfrak{F}(2\operatorname{arcsinh}p).\tag{3.70}$$

For $\ell=\log(1+\sqrt{2})$ and $r=3-2\sqrt{2}$, this gives the exact identity

$$\boxed{\int_0^{\pi/2}\frac{\arctan^2(\sin x)}{\sin x}\,dx=\frac{\pi^2\ell}{4}-\frac{7}{4}\zeta(3)+2\ell\chi_2(r)+2\chi_3(r).}\tag{3.71}$$

Its numerical value is $0.71806431974110340703632871069 3\ldots$; the decimal only identifies the normalization, while the theorem proves the identity.

An exact weighted counterpart of the manuscript's Cleo integral also follows. Set $\alpha=\log(1+\sqrt{2})$ and $\beta=\log\bigl((1+\sqrt{5})/2\bigr)$. Then

$$\begin{aligned}&\int_0^{\pi/2}\frac{1}{\sin x}\arctan^2\!\left(\frac{6\sin x}{3+\cos 2x}\right)dx\\&\qquad=\mathfrak{F}(2\alpha)+2\mathfrak{F}(\alpha+\beta)-2\mathfrak{F}(\alpha-\beta)+\mathfrak{F}(2\beta).\end{aligned}\tag{3.72}$$

To verify the reduction put $s=\sin x\in[0,1]$. Since $\arctan s+\arctan(s/2)<\pi/2$, the tangent addition formula gives

$$\arctan\frac{6\sin x}{3+\cos 2x}=\arctan(\sin x)+\arctan(\tfrac{1}{2}\sin x).$$

Squaring and integrating yields $K(1,1)+2K(1,1/2)+K(1/2,1/2)$, proving (3.72) by (3.68).

3.29.2 The weighted inverse-hyperbolic companion

The same parameter method has a trigonometric form. It gives a second exact family, including convergent improper endpoint values.

Theorem 3.27 (An inverse-hyperbolic bilinear evaluation)**.** *For $|r|, |s| \le 1$ define*

$$L(r,s) = \int_0^{\pi/2} \frac{\operatorname{arctanh}(r\sin x)\operatorname{arctanh}(s\sin x)}{\sin x}\,dx,$$

using real inverse hyperbolic tangents and interpreting an endpoint with $|r| = 1$ or $|s| = 1$ as an improper integral. Put

$$\mathfrak{G}(t) = \frac{7}{4}\zeta(3) - t\operatorname{Im}\chi_2(e^{it}) - 2\operatorname{Re}\chi_3(e^{it}), \qquad |t| \le \pi. \tag{3.73}$$

The chi values on the unit circle are defined by their absolutely convergent series. Then $\mathfrak{G}$ is even, continuous on $[-\pi,\pi]$, and real analytic on $(-\pi,\pi)$, with

$$\mathfrak{G}''(t) = \frac{t}{2\sin t}, \qquad \mathfrak{G}''(0) = \frac{1}{2}, \qquad \mathfrak{G}(0) = \mathfrak{G}'(0) = 0. \tag{3.74}$$

For $a = \arcsin r$, $b = \arcsin s$, including either sign of either parameter,

$$\boxed{L(r,s) = \mathfrak{G}(a+b) - \mathfrak{G}(a-b).} \tag{3.75}$$

In particular,

$$\boxed{\int_0^{\pi/2} \frac{\operatorname{arctanh}^2(\sin x)}{\sin x}\,dx = \frac{7}{2}\zeta(3).} \tag{3.76}$$

Proof. Conjugation of the chi series proves evenness and endpoint continuity of (3.73). For $0 < t < \pi$, polylogarithm differentiation gives

$$\mathfrak{G}'(t) = \operatorname{Im}\chi_2(e^{it}) - t\operatorname{Re}\chi_1(e^{it}), \qquad \mathfrak{G}''(t) = t\operatorname{Im}\chi_0(e^{it}) = \frac{t}{2\sin t}.$$

Here $\operatorname{Re}\chi_1(e^{it}) = \frac{1}{2}\log\cot(t/2)$, so the limit of the first derivative at zero is zero. Also $\mathfrak{G}(0) = 0$ by $\chi_3(1) = 7\zeta(3)/8$. The normalized even primitive of $t/(2\sin t)$ is real analytic on $(-\pi,\pi)$, proving (3.74).

For $|r|, |s| < 1$, differentiation and partial fractions give

$$L_{rs}(r,s) = \frac{r\arcsin r/\sqrt{1-r^2} - s\arcsin s/\sqrt{1-s^2}}{r^2 - s^2} \qquad (r^2 \ne s^2).$$

The underlying elementary integral is $\int_0^{\pi/2} \sin x/(1 - r^2\sin^2 x)\,dx = \arcsin r/(r\sqrt{1-r^2})$, again obtained by $v = \cos x$. For $\widetilde{L}(a,b) = L(\sin a, \sin b)$ the sine addition formulas therefore yield

$$\widetilde{L}_{ab}(a,b) = \frac{1}{2}\left\{\frac{a+b}{\sin(a+b)} + \frac{a-b}{\sin(a-b)}\right\}, \qquad |a|, |b| < \pi/2.$$

The zero quotients take their removable value 1. This is the mixed derivative of the right side of (3.75), and both sides vanish on the two axes. Rectangle integration proves the identity in the open parameter square.

For all $|r|, |s| \le 1$ the absolute value of the integrand is bounded by $\operatorname{arctanh}^2(\sin x)/\sin x$. This majorant is $O(x)$ at zero and $O(\log^2(\pi/2 - x))$ at the other endpoint, so it is integrable. Dominated convergence extends the equality to the closed square. Finally $\chi_3(-1) = -7\zeta(3)/8$ and $\operatorname{Im}\chi_2(-1) = 0$ give $\mathfrak{G}(\pi) = 7\zeta(3)/2$, proving (3.76). The endpoint value also follows directly from $u = \operatorname{arctanh}(\sin x)$:

$$\int_0^\infty \frac{u^2}{\sinh u}\,du = 4\sum_{k=0}^{\infty} \frac{1}{(2k+1)^3} = \frac{7}{2}\zeta(3).$$

□

For algebraic $r, s \in [-1, 1]$, the chi arguments in this theorem are algebraic points on the unit circle, since $e^{ia} = \sqrt{1-r^2} + ir$ and $e^{ib} = \sqrt{1-s^2} + is$. The theorem has been proved directly over real parameters; it does not depend on an unstated continuation of the arctangent formula.

3.29.3 An exact obstruction to the one-product ansatz

The distinction from the unweighted factorization has a short exact proof. Set

$$u(p) = \frac{p}{1+\sqrt{1+p^2}}, \qquad u(0) = 0.$$

The unweighted integral is a function of $u(p)u(q)$. The weighted integral has no such factorization into any one-variable function on a real neighborhood of zero; no regularity assumption on that function is needed.

Proposition 3.28. *There is no function G defined on a real neighborhood of zero for which $K(p,q) = G(u(p)u(q))$ holds for all sufficiently small real p, q.*

Proof. Suppose such a function exists. Fix a sufficiently small $q_0 \neq 0$ and put $c = u(q_0) \neq 0$. The local inverse of u is $p = 2u/(1-u^2)$. The asserted identity therefore forces

$$G(t) = K\left(\frac{2(t/c)}{1-(t/c)^2}, q_0\right)$$

for every sufficiently small real t. The integral K is real analytic near the origin: on compact subdisks of $|p|, |q| < 1$, the product of the arctangent series divided by $\sin x$ converges normally and is dominated by $C \sin x$ for a parameter-independent constant C. Integration of this normally convergent series proves analyticity despite the apparent endpoint singularity. Hence the displayed expression forces G to be real analytic near zero.

For $|p|, |q| < 1$, integration of the absolutely convergent arctangent series gives

$$K(p,q) = \sum_{j,k\geq 0} \frac{(-1)^{j+k} p^{2j+1} q^{2k+1}}{(2j+1)(2k+1)} \frac{4^{j+k}}{(2j+2k+1)\binom{2j+2k}{j+k}}. \tag{3.77}$$

In particular,

$$K(p,q) = pq - \frac{2}{9}(p^3q + pq^3) + \text{terms of total degree at least six},$$

whereas

$$u(p)u(q) = \frac{pq}{4} - \frac{p^3q + pq^3}{16} + \text{terms of total degree at least six}.$$

The coefficient of pq would force $G'(0) = 4$, and hence the coefficient of p^3q would have to be $-1/4$, contradicting $-2/9$. Terms of order at least two in G cannot contain p^3q. □

The obstruction concerns this functional ansatz and implies no arithmetic independence. Theorem 3.26 gives a short alternative in hyperbolic sums and differences. Independent exact checks and signed, diagonal and endpoint quadratures audit the normalization; the analytic proofs establish the identities. The ordinary primitive and higher Euler ladder remain separate integration targets in the editorial ledger.

Chapter 4

Gaussian and Eisenstein depth

The classical transformations suggest a sharper question: how much nesting is necessary after known identities have been used? We first prove an infinite family of reductions. The shuffle and parity proofs below settle several of the source's numerical relations. The other families illustrate how a successful experimental identity gives an upper bound on depth, while a failed search leaves open both a large-coefficient reduction and a missing argument family.

4.1 Depth conventions and Gaussian generators

Write $G = \beta(2)$, $\beta(s) = \sum_{n\geq 0}(-1)^n/(2n+1)^s$, and

$$g_{a,b} = \Im \operatorname{Li}_{a,b}(i,1), \qquad h_{a,b} = \Re \operatorname{Li}_{a,b}(i,1), \qquad \lambda_n = \Im \operatorname{Li}_n((1+i)/2).$$

Subscripts such as g_{41} mean $g_{4,1}$. Series depth counts nested indices. In the experimental tables, a "depth-one reduction" means a polynomial in the stated single-value basket, which is a different convention from the additive depth filtration on a motivic algebra. Products of two depth-two factors have additive depth at most four; their membership in an algebra generated by doubles does not make them depth at most two in that additive filtration.

4.2 An infinite even-weight reduction

Define

$$S_p = \sum_{n\geq 0} \frac{(-1)^n H_n}{(2n+1)^p}, \qquad p \geq 1.$$

For $p = 1$ the sum is conditionally convergent; for $p > 1$ it is absolute.

Theorem 4.1 (Odd-denominator harmonic sums at even weight)**.** *For every integer $m \geq 0$,*

$$S_{2m+1} = (2m+1)\beta(2m+2) - [t^{2m}]\frac{\pi}{4\cos(\pi t/2)}\left(2\log 2 + \sum_{j\geq 1}(2-2^{-2j})\zeta(2j+1)t^{2j}\right). \tag{4.1}$$

In particular, each sum of even weight $2m+2$ belongs to the algebra generated by $\pi, \log 2$, odd zeta values and even beta values.

Proof. Put

$$I(t)=\int_0^1 \frac{x^t\log(1+x^2)}{1+x^2}\,dx,\quad J(t)=\int_0^1 \frac{x^t}{1+x^2}\,dx,\quad A(t)=\int_0^\infty \frac{x^t\log(1+x^2)}{1+x^2}\,dx.$$

These are analytic near $t=0$. Splitting A at one and substituting $x=1/u$ in the second part gives $A(t)=I(t)+I(-t)-2J'(-t)$. The generating function $\sum H_n z^n=-\log(1-z)/(1-z)$, used first with an Abel factor and then by dominated convergence, gives $I^{(2m)}(0)=-(2m)!S_{2m+1}$. Also $J^{(2m+1)}(0)=-(2m+1)!\beta(2m+2)$, by the geometric series for $1/(1+x^2)$; its integrated absolute majorant is summable. Consequently

$$S_{2m+1}=(2m+1)\beta(2m+2)-\frac{[t^{2m}]A(t)}{2}.$$

Differentiating the beta integral in its second parameter yields

$$A(t)=\frac{\pi}{2\cos(\pi t/2)}\left[\psi(1)-\psi\left(\frac{1-t}{2}\right)\right],\qquad |\Re t|<1.$$

Its even part follows from $\psi(1)-\psi(1/2)=2\log 2$ and $\psi^{(2j)}(1/2)=-(2j)!(2^{2j+1}-1)\zeta(2j+1)$. Extracting the coefficient proves the formula. □

The first cases are

$$S_1=G-\frac{\pi}{2}\log 2, \tag{4.2}$$

$$S_3=3\beta(4)-\frac{\pi^3}{16}\log 2-\frac{7\pi}{16}\zeta(3), \tag{4.3}$$

$$768S_5=3840\beta(6)-5\pi^5\log 2-42\pi^3\zeta(3)-372\pi\zeta(5), \tag{4.4}$$

$$92160S_7=645120\beta(8)-61\pi^7\log 2-525\pi^5\zeta(3)-5580\pi^3\zeta(5)-45720\pi\zeta(7). \tag{4.5}$$

The source's high-precision experiments suggested the last two identities; the generating integral proves them and every higher odd-index case at once. The proof gives a sufficient reduction. It does not prove that the even beta values are independent, nor that the odd-weight sums have minimal depth two. No such conclusion follows merely from the absence of a new odd beta value.

Literature and normalization. The odd-index formulas specialize the established Euler-sum theory of Xu and Wang [59]. In their notation

$$R_{1,q}^{1,-1}=\sum_{n\ge 1}\frac{(-1)^{n-1}H_{n-1}}{(n-1/2)^q}=2^qS_q.$$

The independent beta-integral and reflection proofs are retained because they make the normalization and the later certified generalizations transparent. No claim of worldwide priority is made for this evaluation.

4.3 Generators and their exact singularities

For this section, put $\mathfrak{q}(x)=\log(1+x^2)/(1+x^2)$. The beta-integral calculation above gives

$$\mathcal{M}(s):=\int_0^\infty x^{s-1}\mathfrak{q}(x)\,dx=\frac{\pi}{2\sin(\pi s/2)}\left[\psi(1)-\psi(1-s/2)\right],\qquad 0<\Re s<2. \tag{4.6}$$

The harmonic generating function also gives, for integer $p \geq 1$,

$$S_p = -\frac{1}{(p-1)!}\int_0^1 (-\log x)^{p-1}\mathfrak{q}(x)\,dx. \tag{4.7}$$

The first-omitted-term theorem below gives $|S_p| \leq 3^{-p}$. These facts make the actual poles visible even when a special-function expression contains removable singularities.

Theorem 4.2 (Even generator)**.** *The power series*

$$\mathcal{S}_e(t) = \sum_{m\geq 0} S_{2m+1}t^{2m}$$

has radius of convergence exactly 3*. It extends to an even meromorphic function on* $\mathbb{C}$*, with the closed form*

$$\boxed{\mathcal{S}_e(t) = \frac{\psi_1((1+t)/4) - \psi_1((3+t)/4)}{16} + \frac{\pi\,[\psi((1+t)/2) - \psi(1)]}{4\cos(\pi t/2)},} \tag{4.8}$$

where $\psi_1 = \psi'$*, and apparent singularities are interpreted by meromorphic continuation. Its only poles are simple and occur at* $t = \pm(2n+1)$*,* $n \geq 1$*, with*

$$\operatorname{Res}_{t=2n+1}\mathcal{S}_e(t) = \frac{(-1)^{n+1}H_n}{2}, \qquad \operatorname{Res}_{t=-(2n+1)}\mathcal{S}_e(t) = \frac{(-1)^n H_n}{2}. \tag{4.9}$$

In particular, the apparent singularities at $t = \pm 1$ *are removable, and*

$$\mathcal{S}_e(1) = \mathcal{S}_e(-1) = -\frac{\pi^2}{48}. \tag{4.10}$$

Proof. The Mellin representation (4.7) and the bound $|S_p| \leq 3^{-p}$ first define E on $|t| < 3$. Summing the even exponential series under the Mellin integral gives

$$\mathcal{S}_e(t) = -\frac{1}{2}\int_0^1 (x^t + x^{-t})\mathfrak{q}(x)\,dx. \tag{4.11}$$

For $|t| < 3$, absolute values of the summed integrand are bounded by $\mathfrak{q}(x)\cosh(|t|\log x)$, which is integrable at zero since $\mathfrak{q}(x) = O(x^2)$. The right-hand side of (4.11) is holomorphic on the larger vertical strip $|\operatorname{Re} t| < 3$: on each compact subset, the same argument applies with a fixed exponent strictly smaller than 3, and all parameter derivatives add only powers of $|\log x|$. It also proves evenness there.

To evaluate the integral, let

$$J(t) = \int_0^1 x^t\mathfrak{q}(x)\,dx, \qquad K(t) = \int_0^1 \frac{x^t\log x}{1+x^2}\,dx.$$

For $|\operatorname{Re} t| < 1$, splitting the full Mellin integral at one and inverting the second part gives

$$\mathcal{M}(1-t) = J(-t) + J(t) - 2K(t).$$

In this strip, expansion of $(1+x^2)^{-1}$ and absolute summability after integration give

$$\begin{aligned} K(t) &= -\sum_{n\geq 0}\frac{(-1)^n}{(2n+1+t)^2} \\ &= -\frac{1}{16}\big[\psi_1((1+t)/4) - \psi_1((3+t)/4)\big]. \end{aligned}$$

Since $\mathcal{S}_e(t) = -\mathcal{M}(1-t)/2 - K(t)$, substitution of (4.6) proves (4.8) in $|\operatorname{Re} t| < 1$.

The exact global singularity statement should not be inferred merely from the individual terms of that formula, because they have cancelling poles. Instead, retain the conditionally convergent coefficient S_1 separately and sum all the absolutely convergent higher coefficients. For $|t| < 3$, this gives

$$\mathcal{S}_e(t) = S_1 + \sum_{n\ge1} \frac{(-1)^n H_n t^2}{(2n+1)((2n+1)^2 - t^2)}. \tag{4.12}$$

The interchange is valid because

$$\sum_{n\ge1} \frac{H_n}{2n+1} \sum_{m\ge1} \left(\frac{|t|}{2n+1}\right)^{2m} = \sum_{n\ge1} \frac{H_n |t|^2}{(2n+1)((2n+1)^2 - |t|^2)} < \infty.$$

For any compact set avoiding the displayed poles in (4.12), the summands for sufficiently large n are uniformly $O((1+\log n)/n^3)$. Thus the series converges normally there and defines a meromorphic function on $\mathbb{C}$. At a point $2n+1$ or $-(2n+1)$, precisely one summand is singular; its elementary residue is (4.9). These residues are nonzero, and there are no further poles. The right-hand side of (4.8) is also meromorphic on $\mathbb{C}$ and agrees on an open strip, so the meromorphic identity theorem proves that the two representations agree everywhere. In particular, all additional apparent poles in the polygamma expression cancel. The nearest actual poles to zero are 3 and -3, proving that the Taylor radius is exactly 3.

Finally, (4.11) can be evaluated directly at $t = 1$, inside its holomorphy strip:

$$\mathcal{S}_e(1) = -\frac{1}{2}\int_0^1 \frac{\log(1+x^2)}{x}\,dx = \frac{1}{4}\operatorname{Li}_2(-1) = -\frac{\pi^2}{48}.$$

Evenness gives the same value at -1, proving (4.10) without a delicate cancellation of separate Laurent expansions. □

4.3.1 The full generator and the unresolved parity direction

The symmetry can be stated more precisely by keeping all indices.

Proposition 4.3. *The full generating function*

$$\mathcal{S}(t) = \sum_{p\ge1} S_p t^{p-1}$$

has Taylor radius 3*, integral representation*

$$\mathcal{S}(t) = -\int_0^1 x^{-t}\mathfrak{q}(x)\,dx, \qquad \operatorname{Re} t < 3, \tag{4.13}$$

and global meromorphic representation

$$\mathcal{S}(t) = S_1 + \sum_{n\ge1} \frac{(-1)^n H_n t}{(2n+1)(2n+1-t)}. \tag{4.14}$$

Its only poles are simple, at $t = 2n+1$, $n \ge 1$, with residue $(-1)^{n+1}H_n$. Moreover,

$$\frac{\mathcal{S}(t) + \mathcal{S}(-t)}{2} = \mathcal{S}_e(t), \qquad \frac{\mathcal{S}(t) - \mathcal{S}(-t)}{2} = \sum_{m\ge1} S_{2m} t^{2m-1} \quad (|t| < 3). \tag{4.15}$$

Proof. For $|t| < 3$, insert (4.7) and sum the exponential series $\sum_{p\ge1} t^{p-1}(-\log x)^{p-1}/(p-1)! = x^{-t}$. The absolute sum is bounded by $x^{-|t|}\mathfrak{q}(x)$, which is integrable. The resulting integral is holomorphic for $\operatorname{Re} t < 3$, by the same compact-set bounds as above. Separating $p = 1$ and summing the geometric series for $p \ge 2$ gives (4.14); the needed absolute bound is

$$\sum_{n\ge1} \frac{H_n|t|}{(2n+1)(2n+1-|t|)} < \infty, \qquad |t| < 3.$$

On a compact set avoiding positive odd integers at least three, the resulting rational summands are uniformly $O((1+\log n)/n^2)$. Normal convergence gives the stated global continuation. The single summand singular at $t = 2n+1$ has residue $(-1)^{n+1}H_n$, and that residue is nonzero. This proves both the complete pole list and the exact radius. Taking even and odd parts of the Taylor series proves (4.15). □

The inversion $x \mapsto 1/x$ determines the even part of A through the full Mellin transform. It leaves the odd part as a different analytic quantity. The absence of the same reduction for that part is not a proof that its coefficients lie outside the algebra generated by single polylogarithmic values. In particular, no result here proves nonmembership for S_{2m}, which have odd total weight. Such a claim requires an additional arithmetic or motivic argument for the specified constants and specified algebra. The presence of an even beta value likewise supplies no independence certificate: even the irrationality of $G = \beta(2)$ is not known. The correction to the draft is therefore precise: replace the proposed two-sided alternation theorem by (4.1) and its proved even-weight membership statement, and retain the opposite direction as a research question.

4.4 Exact shuffle quotients and a two-product proof at weight five

4.4.1 The quotient being computed

Let $\mathcal{H}^1$ be the rational vector space on the empty word and all binary words ending in 1, with shuffle multiplication. It is bigraded by word length w and number of ones d. Let $\mathcal{J}$ be its augmentation ideal, spanned by nonempty words. The indecomposable quotient

$$Q_{w,d} = (\mathcal{J}/\mathcal{J}^2)_{w,d}$$

sets products of two positive-weight elements equal to zero. This is a formal combinatorial construction. Shuffle equalities evaluate to analytic identities at z; the corresponding map on indecomposable quotients requires also quotienting the evaluated algebra by products. Evaluation can introduce further relations. In particular, the formal dimension is not the dimension of a space of numerical periods.

Theorem 4.4 (The exact formal shuffle count)**.** *For $w \ge 1$ and $1 \le d \le w$,*

$$\dim_{\mathbb{Q}} Q_{w,d} = \ell_{w,d} := \frac{1}{w} \sum_{k \mid \gcd(w,d)} \mu(k) \binom{w/k}{d/k}.$$

A basis is represented by the binary Lyndon words of length w with d ones. Every such word belongs to $\mathcal{H}^1$. Every other word has a finite, exact polynomial expression in the full family of Lyndon generators, each monomial having the original total bidegree. The number of independent product relations in this component is

$$\binom{w-1}{d-1} - \ell_{w,d}.$$

Proof. Order the alphabet by $0 < 1$ and words lexicographically. A Lyndon word is strictly smaller than every nontrivial rotation. Every word has a unique Chen–Fox–Lyndon factorization into a nonincreasing concatenation of Lyndon words. In characteristic zero, shuffle monomials in Lyndon words form a polynomial basis [60]. One can see the constructive point directly: if the factorization is $u_1^{m_1} \cdots u_r^{m_r}$ with distinct $u_1 > \cdots > u_r$, then the shuffle product $u_1^{\amalg m_1} \amalg \cdots \amalg u_r^{\amalg m_r}$ has this concatenation as its lexicographically largest word, with coefficient $m_1! \cdots m_r!$. All other words are smaller. Induction therefore gives a triangular polynomial expression, and the same triangularity proves linear independence of the monomials.

Every Lyndon word of length greater than one that contains both letters starts with 0 and ends with 1. The single-letter words are 0 and 1. The factorization of a word ending in 1 cannot contain the factor 0, because that smallest factor would be last, forcing the concatenation to end in 0. Thus $\mathcal{H}^1$ is exactly the polynomial algebra on all Lyndon words other than 0. Passing to its augmentation quotient leaves one class for each generator.

It remains to count. A primitive binary word of length w with d ones has w distinct rotations and exactly one Lyndon rotation. Every binary word is a power of a unique primitive word. Möbius inversion on this power decomposition gives $\sum_{k \mid \gcd(w,d)} \mu(k) \binom{w/k}{d/k}$ primitive words, proving the formula. Finally, there are $\binom{w-1}{d-1}$ words ending in 1 in the component, which gives the claimed product rank. □

Weight w	$d=1$	$d=2$	$d=3$	$d=4$	$d=5$
3	1	1	0	–	–
4	1	1	1	0	–
5	1	2	2	1	0
6	1	2	3	2	1
7	1	3	5	5	3
8	1	3	7	8	7
9	1	4	9	14	14

Table 4.1: The exact formal indecomposable counts $\ell_{w,d}$. They count Lyndon generators, not independent numerical constants.

For $w = 6, d = 3$, the ten words reduce modulo products to three Lyndon words:

$$000111, \qquad 001011, \qquad 001101,$$

corresponding respectively to $F_{4,1,1}$, $F_{3,2,1}$, and $F_{3,1,2}$. This explains the manuscript's finite rank-seven observation and gives its uniform extension. The companion code records the full polynomial normal forms of all ten triples. For a compact view of the quotient, put $X = [F_{4,1,1}], Y = [F_{3,2,1}], Z = [F_{3,1,2}]$, where brackets mean classes modulo products. Exact triangular elimination gives

	X	Y	Z		X	Y	Z
$[F_{4,1,1}]$	1	0	0	$[F_{1,4,1}]$	6	3	1
$[F_{3,2,1}]$	0	1	0	$[F_{1,3,2}]$	−12	−6	−2
$[F_{3,1,2}]$	0	0	1	$[F_{1,2,3}]$	12	6	2
$[F_{2,3,1}]$	−9	−4	−1	$[F_{1,1,4}]$	−6	−3	−1
$[F_{2,2,2}]$	12	4	0	$[F_{2,1,3}]$	−3	−1	0

For example, one full identity before taking the quotient is

$$F_{2,2,2}(z) = 12F_{4,1,1}(z) + 4F_{3,2,1}(z) - 2\,\mathrm{Li}_2(z)F_{3,1}(z) + \frac{\mathrm{Li}_2(z)^3}{6}. \tag{4.16}$$

Re-expanding its right side into words gives the single word 010101 with coefficient one. This supplies a finite rational certificate, including the product terms that a quotient table suppresses.

4.4.2 An elementary all-weight double elimination

At depth two the count simplifies to

$$\ell_{w,2} = \left\lfloor \frac{w-1}{2} \right\rfloor.$$

There is also a direct triangular proof that avoids general shuffle algebra. For $p+q=w$, shuffling the two single-polylogarithm words gives

$$\mathrm{Li}_p(z)\,\mathrm{Li}_q(z) = \sum_{j=1}^{p} \binom{w-j-1}{q-1} F_{w-j,j}(z) + \sum_{j=1}^{q} \binom{w-j-1}{p-1} F_{w-j,j}(z). \tag{4.17}$$

The binomial coefficient counts the interleavings of the zero blocks before the second nonzero letter.

Proposition 4.5 (Independent double product rows)**.** *For fixed w, the rows of (4.17) with $1 \le p \le \lfloor w/2 \rfloor$ have rank $\lfloor w/2 \rfloor$ on the $w-1$ double coordinates. They express the entire family through products of singles and the $\lfloor (w-1)/2 \rfloor$ coordinates*

$$\{F_{w-j,j} : 1 \le j \le \lfloor (w-1)/2 \rfloor\}.$$

This set is empty when $w = 2$.

Proof. In the row indexed by p, the largest inner index appearing is $q = w-p$. Its coefficient is 1 when $p < q$, and 2 when $p = q$. No larger inner index occurs. Order the rows with p decreasing and use the columns with $j = w-p$; the corresponding square block is triangular with nonzero diagonal. All products of two depth-one words have already been included, since swapping p, q repeats a row. This proves the rank and the elimination statement. □

At $z = i$, taking imaginary parts of these equations gives an unconditional spanning bound for the $g_{a,b}$, with explicit single-value terms. It can be sharpened in the parity sector, as the next section shows.

4.4.3 A normalized Pascal inverse for the shuffle rows

The triangular elimination can be made explicit in outer-index coordinates. Let $m = \lfloor w/2 \rfloor$ and let $A_{p,a} = \binom{a-1}{p-1} + \binom{a-1}{w-p-1}$, for $1 \le p \le m$, $1 \le a < w$. Divide row p by $c_p = 1 + \mathbf{1}_{2p=w}$. Its first m columns are $P_{p,a} = \binom{a-1}{p-1}$, with inverse

$$(P^{-1})_{i,j} = (-1)^{j-i}\binom{j-1}{i-1} \quad (j \ge i), \qquad (P^{-1})_{i,j} = 0 \quad (j < i).$$

Indeed the matrix product reduces to $\binom{j-1}{i-1}\sum_{k=i}^{j}(-1)^{j-k}\binom{j-i}{k-i} = \delta_{ij}$. Writing $R_p = \operatorname{Im}(\operatorname{Li}_p(\mathrm{i})\operatorname{Li}_{w-p}(\mathrm{i}))$, the constructive solution is

$$g_{a,w-a} = \sum_{p=a}^{m}(-1)^{p-a}\binom{p-1}{a-1}\left[\frac{R_p}{c_p} - \sum_{j=m+1}^{w-1}\frac{A_{p,j}}{c_p}g_{j,w-j}\right], \qquad 1 \le a \le m. \tag{4.18}$$

This explicitly leaves the same $\lfloor (w-1)/2 \rfloor$ formal coordinates as Proposition 4.5; evaluation can impose further relations.

4.4.4 The weight-five relation is two shuffles

The words for $\operatorname{Li}_1\operatorname{Li}_4$ and $\operatorname{Li}_2\operatorname{Li}_3$ give

$$\operatorname{Li}_1(z)\operatorname{Li}_4(z) = 2F_{4,1}(z) + F_{3,2}(z) + F_{2,3}(z) + F_{1,4}(z), \tag{4.19}$$
$$\operatorname{Li}_2(z)\operatorname{Li}_3(z) = 6F_{4,1}(z) + 3F_{3,2}(z) + F_{2,3}(z). \tag{4.20}$$

The exact single values are

$$\operatorname{Li}_1(i) = -\frac{L}{2} + \frac{\pi i}{4}, \quad \operatorname{Li}_2(i) = -\frac{\pi^2}{48} + iG, \quad \operatorname{Li}_3(i) = -\frac{3\zeta(3)}{32} + \frac{\pi^3 i}{32},$$

$$\operatorname{Li}_4(i) = -\frac{7\pi^4}{11520} + i\beta(4).$$

Consequently

$$2g_{41} + g_{32} + g_{23} + g_{14} = -\frac{1}{2}\beta(4)L - \frac{7\pi^5}{46080}, \tag{4.21}$$
$$6g_{41} + 3g_{32} + g_{23} = -\frac{\pi^5}{1536} - \frac{3G\zeta(3)}{32}. \tag{4.22}$$

Here g_{41} means $g_{4,1}$, and similarly for other subscripts.

Corollary 4.6 (Proof and strengthening of the recorded relation)**.** *The experimentally discovered identity*

$$576g_{41} + 288g_{32} + 736g_{23} + 960g_{14} = 21G\zeta(3) - 480\beta(4)L \tag{4.23}$$

holds. Moreover the four g-values are spanned by g_{41}, g_{32} and the single-value products $\pi^5, G\zeta(3), \beta(4)L$. Explicitly,

$$g_{23} = -6g_{41} - 3g_{32} - \frac{\pi^5}{1536} - \frac{3G\zeta(3)}{32}, \tag{4.24}$$
$$g_{14} = 4g_{41} + 2g_{32} + \frac{23\pi^5}{46080} + \frac{3G\zeta(3)}{32} - \frac{1}{2}\beta(4)L. \tag{4.25}$$

Proof. Multiply (4.21) by 960, multiply (4.22) by -224, and add. The π^5 coefficients cancel exactly. Solving the two independent equations instead gives (4.24) and (4.25). □

This proof explains why a search can return an apparently complicated relation while missing the simpler structure: eliminating a chosen single-value coordinate can enlarge coefficients. The mathematical content is visible before the elimination.

4.4.5 An infinite odd-weight shuffle family

Theorem 4.7 (An infinite odd-weight family with the pure π term removed)**.** *For $m \geq 2$, set $W = 2m + 1$ and*

$$A_W = 2g_{W-1,1} + \sum_{a=1}^{W-2} g_{a,W-a},$$

$$B_W = 2(W-2)g_{W-1,1} + (W-2)g_{W-2,2} + \sum_{a=2}^{W-3} (a-1)g_{a,W-a}, \tag{4.26}$$

$$\kappa_m = \frac{12(2^{2m-1}-1)B_{2m}}{(2m)(2m-1)E_{2m-2}}. \tag{4.27}$$

Then κ_m is a positive rational number and

$$A_W - \kappa_m B_W = -\frac{1}{2}\beta(2m)L + \kappa_m 2^{-(2m-1)}(1-2^{2-2m})G\zeta(2m-1). \tag{4.28}$$

Proof. Equation (4.17) identifies A_W and B_W with $\mathrm{Im}(\mathrm{Li}_1(\mathrm{i})\,\mathrm{Li}_{2m}(\mathrm{i}))$ and $\mathrm{Im}(\mathrm{Li}_2(\mathrm{i})\,\mathrm{Li}_{2m-1}(\mathrm{i}))$ respectively. Writing $c_k = 2^{-k}(1-2^{1-k})$, these products are

$$A_W = -\tfrac{1}{2}L\beta(2m) - \tfrac{\pi}{4}c_{2m}\zeta(2m),$$
$$B_W = -c_{2m-1}G\zeta(2m-1) - \tfrac{\pi^2}{48}\beta(2m-1).$$

Their pure-π^W terms have ratio

$$\frac{12c_{2m}\zeta(2m)}{\pi\beta(2m-1)}.$$

Euler's even-zeta formula and the classical odd-beta evaluation turn this ratio into (4.27). The signs of B_{2m} and E_{2m-2} agree, so the ratio is positive. Subtraction proves (4.28). □

The first three coefficients are

$$\kappa_2 = \frac{7}{30}, \qquad \kappa_3 = \frac{31}{525}, \qquad \kappa_4 = \frac{127}{8540}.$$

For example, at weight seven the identity can be written without fractions as

$$268800A_7 - 15872B_7 = 465G\zeta(5) - 134400\beta(6)L. \tag{4.29}$$

This theorem does not reduce every odd-weight $g_{a,b}$ individually. It supplies one explicit family of linear combinations in the parity component not individually reduced by Theorem 4.12.

At weight seven a second combination complements the pure-π-free family above:

$$430g_{6,1} + 215g_{5,2} + 79g_{4,3} + 11g_{3,4} - 7g_{2,5} = \frac{105}{512}G\zeta(5) - \frac{75}{32}\beta(4)\zeta(3). \tag{4.30}$$

Its exact certificate is -7 times the $(2,5)$ shuffle row plus 25 times the $(3,4)$ row, after evaluating their single values. The exact row calculation fixes the homogeneous weight-seven products.

4.5 The mixed Gaussian weight-five identity and its exact proof

Put $G = \beta(2)$ and

$$g_{a,b} = \operatorname{Im} \operatorname{Li}_{a,b}(i, 1), \qquad S_p = \sum_{n=0}^{\infty} \frac{(-1)^n H_n}{(2n+1)^p}.$$

The manuscript's equation `gauss:eq:S4-closed` was explicitly classified as a numerically supported identity. We prove precisely that identity. The contribution is a reduction of its mixed Gaussian term to three specified single-color doubles and elementary products. It is not a claim that the remaining doubles are independent or irreducible, nor a claim that S_4 previously lacked every representation in special functions. Indeed the 2021 discussion by Hintze and pisco already records the representation by two colored double values and relates it to hypergeometric evaluations [70, 71].

Theorem 4.8 (Certified mixed Gaussian reduction)**.** *With the series convention* $\operatorname{Li}_{a,b}(x, y) = \sum_{n>m\geq 1} x^n y^m/(n^a m^b)$*,*

$$S_4 = \frac{4g_{4,1} - 3g_{3,2} - 9g_{2,3}}{7} + \frac{\pi^5}{224} - \frac{27}{224} G\zeta(3) - 2\beta(4)\log 2. \tag{4.31}$$

Equivalently,

$$\operatorname{Im} \operatorname{Li}_{4,1}(i, -1) = -\frac{3g_{4,1} + 3g_{3,2} + 9g_{2,3}}{7} + \frac{\pi^5}{224} - \frac{27}{224} G\zeta(3) - 2\beta(4)\log 2. \tag{4.32}$$

The proof consists of convergent changes of variables, standard shuffle/stuffle identities with the explicitly described single-divergence regularization, and a finite rational linear combination supplied as an independently replayable certificate.

4.5.1 The arithmetic translation

For $n = 2k + 1$,

$$H_k = \sum_{m=1}^{n-1} \frac{1 + (-1)^m}{m}.$$

Taking imaginary parts removes even n, while $\operatorname{Im}(i^{2k+1}) = (-1)^k$. Consequently, for $p \geq 1$,

$$S_p = \operatorname{Im}\left(\operatorname{Li}_{p,1}(i, 1) + \operatorname{Li}_{p,1}(i, -1)\right). \tag{4.33}$$

For $p > 1$ the interchange is justified by $\sum_{n\geq 2} H_{n-1}/n^p < \infty$. At $p = 1$, use the common radial Abel limit and the ordinary convergence proved by summation by parts. Thus (4.31) and (4.32) are equivalent.

4.5.2 A convergent octahedral change of variables

Write $\omega_a = dt/(t - a)$ and

$$\mathcal{I}(a_1 \cdots a_w) = \int_{0<t_w<\cdots<t_1<1} \omega_{a_1}(t_1) \cdots \omega_{a_w}(t_w),$$

extended multilinearly in the letters. A convergent word has first letter different from 1 and last letter different from 0. For colors $c_j \in \mathbb{Z}/4\mathbb{Z}$, set

$$Z((s_1, c_1), \ldots, (s_d, c_d)) = \operatorname{Li}_{s_1,\ldots,s_d}(i^{c_1}, \ldots, i^{c_d}).$$

Its integral word is

$$Z(\boldsymbol{s};\boldsymbol{c}) = (-1)^d \mathcal{I}\left(0^{s_1-1} i^{-c_1}\, 0^{s_2-1} i^{-(c_1+c_2)} \cdots 0^{s_d-1} i^{-(c_1+\cdots+c_d)}\right). \tag{4.34}$$

This fixes both the color convention and every depth-dependent sign.

Consider the involution

$$\phi(t) = \frac{1-t}{1+t}.$$

It preserves the puncture set $\{0, \infty, 1, -1, i, -i\}$ and reverses the real interval. Direct differentiation gives

$$\phi^*\omega_a = \omega_{\phi(a)} - \omega_{-1} \quad (a \neq -1), \qquad \phi^*\omega_{-1} = -\omega_{-1},$$

where $\phi(0) = 1$, $\phi(1) = 0$, $\phi(i) = -i$, and $\phi(-i) = i$. Denote this linear map on letters by τ. Changing variables and reversing the order of the simplex proves

$$\mathcal{I}(a_1 \cdots a_w) = (-1)^w \mathcal{I}\big(\tau(a_w) \cdots \tau(a_1)\big). \tag{4.35}$$

No tangential-basepoint correction occurs here: an original last letter other than 0 transforms into first letters other than 1, and an original first letter other than 1 transforms into last letters other than 0. Every term of the expansion therefore remains convergent. The broader use of octahedral symmetries to supplement standard level-four relations goes back to Zhao [72, 69]; the elementary involution above is sufficient for the present certificate.

Only one instance of (4.35) is needed. The word of $\mathrm{Li}_{1,4}(i, 1)$ is $(-i)000(-i)$, and hence

$$D := \mathrm{Im}\left\{\mathcal{I}(e_{-i} e_0^3 e_{-i}) + \mathcal{I}\big((e_i - e_{-1})(e_1 - e_{-1})^3 (e_i - e_{-1})\big)\right\} = 0. \tag{4.36}$$

Here e_a means the letter at a, multiplication is concatenation, and powers are powers in the noncommutative word algebra. The transformed product expands into exactly 32 integral words, with coefficients ± 1. The first integral is the single-color double; the second group is converted to colored polylogarithms using (4.34).

4.5.3 The standard relation rows

The certificate uses two schemas, interpreted over $\mathbb{Q}$ before imaginary parts are taken. First, for two admissible multiple-polylogarithm index/color lists u, v, let $\mathsf{H}(u, v)$ be the integral shuffle product and $\mathsf{S}(u, v)$ the series stuffle product. Then

$$R_{u,v} := \mathrm{Im}\big(\mathsf{H}(u, v) - \mathsf{S}(u, v)\big) = 0. \tag{4.37}$$

The same row is used when $u = ((1, 0))$ and v is admissible, with the standard single-divergence regularization. This point must not be suppressed: $\mathrm{Li}_1(1)$ itself is divergent. Its shuffle and stuffle regularizations are both denoted by a formal variable T. A single leading $(1, 0)$ gives a polynomial of degree at most one in T, and the regularization comparison is the identity in degrees zero and one. Explicitly, its comparison operator satisfies

$$\rho(e^{Tu}) = \exp\left(\sum_{n \geq 2} \frac{(-1)^n \zeta(n)}{n} u^n\right) e^{Tu},$$

so $\rho(1) = 1$ and $\rho(T) = T$ [69, Theorem 2.9]. Subtracting the two product expansions cancels their common divergent word and leaves the convergent identity (4.37). This is the single-divergence case

of regularized double shuffle [73, 69]. The verifier rejects all other divergent input patterns and checks that every surviving word is admissible.

Second, for a list $z = ((s_1, c_1), \dots, (s_d, c_d))$ with $c_j \in \{0, 1\}$ and with all terms below convergent, ordinary distribution is

$$Z(\boldsymbol{s}; 2\boldsymbol{c}) - 2^{w-d} \sum_{\boldsymbol{\epsilon} \in \{0,2\}^d} Z(\boldsymbol{s}; \boldsymbol{c} + \boldsymbol{\epsilon}) = 0, \qquad w = s_1 + \cdots + s_d. \tag{4.38}$$

Indeed summing signs forces every nested summation index to be even, so the factor is exactly 2^{w-d}. The certificate multiplies some such relations by an admissible value of complementary weight and expands that product by integral shuffle. These are *lifted convergent distribution* rows; no regularized distribution formula is assumed. At weight one this includes the elementary seed $\mathrm{Li}_1(-1) = \mathrm{Li}_1(i) + \mathrm{Li}_1(-i)$.

4.5.4 The exact certificate identity

In the formal real vector space whose generators are imaginary parts of convergent colored values of weight five, define

$$\begin{aligned} F = {} & 1120 \operatorname{Im} \mathrm{Li}_{4,1}(i, -1) + 480 g_{4,1} + 480 g_{3,2} + 1440 g_{2,3} - 1536 \operatorname{Im} \mathrm{Li}_5(i) \\ & + 135 \operatorname{Im}\big(\mathrm{Li}_2(i) \mathrm{Li}_3(1)\big) - 2240 \operatorname{Im}\big(\mathrm{Li}_4(i) \mathrm{Li}_1(-1)\big). \end{aligned} \tag{4.39}$$

The two products in this expression are expanded by integral shuffle. Complex conjugation is imposed only by $\operatorname{Im} Z(\boldsymbol{s}; -\boldsymbol{c}) = -\operatorname{Im} Z(\boldsymbol{s}; \boldsymbol{c})$.

Lemma 4.9 (Finite rational certificate)**.** *The accompanying file* `S4_certificate.json` *gives* 911 *rational numbers* q_r *and* 911 *standard-relation descriptors* R_r *such that the following is an exact equality of formal vectors:*

$$F = -1120 D + \sum_{r=1}^{911} q_r R_r. \tag{4.40}$$

The rows consist of 713 *convergent double-shuffle rows,* 181 *rows with one regularized weight-one factor, and* 17 *lifted convergent distribution rows.*

Proof. The supplied program `verify_s4.py` rebuilds every row from the indices and colors in its descriptor. Integral shuffles are generated by choosing interleaving positions, series stuffles by the three defining recursive cases, and distribution by sign enumeration. The program also reconstructs the single duality seed and the target independently of their stored expansions. All coefficients are integers or instances of Python's exact `Fraction` type. Summing the left side minus the right side of (4.40) yields the empty coefficient dictionary, so every formal-word coefficient is exactly zero. The verifier neither reruns a rank calculation nor uses numerical polylogarithms, integer-relation searches, cached matrices, or modular arithmetic. Thus the certificate is a finite rational proof of the stated formal equality. □

Proof of Theorem 4.8. Every R_r vanishes by (4.37) or (4.38), and $D = 0$ by (4.36). The certificate therefore gives $F = 0$. Now

$$\operatorname{Im} \mathrm{Li}_5(i) = \beta(5) = \frac{5\pi^5}{1536}, \qquad \mathrm{Li}_3(1) = \zeta(3), \qquad \mathrm{Li}_1(-1) = -\log 2.$$

Substituting these evaluations into (4.39) gives

$$1120 \operatorname{Im} \mathrm{Li}_{4,1}(i, -1) + 480 g_{4,1} + 480 g_{3,2} + 1440 g_{2,3} - 5\pi^5 + 135 G\zeta(3) + 2240 \beta(4) \log 2 = 0.$$

Division by 1120 proves (4.32), and (4.33) proves (4.31). □

Independent numerical audit. An independent real-quadrature evaluation uses

$$S_4 = \frac{1}{6}\int_0^1 \frac{\log^3 x\,\log(1+x^2)}{1+x^2}\,dx, \qquad g_{a,b} = \frac{1}{(a-1)!}\int_0^1 (-\log x)^{a-1}\,\mathrm{Im}\,\frac{i\,\mathrm{Li}_b(ix)}{1-ix}\,dx.$$

At 70 decimal working digits it gives

$$S_4 = -0.010493456297335043453356751006201751963510618877319761513129 4,$$

with discrepancy about 4.6×10^{-71} between the two sides. This numerical agreement is an audit of conventions, not a step in the proof; the proof is the exact certificate above.

Scope and further work. The existing two relations among $g_{4,1}, g_{3,2}, g_{2,3}, g_{1,4}$ may be used to shorten (4.31); they are prior results and are not counted as new here. The present proof establishes an upper bound on a specific constant's expression complexity. It says nothing about a lower bound, linear independence, or the impossibility of a classical-polylogarithm reduction. A useful next problem is to compress the 911 standard rows into a short conceptual derivation, and to apply the same convergent involution to the analogous odd-weight mixed sums $S_6, S_8, \ldots$.

Corollary 4.10 (The two-coordinate form)**.**

$$S_4 = \frac{58g_{4,1}+24g_{3,2}}{7} + \frac{19\pi^5}{3584} - 2\beta(4)\log 2. \tag{4.41}$$

Proof. Use the two weight-five shuffle eliminations (4.24)–(4.25) in Theorem 4.8. No independence assumption enters. □

4.5.5 A weight-three companion with its full small certificate

Theorem 4.11 (The first even-index harmonic sum)**.** *With $G=\beta(2)$ and $L=\log 2$,*

$$\begin{aligned} S_2 &= -2g_{2,1} + \frac{\pi^3}{32} - 2GL \\ &= 2\,\mathrm{Im}\,\mathrm{Li}_3\!\left(\frac{1+i}{2}\right) - GL - \frac{\pi L^2}{16} - \frac{\pi^3}{64}. \end{aligned}$$

This is the first even-index companion of the weight-five reduction. It is presented as a proved addition, without a claim of priority in the broader Euler-sum literature.

Proof. Use the convergent integral word $J(w)=\mathcal{I}(w)$ of the preceding S_4 proof, with forms $dt/(t-a)$. The short certificate makes the weight-three argument fully explicit. Code the letters by

$$a=-i, \qquad b=-1, \qquad c=1, \qquad d=i, \qquad o=0.$$

Let $\mathcal{D}(U,V)$ be signed stuffle minus shuffle in this integral-word convention. Let P be the letter pullback of the preceding octahedral involution, so $P(a)=d-b$, $P(b)=-b$, $P(c)=o-b$, $P(d)=a-b$ and $P(o)=c-b$. Put $\mathcal{O}(w)=w+\mathrm{rev}\,P(w)$ at weight three. Both evaluate to zero, with the degree-one regularization used when $U=c$, as justified by the comparison operator in the preceding S_4 proof. Every final word is convergent. Let $\mathfrak{I}$ denote projection to the formal imaginary word space: $\mathfrak{I}(\bar{w})=-\mathfrak{I}(w)$, and entirely real words are zero. Here the space has $(80-12)/2=34$ coordinates, with no assertion of numerical independence.

The target is

$$T_3 = 12oaa + 4oad + (d-a)^{\Pi 3} - 8(oa)\,\Pi\, b.$$

The complete certificate is

$$\mathfrak{I}(T_3) = \sum_{(U,V,k) \text{ in Table } 4.2} k\mathfrak{I}(\mathcal{D}(U,V)) + 8\mathfrak{I}(\mathcal{O}(aab)) - 8\mathfrak{I}(\mathcal{O}(aaa)). \tag{4.42}$$

This is a finite rational word identity: expand each shuffle and signed stuffle, apply the five-letter pullback, and identify conjugates. Every coefficient agrees exactly. Both provided implementations reconstruct and verify the identity using rational arithmetic; the second imports none of the first implementation's word code.

U	V	k	U	V	k
a	aa	-12	b	ba	24
a	ab	-6	c	ba	-48
a	ad	12	a	dc	-14
a	bb	-48	a	db	10
a	bc	32	c	ad	24
a	oa	6	c	aa	-24
a	da	12	b	ad	-4
a	dd	12	a	bd	-4
a	ac	6	a	od	-10
b	ab	16	a	oc	-26
a	ob	22	c	oa	48
b	ac	8			

Table 4.2: All 23 double-shuffle coefficients in the S_2 certificate. The four rows with first factor c use one degree of regularization.

The two additional octahedral rows are completely explicit:

w	$\mathcal{O}(w)$	coefficient
aab	$aab - b(d-b)^2$	8
aaa	$aaa + (d-b)^3$	-8

All products inside these two word expressions are concatenations, not shuffle products. For example, $P(a) = d - b$ and $P(b) = -b$; reversal therefore gives $\operatorname{rev} P(aab) = -b(d-b)^2$.

To translate the target, the elementary harmonic parity split yields

$$S_2 = g_{2,1} + u_{2,1}, \qquad u_{2,1} = \operatorname{Im} \operatorname{Li}_{2,1}(i,-1).$$

Furthermore,

$$J(d) - J(a) = \frac{i\pi}{2}, \qquad J(b) = L, \qquad J(oa) = -\operatorname{Li}_2(i).$$

Consequently

$$\operatorname{Im} J(T_3) = 4\left(S_2 + 2g_{2,1} - \frac{\pi^3}{32} + 2GL\right).$$

All rows on the right of (4.42) evaluate to zero, proving the first formula. The height-one reduction below, obtained from (4.58), gives

$$g_{2,1} = -\frac{GL}{2} + \frac{\pi L^2}{32} - \operatorname{Im} \operatorname{Li}_3\left(\frac{1+i}{2}\right) + \frac{3\pi^3}{128}$$

then gives the second formula by substitution. □

4.6 An explicit binomial parity reduction for every double weight

Throughout this section the convention is

$$F_{a,b}(z) = \mathrm{Li}_{a,b}(z,1) = \sum_{n>m>0} \frac{z^n}{n^a m^b}, \qquad U_n(z) = \mathrm{Re}\,\mathrm{Li}_n(z), \quad V_n(z) = \mathrm{Im}\,\mathrm{Li}_n(z).$$

We take $|z| = 1$, $z \neq 1$, and use radial boundary values. These are also the ordinary convergent series values: summation by parts applies to $H_{n-1}^{(b)}/n^a$, whose total variation is finite and which tends to zero. The result below is an elementary explicit specialization of the established multiple-polylogarithm parity theorem. It should not be presented as a new general parity theorem: Panzer proved the general theorem in 2017, including explicit formulas at depths two and three; Umezawa supplied a general-depth explicit formula in 2025 [33, 61]. Its contribution here is a short independently checkable formula in the manuscript's convention, including the boundary cases with an index equal to one.

For $w \geq 2$ define the parity component

$$E_{a,b}(z) = \begin{cases} \mathrm{Im}\,F_{a,b}(z), & w = a+b \text{ even}, \\ \mathrm{Re}\,F_{a,b}(z), & w = a+b \text{ odd}, \end{cases} \qquad W_n(z) = \begin{cases} V_n(z), & w \text{ even}, \\ U_n(z), & w \text{ odd}. \end{cases}$$

The choice of W_n depends on the fixed total weight w, not on n.

Theorem 4.12 (Explicit double parity formula)**.** *Let $w \geq 2$, $1 \leq p \leq w-1$, and $q = w-p$. Put*

$$A_p(z) = \sum_{j=1}^{p} \binom{w-j-1}{q-1} E_{w-j,j}(z),$$

$$P_p(z) = \begin{cases} U_q(z)V_p(z), & w \text{ even}, \ p \text{ odd}, \\ U_p(z)V_q(z), & w \text{ even}, \ p \text{ even}, \\ -V_p(z)V_q(z), & w \text{ odd}, \ p \text{ odd}, \\ U_p(z)U_q(z), & w \text{ odd}, \ p \text{ even}. \end{cases}$$

Then

$$A_p(z) = P_p(z) + \sum_{\substack{3 \leq j \leq p \\ j \text{ odd}}} \binom{w-j-1}{q-1} \zeta(j) W_{w-j}(z) - \sum_{\substack{3 \leq j \leq q \\ j \text{ odd}}} \binom{w-j-1}{p-1} \zeta(j) W_{w-j}(z)$$
$$- \frac{1}{2}\left[\binom{w-1}{q} - \binom{w-1}{p}\right] W_w(z) - \mathbf{1}_{w \text{ odd}} \frac{(-1)^p}{2} \zeta(w). \tag{4.43}$$

Each individual double is recovered by the explicit inverse transform

$$E_{w-b,b}(z) = \sum_{p=1}^{b} (-1)^{b-p} \binom{w-p-1}{b-p} A_p(z), \qquad 1 \leq b < w. \tag{4.44}$$

Consequently every imaginary part of even total weight and every real part of odd total weight in the same-argument double family is an explicit polynomial in single polylogarithms and ordinary zeta values.

Proof. We supply the analytic identity responsible for the reduction, so that no rank computation or numerical recognition enters the proof. Define

$$T_{p,q}(k) = \sum_{m=1}^{\infty} \frac{1}{(m+k)^p m^q}, \qquad k \geq 1.$$

Partial fractions at $m = 0$ and $m = -k$ give

$$\begin{aligned} \frac{1}{(m+k)^p m^q} = &\sum_{j=1}^{q} (-1)^{q-j} \binom{w-j-1}{p-1} \frac{1}{k^{w-j} m^j} \\ &+ (-1)^q \sum_{j=1}^{p} \binom{w-j-1}{q-1} \frac{1}{k^{w-j}(m+k)^j}. \end{aligned} \tag{4.45}$$

The two $j = 1$ divergences cancel. Summing first to a finite cutoff and then letting it tend to infinity yields

$$\begin{aligned} T_{p,q}(k) = &\sum_{j=2}^{q} (-1)^{q-j} \binom{w-j-1}{p-1} \frac{\zeta(j)}{k^{w-j}} + (-1)^q \sum_{j=2}^{p} \binom{w-j-1}{q-1} \frac{\zeta(j)}{k^{w-j}} \\ &+ (-1)^{q+1} \sum_{j=1}^{p} \binom{w-j-1}{q-1} \frac{H_k^{(j)}}{k^{w-j}}. \end{aligned} \tag{4.46}$$

Write

$$\mathcal{A}_p(z) = \sum_{j=1}^{p} \binom{w-j-1}{q-1} F_{w-j,j}(z), \qquad C_p = \binom{w-1}{q},$$

and

$$\mathcal{U}_{p,q}(z) = \sum_{j=2}^{q} (-1)^{q-j} \binom{w-j-1}{p-1} \zeta(j) \operatorname{Li}_{w-j}(z) + (-1)^q \sum_{j=2}^{p} \binom{w-j-1}{q-1} \zeta(j) \operatorname{Li}_{w-j}(z).$$

Using $H_k^{(j)} = H_{k-1}^{(j)} + k^{-j}$ and the hockey-stick identity, equation (4.46) becomes

$$R_{p,q}(z) := \sum_{k \geq 1} z^k T_{p,q}(k) = \mathcal{U}_{p,q}(z) + (-1)^{q+1} \big(\mathcal{A}_p(z) + C_p \operatorname{Li}_w(z) \big). \tag{4.47}$$

There are two product identities. The familiar same-argument shuffle is

$$\operatorname{Li}_p(z) \operatorname{Li}_q(z) = \mathcal{A}_p(z) + \mathcal{A}_q(z). \tag{4.48}$$

The Fourier-convolution identity is

$$\operatorname{Li}_p(z) \operatorname{Li}_q(z^{-1}) = \zeta(w) + R_{p,q}(z) + R_{q,p}(z^{-1}). \tag{4.49}$$

For completeness, this remains valid when $p = 1$ or $q = 1$. Each $\operatorname{Li}_p(e^{i\theta})$ belongs to L^2 because $\sum n^{-2p} < \infty$. The Fourier coefficients of their product are therefore the absolutely convergent convolutions, equal to $\zeta(w)$ at index zero and to $T_{p,q}(k)$ and $T_{q,p}(k)$ at positive and negative indices. Both sequences $T_{p,q}(k)$ decrease to zero. Dirichlet's test gives convergence of the two Fourier series, uniform on closed subarcs away from $z = 1$. Abel–Poisson summation of the L^1 product converges

to the continuous product at every such point, proving (4.49) there. Thus the derivation has never multiplied divergent zeta series.

If w is even, take imaginary parts in (4.48)–(4.49). Since p and q have the same parity, these give the sum and difference of A_p, A_q. Moreover,

$$\begin{aligned}&\operatorname{Im}\bigl(\mathcal{U}_{p,q}(z)+\mathcal{U}_{q,p}(z^{-1})\bigr)\\&\quad=2(-1)^p\left[\sum_{\substack{3\le j\le p\\ j\text{ odd}}}\binom{w-j-1}{q-1}\zeta(j)V_{w-j}(z)-\sum_{\substack{3\le j\le q\\ j\text{ odd}}}\binom{w-j-1}{p-1}\zeta(j)V_{w-j}(z)\right].\end{aligned}$$

Solving the two elementary equations gives (4.43) for even w. If w is odd, take real parts. Now p, q have opposite parity and the same two equations again give the sum and difference. The analogous last display has $-2(-1)^p$ in place of $2(-1)^p$, with U in place of V. The constant term $\zeta(w)$ contributes $-(-1)^p\zeta(w)/2$. This proves (4.43) in the odd case too.

Finally the coefficient matrix in the definition of A_p is lower triangular with diagonal one. Its inverse follows from

$$\binom{w-j-1}{p-j}\binom{w-p-1}{b-p}=\binom{w-j-1}{b-j}\binom{b-j}{p-j}$$

and $\sum_{p=j}^{b}(-1)^{b-p}\binom{b-j}{p-j}=\delta_{b,j}$. This proves (4.44). □

4.6.1 Consequences for Gaussian and Eisenstein values

At $z=i$ we have $V_n(i)=\beta(n)$, $U_1(i)=-\log 2/2$, and

$$U_n(i)=-2^{-n}(1-2^{1-n})\zeta(n),\qquad n\ge 2.$$

The odd beta values are rational multiples of powers of π. Thus Theorem 4.12 supplies an explicit all-weight reduction to the Gaussian single-value algebra in the appropriate parity. It proves all five previously numerical weight-six equations:

$$2048\operatorname{Im}\operatorname{Li}_{5,1}(i,1)=-64\pi^3\zeta(3)-527\pi\zeta(5)+4096\beta(6), \tag{4.50}$$

$$1536\operatorname{Im}\operatorname{Li}_{4,2}(i,1)=96\pi^3\zeta(3)-32\pi^2\beta(4)+1581\pi\zeta(5)-8448\beta(6), \tag{4.51}$$

$$1024\operatorname{Im}\operatorname{Li}_{3,3}(i,1)=-3\pi^3\zeta(3)+64\pi^2\beta(4)-1581\pi\zeta(5)+4608\beta(6), \tag{4.52}$$

$$23040\operatorname{Im}\operatorname{Li}_{2,4}(i,1)=-14\pi^4\beta(2)+135\pi^3\zeta(3)-1440\pi^2\beta(4) \tag{4.53}$$

$$+23715\pi\zeta(5)-69120\beta(6), \tag{4.54}$$

$$92160\operatorname{Im}\operatorname{Li}_{1,5}(i,1)=-150\pi^5\log 2+56\pi^4\beta(2)-270\pi^3\zeta(3) \tag{4.55}$$

$$+1920\pi^2\beta(4)-675\pi\zeta(5). \tag{4.56}$$

An especially simple all-order subfamily is

$$\operatorname{Im}\operatorname{Li}_{2m-1,1}(z,1)=(m-1)V_{2m}(z)+U_{2m-1}(z)V_1(z)-\sum_{k=1}^{m-1}\zeta(2k+1)V_{2m-2k-1}(z),\qquad m\ge 1. \tag{4.57}$$

For example, this gives the further proved weight-eight identity

$$\operatorname{Im}\operatorname{Li}_{7,1}(i,1)=3\beta(8)-\frac{5\pi^5}{1536}\zeta(3)-\frac{\pi^3}{32}\zeta(5)-\frac{8255\pi}{32768}\zeta(7).$$

At $z=\rho=e^{2\pi i/3}$ write $C_n=\operatorname{Im}\operatorname{Li}_n(\rho)$. The same theorem applies unchanged; in particular

$$\operatorname{Im}\operatorname{Li}_{5,1}(\rho,1)=2C_6-\frac{2\pi^3}{81}\zeta(3)-\frac{121\pi}{486}\zeta(5),\qquad \operatorname{Im}\operatorname{Li}_{3,1}(\rho,1)=C_4-\frac{13\pi}{54}\zeta(3).$$

These statements assert reduction identities, not numerical or motivic independence of the remaining single values.

4.7 Classical reductions for every index with one two

This section uses the one-variable abbreviation $\operatorname{Li}_{k_1,\dots,k_d}(z)=\operatorname{Li}_{k_1,\dots,k_d}(z,1,\dots,1)$. Let $\{1\}^r$ denote r consecutive ones. Work first on $0<z<1$, and then continue analytically along paths avoiding the cuts of the displayed functions. In particular, the formulas hold at $z=i$ by continuation through the upper half-plane, with the principal values used below.

Theorem 4.13 (One two at arbitrary depth)**.** *For $d\ge 1$, putting $a=\log(1-z)$ gives*

$$\operatorname{Li}_{2,\{1\}^{d-1}}(z)=\zeta(d+1)+\frac{(-1)^d a^d\log z}{d!}+\sum_{k=0}^{d-1}\frac{(-1)^{k+1}a^k}{k!}\operatorname{Li}_{d+1-k}(1-z). \tag{4.58}$$

For all $r,s\ge 0$, put $\alpha=-\log(1-z)$. Then

$$\operatorname{Li}_{\{1\}^r,2,\{1\}^s}(z)=\sum_{j=0}^{r}(-1)^j\binom{s+j+1}{j}\frac{\alpha^{r-j}}{(r-j)!}\operatorname{Li}_{2,\{1\}^{s+j}}(z). \tag{4.59}$$

Consequently every value with exactly one index equal to two and all other indices equal to one reduces to classical polylogarithms and logarithms.

The height-one functional equation is classical; a modern treatment and arithmetic analogue appear in [63]. The proof below also records the shuffle reduction when the two occupies any position.

Proof. The iterated integral for d ones is $\operatorname{Li}_{\{1\}^d}(z)=\alpha^d/d!$. Hence

$$\operatorname{Li}_{2,\{1\}^{d-1}}(z)=\frac{(-1)^d}{d!}\int_0^z\frac{\log^d(1-t)}{t}\,dt.$$

Set $s=1-t$ and integrate by parts d times. The antiderivative

$$-\log^d s\log(1-s)+\sum_{j=1}^{d}(-1)^j\frac{d!}{(d-j)!}\log^{d-j}s\operatorname{Li}_{j+1}(s)$$

has derivative $\log^d(s)/(1-s)$. At $s=1$ its limiting value is $(-1)^d d!\zeta(d+1)$. This gives (4.58) including the constant term and its sign.

For the second identity, locate the unique zero letter in the iterated integral word $1^r0\,1^{s+1}$. Its variable is t. Integrating the r outer and $s+1$ inner identical letters gives

$$\operatorname{Li}_{\{1\}^r,2,\{1\}^s}(z)=\frac{1}{r!(s+1)!}\int_0^z\big(\alpha(z)-\alpha(t)\big)^r\alpha(t)^{s+1}\frac{dt}{t}.$$

Expanding the first factor and using the preceding integral formula proves (4.59). Analytic continuation then gives the asserted complex values. □

4.7.1 Proof of all three weight-four Gaussian triple formulas

Put $L = \log 2$, $G = \beta(2)$, $r = (1+i)/2$, and $\lambda_n = \operatorname{Im}\operatorname{Li}_n(r)$, $\mu_n = \operatorname{Re}\operatorname{Li}_n(r)$. The elementary dilogarithm reflection, Landen transformation, and trilogarithm reflection give

$$\begin{aligned} \mu_2 &= \frac{5\pi^2}{96} - \frac{L^2}{8}, & \lambda_2 &= G - \frac{\pi L}{8}, \\ \mu_3 &= \frac{35}{64}\zeta(3) - \frac{5\pi^2 L}{192} + \frac{L^3}{48}. \end{aligned} \tag{4.60}$$

For clarity, the trilogarithm identity used for the last formula is

$$\begin{aligned} &\operatorname{Li}_3(z) + \operatorname{Li}_3(1-z) + \operatorname{Li}_3\!\left(-\frac{z}{1-z}\right) \\ &\quad = \zeta(3) + \zeta(2)\log(1-z) - \frac{1}{2}\log z \log^2(1-z) + \frac{1}{6}\log^3(1-z). \end{aligned}$$

It follows by differentiation using the dilogarithm identities and fixing the constant as $z \to 0$; it is then continued to $z = r$, where $1-r = \bar{r}$ and $-r/(1-r) = -i$. Taking real parts proves the displayed μ_3 evaluation. The first two formulas follow respectively from $\operatorname{Li}_2(r) + \operatorname{Li}_2(\bar{r}) = \zeta(2) - \log r \log \bar{r}$ and $\operatorname{Li}_2(i) + \operatorname{Li}_2(\bar{r}) = -\frac{1}{2}\log^2(1-i)$.

To evaluate the height-one formulas at i, write

$$a = \log(1-i) = \frac{L}{2} - \frac{\pi i}{4}, \qquad c = \log(-(1-i)) = \frac{L}{2} + \frac{3\pi i}{4}.$$

Since $(1-i)^{-1} = r$, the needed principal-branch inversion formulas are

$$\begin{aligned} \operatorname{Li}_2(1-i) &= -\operatorname{Li}_2(r) - \frac{\pi^2}{6} - \frac{c^2}{2}, \\ \operatorname{Li}_3(1-i) &= \operatorname{Li}_3(r) - \frac{\pi^2 c}{6} - \frac{c^3}{6}, \\ \operatorname{Li}_4(1-i) &= -\operatorname{Li}_4(r) - \frac{7\pi^4}{360} - \frac{\pi^2 c^2}{12} - \frac{c^4}{24}. \end{aligned}$$

Substitution into (4.58) with $d = 2, 3$, using $\log i = \pi i/2$, yields

$$\begin{aligned} \operatorname{Re}\operatorname{Li}_{2,1}(i) &= \frac{29}{64}\zeta(3) - \frac{\pi G}{4}, \\ \operatorname{Im}\operatorname{Li}_{2,1}(i) &= -\frac{GL}{2} + \frac{\pi L^2}{32} - \lambda_3 + \frac{3\pi^3}{128}, \\ \operatorname{Im}\operatorname{Li}_{2,1,1}(i) &= \lambda_4 + \frac{L\lambda_3}{2} - \frac{G(\pi^2 - 4L^2)}{32} - \frac{\pi(8L^3 + 105\zeta(3))}{768}. \end{aligned}$$

The real-part evaluation of $\operatorname{Li}_{2,1}(i)$ is also printed in Panzer [33], after reversing his index convention. The remaining two identities follow from exact shuffle relations:

$$\begin{aligned} \operatorname{Li}_1(z)\operatorname{Li}_{2,1}(z) &= \operatorname{Li}_{1,2,1}(z) + 3\operatorname{Li}_{2,1,1}(z), \\ \operatorname{Li}_2(z)\operatorname{Li}_{1,1}(z) &= 3\operatorname{Li}_{2,1,1}(z) + 2\operatorname{Li}_{1,2,1}(z) + \operatorname{Li}_{1,1,2}(z). \end{aligned}$$

Using $\operatorname{Li}_1(i) = -a$, $\operatorname{Li}_{1,1}(i) = a^2/2$, and $\operatorname{Li}_2(i) = -\pi^2/48 + iG$, one obtains

$$\begin{aligned} \operatorname{Im}\operatorname{Li}_{1,2,1}(i) &= -3\lambda_4 - L\lambda_3 + \frac{G(\pi^2 - 4L^2)}{32} - \frac{3\pi^3 L}{256} + \frac{\pi(2L^3 + 67\zeta(3))}{128}, \\ \operatorname{Im}\operatorname{Li}_{1,1,2}(i) &= 3\lambda_4 + \frac{L\lambda_3}{2} + \frac{5\pi^3 L}{192} - \frac{163\pi\zeta(3)}{256}. \end{aligned}$$

These are exactly the three numerical formulas in the original experimental draft, now proved. The arbitrary-depth theorem proves additional such identities without a new integer-relation search. It does not address the independent status of the weight-six triples $(3,1,2),(3,2,1),(4,1,1)$, which contain more than one zero letter and lie outside this reduction mechanism.

4.8 Classical reductions with one index equal to two

This section settles the three Gaussian weight-four triple identities printed in Chapter 4 of the source manuscript. The proof gives a uniform formula for every multiple polylogarithm whose indices consist of one 2 and any number of 1's. The general phenomenon is classical: Borwein and Straub give the complementary-argument formula for $\mathrm{Li}_{2,\{1\}^{p-1}}$ in [66, Eq. (33)], including the weight-four instance in their Eq. (34). The results below reorganize that reduction at the reciprocal complementary argument, make the dependence on the position of the 2 explicit, and prove the specific Gaussian formulas in the manuscript. We do not claim that classical reducibility of this family is a new phenomenon.

Throughout this section, a single displayed argument abbreviates the remaining arguments 1:

$$\mathrm{Li}_{s_1,\dots,s_d}(z) := \mathrm{Li}_{s_1,\dots,s_d}(z,1,\dots,1).$$

The defining series has strictly decreasing summation indices. Set

$$U_{a,b}(z) = \mathrm{Li}_{\{1\}^a,2,\{1\}^b}(z), \qquad H_p(z) = \mathrm{Li}_{2,\{1\}^{p-1}}(z),$$

where $a,b \geq 0$ and $p \geq 1$. The weight of $U_{a,b}$ is $N = a+b+2$, and its series depth is $N-1$.

4.8.1 A logarithmic moment and its branch conventions

We use principal branches and first work on

$$\mathcal{D} = \mathbb{C} \setminus [0,\infty), \qquad w = \frac{1}{1-z}, \qquad L = \operatorname{Log} w = -\operatorname{Log}(1-z).$$

On this domain, w avoids the standard polylogarithm cut $[1,\infty)$ and the logarithm cut $(-\infty,0]$. All terms in the formulas below are therefore single-valued analytic functions. The formulas also extend to the usual slit plane by consistent continuation; no such extension is needed at $z = i$ or at a nonreal root of unity.

Lemma 4.14 (Logarithmic moment)**.** *Let $q(t) = -\operatorname{Log}(1-t)$. For an integration path from 0 to z avoiding $\{0,1\}$ apart from its initial point 0, with the branches continued along that path,*

$$U_{a,b}(z) = \frac{1}{a!(b+1)!} \int_0^z \frac{(q(z)-q(t))^a q(t)^{b+1}}{t}\, dt. \tag{4.61}$$

Consequently,

$$U_{a,b}(z) = \sum_{j=0}^{a} (-1)^j \binom{b+j+1}{j} \frac{q(z)^{a-j}}{(a-j)!}\, H_{b+j+1}(z). \tag{4.62}$$

Proof. The iterated-integral word of $U_{a,b}$ is $1^a 0\, 1^{b+1}$, with forms $dt/(1-t)$ and dt/t attached to 1 and 0, respectively. The innermost $b+1$ identical forms contribute $q(t)^{b+1}/(b+1)!$. The a identical forms between t and z contribute $(q(z)-q(t))^a/a!$. Integrating the single remaining 0 form proves

(4.61). Equivalently, one may verify the same formula by differentiating successively with respect to z and using the zero initial values at $z = 0$. Expanding the binomial and using

$$H_p(z) = \frac{1}{p!} \int_0^z \frac{q(t)^p}{t}\, dt$$

proves (4.62). □

Theorem 4.15 (Explicit reduction for every position of the 2)**.** *For $a, b \geq 0$, $N = a + b + 2$, and $z \in \mathcal{D}$,*

$$U_{a,b}(z) = \sum_{k=a+1}^{N} (-1)^{a+k} \binom{k-1}{a} \frac{L^{N-k}}{(N-k)!} \operatorname{Li}_k(w) - \frac{L^N}{N!} + (-1)^{b+1} \sum_{j=0}^{a} \binom{b+j+1}{j} \frac{L^{a-j}}{(a-j)!} \zeta(b+j+2). \tag{4.63}$$

In particular,

$$H_p(z) = (-1)^p \zeta(p+1) - \frac{L^{p+1}}{(p+1)!} + \sum_{k=1}^{p+1} (-1)^k \frac{L^{p+1-k}}{(p+1-k)!} \operatorname{Li}_k(w). \tag{4.64}$$

Proof. We first prove (4.64) for real $z < 0$, so that $0 < w < 1$ and every intermediate integral is real. Substitute $v = 1/(1-t)$ in the logarithmic moment. Then

$$\frac{dt}{t} = \frac{dv}{v(v-1)}, \qquad q(t) = \log v,$$

and

$$H_p(z) = \frac{1}{p!} \int_1^w \frac{\log^p v}{v(v-1)}\, dv.$$

The differential identity

$$\frac{d}{dv} \left[\sum_{k=1}^{p+1} \frac{(-1)^{k-1} p!}{(p+1-k)!} \log^{p+1-k} v \, \operatorname{Li}_k(v) \right] = \frac{\log^p v}{1-v} \tag{4.65}$$

follows by cancellation of adjacent terms, using $\operatorname{Li}_1(v) = -\log(1-v)$ and $\operatorname{Li}_k'(v) = \operatorname{Li}_{k-1}(v)/v$ for $k \geq 2$. Since $1/[v(v-1)] = -1/v - 1/(1-v)$, integration gives (4.64). At $v = 1$, every term in (4.65) with a positive power of $\log v$ tends to zero. This includes the apparently singular term $\log^p v \, \operatorname{Li}_1(v)$, which is $O((v-1)^p \log(1-v))$ and tends to zero because $p \geq 1$. The sole nonzero endpoint term is $(-1)^p p! \zeta(p+1)$. Thus the constant in (4.64) is fixed, not inferred from a differential identity alone. Analytic continuation on the connected domain $\mathcal{D}$ proves that identity throughout the stated domain.

Insert (4.64) in (4.62), using $q(z) = L$. The zeta terms immediately give the second line of (4.63). The coefficient of $(-1)^k L^{N-k} \operatorname{Li}_k(w)$ is

$$\sum_{j=0}^{a} \frac{(-1)^j \binom{b+j+1}{j}}{(a-j)!(b+j+2-k)!} = \frac{(-1)^a \binom{k-1}{a}}{(N-k)!}.$$

Terms containing a factorial of a negative integer are omitted. To verify this finite identity, write its left side as

$$\frac{(k-1)!}{(b+1)!a!}\sum_{j=0}^{a}(-1)^j\binom{a}{j}\binom{b+j+1}{k-1}$$

and extract the coefficient of x^{k-1} in

$$(1+x)^{b+1}\sum_{j=0}^{a}(-1)^j\binom{a}{j}(1+x)^j=(-x)^a(1+x)^{b+1}.$$

The coefficients vanish when $k\le a$, explaining the lower endpoint $a+1$ in the sum. Finally, the coefficient of $-L^N$ equals

$$\begin{aligned}\sum_{j=0}^{a}\frac{(-1)^j\binom{b+j+1}{j}}{(a-j)!(b+j+2)!}&=\frac{1}{a!(b+1)!}\int_0^1 t^{b+1}(1-t)^a\,dt\\&=\frac{1}{N!}.\end{aligned}$$

This proves the complete formula. □

For the Gaussian endpoint the path itself can be inspected explicitly:

$$z=ix\quad(0\le x\le 1),\qquad w(x)=\frac{1+ix}{1+x^2}.$$

For $x>0$ this path lies in the upper half-plane and does not meet the polylogarithm cut. Its initial point is $w(0)=1$, where the endpoint limits in the proof apply, and its final point is $(1+i)/2$. This verifies the branch choices directly, without using an inversion formula with an unspecified logarithmic correction.

The boundary series at $z=i$ agree with these analytic values. Indeed the finite nested harmonic sums in a coefficient satisfy a bound $O((1+\log n)^{d-1})$. After division by n^{s_1} their coefficients tend to zero and have summable successive differences. Summation by parts against the bounded partial sums of i^n proves convergence, even when $s_1=1$; Abel's theorem then identifies the series with the radial analytic limit.

Corollary 4.16 (A short formula when the 2 is last)**.** *For every $N\ge 2$,*

$$\begin{aligned}U_{N-2,0}(z)=(N-1)\operatorname{Li}_N(w)-L\operatorname{Li}_{N-1}(w)-\frac{L^N}{N!}\\-\sum_{m=0}^{N-2}(N-1-m)\zeta(N-m)\frac{L^m}{m!}.\end{aligned}\tag{4.66}$$

Thus only two nonconstant classical polylogarithms are needed in this subfamily, regardless of series depth.

Proof. Set $a=N-2$, $b=0$ in (4.63) and reindex the zeta sum. □

Corollary 4.17 (The top two series-depth layers)**.** *Every single-endpoint multiple polylogarithm of weight $N\ge 2$ and series depth at least $N-1$ has an explicit expression in classical polylogarithms at $(1-z)^{-1}$, zeta values, and logarithms, on the domain $\mathcal{D}$.*

Proof. Positive indices with sum N and length N are all 1, and $\mathrm{Li}_{\{1\}^N}(z) = L^N/N!$. Those with length $N-1$ have exactly one index 2 and all other indices 1, so Theorem 4.15 applies. These exhaust the possibilities. □

This statement concerns the literal depth of the defining series. It does not identify that depth with a motivic filtration or prove a minimal depth for the resulting numerical constants.

4.8.2 One possible new direction at each weight

Put

$$\ell = \log 2, \qquad w = \frac{1+i}{2}, \qquad \mu_k = \Re\,\mathrm{Li}_k(w), \qquad \lambda_k = \Im\,\mathrm{Li}_k(w).$$

For $N \geq 2$, let $\mathcal{P}_N$ be the rational span of products of at least two positive-weight atoms selected from

$$\pi, \ell, \quad \zeta(k), \mu_k, \lambda_k \quad (2 \leq k < N),$$

whose assigned weights sum to N; the assigned weights are 1 for π, ℓ and k for the atoms with subscript or argument k. This definition is a concrete span of numerical products. It does not assume that numerical periods form a direct sum by weight.

Corollary 4.18 (Signed binomial coefficients modulo products)**.** *For $a+b+2 = N$, set $c_{N,a} = (-1)^{N+a}\binom{N-1}{a}$. Then*

$$U_{a,b}(i) \equiv c_{N,a}\big(\mathrm{Li}_N(w) - \zeta(N)\big) \pmod{\mathcal{P}_N + i\mathcal{P}_N}, \tag{4.67}$$

$$\Re U_{a,b}(i) \equiv c_{N,a}\big(\mu_N - \zeta(N)\big) \pmod{\mathcal{P}_N}, \tag{4.68}$$

$$\Im U_{a,b}(i) \equiv c_{N,a}\lambda_N \pmod{\mathcal{P}_N}. \tag{4.69}$$

In particular, the imaginary parts of this entire family contribute at most one direction after products of the stated lower-weight atoms have been factored out.

Proof. At $z = i$, $L = -\ell/2 + i\pi/4$, and $\mathrm{Li}_1(w) = -\overline{L}$. In (4.63), every term is a product of lower-weight atoms except the term $k = N$ and the term $j = a$ in the zeta sum. Their respective coefficients are $c_{N,a}$ and $-c_{N,a}$. Taking real and imaginary parts proves the statements. □

The coefficients of the highest-weight half-point imaginary value are

N	$c_{N,0}, \ldots, c_{N,N-2}$
4	$1, -3, 3$
5	$-1, 4, -6, 4$
6	$1, -5, 10, -10, 5.$

The corollary is an upper bound, and proves neither that λ_N is nonzero in the quotient nor that it is independent of constants from other argument families. It also does not identify series depth with an additive motivic depth filtration.

4.8.3 A sixth-root specialization in every position

The special role of sixth roots in relations for Nielsen polylogarithms is studied by Borwein and Straub [66]. The preceding position formula gives a direct specialization in that established setting, including an explicit rational coefficient for one component at every weight. Throughout this subsection,

$$\omega = e^{i\pi/3}, \qquad N = a+b+2, \qquad a, b \geq 0,$$

and all logarithms and polylogarithms use their principal branches.

Corollary 4.19 (Sixth-root closure and a pure parity component)**.** *At $z=\omega$, formula (4.63) reduces $U_{a,b}(\omega)$ to classical values $\mathrm{Li}_k(\omega)$, zeta values, and powers of $i\pi$, since*

$$\frac{1}{1-\omega}=\omega, \qquad -\operatorname{Log}(1-\omega)=\operatorname{Log}\omega=\frac{i\pi}{3}. \tag{4.70}$$

Moreover,

$$U_{a,b}(\omega)+(-1)^N\overline{U_{a,b}(\omega)}=R_{N,a}(i\pi)^N, \qquad R_{N,a}\in\mathbb{Q}. \tag{4.71}$$

Thus $\Re U_{a,b}(\omega)$ is a rational multiple of π^N when N is even, and $\Im U_{a,b}(\omega)$ is a rational multiple of π^N when N is odd. Explicitly, with Bernoulli polynomials defined by

$$\frac{te^{xt}}{e^t-1}=\sum_{k\ge 0}B_k(x)\frac{t^k}{k!}, \qquad B_k=B_k(0),$$

one may take

$$\begin{aligned} R_{N,a}=&-\frac{2}{3^N N!}-\sum_{k=a+1}^{N}\frac{(-1)^{a+k}\binom{k-1}{a}\,2^kB_k(1/6)}{3^{N-k}(N-k)!k!}\\ &+(-1)^{N-a}\sum_{\substack{N-a\le m\le N\\ m\ \text{even}}}\frac{\binom{m-1}{N-a-1}\,2^mB_m}{3^{N-m}(N-m)!m!}. \end{aligned} \tag{4.72}$$

The same formulas at $\overline{\omega}$ follow by complex conjugation.

Proof. The identities in (4.70) follow from $1-\omega=\overline{\omega}$. Along the radial segment $z=r\omega$, $0\le r\le 1$, the real part of $1-z$ is at least $1/2$, so the indicated logarithm has no branch crossing. The transformed path lies in the upper half-plane for $r>0$, starts at 1, and ends at ω; the endpoint limits already used to prove (4.63) apply. Consequently that formula has exactly the stated principal values. The ordinary boundary multiple series agree with their radial limits by the same summation-by-parts argument used at $z=i$.

Put $T=i\pi$ and $L=T/3$. The classical Bernoulli Fourier identity is

$$\mathrm{Li}_k(e^{2\pi ix})+(-1)^k\,\mathrm{Li}_k(e^{-2\pi ix})=-\frac{(2\pi i)^k}{k!}B_k(x), \qquad 0<x<1, \quad k\ge 1. \tag{4.73}$$

For $k=1$, this follows directly from $\mathrm{Li}_1(z)=-\operatorname{Log}(1-z)$ on the stated arc. Successive integration, with the zero average of both sides fixing the constants, proves every higher case. Since $\overline{L}=-L$, the parity projection of a classical term in (4.63) is

$$\begin{aligned} &L^{N-k}\,\mathrm{Li}_k(\omega)+(-1)^N\overline{L^{N-k}\,\mathrm{Li}_k(\omega)}\\ &\qquad=L^{N-k}\big(\mathrm{Li}_k(\omega)+(-1)^k\,\mathrm{Li}_k(\overline{\omega})\big)=-\frac{2^kB_k(1/6)}{3^{N-k}k!}\,T^N. \end{aligned}$$

The polynomial term $-L^N/N!$ contributes $-2T^N/(3^NN!)$. A zeta term with zeta weight m contributes zero when m is odd. For even $m\ge 2$, use

$$\frac{2\zeta(m)}{(i\pi)^m}=-\frac{2^mB_m}{m!}.$$

In the last sum of (4.63), write $m=b+j+2=N-a+j$. Its binomial coefficient becomes $\binom{m-1}{N-a-1}$, while its sign after the even-zeta evaluation is $(-1)^b=(-1)^{N-a}$. Combining these three contributions proves (4.72) and hence (4.71). □

There is a precise geometric reason that this specialization closes on a unit-circle argument. If $|z| = 1$ and $z \neq 1$, then

$$\left|\frac{1}{1-z}\right| = 1 \quad \Longleftrightarrow \quad |1-z|^2 = 2 - 2\Re z = 1 \quad \Longleftrightarrow \quad z \in \{\omega, \overline{\omega}\}.$$

Both of these points are fixed by $z \mapsto (1-z)^{-1}$. This classifies the unit-circle endpoint geometry of this particular transformation.

Corollary 4.20 (The two weight-three values)**.** *Define the ordinary Clausen value by its absolutely convergent series*

$$C = \mathrm{Cl}_2(\pi/3) = \sum_{n \geq 1} \frac{\sin(n\pi/3)}{n^2}.$$

Then

$$U_{0,1}(\omega) = \mathrm{Li}_{2,1}(\omega) = \frac{2}{3}\zeta(3) - \frac{\pi}{3}C + \frac{i\pi^3}{324}, \tag{4.74}$$

$$U_{1,0}(\omega) = \mathrm{Li}_{1,2}(\omega) = -\frac{4}{3}\zeta(3) + \frac{\pi}{3}C + \frac{i\pi^3}{324}. \tag{4.75}$$

Proof. The classical values needed are

$$\mathrm{Li}_1(\omega) = \frac{i\pi}{3}, \qquad \mathrm{Li}_2(\omega) = \frac{\pi^2}{36} + iC, \qquad \mathrm{Li}_3(\omega) = \frac{\zeta(3)}{3} + \frac{5i\pi^3}{162}.$$

The real dilogarithm component and imaginary trilogarithm component follow from (4.73). The real trilogarithm component follows by grouping the defining series into residue classes, or from the distribution identity

$$\mathrm{Li}_s(\omega) + \mathrm{Li}_s(\overline{\omega}) = (1 - 2^{1-s})(1 - 3^{1-s})\zeta(s), \qquad s > 1.$$

For example, summing over the three cube roots gives $(3^{1-s} - 1)\zeta(s)$ for the two nontrivial cube roots; applying the two-term distribution formula to the pair of sixth roots multiplies that sum by $2^{1-s} - 1$. Now (4.64) at $p = 2$ gives, with $L = i\pi/3$,

$$U_{0,1}(\omega) = \zeta(3) - \mathrm{Li}_3(\omega) + L\,\mathrm{Li}_2(\omega) - \frac{2}{3}L^3.$$

Substitution yields (4.74). The shuffle identity

$$\mathrm{Li}_1(\omega)\,\mathrm{Li}_2(\omega) = U_{1,0}(\omega) + 2U_{0,1}(\omega)$$

then proves (4.75). In both positions, (4.72) gives $R_{3,a} = -1/162$. □

The independent script `code/independent/one_two/sixth_root.py` computes (4.72) with exact rational arithmetic and compares the specialized closed formula, its pure parity component, and the displayed weight-three examples with the logarithmic moment. It covers every position of the 2 at weights 2 through 8, at both ω and $\overline{\omega}$, using 100 working decimal digits. Across these 56 cases, the largest closed-form versus moment residual is below $6.59 \cdot 10^{-100}$; the largest parity-projection residual is below $7.15 \cdot 10^{-102}$. The four checks of the two explicit examples and their conjugates have residual below $2.16 \cdot 10^{-101}$. These are numerical checks of the proved identities, rather than rigorous quadrature error bounds.

4.8.4 Further exact identities at weights five and six

The same calculation gives formulas beyond the three displayed manuscript rows. They involve the unavoidable output of this particular reduction, namely higher classical values at w; no independence claim about those values is implied.

Corollary 4.21 (Two higher-weight Gaussian reductions)**.** *With* $H_4 = \mathrm{Li}_{2,1,1,1}$ *and* $H_5 = \mathrm{Li}_{2,1,1,1,1}$,

$$\begin{aligned}\Im H_4(i) &= -\lambda_5 - \frac{\ell\lambda_4}{2} + \frac{\pi\mu_4}{4} + \frac{\pi^2 - 4\ell^2}{32}\lambda_3 \\ &\quad + \frac{G\ell(3\pi^2 - 4\ell^2)}{192} + \frac{35\pi\ell\zeta(3)}{512} + \frac{\pi\ell^2(\ell^2 - \pi^2)}{384} - \frac{17\pi^5}{92160}, \qquad (4.76)\\ \Im H_5(i) &= \lambda_6 + \frac{\ell\lambda_5}{2} - \frac{\pi\mu_5}{4} + \frac{4\ell^2 - \pi^2}{32}\lambda_4 - \frac{\pi\ell\mu_4}{8} \\ &\quad + \frac{\ell(4\ell^2 - 3\pi^2)}{192}\lambda_3 + \frac{G(16\ell^4 - 24\ell^2\pi^2 + \pi^4)}{6144} \\ &\quad + \frac{35\pi(\pi^2 - 12\ell^2)\zeta(3)}{24576} + \frac{\pi\ell^3(5\pi^2 - 3\ell^2)}{5760}. \qquad (4.77)\end{aligned}$$

Proof. Use $p = 4$ and $p = 5$ in (4.64), substitute $L = -\ell/2 + i\pi/4$ and Lemma (4.60), and take imaginary parts. The accompanying coefficient generator performs these polynomial operations over $\mathbb{Q}$ and also emits every position of the 2 at weights four, five, and six. □

The real parts can be treated by the same formula. For example,

$$\Re\,\mathrm{Li}_{2,1,1}(i) = \mu_4 + \frac{\pi\lambda_3}{4} + \frac{\pi G\ell}{8} + \frac{35\ell\zeta(3)}{128} + \frac{\ell^4}{384} - \frac{11\pi^2\ell^2}{768} - \frac{1249\pi^4}{92160}. \qquad (4.78)$$

The machine-readable tables include both real and imaginary parts, so this companion family can be reproduced without manual transcription.

4.9 From one-two reductions to general complementary depth

For positive indices we abbreviate

$$\mathrm{Li}_{s_1,\dots,s_d}(z) := \mathrm{Li}_{s_1,\dots,s_d}(z,1,\dots,1) = \sum_{n_1>\cdots>n_d\geq 1} \frac{z^{n_1}}{n_1^{s_1}\cdots n_d^{s_d}}, \qquad |z| < 1. \qquad (4.79)$$

The indices decrease in the summation order, as in the manuscript. Weight is $w = s_1 + \cdots + s_d$ and series depth is d. An empty index denotes the multiplicative identity. We use convergent multiple zeta values $\zeta(\boldsymbol{s}) = \mathrm{Li}_{\boldsymbol{s}}(1)$ when $s_1 \geq 2$.

Let

$$\omega_0(t) = \frac{dt}{t}, \qquad \omega_1(t) = \frac{dt}{1-t}.$$

Words are written *outer letter first*. Thus

$$H_{0^{s_1-1}1\cdots 0^{s_d-1}1}(z) = \mathrm{Li}_{s_1,\dots,s_d}(z). \qquad (4.80)$$

In particular $H_{01} = \mathrm{Li}_2$, not its negative or a reversed word. The recursion is $H_{av}(z) = \int_0^z \omega_a(t)H_v(t)$ whenever this is an ordinary convergent integral. Pure-zero words are defined by $H_{0^r}(z) = \log^r z/r!$,

and trailing-zero words are obtained by shuffle regularization. Every word ending in 1 is convergent at its initial endpoint and has the ordinary interpretation in (4.80).

Throughout the functional identities we use

$$D = \mathbb{C} \setminus [0, \infty), \qquad q = \frac{1}{1-z}, \qquad L = \log q = -\log(1-z), \qquad z \in D, \tag{4.81}$$

with principal logarithms and principal polylogarithms. The image of D avoids both $(-\infty, 0]$ and $[1, \infty)$ in the q-plane, so the hyperlogarithms with possible trailing zeros have unambiguous branches. The origin is a limiting point, not an included point of D.

We first prove formulas on $z < 0$, where $0 < q < 1$ and all logarithms in q are real. Analytic continuation on the connected slit domain D then proves the stated complex identities. This method avoids an unstated choice of a value on the polylogarithm cut. It includes both $z = i$ and $z = \rho = e^{2\pi i/3}$:

$$z = i: \quad q = \frac{1+i}{2}, \qquad L = -\frac{1}{2}\log 2 + \frac{i\pi}{4}, \tag{4.82}$$

$$z = \rho: \quad q = \frac{1}{2} + \frac{i\sqrt{3}}{6}, \qquad L = -\frac{1}{2}\log 3 + \frac{i\pi}{6}. \tag{4.83}$$

Their transformed moduli are $2^{-1/2}$ and $3^{-1/2}$ respectively.

Definition 4.22 (Depth with logarithmic coefficients)**.** *A term $L^j \operatorname{Li}_{\boldsymbol{a}}(q)\zeta(\boldsymbol{b})$ has* reduced depth $\operatorname{dep}(\boldsymbol{a}) + \operatorname{dep}(\boldsymbol{b})$. *The factor L^j is a coefficient, not an additional nested sum.*

This is an explicit convention for the following reduction. It is not an identification with every motivic depth filtration, and does not turn arbitrary products of deep factors into depth-one objects.

4.10 An explicit complementary-depth theorem

4.10.1 Endpoint transport with convergent constants

For a word v ending in 0, write $J_v(q)$ for the iterated integral from 1 to q using the same forms and the same outer-first convention. These integrals converge at 1: the innermost form is ω_0, which is regular there; preceding ω_1 integrations do not destroy integrability. Set $J_\varnothing = 1$.

Lemma 4.23 (Connection formula)**.** *For every nonempty word v ending in 0 and $0 < q < 1$,*

$$J_v(q) = \sum_{v = uv'} (-1)^{|v'|} H_u(q) H_{\operatorname{rev}(v')}(1). \tag{4.84}$$

The sum includes empty prefixes and suffixes. Every nonempty endpoint word $\operatorname{rev}(v')$ starts in 0, so its value at 1 is convergent after the usual trailing-zero normalization at 0.

Proof. We give the endpoint justification rather than silently substituting into a divergent path identity. Fix a maximum word length N and work in the truncated noncommutative algebra on e_0, e_1. Let $F(q) = \sum_{|u| \le N} H_u(q) u$. It satisfies

$$F'(q) = \left(\frac{e_0}{q} + \frac{e_1}{1-q} \right) F(q).$$

Put $\ell = -\log(1-q)$. The function $e^{-\ell e_1} F(q)$ has a finite coefficientwise limit Z as $q \to 1^-$: its differential equation has coefficient $e^{-\ell e_1} e_0 e^{\ell e_1}/q$, a finite polynomial in ℓ at each truncated weight,

and this is integrable at 1. Consequently $F(q) = e^{\ell e_1} Z + O((1-q)\log^M(1-q))$ coefficientwise, for a sufficiently large finite M.

Shuffle identities say that F is group-like; the same holds for Z. The inverse coefficient in a group-like word series is $(Z^{-1})_{v'} = (-1)^{|v'|} Z_{\mathrm{rev}(v')}$, by the shuffle antipode, or equivalently by reversal of an iterated path. The coefficients of $R(q) = F(q)Z^{-1}$ solve the same differential recursion and tend to zero at 1 for every nonempty word ending in 0. Induction on word length therefore identifies them with the convergent integrals $J_v(q)$.

If v' is a nonempty suffix of v, its reverse begins in 0. The corresponding coefficient of Z is simply $H_{\mathrm{rev}(v')}(1)$: multiplication by $e^{-\ell e_1}$ only changes words beginning with 1. Taking the coefficient of v in $F(q)Z^{-1}$ proves (4.84). Since N was arbitrary, this proves every finite word identity. □

4.10.2 The word formula and the depth bound

For a word w ending in 1, let K be its number of zeros. Replace each original 1 by 0, and replace each original 0 independently by either 0 or 1. Denote the resulting 2^K words by $\tau_A(w)$, where A records the zero positions replaced by 1. Every $\tau_A(w)$ ends in 0.

Theorem 4.24 (Complementary depth)**.** *For $z \in D$ and every nonempty word w ending in 1,*

$$H_w(z) = (-1)^K \sum_A \sum_{\tau_A(w)=uv} (-1)^{|v|} H_u(q) H_{\mathrm{rev}(v)}(1). \tag{4.85}$$

After trailing-zero normalization, this is a finite rational linear combination of terms

$$L^j \operatorname{Li}_{\boldsymbol{a}}(q)\zeta(\boldsymbol{b}), \qquad j + \mathrm{wt}(\boldsymbol{a}) + \mathrm{wt}(\boldsymbol{b}) = |w|, \quad \mathrm{dep}(\boldsymbol{a}) + \mathrm{dep}(\boldsymbol{b}) \le K. \tag{4.86}$$

Nonempty zeta indices are convergent. Thus a one-variable polylogarithm of weight w and depth d has the stated reduction with $K = w - d$.

Proof. On $z < 0$, make the substitution $u = (1-t)^{-1}$ in every integration. Its pullbacks are

$$\frac{dt}{t} = -\frac{du}{u} - \frac{du}{1-u}, \qquad \frac{dt}{1-t} = \frac{du}{u}. \tag{4.87}$$

The initial endpoint $t = 0$ becomes $u = 1$. Multilinearity gives $H_w(z) = (-1)^K \sum_A J_{\tau_A(w)}(q)$. Lemma 4.23 proves (4.85).

It remains to check the claimed normal form. The identity $H_v H_0 = H_{v\Pi 0}$ removes a trailing zero. If a word ends in exactly r zeros, its coefficient in the shuffle of its prefix with 0 is r; all other words in that shuffle have fewer trailing zeros. Induction therefore expresses each $H_u(q)$ as a rational combination of $L^j H_{u'}(q)$ with u' empty or ending in 1. The operation preserves the number of 1 letters.

For an endpoint word beginning in 0, the same normalization preserves its initial 0 in every nonzero term with no logarithmic factor. Terms containing a positive power of $\log 1$ vanish. The surviving nonempty terms begin in 0 and end in 1, hence are convergent MZVs. In each split of $\tau_A(w)$ the total number of 1 letters in the two factors is $|A| \le K$. This proves the depth bound and the weight identity. Finally continue analytically from $z < 0$ to D. □

Corollary 4.25 (An argument-enlarged depth bound)**.** *For $w \ge 2$, every one-variable MPL of weight w can be represented using nonlogarithmic factors of depth at most $\lfloor w/2 \rfloor$, allowing both the original argument z and $q = (1-z)^{-1}$, MZVs, and powers of L as coefficients. More precisely, use the original expression when $d \le w/2$, and Theorem 4.24 otherwise. The all-one case is $L^w/w!$.*

This corollary is deliberately not phrased as a theorem about arbitrary multivariable MPLs, or as a reduction inside an unchanged root-of-unity basket. At $z = i$ the half-point q is not a root of unity. The theorem explains an argument-enlarged upper bound; it does not prove the absence or presence of new cyclotomic periods in a smaller comparison algebra.

For the next family put $F_r(z) = \mathrm{Li}_{2,\{1\}^{r-1}}(z)$, as in the one-two formulas above. Their reciprocal-complementary form is

$$F_r(z) = \sum_{k=1}^{r+1} \frac{(-1)^k L^{r+1-k}}{(r+1-k)!} \mathrm{Li}_k(q) + (-1)^r \zeta(r+1) - \frac{L^{r+1}}{(r+1)!}.$$

4.11 A double-polylogarithm formula for an all-weight family

The next formula addresses the first case beyond the classical ladder. It reduces the height-one family with initial index 3 to singles and doubles at q, irrespective of its original depth. Let

$$C_1(q) = 0, \qquad C_k(q) = \sum_{a=1}^{k-1} \mathrm{Li}_{a,k-a}(q) \quad (k \geq 2). \tag{4.88}$$

All MPLs on the right have the one-variable meaning (4.79).

Theorem 4.26 (Two-zero height-one identity)**.** *For every $w \geq 3$ and $z \in D$,*

$$\begin{aligned}\mathrm{Li}_{3,\{1\}^{w-3}}(z) = \frac{L^w}{w!} + \sum_{k=1}^{w} \frac{(-1)^k L^{w-k}}{(w-k)!} \big((k-2)\,\mathrm{Li}_k(q) + C_k(q)\big) \\ + (-1)^{w-1} \Big((L + \mathrm{Li}_1(q))\zeta(w-1) + (w-2)\zeta(w) - \zeta(w-1,1) \Big).\end{aligned} \tag{4.89}$$

The right side has reduced depth at most two.

Proof. Write $D_q = q\,d/dq = d/dL$. Direct differentiation of the nested series, followed by continuation, gives

$$D_q C_k = C_{k-1} + \frac{q}{1-q}\,\mathrm{Li}_{k-1} \quad (k \geq 2).$$

Put $V_k = (k-2)\,\mathrm{Li}_k + C_k$, with $V_1 = -\,\mathrm{Li}_1$. Then

$$D_q V_k = V_{k-1} + \frac{\mathrm{Li}_{k-1}}{1-q} \quad (k \geq 2), \qquad D_q V_1 = -\frac{q}{1-q}.$$

Differentiating the finite sum in (4.89) telescopes and yields

$$\frac{F_{w-2}(z) - (-1)^{w-2}\zeta(w-1)}{q-1}.$$

The derivative of the explicit $(L + \mathrm{Li}_1(q))\zeta(w-1)$ term supplies the missing constant term, because $D_q(L + \mathrm{Li}_1(q)) = 1/(1-q)$. The derivative of the entire right side is therefore $F_{w-2}(z)/(q-1)$. The left side has the same derivative, since $d\,\mathrm{Li}_{3,\{1\}^{w-3}}(z)/dz = F_{w-2}(z)/z$ and $dz/z = dL/(q-1)$.

For the endpoint constant, recall the convergent depth-two sum identity

$$\sum_{a=2}^{w-1} \zeta(a, w-a) = \zeta(w), \qquad w \geq 3. \tag{4.90}$$

It follows, for example, by subtracting the shuffle and stuffle expansions of the once-regularized product $\zeta(1)\zeta(w-1)$: its common divergent term $\zeta(1,w-1)$ cancels, leaving precisely (4.90). Only one logarithmic divergence is involved, so no higher regularization correction enters. Equivalently one may use finite cutoffs and subtract the common harmonic-logarithmic part before taking the limit.

The convergent colored stuffle identity, for $0<q<1$, gives

$$\mathrm{Li}_{1,w-1}(q)-\mathrm{Li}_1(q)\zeta(w-1)=-\mathrm{Li}_{w-1,1}(1,q)-\mathrm{Li}_w(q).$$

The right side tends to $-\zeta(w-1,1)-\zeta(w)$ by dominated convergence. Combining this with (4.90) yields

$$\lim_{q\to1^-}\big(C_w(q)-\mathrm{Li}_1(q)\zeta(w-1)\big)=-\zeta(w-1,1).$$

All positive powers of L multiplying at most logarithmically singular terms vanish. The remaining constants in (4.89) cancel, so its right side tends to zero as $z\to0^-$. The left side does too. Equality of derivatives and the endpoint limit proves the result; continue to D. □

Remark 4.27 (The sum identity and the proof dependency). Equation (4.90) is the classical depth-two Euler sum formula, not a new conjecture assumed here. The same two-zero formula can also be obtained directly from Theorem 4.24 by expanding its two zero positions and normalizing trailing zeros. The supplied code verifies the agreement of these two independently specified formulas through weight twelve.

At weight six, (4.89) is a complete depth-two representation of $\mathrm{Li}_{3,1,1,1}(i)$ and of $\mathrm{Li}_{3,1,1,1}(\rho)$. In particular a depth-four index does not by itself identify a new depth-four period in the enlarged comparison algebra. The endpoint double is harmless as a named convergent constant; standard Euler reductions can simplify it further. For example,

$$\zeta(4,1)=2\zeta(5)-\zeta(2)\zeta(3),\qquad \zeta(5,1)=\frac{3}{4}\zeta(6)-\frac{1}{2}\zeta(3)^2.$$

These last simplifications are not required for the theorem.

4.12 The colored stuffle identity

Whenever the defining sums are absolutely convergent, sorting the two indices in a product proves

$$\mathrm{Li}_a(x)\,\mathrm{Li}_b(y)=\mathrm{Li}_{a,b}(x,y)+\mathrm{Li}_{b,a}(y,x)+\mathrm{Li}_{a+b}(xy). \tag{4.91}$$

At conditionally convergent boundary points, Abel limits give the same identity when the limiting products and constituent series exist.

4.13 Gaussian doubles: closed forms

Write $\mathrm{Li}_{a,b}$ at the three Gaussian base points beyond $(\mathrm{i},1)$:

$$(\mathrm{i},\mathrm{i}),\qquad(-1,\mathrm{i}),\qquad(\mathrm{i},-1),$$

all generated to 700 digits by an accelerated evaluator (Section 12) and agreeing with Wolfram 15 to full precision. Under the level-4 complex conjugation $\mathrm{i}\mapsto-\mathrm{i}$ the imaginary parts are odd and the real parts even.

4.13.1 Family 1: the stuffle closed forms

Specializing (4.91) at the Gaussian points ($uv = -1$ at (i,i); $uv = -\mathrm{i}$ at the mixed pair) gives a complete set of exact closed forms for the symmetric combinations:

$$\mathrm{Li}_{a,b}(\mathrm{i},\mathrm{i}) + \mathrm{Li}_{b,a}(\mathrm{i},\mathrm{i}) = \mathrm{Li}_a(\mathrm{i})\,\mathrm{Li}_b(\mathrm{i}) - \mathrm{Li}_{a+b}(-1), \tag{4.92}$$

$$\mathrm{Li}_{a,b}(-1,\mathrm{i}) + \mathrm{Li}_{b,a}(\mathrm{i},-1) = \mathrm{Li}_a(-1)\,\mathrm{Li}_b(\mathrm{i}) - \mathrm{Li}_{a+b}(-\mathrm{i}), \tag{4.93}$$

$$\mathrm{Li}_{a,b}(\mathrm{i},-1) + \mathrm{Li}_{b,a}(-1,\mathrm{i}) = \mathrm{Li}_a(\mathrm{i})\,\mathrm{Li}_b(-1) - \mathrm{Li}_{a+b}(-\mathrm{i}),$$

with the clean diagonal corollary

$$\mathrm{Li}_{a,a}(\mathrm{i},\mathrm{i}) = \tfrac{1}{2}\big(\mathrm{Li}_a(\mathrm{i})^2 - \mathrm{Li}_{2a}(-1)\big)\,. \tag{4.94}$$

Every single polylogarithm on the right is elementary in the level-4 vocabulary: $\operatorname{Im}\mathrm{Li}_n(\mathrm{i}) = \beta(n)$ is the Dirichlet beta value, $\operatorname{Re}\mathrm{Li}_n(\mathrm{i})$ and $\mathrm{Li}_n(-1) = -\eta(n)$ are rational multiples of $\zeta(n)$ for $n \geq 2$ (or π^n at even n), and $\mathrm{Li}_n(-\mathrm{i}) = \overline{\mathrm{Li}_n(\mathrm{i})}$. At $n = 1$, the real values are $\operatorname{Re}\mathrm{Li}_1(\mathrm{i}) = -\log 2/2$ and $\mathrm{Li}_1(-1) = -\log 2$. So Family 1 reduces every symmetric and diagonal Gaussian double to single L-values in closed form.

Remark 4.28. The three stuffle identities (4.92)–(4.93) were checked against the stored 700-digit values to $\sim 10^{-692}$ for every admissible a, b, and the closed forms were confirmed symbolically and numerically by Wolfram 15 at low weight, including the weight-6 diagonal. Its exact homogeneous expression follows directly from stuffle:

$$\mathrm{Li}_{3,3}(\mathrm{i},\mathrm{i}) = \frac{9}{2048}\zeta(3)^2 + \frac{47\pi^6}{1935360} - \frac{3\mathrm{i}\pi^3\zeta(3)}{1024}.$$

Indeed $\mathrm{Li}_3(\mathrm{i}) = (-3\zeta(3) + \mathrm{i}\pi^3)/32$ and $\mathrm{Li}_6(-1) = -31\pi^6/30240$.

After the symmetric stuffle combinations have been removed, the remaining information is carried by $\mathrm{Li}_{a,b}(x,y) - \mathrm{Li}_{b,a}(y,x)$, exchanging both indices and colors. Equal indices alone do not force it to vanish at mixed points. The same-color diagonal is a special case.

Here is an exact counterexample entirely inside the manuscript's alphabet:

$$\Im\big(\mathrm{Li}_{1,1}(-1,i) - \mathrm{Li}_{1,1}(i,-1)\big) = \frac{\pi}{2}\log 2 - G = \int_0^1 \frac{\log(1+t^2)}{1+t^2}\,dt > 0. \tag{4.95}$$

To prove the first equality, differentiating the convergent defining series in the disk gives

$$\frac{\partial}{\partial x}\,\mathrm{Li}_{1,1}(x,y) = \frac{\mathrm{Li}_1(xy)}{1-x}.$$

Continuation along the segment to $x = -1$ is nonsingular, and hence

$$\Im\,\mathrm{Li}_{1,1}(-1,i) = \int_0^1 \frac{\arctan t}{1+t}\,dt.$$

Put $K = \int_0^1 \log(1+t)/(1+t^2)\,dt$. With $t = \tan\theta$,

$$K = \int_0^{\pi/4} \big(\tfrac{1}{2}\log 2 + \log\sin(\theta+\pi/4) - \log\cos\theta\big)\,d\theta = \frac{\pi}{8}\log 2.$$

The last two integrals cancel after reflection about $\pi/2$. Integration by parts now gives $\Im\,\mathrm{Li}_{1,1}(-1,i) = \pi\log 2/4 - K = \pi\log 2/8$. Stuffle and

$$\mathrm{Li}_1(-1) = -\log 2, \qquad \mathrm{Li}_1(i) = -\tfrac{1}{2}\log 2 + \tfrac{\pi i}{4}, \qquad \Im\,\mathrm{Li}_2(-i) = -G$$

give

$$\Im\big(\mathrm{Li}_{1,1}(-1,i)+\mathrm{Li}_{1,1}(i,-1)\big)=G-\frac{\pi}{4}\log 2.$$

Subtracting proves the first equality in (4.95); the second also follows directly by setting $t=\tan\theta$. Indeed,

$$\int_0^{\pi/4}\log\cos\theta\,d\theta=-\frac{\pi}{4}\log 2+\frac{G}{2},$$

which follows from the sum and difference of the corresponding log-sine and log-cosine integrals: their difference is G, and their sum is $-\pi\log 2/2$ by the double-angle formula. Thus the displayed integral is $-2\int_0^{\pi/4}\log\cos\theta\,d\theta$. Its integrand is positive away from zero. The difference is approximately 0.17282745097458205019574093418642286289514247590297.

4.13.2 Family 2: antisymmetric reductions to the $(\mathrm{i},1)$ algebra

Against the basis of single-polylog monomials together with the $(\mathrm{i},1)$ generators $g_{a,b}=\operatorname{Im}\mathrm{Li}_{a,b}(\mathrm{i},1)$, $h_{a,b}=\operatorname{Re}\mathrm{Li}_{a,b}(\mathrm{i},1)$, the *low-weight* antisymmetric parts reduce, each reduction a new closed-form identity found by integer-relation search and confirmed in Wolfram 15. A representative instance, with $\mu_n=\operatorname{Re}\mathrm{Li}_n(\frac{1+\mathrm{i}}{2})$ and $\lambda_n=\operatorname{Im}\mathrm{Li}_n(\frac{1+\mathrm{i}}{2})$:

$$\operatorname{Re}[\mathrm{Li}_{1,2}(\mathrm{i},\mathrm{i})-\mathrm{Li}_{2,1}(\mathrm{i},\mathrm{i})]=3\zeta(3)-4\log 2\,\mu_2-\pi\lambda_2-4h_{1,1}\log 2+3h_{1,2}. \tag{4.96}$$

Such reductions were found at weights 2–4 at the mixed points and at weight 3 at (i,i). At higher weights the reported searches did not find reductions against the specified $(\mathrm{i},1)$ basket. This records the outcome of those searches; it does not prove numerical nonreduction. Exact shuffle relations and a precisely specified formal quotient determine which directions survive the available relation vocabulary.

4.14 Gaussian doubles: the double-shuffle dimension

4.14.1 A cautionary integer-relation count

A first attempt to count the genuinely-new depth-2 directions proceeds by integer-relation search: test each antisymmetric part for membership in the span of the $(\mathrm{i},1)$ generators and single-polylog products, and tabulate those that fail. Carried out at 700 digits, this reports **30** independent new directions through weight 6. The number is wrong. It is an artifact of the spurious scale: the genuine reductions are of large height, and against a basis of a hundred-odd atoms the search cannot see them. (Three searches against bases of 46, 55, and the $(\mathrm{i},1)$-plus-single set returned "dimensions" of 3, 6, and 30 respectively—all different, all wrong, the count *decreasing* as the basis grew and the ceiling fell.)

4.14.2 The shuffle, and the correct count

The shuffle settles it. The key relation is that the shuffle of a (-1)-polylogarithm with an i-polylogarithm produces (i,i) and $\overline{(-1,\mathrm{i})}$ doubles, for instance

$$\mathrm{Li}_2(-1)\,\mathrm{Li}_2(\mathrm{i})=\mathrm{Li}_{2,2}(\mathrm{i},\mathrm{i})+2\,\mathrm{Li}_{3,1}(\mathrm{i},\mathrm{i})+\overline{\mathrm{Li}_{2,2}(-1,\mathrm{i})}+2\,\overline{\mathrm{Li}_{3,1}(-1,\mathrm{i})} \tag{4.97}$$

(verified to $\sim 10^{-694}$), whereas the *stuffle* of the same product produces the $(-1,\mathrm{i}),(\mathrm{i},-1)$ doubles. Equating ties all three Gaussian points together. Collecting all 240 stuffle and shuffle relations

from the products $\mathrm{Li}_p(s)\,\mathrm{Li}_q(t)$, $s,t \in \{\mathrm{i},-1,-\mathrm{i}\}$, $2 \le p+q \le 6$, verifying each to $\sim 10^{-600}$, and computing the exact rational rank of the linear system on the doubles, modulo single polylogarithms and the $(\mathrm{i},1)$ generators, gives:

weight	unknowns (Re+Im)	dim (stuffle only)	dim (double shuffle)
4	18	8	**1**
5	24	12	**2**
6	30	14	**2**
total		34	**5**

Observation 4.29. Modulo single polylogarithms and the $(\mathrm{i},1)$ generators, the Gaussian double points $(\mathrm{i},\mathrm{i}),(-1,\mathrm{i}),(\mathrm{i},-1)$ contribute exactly **5** candidate depth-two generators through weight 6—one at weight 4, two at weight 5, two at weight 6. A representative basis is

$$\mathrm{Re}\,\mathrm{Li}_{3,1}(\mathrm{i},-1); \quad \mathrm{Im}\,\mathrm{Li}_{3,2}(\mathrm{i},-1),\ \mathrm{Im}\,\mathrm{Li}_{4,1}(\mathrm{i},-1); \quad \mathrm{Re}\,\mathrm{Li}_{4,2}(\mathrm{i},-1),\ \mathrm{Re}\,\mathrm{Li}_{5,1}(\mathrm{i},-1).$$

Every other Gaussian double of weight ≤ 6 is an explicit combination of these five, the $(\mathrm{i},1)$ generators, and products of single L-values.

The rank is verified two ways: each relation reproduces its value to 10^{-600}, and the exact-rational rank is precision-independent. As a final gold-check at weight 4, fixing the single survivor $\mathrm{Re}\,\mathrm{Li}_{3,1}(\mathrm{i},-1)$ to its true value and solving the system predicts all other 17 weight-4 components to $\sim 10^{-692}$.

Remark 4.30 (the methodological moral). Observation 4.29 corrects the integer-relation count of 30 by a factor of six. The reductions are exact consequences of the small-height double-shuffle generators, but the *net* reductions are of large height—so a search at a specified precision and coefficient bound can miss them. This is the recurring trap of high-precision experimental mathematics, here biting in the direction of *over*-stating independence. The exact-rank calculation determines the formal quotient by the supplied valid relations. It does not make an arithmetic-independence claim.

4.15 Triple sums and the scope of formal quotient dimensions

The word for $\mathrm{Li}_{a,b,c}(i,1,1)$ is $0^{a-1}(-i)0^{b-1}(-i)0^{c-1}(-i)$ in the outer-letter-first convention. Shuffle relations express products of lower-weight functions as sums of such triples. At weight six the source reports rank seven for thirteen relations on ten triple coordinates, leaving three formal coordinates represented by $(3,1,2),(3,2,1),(4,1,1)$. Rank seven means that seven coordinates can be eliminated using the chosen relations and the three remaining coordinates. It does not mean that seven particular triples are individually products, or that the remaining three numerical values are independent.

The source also claimed binomial dimensions $\binom{n}{p}$ for the actual depth filtration from a total Hilbert series $1/(1-2t)$. A total Hilbert series alone cannot determine a depth grading. A grading by the length of a chosen motivic word basis must first be identified with the filtration under discussion; moreover a motivic lower bound does not give a numerical lower bound without injectivity of the period map. We therefore make no unconditional claim that genuine numerical depth three first occurs at weight six. A rigorous reduction or obstruction for these specific triples is an open problem. Panzer's parity theorem [33] supplies exact lower-depth conjugation combinations, but does not assert independence of the complementary components.

4.16 The Eisenstein analogue

Everything above has a level-3 counterpart at the cube roots of unity. Write $\rho = e^{2\pi i/3}$, so $\rho^2 = \bar{\rho}$.

4.16.1 Family 1, with an arithmetic twist

The stuffle closed forms read

$$\mathrm{Li}_{a,b}(\rho,\rho) + \mathrm{Li}_{b,a}(\rho,\rho) = \mathrm{Li}_a(\rho)\,\mathrm{Li}_b(\rho) - \mathrm{Li}_{a+b}(\rho^2), \tag{4.98}$$

$$\mathrm{Li}_{a,b}(\rho^2,\rho) + \mathrm{Li}_{b,a}(\rho,\rho^2) = \mathrm{Li}_a(\rho^2)\,\mathrm{Li}_b(\rho) - \zeta(a+b), \tag{4.99}$$

with the diagonal corollary $\mathrm{Li}_{a,a}(\rho,\rho) = \frac{1}{2}(\mathrm{Li}_a(\rho)^2 - \mathrm{Li}_{2a}(\rho^2))$. The arithmetic twist is in (4.99): because $\rho^2 \cdot \rho = \rho^3 = 1$, the colored tail degenerates to the *ordinary* zeta value $\zeta(a+b)$, not a level-3 value. The single cube-root L-values at $n \geq 2$ split cleanly under the distribution relation $\mathrm{Li}_n(\rho) + \mathrm{Li}_n(\rho^2) = (3^{1-n} - 1)\zeta(n)$:

$$\mathrm{Re}\,\mathrm{Li}_n(\rho) = \tfrac{1}{2}(3^{1-n} - 1)\,\zeta(n), \qquad \mathrm{Im}\,\mathrm{Li}_n(\rho) = \mathrm{Sl}_n(2\pi/3), \tag{4.100}$$

At weight one, $\mathrm{Li}_1(\rho) = -\frac{1}{2}\log 3 + i\pi/6$. For the displayed higher-weight formula, $\mathrm{Sl}_n(\theta) = \sum_{k\geq 1} \sin(k\theta)/k^n$. The real part is a rational multiple of $\zeta(n)$; the sine series agrees with Cl_n at even weight and is elementary at rational angles at odd weight. Also $\mathrm{Cl}_2(2\pi/3) = 2V/3$, where $V = \mathrm{Cl}_2(\pi/3)$ is the Gieseking constant. All of (4.98)–(4.100) are confirmed in Wolfram 15.

4.16.2 The (ρ,ρ) frontier: one constant

Running the level-3 double shuffle as in Section 4.14—90 stuffle and shuffle relations from products $\mathrm{Li}_p(s)\,\mathrm{Li}_q(t)$, $s,t \in \{\rho,\rho^2\}$, each verified to $\sim 10^{-600}$, exact rational rank—collapses the (ρ,ρ) doubles still more sharply than the Gaussian:

weight	unknowns (Re+Im)	dim (stuffle only)	dim (double shuffle)
2	2	0	0
3	4	2	0
4	6	2	0
5	8	4	**1**
6	10	4	0
total		12	**1**

Observation 4.31. Modulo single polylogarithms and the $(\rho,1)$ generators, the entire (ρ,ρ) family through weight 6 reduces to a *single* candidate depth-two generator—$\mathrm{Im}\,\mathrm{Li}_{3,2}(\rho,\rho)$, at weight 5—with weights $2,3,4$ and 6 collapsing completely.

A gold-check at weight 5 fixes the survivor and predicts all seven other weight-5 components to $\sim 10^{-696}$. The pattern mirrors the Gaussian: there too the weight-5 survivors are imaginary $\mathrm{Li}_{3,2}/\mathrm{Li}_{4,1}$ directions.

Remark 4.32 (a genuine cube-root obstacle). The decisive shuffle $\mathrm{Li}_p(\rho^2)\,\mathrm{Li}_q(\rho) \to (\rho,\rho) + (\rho^2,\rho^2)$ lands only on points with $w = z_1 z_2 \neq 1$, so the computation never needs the mixed points $(\rho^2,\rho),(\rho,\rho^2)$, where $w = \rho^3 = 1$. Those points presented a numerical obstacle in the earlier evaluator: both the Lerch-plus-Wynn evaluator and a direct cumulative-sum evaluator stall there at $\sim$8 digits—the period-three, non-oscillating tail defeats Wynn's ε-algorithm, unlike the even-period

i, −1 cases, which converge to full precision. Folding the mixed cube-root points in (for the *full* level-3 frontier) requires a period-three-aware accelerated evaluator, or Wolfram; the following mixed-point method addresses that computational gap.

4.17 Why the standard evaluator degenerates at $w = 1$

With the convention the nested-series definition, the accelerated evaluator of the article writes the inner tail as a Lerch transcendent and sums the *outer* series with an acceleration factor $w^n = (z_1 z_2)^n$. At the mixed points $w = 1$, that factor is identically 1: the oscillation Wynn's ε-algorithm relies on disappears, leaving a bare period-three power-law tail that the algorithm cannot resolve past a handful of digits. Summing over the *larger* index instead keeps an oscillating factor $z_1^{n_1}$, but for $z_1 = \rho^2$ this is a *period-three* oscillation, and a plain ε-algorithm (tuned for geometric tails) again stalls.

4.18 A robust $w = 1$ evaluator

Two ingredients fix it.

(a) A smooth correction identity. Writing $l = n_1$ (the larger index) and splitting the inner partial sum $P(l) = \sum_{n<l} z_2^n/n^b = \mathrm{Li}_b(z_2) - R(l)$ with tail $R(l) = \sum_{n\geq l} z_2^n/n^b = z_2^l\,\Phi(z_2, b, l)$ (Lerch Φ), one gets the exact identity

$$\boxed{\mathrm{Li}_{a,b}(z_1, z_2) = \mathrm{Li}_a(z_1)\;\mathrm{Li}_b(z_2) \;-\; \sum_{l\geq 1} \frac{w^l}{l^a}\,\Phi(z_2, b, l)} \tag{4.101}$$

At $w = 1$ the correction summand $\Phi(z_2, b, l)/l^a$ is *smooth*—every oscillating factor $z_1^l z_2^l = w^l$ collapses to 1—with the clean asymptotic $\Phi(z_2, b, l) \sim \sum_{j\geq 0} \binom{-b}{j} c_j(z_2)\, l^{-b-j}$, where $c_0 = \frac{1}{1-z_2}$ and $c_{j\geq 1} = \mathrm{Li}_{-j}(z_2)$ (rational in z_2). The subtle point—the source of a factor-of-one bug that cost five digits in a first draft—is that $c_0 = \sum_{k\geq 0} z_2^k = \frac{1}{1-z_2}$ is *not* $\mathrm{Li}_0(z_2) = \frac{z_2}{1-z_2}$; they differ by exactly 1. This gives an Euler–Maclaurin tail in Hurwitz zetas, $\sum_{l>M} \Phi/l^a = \sum_j \binom{-b}{j} c_j\, \zeta(a + b + j, M + 1)$, a fast and accurate cross-check at low to medium precision.

(b) Period-aware ρ-acceleration for production. For high precision the production evaluator sums the outer cumulative series directly, samples the partial sums at the common period $N = \mathrm{lcm}(\mathrm{ord}\, z_1, \mathrm{ord}\, z_2)$ (so every oscillating factor realigns to 1 at the sample points), and accelerates the resulting smooth, power-law-convergent sequence with **Wynn's ρ-algorithm** —the algebraic-convergence counterpart of the ε-algorithm. The ρ-algorithm yields empirical convergence on the tested families; precision inflation and independent comparison are needed to assess its conditioning (`mpl_w1.py`).

The general certified evaluators and their geometric bounds are developed in Chapter 5. The historical Lerch evaluator above remains a useful independent comparison.

4.19 Validation

The evaluator is checked by three independent, Wolfram-free routes plus one Wolfram cross-check.

1. **Exact symmetric closed form** (single polylogs only, an entirely separate code path):

$$\mathrm{Li}_{a,b}(\rho^2,\rho)+\mathrm{Li}_{b,a}(\rho,\rho^2)=\mathrm{Li}_a(\rho^2)\,\mathrm{Li}_b(\rho)-\zeta(a+b), \tag{4.102}$$

   and the Gaussian analogue $\mathrm{Li}_{a,b}(\mathrm{i},-\mathrm{i})+\mathrm{Li}_{b,a}(-\mathrm{i},\mathrm{i})=\mathrm{Li}_a(\mathrm{i})\,\mathrm{Li}_b(-\mathrm{i})-\zeta(a+b)$. (The collision term is $\mathrm{Li}_{a+b}(z_1z_2)=\mathrm{Li}_{a+b}(1)=\zeta(a+b)$.)

2. **Conjugation** $\mathrm{Li}_{a,b}(\rho,\rho^2)=\overline{\mathrm{Li}_{a,b}(\rho^2,\rho)}$.

3. **Two independent summation methods**—the ρ-accelerated cumulative sum and the Euler–Maclaurin smooth-correction series—agree to full working precision.

4. **Wolfram 15** `MultiplePolyLog` at moderate precision.

All four agree; the symmetric closed form (4.102) holds to the full generated precision ($\sim 10^{-500}$) for every (a,b).

4.20 Closed forms for the Eisenstein mixed-point values

With 500-digit values in hand, an integer-relation (PSLQ/LLL) search against the level-3 basis —monomials in the single L-values $\{\pi,\log 3,\zeta(3),\zeta(5),\mathrm{Cl}_2(\frac{2\pi}{3}),\mathrm{Cl}_4(\frac{2\pi}{3}),\mathrm{Cl}_6(\frac{2\pi}{3})\}$ times the $(\rho,1)$ generators $g_{a,b}=\mathrm{Im}\,\mathrm{Li}_{a,b}(\rho,1)$, $h_{a,b}=\mathrm{Re}\,\mathrm{Li}_{a,b}(\rho,1)$, and the lone (ρ,ρ) double-shuffle survivor $\mathrm{Im}\,\mathrm{Li}_{3,2}(\rho,\rho)$—**reduces every mixed value through weight 4, and every weight-5 imaginary part**, to explicit closed forms (each canary-clean, height ≤ 10, re-verified to 10^{-425} against the *complete* weight-graded basis). The two simplest:

$$\mathrm{Im}\,\mathrm{Li}_{1,1}(\rho^2,\rho)=-\big(\mathrm{Cl}_2(\tfrac{2\pi}{3})+\mathrm{Im}\,\mathrm{Li}_{1,1}(\rho,1)\big),\qquad \mathrm{Re}\,\mathrm{Li}_{1,1}(\rho^2,\rho)=5\,\mathrm{Re}\,\mathrm{Li}_{1,1}(\rho,1)-\tfrac{1}{2}\log^2 3. \tag{4.103}$$

(The full list of $\sim$16 reductions is in `harvest_eis_mixed.py`.) The original weight-five real and weight-six searches failed in their selected baskets. The exact inverse-color reduction below supersedes the interpretation of that failure as a new mixed-point obstruction.

4.21 Inverse colors: the exact reduction that the searches missed

The numerical evaluator above made the mixed points accessible, but its failed relation searches did not identify the correct quotient. An exact partial-fraction argument now proves that inverse-color doubles add no directions relative to the one-direction family and single-value products. This supersedes the earlier mixed-point candidate counts at weights five and six, in both the Gaussian and Eisenstein cases.

Put $M_{a,b}(z)=\mathrm{Li}_{a,b}(z,z^{-1})$ and $w=a+b$. We allow $0<|z|<1$, and $|z|=1, z\neq 1$, with positive integers a,b. Boundary sums use the stated outer-index order. They converge absolutely if $a\ge 2$. For $a=1$, the inner colored harmonic sum equals its limit plus $O(n^{-b})$; the outer sum is therefore a convergent oscillatory series plus an absolutely convergent remainder.

Theorem 4.33 (All-weight inverse-argument reduction)**.** *In this domain,*

$$\begin{aligned} M_{a,b}(z) = (-1)^{b+1}\sum_{j=1}^{a}\binom{w-j-1}{b-1}\operatorname{Li}_{w-j,j}(z,1) + (-1)^{b+1}\binom{w-1}{b}\operatorname{Li}_w(z) \\ + \sum_{j=2}^{b}(-1)^{b-j}\binom{w-j-1}{a-1}\zeta(j)\operatorname{Li}_{w-j}(z) \\ + (-1)^b\sum_{j=2}^{a}\binom{w-j-1}{b-1}\zeta(j)\operatorname{Li}_{w-j}(z). \end{aligned} \tag{4.104}$$

Empty sums are zero. In particular, $F_a = M_{a,1}$ *satisfies*

$$F_a(z) = a\operatorname{Li}_{a+1}(z) + \sum_{j=1}^{a}\operatorname{Li}_{a+1-j,j}(z,1) - \sum_{j=2}^{a}\zeta(j)\operatorname{Li}_{a+1-j}(z). \tag{4.105}$$

Proof. Set $h = n - m$. Principal parts at $m = 0, -h$ give

$$\begin{aligned} \frac{1}{m^b(m+h)^a} = \sum_{j=1}^{b}\frac{(-1)^{b-j}\binom{w-j-1}{a-1}}{h^{w-j}m^j} \\ + (-1)^b\sum_{j=1}^{a}\frac{\binom{w-j-1}{b-1}}{h^{w-j}(m+h)^j}. \end{aligned} \tag{4.106}$$

Indeed, the difference has no poles and tends to zero at infinity. The $j = 1$ coefficients are opposite; their sum over m is finite because $\sum_{m\geq 1}(1/m - 1/(m+h)) = H_h$. For $j \geq 2$, the shifted sum is $\zeta(j) - H_h^{(j)}$. Consequently

$$\begin{aligned} c_h := \sum_{m\geq 1}\frac{1}{m^b(m+h)^a} = (-1)^{b+1}\sum_{j=1}^{a}\binom{w-j-1}{b-1}\frac{H_h^{(j)}}{h^{w-j}} \\ + \sum_{j=2}^{b}(-1)^{b-j}\binom{w-j-1}{a-1}\frac{\zeta(j)}{h^{w-j}} + (-1)^b\sum_{j=2}^{a}\binom{w-j-1}{b-1}\frac{\zeta(j)}{h^{w-j}}. \end{aligned}$$

Multiply by z^h, sum, and use

$$\sum_{h\geq 1}\frac{z^h H_h^{(j)}}{h^{w-j}} = \operatorname{Li}_{w-j,j}(z,1) + \operatorname{Li}_w(z), \qquad \sum_{j=1}^{a}\binom{w-j-1}{b-1} = \binom{w-1}{b}.$$

For $|z| < 1$ these operations are absolute.

To justify the boundary even when $a = 1$, let O_N be the original sum over $n \leq N$ and $D_N = \sum_{h=1}^{N-1} z^h c_h$. Partial summation gives

$$\left|\sum_{h=h_0}^{h_1}\frac{z^h}{(m+h)^a}\right| \leq \frac{4}{|1-z|(m+h_0)^a}.$$

Splitting the discrepancy at $m = N$ now yields

$$|D_N - O_N| \leq \frac{4}{|1-z|}\left(N^{-a}H_N^{(b)} + \sum_{m>N} m^{-a-b}\right) \longrightarrow 0.$$

The coefficients c_h decrease to zero, the latter by dominated convergence with majorant m^{-a-b}. Dirichlet's test proves convergence of $\sum z^h c_h$. The one-direction doubles on the right also converge. This proves the identity in the original boundary convention. □

Corollary 4.34 (The mixed quotient is zero)**.** *Let $\mathcal{A}_w(z)$ be the $\mathbb{Q}$-span of*

$$\mathrm{Li}_{r,w-r}(z,1)\ (1 \le r < w), \quad \mathrm{Li}_w(z), \quad \zeta(j)\,\mathrm{Li}_{w-j}(z)\ (2 \le j < w).$$

Every $M_{a,b}(z)$ of weight w belongs to $\mathcal{A}_w(z)$. Adding the mixed values contributes zero dimension modulo this space.

At $z = \mathrm{i}$, this is the Gaussian family already used in the report. At $z = \omega^2$, the one-direction values are conjugates of its $(\omega, 1)$ values, whose real and imaginary parts it includes. Thus the proposed additional mixed directions disappear at every weight. The result concerns inverse-color pairs, and does not assert that all colored double polylogarithms reduce to one direction.

4.21.1 An independent integral representation

Proposition 4.35. *In the same domain,*

$$M_{a,b}(z) = \frac{z}{(a-1)!}\int_0^1 \frac{(-\log t)^{a-1}\,\mathrm{Li}_b(t)}{1-zt}\,dt, \tag{4.107}$$

$$\mathrm{Li}_{a,b}(z,1) = \frac{z}{(a-1)!}\int_0^1 \frac{(-\log t)^{a-1}\,\mathrm{Li}_b(zt)}{1-zt}\,dt. \tag{4.108}$$

Proof. Insert $n^{-a} = (a-1)!^{-1}\int_0^1 (-\log t)^{a-1} t^{n-1}\,dt$. The $h = n-m$ geometric sum gives $zt/(1-zt)$; the m-sum gives $\mathrm{Li}_b(t)$ or $\mathrm{Li}_b(zt)$. At a fixed boundary point $z \neq 1$, the denominator is bounded away from zero on $[0,1]$, and the possible logarithmic endpoint singularities are integrable. Dominated convergence and the preceding boundary argument complete the proof. □

These formulas supply independent numerical checks without the power-law acceleration problem at $z_1 z_2 = 1$. Quadrature itself is not a rigorous certificate unless its numerical error is enclosed.

4.21.2 The mixed parity component reduces to single values

Panzer's theorem predicts lower depth for the real component at odd weight and the imaginary component at even weight [33]. Here is a direct proof for the mixed F_a family.

Theorem 4.36 (Mixed parity formulas)**.** *For $|z| = 1, z \neq 1$ and $m \ge 1$,*

$$\begin{aligned}\operatorname{Re} F_{2m}(z) = {}& \frac{2m+1}{2}\operatorname{Re}\mathrm{Li}_{2m+1}(z) - \sum_{j=1}^{m}\zeta(2j)\operatorname{Re}\mathrm{Li}_{2m+1-2j}(z)\\ & - \frac{1}{2}\zeta(2m+1) + \operatorname{Re}\mathrm{Li}_{2m}(z)\operatorname{Re}\mathrm{Li}_1(z).\end{aligned} \tag{4.109}$$

For $m \ge 0$,

$$\begin{aligned}\operatorname{Im} F_{2m+1}(z) = {}& (m+1)\operatorname{Im}\mathrm{Li}_{2m+2}(z) - \sum_{j=1}^{m}\zeta(2j)\operatorname{Im}\mathrm{Li}_{2m+2-2j}(z)\\ & + \operatorname{Im}\mathrm{Li}_{2m+1}(z)\operatorname{Re}\mathrm{Li}_1(z).\end{aligned} \tag{4.110}$$

Proof. For $a \geq 2$, set $\mathcal{B}_a(z) = -\operatorname{Li}_a(z) - (-1)^a \operatorname{Li}_a(z^{-1})$. The Fourier series $-\mathcal{B}_a(z) = \sum_{n\neq 0} z^n/n^a$ is absolutely summable, while $\operatorname{Li}_1(z^{-1}) = \sum_{\ell\geq 1} z^{-\ell}/\ell$ belongs to L^2 on the circle. Their product has positive h-th coefficient $c_h = \sum_{\ell\geq 1}[\ell(\ell+h)^a]^{-1}$, constant coefficient $\zeta(a+1)$, and negative h-th coefficient

$$q_{-h} = \sum_{\ell\geq 1,\ \ell\neq h} \frac{1}{\ell(\ell-h)^a}.$$

The difference $q_{-h} - (-1)^a c_h$ is

$$\sum_{\ell\in\mathbb{Z}\setminus\{0,h\}} \frac{1}{\ell(\ell-h)^a} = (-1)^a \left[2 \sum_{j=1}^{\lfloor a/2\rfloor} \frac{\zeta(2j)}{h^{a+1-2j}} - \frac{a+1}{h^{a+1}} \right]. \tag{4.111}$$

To check this absolutely convergent bilateral sum, use

$$\frac{1}{\ell(\ell-h)^a} = \frac{1}{(-h)^a \ell} - \sum_{r=1}^{a} \frac{1}{(-h)^{a+1-r}(\ell-h)^r}.$$

Symmetric summation of the $r=1$ terms gives respectively $-1/h$ and $1/h$. For $r\geq 2$, the shifted sum is $(1+(-1)^r)\zeta(r) - (-h)^{-r}$. Substitution gives (4.111). Comparing all Fourier coefficients now gives

$$\begin{aligned} F_a(z) + (-1)^a F_a(z^{-1}) = (-1)^a &\left[(a+1)\operatorname{Li}_{a+1}(z^{-1}) - 2\sum_{j=1}^{\lfloor a/2\rfloor} \zeta(2j)\operatorname{Li}_{a+1-2j}(z^{-1}) \right] \\ &- \zeta(a+1) - \operatorname{Li}_1(z^{-1})\mathcal{B}_a(z). \end{aligned} \tag{4.112}$$

The equality holds first in L^2, then pointwise by continuity away from $z=1$. Real parts for even a, and imaginary parts for odd a, give the claims. The remaining case $a=1$ follows from $F_1(z) = \operatorname{Li}_2(z) + \frac{1}{2}\operatorname{Li}_1(z)^2$, itself a consequence of (4.105) and the elementary shuffle identity. □

4.21.3 Four explicit replacements for the claimed survivors

Corollary 4.37. *The four highlighted mixed-point constants satisfy*

$$\operatorname{Re}\operatorname{Li}_{4,1}(\mathrm{i},-\mathrm{i}) = -\frac{587}{1024}\zeta(5) + \frac{3}{32}\zeta(2)\zeta(3) + \frac{135}{256}\zeta(4)\log 2, \tag{4.113}$$

$$\operatorname{Im}\operatorname{Li}_{5,1}(\mathrm{i},-\mathrm{i}) = 3\beta(6) - \zeta(2)\beta(4) - \zeta(4)G - \frac{5\pi^5\log 2}{3072}, \tag{4.114}$$

$$\operatorname{Re}\operatorname{Li}_{4,1}(\omega^2,\omega) = -\frac{281}{162}\zeta(5) + \frac{4}{9}\zeta(2)\zeta(3) + \frac{20}{27}\zeta(4)\log 3, \tag{4.115}$$

$$\operatorname{Im}\operatorname{Li}_{5,1}(\omega^2,\omega) = -3C_6 + \zeta(2)C_4 + \zeta(4)C_2 + \frac{\pi^5\log 3}{729}. \tag{4.116}$$

Proof. Use Theorem 4.36 with $m=2$, together with

$$\operatorname{Re}\operatorname{Li}_s(\mathrm{i}) = -2^{-s}(1-2^{1-s})\zeta(s), \qquad \operatorname{Re}\operatorname{Li}_s(\omega) = \frac{3^{1-s}-1}{2}\zeta(s) \quad (s>1),$$

and

$$\operatorname{Re}\operatorname{Li}_1(\mathrm{i}) = -\tfrac{1}{2}\log 2, \quad \operatorname{Re}\operatorname{Li}_1(\omega) = -\tfrac{1}{2}\log 3, \quad \operatorname{Im}\operatorname{Li}_5(\mathrm{i}) = \frac{5\pi^5}{1536}, \quad \operatorname{Im}\operatorname{Li}_5(\omega) = \frac{2\pi^5}{729}.$$

The real identities follow by residue-class splitting; the imaginary fifth-order values follow from the odd-beta and Bernoulli Fourier formulas, with $B_5(1/3) = -5/243$. Conjugation changes the sign of imaginary parts at ω^2. □

The four values, in the displayed order, are approximately

$$\begin{aligned} &-0.01342026735728779, && 0.03269424445221378,\\ &-0.03903921845650667, && -0.02533938136807792. \end{aligned}$$

Their right sides already belong to the report's listed single-value bases. Its `thm:parallel` is therefore false in the printed convention. The evaluator scripts and value stores cited by that report were not imported into ProveIt, so their precise failure cannot be diagnosed from the available files. Replace the survivor claims by Theorem 4.33 and Corollary 4.37, and recompute any dimension conclusions that depended on the proposed new directions.

4.22 Mixed endpoints and all-weight cyclotomic product quotients

Put $A_{a,b} = \operatorname{Im}\operatorname{Li}_{a,b}(\mathrm{i},\mathrm{i})$ and $K_{a,b} = \operatorname{Im}\operatorname{Li}_{a,b}(\mathrm{i},-1)$; retain $g_{a,b}$ and S_p as above. The all-exponent endpoint identity relates the mixed family to the same-color family before any period-independence question is asked.

4.22.1 An all-exponent mixed endpoint identity

Theorem 4.38 (Gaussian mixed endpoint)**.** *For every integer $p \geq 1$,*

$$K_{p,1} + (-1)^{p-1}A_{1,p} = \beta(p+1) - 2\log 2\,\beta(p) + \sum_{j=2}^{p}(-1)^j\eta(j)\beta(p+1-j). \tag{4.117}$$

Moreover,

$$S_p = g_{p,1} + K_{p,1}. \tag{4.118}$$

Proof. All the boundary values used here converge. First-index-one values with first argument i can be obtained by Dirichlet's test or radial Abel limits. The shuffle and stuffle identities may therefore be justified by convergent iterated integrals and Abel limiting.

Fix $w = p+1$, and define homogeneous generating polynomials of degree $w-2$:

$$\mathcal{A}(X,Y) = \sum_{a+b=w} A_{a,b}X^{a-1}Y^{b-1}, \quad \mathcal{K}(X,Y) = \sum_{a+b=w} K_{a,b}X^{a-1}Y^{b-1}.$$

Let $\mathcal{V}$ denote the same polynomial for $\operatorname{Im}\operatorname{Li}_{a,b}(-1,\mathrm{i})$, and put

$$P(X,Y) = \sum_{a+b=w} \operatorname{Im}[\operatorname{Li}_a(-1)\operatorname{Li}_b(\mathrm{i})]X^{a-1}Y^{b-1}, \qquad H(X,Y) = \sum_{a+b=w} X^{a-1}Y^{b-1}.$$

Stuffle and shuffle respectively give

$$\mathcal{V}(X,Y) + \mathcal{K}(Y,X) = P(X,Y) + \beta(w)H(X,Y),$$

$$\mathcal{A}(X+Y,X) - \mathcal{V}(X+Y,Y) = P(X,Y).$$

The collision term in the first equation is $-\operatorname{Im}\operatorname{Li}_w(-\mathrm{i}) = \beta(w)$. Eliminating $\mathcal{V}$ yields

$$\mathcal{K}(U,V) = P(V-U,U) + P(V,U) + \beta(w)H(V,U) - \mathcal{A}(V,V-U).$$

Set $V = 0$ and compare coefficients of U^{w-2}. The evaluation $\operatorname{Li}_j(-1) = -\eta(j)$ gives (4.117), including the two occurrences of $-\log 2\,\beta(p)$.

For (4.118), taking imaginary parts of the outer index in $\operatorname{Li}_{p,1}(\mathrm{i},1) + \operatorname{Li}_{p,1}(\mathrm{i},-1)$ retains indices $2n+1$, with sign $(-1)^n$. Their inner sum is

$$\sum_{m=1}^{2n} \frac{1+(-1)^m}{m} = H_n.$$

Absolute convergence proves the result directly for $p > 1$. An Abel limit gives the endpoint $p = 1$. □

At $p = 4$ this specializes to

$$K_{4,1} = A_{1,4} + \frac{191\pi^5}{23040} - \frac{3}{4}G\zeta(3) - 2\beta(4)\log 2. \tag{4.119}$$

4.22.2 A precisely defined formal presentation

Fix a level $L = 3$ or 4 and a weight $w \geq 2$. Let $\omega = e^{2\pi\mathrm{i}/L}$. Introduce rational formal symbols

$$D_a(r,s) \quad (1 \leq a < w, \quad r,s \in \mathbb{Z}/L\mathbb{Z})$$

for imaginary parts of $\operatorname{Li}_{a,w-a}(\omega^r,\omega^s)$. Impose conjugation $D_a(-r,-s) = -D_a(r,s)$. Symbols fixed by this involution vanish. Put all single values and imaginary parts of products of two single values in the background, and also set $D_a(1,0) = 0$. The latter is precisely quotienting by the chosen g-family.

The only further rows imposed are, for $p+q = w$,

$$D_p(r,s) + D_q(s,r) = 0, \tag{4.120}$$

$$\sum_{j=0}^{p-1}\binom{q+j-1}{j} D_{q+j}(s,r-s) + \sum_{j=0}^{q-1}\binom{p+j-1}{j} D_{p+j}(r,s-r) = 0. \tag{4.121}$$

They are the stuffle and shuffle rows of two single values, with their one-leading-one regularizations. A regularized product involving the real parameter T belongs to the background. At $w = 2$, the only two-divergent-factor correction is real and disappears on taking imaginary parts. Call the resulting quotient $V_{L,w}$.

This is *not* the full cyclotomic double-shuffle algebra. Products involving a depth-two factor, relations lifted through higher depths, additional functional equations, and motivic period assertions have not been imposed. In particular, Zhao's standard relations [81] are substantially broader than this presentation. The theorem below computes $V_{L,w}$, not a dimension of actual numbers.

For a scalar assignment to the symbols, introduce

$$\mathcal{F}^{r,s}(X,Y) = \sum_{a=1}^{w-1} D_a(r,s)X^{a-1}Y^{w-a-1}.$$

The conditions that the assignment annihilate all rows are exactly

$$\mathcal{F}^{r,s}(X,Y) + \mathcal{F}^{s,r}(Y,X) = 0, \tag{4.122}$$

$$\mathcal{F}^{s,r-s}(X+Y,X) + \mathcal{F}^{r,s-r}(X+Y,Y) = 0. \tag{4.123}$$

These follow by comparing binomial coefficients after substitution. They describe the dual of $V_{L,w}$, so their solution dimension is the required quotient dimension.

4.22.3 All-weight dimension theorems

Theorem 4.39 (Level four)**.** *The formal quotient satisfies*

$$\dim_{\mathbb{Q}} V_{4,w} = \begin{cases} 0, & w \text{ even}, \\ (w-1)/2, & w \text{ odd}. \end{cases} \qquad \sum_{w\geq 2} \dim V_{4,w}\, t^w = \frac{t^3}{(1-t^2)^2}.$$

At odd weight, its dual is parameterized by arbitrary antisymmetric homogeneous polynomials of degree $w-2$.

Proof. Up to conjugation, the six color pairs are $(0,1),(1,0),(1,1),(1,2),(1,3),(2,1)$. Write their polynomials as U, G, A, K, M, V, respectively, and let $n = w-2$. The background condition gives $G = 0$; stuffle gives

$$U = 0, \quad A(Y,X) = -A(X,Y), \quad V(X,Y) = -K(Y,X), \quad M(Y,X) = M(X,Y).$$

The shuffle with colors $(1,0)$ forces $M = 0$. The shuffle with colors $(2,1)$ gives

$$A(X+Y,X) + K(Y,X+Y) = 0,$$

hence $K(X,Y) = -A(Y,Y-X)$. The shuffle with colors $(1,3)$ is $K(X,Y) = K(X,X-Y)$. After substitution and antisymmetry this condition is $A = (-1)^{n+1}A$.

For even n all polynomials therefore vanish. For odd n, any antisymmetric A, with K, V as above and $U = G = M = 0$, satisfies the rows. To check exhaustiveness, the unordered color classes up to conjugation are

$$(0,0),(0,1),(0,2),(1,1),(1,2),(1,3),(2,2).$$

The real classes vanish, and the remaining classes are precisely the conditions just used, together with their transposes. Thus there are no unexamined rows. A basis for A at odd w is

$$X^{a-1}Y^{w-a-1} - X^{w-a-1}Y^{a-1}, \qquad 1 \leq a \leq (w-1)/2.$$

Counting these basis vectors and summing over odd weights proves the formulas. □

Theorem 4.40 (Level three)**.** *The formal quotient satisfies*

$$\dim_{\mathbb{Q}} V_{3,w} = \begin{cases} 0, & w \text{ even}, \\ \lfloor (w+1)/6 \rfloor, & w \text{ odd}, \end{cases} \qquad \sum_{w\geq 2} \dim V_{3,w}\, t^w = \frac{t^5}{(1-t^2)(1-t^6)}.$$

Writing $\Delta = XY(X-Y)$, $Q = X^2 - XY + Y^2$, the dual is parameterized by the homogeneous polynomials

$$A(X,Y) = \Delta\, P(Q, \Delta^2)$$

of degree $w-2$.

Proof. Up to conjugation the four pairs are $(0,1),(1,0),(1,1),(1,2)$. As before the first two vanish and the shuffle with a zeta factor kills the inverse-argument polynomial. The remaining polynomial A satisfies exactly

$$A(Y,X) = -A(X,Y), \qquad A(X,X-Y) = A(X,Y).$$

Antisymmetry implies divisibility by $X-Y$; the second symmetry then forces divisibility by Y, and antisymmetry forces divisibility by X. Hence $A = \Delta B$. The polynomial Δ has the same two transformation signs as A, so B is invariant under swapping X, Y and under $(X,Y) \mapsto (X, X-Y)$.

On the plane $a+b+c=0$, with $(a,b,c)=(X,-Y,Y-X)$, the second transformation swaps b,c; the first is global negation followed by the transposition of a,b. The generated group contains global negation, obtained by cubing the product, and all permutations. The elementary symmetric-polynomial theorem therefore identifies its invariant ring with

$$\mathbb{Q}[e_2,e_3^2]=\mathbb{Q}[Q,\Delta^2].$$

Conversely every $\Delta P(Q,\Delta^2)$ has the required symmetries, and direct substitution verifies all the original color rows. Counting monomials of degrees $3+2a+6b=w-2$ gives the result. Explicitly, at odd $w\geq 5$ a basis is

$$\Delta^{2b+1}Q^{(w-5)/2-3b},\qquad 0\leq b\leq \lfloor (w-5)/6\rfloor.$$

□

For comparison, the exact dimensions at the first few odd weights are

w	3	5	7	9	11	13	15
$\dim V_{3,w}$	0	1	1	1	2	2	2
$\dim V_{4,w}$	1	2	3	4	5	6	7

and all even-weight dimensions are zero. The exact matrix script checks both formulas through weight 16, including that the displayed polynomial bases annihilate every raw row and have the stated rank.

4.22.4 A separating functional and the scope of the S4 proof

At every odd weight $w\geq 3$, choose $A(X,Y)=(Y-X)^{w-2}$ in the level-four theorem. Then $K(X,Y)=X^{w-2}$, $V(X,Y)=-Y^{w-2}$. This gives a rational linear functional ℓ that annihilates every defining row and every background value:

$$\ell(D_a(1,1))=(-1)^{a+1}\binom{w-2}{a-1},$$
$$\ell(D_{w-1}(1,2))=1,\qquad \ell(D_1(2,1))=-1. \tag{4.124}$$

All other entries vanish, apart from conjugates with opposite signs.

At weight five its nonzero entries are particularly transparent:

symbol	$A_{1,4}$	$A_{2,3}$	$A_{3,2}$	$A_{4,1}$	$\operatorname{Im}\operatorname{Li}_{1,4}(-1,\mathrm{i})$	$K_{4,1}$
ℓ	1	−3	3	−1	−1	1

The polynomial proof verifies all weights; the delivered certificate also checks this six-entry vector against every exact weight-five row.

The now-proved identity of Theorem 4.8 has an informative image in this restricted presentation.

The exact identity (4.119) makes this equivalent to

$$161280A_{1,4}+69120(g_{4,1}+g_{3,2}+3g_{2,3})+617\pi^5-101520G\zeta(3)=0. \tag{4.125}$$

The functional ℓ sends its left side to 161280. Consequently this relation does not follow from (4.120)–(4.121) and the stated background alone. This is an exact obstruction to a specific proof strategy. The convergent octahedral relation used in its proof supplies precisely information outside this presentation.

4.22.5 A new weight-seven candidate

Proposition 4.41 (Two weight-seven coordinate baskets)**.** *The Gaussian doubles satisfy the exact identity*

$$\begin{aligned} g_{2,5} = 30g_{6,1} + 15g_{5,2} + 5g_{4,3} - \frac{15}{512}G\zeta(5) \\ + \frac{3}{16}\beta(4)\zeta(3) - \frac{11\pi^7}{368640}. \end{aligned} \tag{4.126}$$

Proof. The convergent same-point shuffles give

$$\begin{aligned} g_{2,5} + 2g_{3,4} + 3g_{4,3} + 5g_{5,2} + 10g_{6,1} &= \operatorname{Im}(\operatorname{Li}_2(i)\operatorname{Li}_5(i)), \\ g_{3,4} + 4g_{4,3} + 10g_{5,2} + 20g_{6,1} &= \operatorname{Im}(\operatorname{Li}_3(i)\operatorname{Li}_4(i)). \end{aligned}$$

Subtract twice the second row from the first and use

$$\operatorname{Im}(\operatorname{Li}_2(i)\operatorname{Li}_5(i)) = -\frac{15}{512}G\zeta(5) - \frac{5\pi^7}{73728},$$

$$\operatorname{Im}(\operatorname{Li}_3(i)\operatorname{Li}_4(i)) = -\frac{3}{32}\beta(4)\zeta(3) - \frac{7\pi^7}{368640}.$$

These product evaluations follow from the ordinary depth-one fourth-root formulas. The coefficient of $g_{3,4}$ cancels, giving the identity. □

The new Cayley report's proposed formula

$$\begin{aligned} S_6 = &-\frac{12334}{527}g_{6,1} - \frac{384}{31}g_{5,2} - \frac{2136}{527}g_{4,3} + \frac{3179\pi^7}{5713920} \\ &- \frac{801}{2108}\beta(4)\zeta(3) - 2\beta(6)\log 2 \end{aligned}$$

is therefore exactly the same conjecture as the following earlier form, written in a different Gaussian basket. If the two residuals mean left side minus right side, their difference is $-128/155$ times the zero relation obtained by moving (4.126) to the left. The exact two-row check has zero rational residual. This proves equivalence of the candidates, not the conjectured vanishing of either one.

Conjecture 4.42 (A reduction for S_6)**.** *With the same definitions,*

$$\begin{aligned} S_6 = \frac{722}{527}g_{6,1} + \frac{40}{527}g_{4,3} - \frac{128}{155}g_{2,5} + \frac{15191}{28569600}\pi^7 \\ - \frac{3}{124}G\zeta(5) - \frac{2373}{10540}\beta(4)\zeta(3) - 2\beta(6)\log 2. \end{aligned} \tag{4.127}$$

The candidate was found in the ordered basket

$$(S_6, g_{6,1}, g_{4,3}, g_{2,5}, \pi^7, G\zeta(5), \beta(4)\zeta(3), \beta(6)\log 2).$$

Its primitive integer relation vector is

$$\begin{aligned} (&485683200, -665395200, -36864000, 401080320, \\ &-258247, 11750400, 109347840, 971366400). \end{aligned}$$

A 100-digit search used coefficient bound 10^{12} and tolerance 10^{-90}. After the coefficients were frozen, reevaluation by Mellin integrals at 220 working digits gave an integer-normalized residual of approximately -2.315×10^{-213}. An independent Euler transformation of the defining alternating series, using 900 terms and 300 working digits, gave approximately 2.854×10^{-263}. The script records the full values and the exact integer vector. These are numerical evidence, not a proof or an interval certificate.

For the second method, the defining series are

$$g_{a,b} = \sum_{n\geq 0} \frac{(-1)^n H_{2n}^{(b)}}{(2n+1)^a}, \qquad H_m^{(b)} = \sum_{j=1}^{m} j^{-b},$$

and the analogous series with H_n for S_6. Thus the second evaluation does not reuse the discovery integral. Last transformed terms are recorded as diagnostics; their small size alone is not asserted to bound the complete tail.

The weight-seven functional (4.124) sends the integer-normalized candidate to 485683200. Its proof therefore also requires information beyond the restricted single-product presentation. The S4 certificate suggests seeking a corresponding weight-seven functional equation for (4.127), with the nonzero functional serving as a check that the missing relation has actually been introduced.

4.22.6 A distinct frozen candidate at weight nine

The later signed/zero continuation supplies a new S_8 candidate in a Gaussian basket with odd inner indices. It is distinct from the vector rigorously rejected in Proposition 12.1.

Conjecture 4.43 (The next mixed Gaussian reduction)**.**

$$\begin{aligned} S_8 = {} & \frac{52334}{37973} g_{8,1} + \frac{2824}{37973} g_{6,3} - \frac{6832}{189865} g_{4,5} - \frac{139760}{265811} g_{2,7} \\ & + \frac{998183\pi^9}{17283022848} - \frac{78615}{19442176} G\zeta(7) - \frac{201021}{4252976} \beta(4)\zeta(5) \\ & - \frac{290505}{1063244} \beta(6)\zeta(3) - 2\beta(8)\log 2. \end{aligned} \tag{4.128}$$

In the order

$$(S_8, g_{8,1}, g_{6,3}, g_{4,5}, g_{2,7}, \pi^9, G\zeta(7), \beta(4)\zeta(5), \beta(6)\zeta(3), \beta(8)\log 2),$$

its primitive integer vector is

$$\begin{aligned} (& -10974719508480, 15125246115840, 816174858240, -394908401664, \\ & -5770366156800, 633846205, -44376595200, -518730670080, \\ & -2998569369600, -21949439016960). \end{aligned}$$

All entries have greatest common divisor one; each displayed right-side coefficient is minus its vector entry divided by the first entry.

The discovery used 720 Euler terms at 250 working digits and then PSLQ at 150 digits, with tolerance 10^{-140} and coefficient bound 10^{20}. After freezing the vector, separate Mellin integrals at 220 digits and a 1400-term Euler evaluation at 500 digits corroborated it. Their residuals are numerical diagnostics and are not used to prove equality.

The fresh standard-library replay reconstructs two rational proximity certificates: the 1100-term source-bound scheme and the 1200-term direct scheme. For R_8 equal to the left side minus the right side of (4.128), the latter proves

$$|R_8| < 10^{-355}.$$

The same scheme improves the normalized S_6 residual to that bound. Each residual interval contains zero. The exact endpoints and frozen vectors are preserved with the signed/zero continuation. These enclosures prove proximity and leave both identities conjectural.

4.23 Harmonic index, reflection and certified asymptotics

The first-index sum S_p is part of a larger family. Here the harmonic index measures the displayed nesting length, not the minimal arithmetic or motivic depth. The same beta kernel gives reflection identities, signed truncation certificates and a joint index–exponent transition. These are analytic consequences of a positive kernel, and therefore remain useful independently of unresolved period-independence questions.

4.24 The harmonic-index family and its integral representation

Definition 4.44. *For integers $n, r \geq 0$, put*

$$e_r(n) = [u^r] \prod_{j=1}^{n} \left(1 + \frac{u}{j}\right), \qquad e_0(n) = 1. \tag{4.129}$$

Thus $e_r(n) = 0$ for $n < r$, $e_1(n) = H_n$, and $e_2(n) = (H_n^2 - H_n^{(2)})/2$. For real $a, p > 0$, define

$$T_{p,r}(a) = \sum_{n \geq 0} \frac{(-1)^n e_r(n)}{(n+a)^p}, \qquad S_{p,r} = 2^{-p} T_{p,r}(1/2). \tag{4.130}$$

The original S_p is $S_{p,1}$, while $S_{p,0} = \beta(p)$.

Lemma 4.45 (Convergence)**.** *For fixed $r \geq 0$ and $a, p > 0$, the series (4.130) converges. It converges absolutely for $p > 1$.*

Proof. The bound $e_r(n) \leq H_n^r/r!$ proves absolute convergence for $p > 1$ and convergence of the summand magnitude to zero for every $p > 0$. For eventual monotonicity, when $r \geq 1$ and $n \geq r$, counting each r-element subset in r ways gives

$$r e_r(n) = \sum_{|A|=r-1} \left(\prod_{j \in A} \frac{1}{j}\right) \left(H_n - \sum_{j \in A} \frac{1}{j}\right) \geq (H_n - H_{r-1}) e_{r-1}(n).$$

Together with $e_r(n+1) = e_r(n) + e_{r-1}(n)/(n+1)$, this shows that the logarithm of the ratio of consecutive magnitudes is at most

$$\frac{r}{(n+1)(H_n - H_{r-1})} - p \log\left(1 + \frac{1}{n+a}\right) < 0$$

for sufficiently large n. The alternating-series test applies. The case $r = 0$ is immediate. □

Proposition 4.46 (Mellin integral)**.** *For the same parameters,*

$$T_{p,r}(a) = \frac{(-1)^r}{r!\Gamma(p)} \int_0^1 t^{a-1}(-\log t)^{p-1} \frac{\log^r(1+t)}{1+t}\,\mathrm{d}t. \tag{4.131}$$

In particular, $(-1)^r T_{p,r}(a) > 0$.

Proof. The binomial theorem and (4.129) give, for $|t| < 1$,

$$\sum_{n\geq 0}(-1)^n e_r(n)t^n = [u^r](1+t)^{-1-u} = \frac{(-1)^r \log^r(1+t)}{r!(1+t)}. \tag{4.132}$$

Insert an Abel factor ρ^n with $0 < \rho < 1$, integrate termwise using $\int_0^1 t^{n+a-1}(-\log t)^{p-1}\,\mathrm{d}t = \Gamma(p)/(n+a)^p$, and let $\rho \uparrow 1$. Dominated convergence applies on the integral side, and Lemma 4.45 identifies the Abel limit with the ordinary sum. The integrand in (4.131) is strictly positive almost everywhere after removing the sign. □

4.24.1 Exact cyclotomic realizations

For integer $p \geq 1$, the series connects directly to the repository's Gaussian multiple polylogarithms:

$$S_{p,r} = \sum_{\epsilon_1,\dots,\epsilon_r \in \{1,-1\}} \Im\,\mathrm{Li}_{p,\underbrace{1,\dots,1}_{r}}(\mathrm{i}, \epsilon_1, \dots, \epsilon_r). \tag{4.133}$$

Indeed, write the outer index as $m_0 = 2n+1$ and the inner indices as $m_j = 2k_j$. The 2^r from $1/k_j = 2/m_j$ cancels the factors $1/2$ in the even-index filters. Finally, $\Im(\mathrm{i}^{m_0})$ is exactly $(-1)^n$ for odd m_0 and zero for even m_0. This proves (4.133), first with an Abel factor and then by convergence. For $r = 1$, both second arguments 1 and -1 occur; S_p is not simply one of the draft's $g_{a,b}$.

More generally, let $a = m/q$ with $0 < m < q$ and $\xi = e^{\pi \mathrm{i}/q}$. Residue-class filters give

$$\begin{aligned} T_{p,r}(m/q) = q^{p-1} \sum_{h=0}^{q-1} \sum_{b_1,\dots,b_r=0}^{q-1} & \xi^{-(2h+1)m} \\ & \cdot \mathrm{Li}_{p,1,\dots,1}(\xi^{2h+1}, \xi^{2b_1}, \dots, \xi^{2b_r}). \end{aligned} \tag{4.134}$$

To check the normalization, the denominator rescaling contributes q^{p+r}, while the $r+1$ residue filters contribute q^{-r-1}. Thus rational parameters lead to level $2q$ values. This is a representation of nesting length at most $r+1$, not a minimal-depth assertion.

4.25 An all-index reflection identity

Define, initially for $\Re(a-z) > 0$,

$$\Phi_a(z,u) = \int_0^1 t^{a-z-1}(1+t)^{-1-u}\,\mathrm{d}t. \tag{4.135}$$

Near $(z,u) = (0,0)$, coefficient extraction in (4.131) yields

$$\Phi_a(z,u) = \sum_{p\geq 1}\sum_{r\geq 0} T_{p,r}(a) z^{p-1} u^r. \tag{4.136}$$

There are no factorials in this ordinary Taylor series. This convention matters for every coefficient below.

Theorem 4.47 (Reflection)**.** *If $0 < a < 1$, then as analytic germs at the origin,*

$$\boxed{\Phi_a(z,u) + \Phi_{1-a}(-z-u,u) = \frac{\Gamma(a-z)\Gamma(1-a+z+u)}{\Gamma(1+u)}.} \tag{4.137}$$

More generally the identity holds whenever $\Re(a-z) > 0$ and $\Re(1-a+z+u) > 0$.

Proof. The beta integral evaluates

$$\int_0^\infty t^{a-z-1}(1+t)^{-1-u}\,\mathrm{d}t = \frac{\Gamma(a-z)\Gamma(1-a+z+u)}{\Gamma(1+u)}.$$

The part over $(0,1)$ is (4.135). In the part over $(1,\infty)$, substitute $t = 1/x$. The transformed integrand is $x^{-a+z+u}(1+x)^{-1-u}$, which is exactly the second term of (4.137). The stated inequalities justify both integrals. □

Let

$$C_{p,r}(a) = [z^{p-1}u^r]\frac{\Gamma(a-z)\Gamma(1-a+z+u)}{\Gamma(1+u)}. \tag{4.138}$$

Corollary 4.48 (Finite triangular reduction)**.** *For integers $p \geq 1, r \geq 0$ and $0 < a < 1$,*

$$\begin{aligned} T_{p,r}(a) + (-1)^{p-1}T_{p,r}(1-a) = C_{p,r}(a) \\ - \sum_{j=1}^{r}(-1)^{p-1+j}\binom{p-1+j}{j}T_{p+j,r-j}(1-a). \end{aligned} \tag{4.139}$$

Every sum on the right has smaller harmonic index.

Proof. The coefficient of $z^{p-1}u^r$ in $\Phi_{1-a}(-z-u,u)$ is

$$\sum_{j=0}^{r}(-1)^{p-1+j}\binom{p-1+j}{j}T_{p+j,r-j}(1-a).$$

Separate $j = 0$ in Theorem 4.47. □

4.25.1 A finite compiler for the gamma coefficient

Put $R_a(z,u)$ equal to the gamma quotient in (4.138), and write $R_a = (\pi/\sin\pi a)\exp(\sum_{d\geq 1}\ell_d)$, with ℓ_d homogeneous of total degree d. Then

$$\ell_d(z,u) = \frac{\psi^{(d-1)}(a)(-z)^d + \psi^{(d-1)}(1-a)(z+u)^d - \psi^{(d-1)}(1)u^d}{d!}. \tag{4.140}$$

Here $\psi^{(0)} = \psi$. If E_n is the homogeneous degree-n part of the exponential, then

$$E_0 = 1, \qquad E_n = \frac{1}{n}\sum_{d=1}^{n} d\ell_d E_{n-d}. \tag{4.141}$$

Only degrees through $p+r-1$ are needed for $C_{p,r}$. These recurrences are a finite identity certificate, not a relation search.

For rational a, the coefficients reduce to ordinary cyclotomic values and their products. For derivatives of positive order, use $\psi^{(k)}(a) = (-1)^{k+1}k!\zeta(k+1,a)$. For $s \geq 2$,

$$\zeta(s, m/q) = q^{s-1} \sum_{h=0}^{q-1} e^{-2\pi \mathrm{i} hm/q} \operatorname{Li}_s(e^{2\pi \mathrm{i} h/q}).$$

The Euler constants cancel in the linear part of (4.140); the remaining rational digamma combinations are expressible in logarithms and trigonometric constants. Thus (4.139) reduces the reflection combination to lower-index sums and products of depth-one values. It does not prove that the complementary combination is irreducible.

4.25.2 The self-reflecting Gaussian case

Set

$$F(z,u) = \tfrac{1}{2}\Phi_{1/2}(z/2,u) = \sum_{p\geq 1, r\geq 0} S_{p,r} z^{p-1} u^r.$$

Then

$$F(z,u) + F(-z-2u,u) = K(z,u), \qquad K(z,u) = \frac{\Gamma((1-z)/2)\Gamma((1+z)/2+u)}{2\Gamma(1+u)}. \tag{4.142}$$

For odd p,

$$2S_{p,r} = [z^{p-1}u^r]K - \sum_{j=1}^{r} (-1)^j 2^j \binom{p-1+j}{j} S_{p+j,r-j}. \tag{4.143}$$

For example,

$$S_{p,2} = pS_{p+1,1} - p(p+1)\beta(p+2) + \tfrac{1}{2}[z^{p-1}u^2]K \qquad (p \text{ odd}).$$

At even p, the coefficient of $S_{p,r}$ cancels. The identity then supplies relations among lower-index quantities rather than an unsupported converse to the odd-p reduction.

4.26 The first harmonic index revisited

At $r = 1$, the preceding reflection identity recovers equation (4.1). Its explicit coefficient form is equivalent to the generating-integral proof at the beginning of this chapter. This is one theorem reached from two useful representations: the full beta integral and the incomplete-beta reflection germ.

4.27 Complete monotonicity and globally enveloping truncations

Define

$$h_r(t) = \frac{1}{1+t}\left(\frac{\log(1+t)}{t}\right)^r, \qquad h_r(0) = 1. \tag{4.144}$$

The elementary representation

$$h_r(t) = \int_{[0,1]^r} \frac{1}{1+t} \prod_{j=1}^{r} \frac{1}{1+x_j t}\, \mathrm{d}x_1 \cdots \mathrm{d}x_r \tag{4.145}$$

is the source of the signed bounds.

Lemma 4.49. *For $t \geq 0$, h_r is completely monotone: $(-1)^k h_r^{(k)}(t) \geq 0$ for all $k \geq 0$. Moreover,*

$$h_r(t) = r! \sum_{j \geq 0} (-1)^j e_r(r+j) t^j \quad (|t| < 1). \tag{4.146}$$

For every integer $M \geq 0$ and $0 \leq t \leq 1$,

$$0 \leq (-1)^M \left(h_r(t) - r! \sum_{j=0}^{M-1} (-1)^j e_r(r+j) t^j \right) \leq r! e_r(r+M) t^M. \tag{4.147}$$

Proof. Every factor in (4.145) has derivatives of alternating sign, and products preserve this property by Leibniz's rule. Integration preserves it as well. Formula (4.146) follows from (4.132) after removing the zero of order r at $t = 0$. Taylor's integral remainder has sign $(-1)^M$. Its magnitude is at most $|h_r^{(M)}(0)| t^M / M!$ because complete monotonicity makes $|h_r^{(M)}(t)|$ nonincreasing. This is (4.147); $M = 0$ means $0 \leq h_r(t) \leq 1$. □

Theorem 4.50 (First-omitted-term certificate)**.** *For every real $p, a > 0$, integer $r \geq 0$, and integer $M \geq 0$,*

$$0 \leq (-1)^{r+M} \left(T_{p,r}(a) - \sum_{n=r}^{r+M-1} \frac{(-1)^n e_r(n)}{(n+a)^p} \right) \leq \frac{e_r(r+M)}{(r+M+a)^p}. \tag{4.148}$$

In the Gaussian normalization, replace every $(n+a)^p$ by $(2n+1)^p$.

Proof. Multiply (4.147) by $t^{a+r-1}(-\log t)^{p-1}/(r!\Gamma(p))$ and integrate. Equation (4.131) gives the middle term, and each monomial integrates explicitly. □

Remark 4.51. The certificate holds even when the first few summand magnitudes increase. Its proof is not an application of the alternating-series remainder theorem at an unjustified starting index. For example r may be large and p small; the positivity of the integral, rather than early term monotonicity, controls the error.

Since $e_r(r) = 1/r!$ and $e_r(r+1) = (r+2)/(2r!)$, we obtain

$$0 \leq 1 - (-1)^r r!(r+a)^p T_{p,r}(a) \leq \frac{r+2}{2} \left(\frac{r+a}{r+a+1} \right)^p. \tag{4.149}$$

The lower expression furnished by this bound may be negative for small p; positivity from Proposition 4.46 remains available. For fixed r, a and $p \to \infty$,

$$T_{p,r}(a) = \frac{(-1)^r}{r!(r+a)^p} + \frac{(-1)^{r+1}(r+2)}{2r!(r+a+1)^p} + R_{p,r}, \quad |R_{p,r}| \leq \frac{e_r(r+2)}{(r+a+2)^p}. \tag{4.150}$$

More generally, every finite number of exponential terms has its own signed explicit remainder. The first-omitted-term constant is asymptotically sharp for fixed r, a, M, by applying the bound once more to the remainder.

4.27.1 An exact symmetric-function certificate

Expanding (4.145) at zero gives

$$r!e_r(r+j) = \sum_{k_0+\cdots+k_r=j} \prod_{i=1}^{r} \frac{1}{k_i+1}. \tag{4.151}$$

Equivalently, it is the integral of the complete homogeneous polynomial $h_j(1, x_1, \ldots, x_r)$ over $[0,1]^r$. This identity is checked using exact rational arithmetic by two different recurrences in the supplied code. In particular, for fixed j,

$$r!e_r(r+j) = \frac{r^j}{2^j j!} + O_j(r^{j-1}). \tag{4.152}$$

The leading contribution comes from j distinct variables among $x_1, \ldots, x_r$; repeated variables and use of the distinguished entry 1 contribute $O_j(r^{j-1})$.

4.27.2 Pole stripping and exact Taylor radius

Let $F_{a,r}(z) = [u^r]\Phi_a(z,u)$. The integral representation shows that it extends holomorphically to $\Re z < a + r$, because the numerator vanishes to order r at the origin. Subtracting finitely many terms of (4.132) then continues it meromorphically to the entire z-plane, with precisely the simple poles

$$z = a + n \quad (n \geq r), \qquad \operatorname{Res}_{z=a+n} F_{a,r}(z) = (-1)^{n+1} e_r(n). \tag{4.153}$$

For clarity, subtract through degree N inside the integral. The remainder is $O(t^{N+1})$ at zero and is holomorphic for $\Re z < a + N + 1$; the subtracted terms are $(-1)^n e_r(n)/(a+n-z)$. Every residue in (4.153) is nonzero. Therefore the Taylor series $F_{a,r}(z) = \sum_{p \geq 1} T_{p,r}(a) z^{p-1}$ has exact radius $a + r$. This explains the first exponential scale in (4.150).

4.28 The joint depth–exponent transition

For $r \geq 1$, set

$$\mathcal{R}_{p,r}(a) = (-1)^r r!(r+a)^p T_{p,r}(a), \qquad L = p/r, \quad c = L - \log r, \quad \lambda = \tfrac{1}{2}e^{-c}. \tag{4.154}$$

At $a = 1/2$, this is $(-1)^r r!(2r+1)^p S_{p,r}$. By the strictly positive integral and $0 < h_r(t) < 1$ for $t > 0$, we have $0 < \mathcal{R}_{p,r}(a) < 1$.

Proposition 4.52 (Gamma expectation)**.** *If Y has gamma distribution with shape p and rate $r + a$, then*

$$\mathcal{R}_{p,r}(a) = \mathbb{E}h_r(e^{-Y}). \tag{4.155}$$

Proof. Set $t = e^{-y}$ in (4.131) and factor e^{-ry} out of $\log^r(1+e^{-y})$. Multiplying by the normalization in (4.154) leaves precisely the gamma density $(r+a)^p y^{p-1} e^{-(r+a)y}/\Gamma(p)$. □

Theorem 4.53 (Uniform transition and first correction)**.** *Let a range over a fixed compact subset of $(0,\infty)$ and let c range over a fixed compact subset of $\mathbb{R}$. Put $p = r(\log r + c)$ and $L = \log r + c$. As the integer $r \to \infty$,*

$$\mathcal{R}_{p,r}(a) = e^{-\lambda}\left[1 + \frac{1}{r}\left\{\left(\frac{\lambda^2}{2} - (a+\tfrac{1}{2})\lambda\right)L + \frac{5\lambda^2}{6} - 2\lambda\right\}\right] + O\left(\frac{L^2}{r^2}\right), \qquad \lambda = \tfrac{1}{2}e^{-c}. \tag{4.156}$$

The implied constant is uniform on those compact sets. In particular,

$$\mathcal{R}_{p,r}(a) \longrightarrow \exp(-e^{-c}/2). \tag{4.157}$$

For integer p, use $c = p/r - \log r$ in the expansion; rounding $r(\log r + c_0)$ to an integer preserves the limit with c_0.

We prove a uniform kernel lemma first. It is also the basis of the all-orders result.

Lemma 4.54 (Uniform kernel expansion)**.** *Write*

$$\log\left(\frac{\log(1+t)}{t}\right) = \sum_{m\geq 1} b_m t^m, \qquad b_1 = -\tfrac{1}{2}, \quad b_2 = \tfrac{5}{24}. \tag{4.158}$$

Define polynomials $\mathcal{U}_j(v)$ by

$$\sum_{j\geq 0} \mathcal{U}_j(v)\varepsilon^j = \exp\left(\sum_{k\geq 1} \varepsilon^k \left(2^{k+1} b_{k+1} v^{k+1} + \frac{(-1)^k 2^k v^k}{k}\right)\right). \tag{4.159}$$

For each fixed $J \geq 0$,

$$\sup_{0\leq t\leq 1} \left| h_r(t) - e^{-rt/2} \sum_{j=0}^{J} \frac{\mathcal{U}_j(rt/2)}{r^j} \right| \leq C_J r^{-J-1}. \tag{4.160}$$

In particular $\mathcal{U}_0 = 1$ and $\mathcal{U}_1(v) = 5v^2/6 - 2v$.

Proof. Put $v = rt/2$. Expanding $r\log(\log(1+t)/t) - \log(1+t)$ at zero gives $-v$ plus the exponent in (4.159) with $\varepsilon = 1/r$. Fix a sufficiently small $\delta > 0$. On $0 \leq t \leq \delta$, Taylor remainders are bounded by r^{-J-1} times a fixed polynomial in v, after expansion through the required order. The perturbation of $-v$ is bounded by a small fixed multiple of v (enlarge the threshold for r if needed). Taylor's remainder for the exponential is consequently dominated by $r^{-J-1}\Theta_J(v)e^{-\eta v}$ for some $\eta > 0$. Its supremum is finite.

On $\delta \leq t \leq 1$, the ratio $\log(1+t)/t$ is bounded above by a number strictly less than one. Hence $h_r(t)$ decays exponentially in r, while every term of the proposed approximation is a polynomial in r times $e^{-r\delta/2}$. Both are $O_J(r^{-J-1})$. Combining the two ranges proves the claim. All constants here depend only on J and the chosen fixed δ. □

Proof of Theorem 4.53. Let $\Delta = Y - L$. Its mean and centered gamma moments give, uniformly in the stated parameter sets,

$$\begin{aligned} \mathbb{E}\Delta &= -\frac{aL}{r} + O(L/r^2), & \mathbb{E}\Delta^2 &= \frac{L}{r} + O(L^2/r^2), \\ \mathbb{E}\Delta^3 &= O(L^2/r^2), & \mathbb{E}\Delta^4 &= O(L^2/r^2). \end{aligned} \tag{4.161}$$

For example $\mathbb{E}Y = rL/(r+a)$, $\operatorname{Var} Y = rL/(r+a)^2$, and the centered third and fourth moments are $2rL/(r+a)^3$ and $3(rL)^2/(r+a)^4 + 6rL/(r+a)^4$. These formulas imply (4.161) by translation.

Define $v = \lambda e^{-\Delta}$, $f(\Delta) = e^{-\lambda e^{-\Delta}}$, and $g(\Delta) = e^{-v}(5v^2/6 - 2v)$. All derivatives of f and g of fixed order are bounded on the real line: each is a polynomial in $v \geq 0$ times e^{-v}. Lemma 4.54 with $J = 1$ gives

$$\mathcal{R}_{p,r}(a) = \mathbb{E}f(\Delta) + r^{-1}\mathbb{E}g(\Delta) + O(r^{-2}).$$

Taylor-expand f through degree three, retaining the fourth-order remainder. Since $f'(0) = \lambda e^{-\lambda}$ and $f''(0) = (\lambda^2 - \lambda)e^{-\lambda}$, (4.161) yields

$$\mathbb{E}f(\Delta) = e^{-\lambda}\left[1 + \frac{L}{r}\left(\frac{\lambda^2}{2} - (a + \tfrac{1}{2})\lambda\right)\right] + O(L^2/r^2).$$

A second-order Taylor bound gives $\mathbb{E}g(\Delta) = e^{-\lambda}(5\lambda^2/6 - 2\lambda) + O(L/r)$. Substitute the two estimates. Since $L \asymp \log r$, this is exactly (4.156). □

4.28.1 A second proof of the limiting profile

Theorem 4.50 and (4.152) provide a useful independent explanation. For fixed j and $p = r(\log r + c)$,

$$r!e_r(r + j)\left(\frac{r + a}{r + a + j}\right)^p \longrightarrow \frac{\lambda^j}{j!}.$$

Every even and odd truncation bounds $\mathcal{R}_{p,r}(a)$ from opposite sides. Taking $r \to \infty$ for a fixed truncation and then increasing the truncation squeezes the limit between the alternating Taylor bounds for $e^{-\lambda}$. This argument does not interchange an uncontrolled infinite sum with a limit. It also gives compact-c uniform convergence by making the finite exponential remainder uniformly small.

4.28.2 The threshold outside the compact transition window

For fixed $a > 0$ (also uniformly on compact a-sets), if $p/r - \log r \to +\infty$, then (4.149) implies $\mathcal{R}_{p,r}(a) \to 1$. One may split into $p/r \le 2\log r$ and $p/r > 2\log r$ to verify the exponential estimate without assuming p/r^2 is small.

If $p/r - \log r \to -\infty$, with $p > 0$, then eventually $L \le \log r$, and $Y - L \to 0$ in probability because its variance and mean shift tend to zero. Thus $re^{-Y} \to \infty$ in probability. The elementary inequality $\log(1 + t)/t \le 1 - t/6$ on $0 \le t \le 1$ gives $h_r(t) \le e^{-rt/6}$. Bounded convergence in probability in (4.155) now gives $\mathcal{R}_{p,r}(a) \to 0$. Therefore $p \sim r\log r$ is the actual threshold for first-pole dominance.

4.28.3 All-orders transition polynomials

Define the operator on polynomials in v

$$\mathcal{D}P(v) = vP(v) - vP'(v), \tag{4.162}$$

and the polynomials

$$d_j(t; a) = \frac{(-1)^j}{j + 1}\left(a^{j+1} - (a - t)^{j+1}\right) \quad (j \ge 1). \tag{4.163}$$

The operator convention follows from $\partial_\Delta(e^{-v}P(v)) = e^{-v}\mathcal{D}P(v)$ for $v = \lambda e^{-\Delta}$.

Theorem 4.55 (All-orders compiler)**.** *For every fixed $J \ge 0$, in the parameter range of Theorem 4.53,*

$$\mathcal{R}_{p,r}(a) = e^{-\lambda}\sum_{j=0}^{J}\frac{\Theta_j(L, \lambda; a)}{r^j} + O_J\left(\frac{L^{J+1}}{r^{J+1}}\right), \tag{4.164}$$

uniformly on compact a, c-sets. The Θ_j have rational coefficients and degree at most j in L. They are determined by the finite formal prescription

$$\sum_{j\ge 0}\Theta_j(L,v;a)\varepsilon^j=\exp\left(L\sum_{k\ge 1}\varepsilon^k d_k(\mathcal{D};a)\right)\left(\sum_{k\ge 0}\varepsilon^k\mathcal{U}_k(v)\right). \tag{4.165}$$

In particular

$$\Theta_1=\left(\frac{v^2}{2}-(a+\tfrac{1}{2})v\right)L+\frac{5}{6}v^2-2v. \tag{4.166}$$

Proof. The exact moment-generating function of Δ is

$$\log\mathbb{E}e^{t\Delta}=-Lt+rL\{\log(1+a/r)-\log(1+(a-t)/r)\}=L\sum_{j\ge 1}r^{-j}d_j(t;a).$$

Consequently expectation, formally through any fixed order in r^{-1}, is the differential operator $\exp(L\sum r^{-j}d_j(\partial_\Delta;a))$ evaluated at $\Delta=0$. Conjugation by e^{-v} turns ∂_Δ into (4.162), proving the coefficient rule.

For the error justification, apply Lemma 4.54 through order J. For the term containing $r^{-k}\mathcal{U}_k$, Taylor-expand $e^{-v}\mathcal{U}_k(v)$ as a function of Δ through degree $2(J-k)+1$. Its fixed-order derivatives are bounded, and the gamma moments imply $\mathbb{E}|\Delta|^{2s}=O_s((L/r)^s)$ for each fixed s, since $L\asymp\log r$. The Taylor remainder therefore contributes $O(r^{-J-1}L^{J-k+1})$. The finitely many retained moments are explicit rational functions of $r+a$ with polynomial dependence on rL; expanding them through order J yields the stated formal operator and an error $O_J(L^{J+1}/r^{J+1})$. This also controls the uniform kernel error. Finally, each factor in the operator exponent carries one L and at least one power of ε, so the coefficient of ε^j has degree at most j in L. □

For reproducibility, write $\Theta_2=A_2L^2+B_2L+C_2$, where

$$\begin{aligned}
A_2&=\frac{v^4}{8}-\left(\frac{a}{2}+\frac{3}{4}\right)v^3+\left(\frac{a^2}{2}+\frac{3a}{2}+\frac{7}{8}\right)v^2-\left(\frac{a^2}{2}+\frac{a}{2}+\frac{1}{8}\right)v,\\
B_2&=\frac{5v^4}{12}-\left(\frac{5a}{6}+\frac{11}{4}\right)v^3+\left(\frac{8a}{3}+\frac{11}{3}\right)v^2+\left(a^2-a-\frac{2}{3}\right)v,\\
C_2&=\frac{25v^4}{72}-\frac{8}{3}v^3+4v^2.
\end{aligned}$$

The supplementary symbolic script produces $\Theta_0,\dots,\Theta_4$ from (4.165); it does not fit them to numerical data.

4.28.4 The complementary fixed-exponent regime

The transition theorem is not uniform down to fixed p. A separate endpoint expansion describes that regime.

Theorem 4.56 (Fixed p, growing harmonic index)**.** *Let $b=\log 2$. For p,a in compact subsets of $(0,\infty)$ and $r\to\infty$,*

$$T_{p,r}(a)=\frac{(-1)^r2^{p-1}b^{r+p}}{r!r^p}\left[1+\frac{(1-2a)bp+(b-1)p(p+1)/2}{r}+O(r^{-2})\right]. \tag{4.167}$$

Thus

$$S_{p,r}=\frac{(-1)^rb^{r+p}}{2r!r^p}\left[1+\frac{(b-1)p(p+1)}{2r}+O_p(r^{-2})\right]. \tag{4.168}$$

Proof. In the $y = -\log t$ form of (4.131), the factor $\log^r(1+e^{-y})$ localizes the integral near $y = 0$. Its logarithm expands as

$$\log\left(\frac{\log(1+e^{-y})}{b}\right) = -\frac{y}{2b} + \frac{b-1}{8b^2}y^2 + O(y^3),$$

while $e^{-ay}/(1+e^{-y}) = \frac{1}{2}(1+(1/2-a)y+O(y^2))$. Set $y = 2bv/r$. The integrand becomes $v^{p-1}e^{-v}$ times

$$1 + \frac{(1-2a)bv + (b-1)v^2/2}{r} + O\big(r^{-2}P(v)\big)$$

in a fixed small-y interval, with an integrable exponential majorant after the usual smaller-interval split. The complementary y-interval is exponentially small relative to $b^r r^{-p}$: the logarithmic ratio is uniformly below one there, and its large-y decay supplies integrability. Integrating the displayed terms uses $\Gamma(p+1)/\Gamma(p) = p$ and $\Gamma(p+2)/\Gamma(p) = p(p+1)$. Uniform Taylor bounds on compact a, p-sets give the stated remainder. Finally divide by 2^p at $a = 1/2$. □

4.29 Geometric evaluation with explicit error bounds

The signed partial-sum bounds are exact but may converge slowly at small p. An incomplete-beta expansion supplies geometric convergence. In (4.135), put $y = t/(1+t)$; then

$$\Phi_a(z,u) = \int_0^{1/2} y^{a-z-1}(1-y)^{u-a+z}\,\mathrm{d}y = 2^{-a+z}\sum_{n\ge 0}\frac{(a-z-u)_n}{n!\,2^n(n+a-z)}. \tag{4.169}$$

This is the usual binomial expansion of the incomplete-beta integral, with all variables and normalizations displayed.

Theorem 4.57 (Explicit Cauchy tail bound)**.** *Let $p \ge 1$ and $r \ge 0$ be integers, and let $a > 0$. Choose $0 < \rho < a$, $\sigma > 0$, and let $A = a + \rho + \sigma$. For an integer $N \ge 1$, put*

$$q_N = \frac{1}{2}\max\left(1, \frac{N+A}{N+1}\right), \qquad B_N = 2^{-a+\rho}\frac{(A)_N}{N!2^N(N+a-\rho)}.$$

If $q_N < 1$, the error after retaining $0 \le n < N$ in (4.169) has modulus at most $B_N/(1-q_N)$ on $|z| \le \rho, |u| \le \sigma$. Consequently the error in $T_{p,r}(a)$ is at most

$$\rho^{-(p-1)}\sigma^{-r}\frac{B_N}{1-q_N}. \tag{4.170}$$

Proof. For every factor of the rising factorial, $|a-z-u+j| \le A+j$. Also $|n+a-z| \ge n+a-\rho$ and $|2^{-a+z}| \le 2^{-a+\rho}$. The resulting positive majorant term has ratio

$$\frac{n+A}{2(n+1)}\frac{n+a-\rho}{n+1+a-\rho} \le q_N \qquad (n \ge N).$$

Sum the geometric majorant. The bivariate Cauchy coefficient estimate gives (4.170). □

Corollary 4.58 (A rational Gaussian error bound)**.** *For $a = 1/2$ take $\rho = \sigma = 1/4$. Let $S_{p,r}^{[N]}$ be the coefficient approximation from (4.169), divided by 2^p. Then*

$$\boxed{|S_{p,r} - S_{p,r}^{[N]}| \le \frac{2^{p+2r-1-N}}{N+1/4}.} \tag{4.171}$$

Proof. Here $A = 1$, $(A)_N/N! = 1$, $q_N = 1/2$, and $2^{-1/4} < 1$. Insert these values into (4.170) and multiply by 2^{-p}. □

4.29.1 How the supplied exact certificates are constructed

For rational $a = 1/2$, the finite approximation has the form

$$S_{p,r}^{[N]} = 2^{-p-1/2} \sum_{\ell=0}^{p-1} c_\ell (\log 2)^\ell, \qquad c_\ell \in \mathbb{Q}. \tag{4.172}$$

The coefficients are computed exactly. Maintain the truncated bivariate polynomial $\Theta_n(z,u) = (1/2 - z - u)_n/n!$ using

$$\Theta_{n+1} = \frac{n + 1/2 - z - u}{n+1} \Theta_n.$$

Multiply by the truncated geometric expansion of $(n + 1/2 - z)^{-1}$, accumulate the 2^{-n} terms, and finally multiply by the Taylor polynomial of $e^{z \log 2}$. All operations up to (4.172) are rational.

The code then encloses $\log 2$ using

$$\log 2 = 2 \sum_{k=0}^{K-1} \frac{1}{(2k+1)3^{2k+1}} + \eta_K, \qquad 0 < \eta_K \le \frac{9}{4(2K+1)3^{2K+1}},$$

and encloses $2^{-1/2}$ by an integer square-root computation on a dyadic scale. Rational interval Horner evaluation controls the error in (4.172), and the rational bound (4.171) controls the infinite tail. Thus the resulting interval endpoints are exact fractions. They do not depend on trusting a floating-point quadrature result or a special-function library.

These certificates establish enclosures for individual constants. Overlap of two intervals would not, by itself, prove an identity between the constants; the analytic proofs in earlier sections perform that task. Conversely, disjoint validated intervals can rigorously disprove a proposed equality. A finite-height exclusion program would require a further certified enumeration or lattice argument, which is not included here.

4.30 Exponential cancellation at proportional harmonic depth

The positive integral representation now leads to an analytic question left open by the earlier alternating-harmonic report Chapter 4: how rapidly does its normalized sum tend to zero below the first-term transition? The case of proportional outer exponent and harmonic depth has an explicit rate function. This is a continuation of that report's results, not another proof of its already established reflection identity.

For integers $h, n \ge 0$, define

$$e_h(n) = [u^h] \prod_{j=1}^{n} (1 + u/j), \qquad T_{p,h}(a) = \sum_{n \ge 0} \frac{(-1)^n e_h(n)}{(n+a)^p}, \quad a, p > 0.$$

Thus $e_0(n) = 1$, $e_1(n) = H_n$, and $e_2(n) = (H_n^2 - H_n^{(2)})/2$. Ordinary convergence holds for $p > 0$; absolute convergence holds for $p > 1$. The Gaussian normalization is $S_{p,h} = 2^{-p} T_{p,h}(1/2)$. The positive Mellin integral

$$T_{p,h}(a) = \frac{(-1)^h}{h!\,\Gamma(p)} \int_0^\infty \frac{x^{p-1} e^{-ax} \log^h(1 + e^{-x})}{1 + e^{-x}}\, dx \tag{4.173}$$

follows from

$$\sum_{n\ge 0} e_h(n) z^n = \frac{(-\log(1-z))^h}{h!(1-z)}$$

by inserting an Abel factor, using the gamma integral, and taking its limit. The exponential decay at infinity and the bound $\log(1+e^{-x}) \le \log 2$ near zero justify the integral limit. The convergence and Abel-limit argument are also proved in the integral representation above. In particular, $(-1)^h T_{p,h}(a) > 0$.

Put

$$R_{p,h}(a) = (-1)^h h!(h+a)^p T_{p,h}(a). \tag{4.174}$$

The earlier report proves $0 < R_{p,h}(a) < 1$ and identifies the threshold $p \sim h\log h$. When $p/h - \log h \to -\infty$, it proves that $R_{p,h}(a) \to 0$. The following theorem quantifies this statement when p/h remains in a fixed positive compact interval.

4.30.1 The unique saddle and the rate function

Write

$$L(x) = \log(1+e^{-x}), \qquad w(x) = \frac{1}{(1+e^x)L(x)} = -\frac{d}{dx}\log L(x), \quad x > 0. \tag{4.175}$$

Lemma 4.59. *The function w is strictly increasing, with*

$$\frac{1}{2\log 2} < w(x) < 1.$$

For every $\alpha > 0$ there is exactly one $x_\alpha > 0$ satisfying

$$x_\alpha w(x_\alpha) = \alpha. \tag{4.176}$$

Moreover $\alpha < x_\alpha < 2\alpha\log 2$.

Proof. Set $t = e^{-x} \in (0,1)$. Then $w = t/((1+t)\log(1+t))$ and

$$\frac{d}{dt}\log w = \frac{\log(1+t) - t}{t(1+t)\log(1+t)} < 0.$$

Consequently w increases with x. Its endpoint limits give the bounds. The function $xw(x)$ has derivative $w + xw' > 0$, tends to zero at zero, and tends to infinity at infinity. This proves existence, uniqueness, and the stated enclosure. □

For $x = x_\alpha$, set

$$\Phi_\alpha(y) = \alpha\log y + \log L(y), \tag{4.177}$$

$$A_\alpha = \frac{\alpha}{x^2} + w'(x) > 0, \tag{4.178}$$

$$I(\alpha) = -\alpha\log(x/\alpha) - \alpha - \log L(x), \tag{4.179}$$

$$C(\alpha, a) = \frac{\sqrt{\alpha/A_\alpha}}{x(1+e^{-x})}\exp\big(a(\alpha - x)\big). \tag{4.180}$$

All of these functions are real analytic for $\alpha > 0$.

Theorem 4.60 (Proportional-depth asymptotics)**.** *Let a and α range over fixed compact subsets of $(0,\infty)$, and put $p=\alpha h$. As the integer $h\to\infty$,*

$$R_{\alpha h,h}(a)=C(\alpha,a)e^{-hI(\alpha)}\big(1+O(h^{-1})\big), \tag{4.181}$$

uniformly on those compact sets. In fact

$$R_{\alpha h,h}(a)=C(\alpha,a)e^{-hI(\alpha)}\left(1+\frac{D(\alpha,a)}{h}+O(h^{-2})\right), \tag{4.182}$$

where, with $x=x_\alpha$, $A=A_\alpha$, and $g(y)=e^{-ay}/(y(1+e^{-y}))$,

$$D(\alpha,a)=\frac{g''(x)}{2Ag(x)}+\frac{\Phi_\alpha'''(x)g'(x)}{2A^2g(x)}+\frac{\Phi_\alpha''''(x)}{8A^2}+\frac{5\Phi_\alpha'''(x)^2}{24A^3}-\frac{\alpha a^2}{2}-\frac{1}{12\alpha}. \tag{4.183}$$

Every fixed further order in h^{-1} exists and is effectively obtained from derivatives of L and g at x_α and the Stirling series. The remainder at every fixed order is uniform on the same compact sets.

Proof. Equation (4.173) gives

$$R_{\alpha h,h}(a)=\frac{(h+a)^{\alpha h}}{\Gamma(\alpha h)}\int_0^\infty e^{h\Phi_\alpha(y)}g(y)\,dy.$$

We have $\Phi_\alpha'(y)=\alpha/y-w(y)$ and $\Phi_\alpha''(y)=-\alpha/y^2-w'(y)<0$. The sole critical point is therefore a strict global maximum. Its location stays in a fixed compact subinterval of $(0,\infty)$ as α varies in the prescribed compact set, and its curvature is bounded away from zero.

Here are explicit tail controls for the application of Laplace's method [78, §2.3(iii)] [78, Section 2.3(iii)]. Near zero, $\exp(h\Phi_\alpha(y))\le y^{h\alpha}(\log 2)^h$, while $g(y)=O(y^{-1})$ uniformly in a. Choosing the endpoint interval small enough makes its integral exponentially smaller than the maximum, uniformly in α. At infinity, $L(y)<e^{-y}$ implies $\Phi_\alpha(y)<\alpha\log y-y$; this supplies an integrable exponential bound uniform in the same compact parameters. On the remaining compact set outside a fixed neighborhood of x_α, strict concavity and compactness give a uniform negative gap from the maximum. These observations justify discarding both tails with an exponentially small relative error.

In the neighborhood, substitute $y=x+v/\sqrt{h}$. Taylor expansion, followed by Gaussian integration, gives

$$\int_0^\infty e^{h\Phi_\alpha(y)}g(y)\,dy=e^{h\Phi_\alpha(x)}g(x)\sqrt{\frac{2\pi}{hA}}\left(1+\frac{E}{h}+O(h^{-2})\right),$$
$$E=\frac{g''}{2Ag}+\frac{\Phi'''g'}{2A^2g}+\frac{\Phi''''}{8A^2}+\frac{5(\Phi''')^2}{24A^3},$$

with all derivatives evaluated at x. Odd Gaussian moments vanish; the even moments $1/A$, $3/A^2$, and $15/A^3$ produce precisely these four terms. The analytic functions have uniformly bounded derivatives on a common neighborhood of the saddle. Taylor remainders there, together with the tail bounds just given, prove the claimed uniform error and also permit expansion to any prescribed order.

Finally Stirling's formula, uniformly for α in its compact set, and $\alpha h\log(1+a/h)=\alpha a-\alpha a^2/(2h)+O(h^{-2})$ give

$$\frac{(h+a)^{\alpha h}}{\Gamma(\alpha h)}=\sqrt{\frac{\alpha h}{2\pi}}e^{h\alpha(1-\log\alpha)+a\alpha}\left(1-\frac{\alpha a^2/2+1/(12\alpha)}{h}+O(h^{-2})\right).$$

Multiplying the two expansions proves (4.181)–(4.183). □

Remark 4.61 (Integer exponents and rounding). For integer p, set $\alpha = p/h$ in the theorem. If instead one fixes α_0 and rounds $\alpha_0 h$ to p, the $O(1/h)$ change of α must remain inside $e^{-hI(\alpha)}$: replacing it by $e^{-hI(\alpha_0)}$ changes the leading constant by a generally nontrivial bounded factor. This is a practical distinction in numerical comparisons.

4.30.2 Monotonicity, convexity, and matching limits

Proposition 4.62. *The rate I is positive, strictly decreasing, and strictly convex on $(0,\infty)$. More explicitly,*

$$I'(\alpha) = \log\frac{\alpha}{x_\alpha} = \log w(x_\alpha) < 0, \qquad I''(\alpha) = \frac{w'(x_\alpha)}{w(x_\alpha)(w(x_\alpha) + x_\alpha w'(x_\alpha))} > 0. \tag{4.184}$$

Its endpoint limits are

$$I(0+) = -\log\log 2, \qquad I(+\infty) = 0.$$

As $\alpha \to \infty$,

$$x_\alpha = \alpha + \frac{\alpha}{2}e^{-\alpha} + O(\alpha^2 e^{-2\alpha}), \tag{4.185}$$

$$I(\alpha) = \frac{1}{2}e^{-\alpha} - \frac{3\alpha+5}{24}e^{-2\alpha} + O(\alpha^2 e^{-3\alpha}), \tag{4.186}$$

$$C(\alpha, a) = 1 - \left(\frac{(a+1/2)\alpha}{2} + 1\right)e^{-\alpha} + O_a(\alpha^2 e^{-2\alpha}). \tag{4.187}$$

Also $C(\alpha, a) \to 1/2$ as $\alpha \downarrow 0$.

Proof. Differentiating the optimized value in (4.179) cancels its x'_α terms by (4.176). This gives the first identity in (4.184). Differentiating $\alpha = xw(x)$ gives the second. Positivity also follows directly: $\log L(y) < -y$, so

$$\max_{y>0}\{\alpha\log(y/\alpha) + \alpha + \log L(y)\} < \max_{y>0}\{\alpha\log(y/\alpha) + \alpha - y\} = 0.$$

The strict inequality is valid because the first maximum is attained; at its maximizing point the second expression is at most zero.

The small-α limits follow from $x_\alpha/\alpha \to 2\log 2$ and $A_\alpha \sim \alpha/x_\alpha^2$. At infinity, put $t = e^{-y}$ and use

$$\log L(y) = -y - \frac{t}{2} + \frac{5t^2}{24} + O(t^3), \qquad w(y) = 1 - \frac{t}{2} + \frac{5t^2}{12} + O(t^3).$$

Substitution in (4.176) first gives (4.185). Taylor expansion of (4.179) around $y = \alpha$, retaining the quadratic displacement, gives (4.186). Expansion of (4.180) and $A_\alpha = \alpha/x_\alpha^2 + w'(x_\alpha)$ gives (4.187). The remainders stated above follow from analyticity of $\log(\log(1+t)/t)$ for t near zero, with $x_\alpha - \alpha = O(\alpha e^{-\alpha})$. □

The rate formula explains the scale of cancellation even when the terms of the original alternating sum are much larger than its value. For example, $I(1) = 0.15091647898554\ldots$ and $C(1, 1/2) = 0.66325765726435\ldots$. Thus

$$R_{h,h}(1/2) \sim 0.66325765726435\ldots\, e^{-0.15091647898554\ldots h}.$$

This is a proved exponential rate, rather than just a limit of zero.

The large-α expansions also explain why the transition parameter in the previous report was $\lambda = he^{-\alpha}/2$. Formally putting $\alpha = \log h + c$ in $hI(\alpha)$ gives the leading value $e^{-c}/2$. This agrees with the independently proved transition $R \to \exp(-e^{-c}/2)$ in Chapter 4. Such matching is a consistency calculation: the compact-α theorem proved here does *not* by itself establish a uniform error for growing α. Obtaining a single effective formula across that expanding range remains a precise further problem.

4.30.3 A probability law at the saddle

There is a useful formulation independent of the alternating series. Normalize the positive integrand in (4.173), with $p = \alpha h$, to a probability density and call the resulting random variable X_h. Then

$$\sqrt{hA_\alpha}(X_h - x_\alpha) \implies N(0,1). \tag{4.188}$$

Indeed, the same change of variables used in the proof of Theorem 4.60, its uniform Gaussian domination on a shrinking neighborhood, and its exponentially negligible tails prove convergence against every bounded continuous test function. The associated rate for X_h is $\Phi_\alpha(x_\alpha) - \Phi_\alpha(y)$. This identifies which part of the positive integral carries the very small alternating sum and gives a stable center for numerical quadrature.

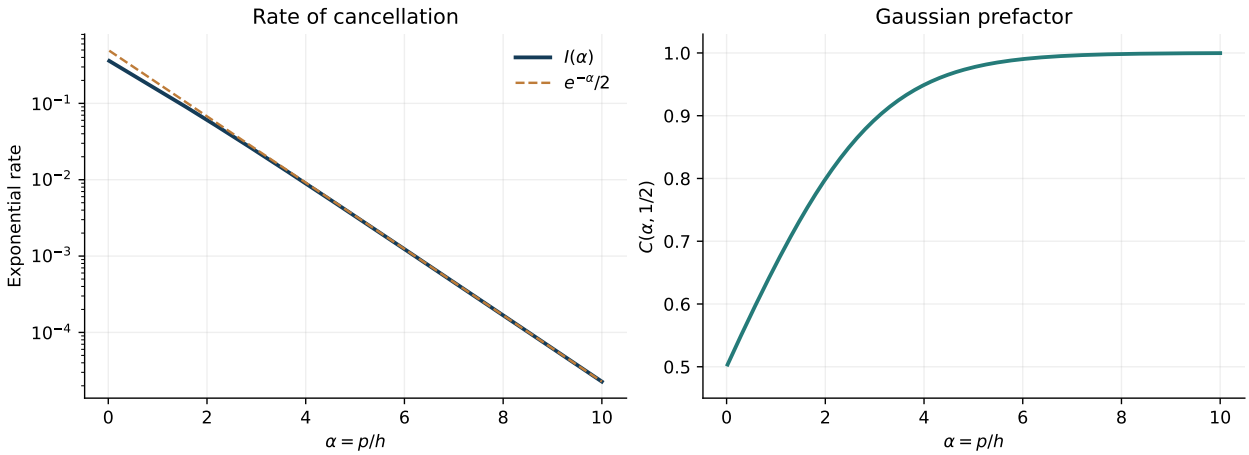


Figure 4.1: The proportional-depth rate and normalized asymptotic. The positive, decreasing and convex rate is proved in the text; sampled values illustrate it. The compact-parameter theorem does not by itself cover a growing $\alpha = \log h + c$ transition.

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[129] ProveIt research continuation, *Exact Resonant Identities Beyond the Diagonal*, 11 October 2026. Original package preserved under `reports/stieltjes-harmonic-resonance/continuations/exact-resonant-identities/`. The complete-Cayley formal obstruction is audited in Chapter 12; the numerical S_6 conjecture remains open.

[130] ProveIt research continuation, *Harmonic Parity and Resolvent Identities*, 11 October 2026. Original package preserved under `reports/stieltjes-harmonic-resonance/continuations/harmonic-parity-resolvents/`. The centered trigamma-square moments and their polylogarithmic generator are audited in Chapter 8; the broader classification and resolvent claims retain separate scopes.

[131] ProveIt research continuations, *Higher Digamma Powers and Ordered Gamma Sums*, *Canonical Digamma Polynomials and the Fifth Harmonic Bridge*, and *Mixed Resolvent Products, Higher Digamma Powers, and Stieltjes Collisions*, 11 October 2026. Original packages are preserved as `reports/stieltjes-harmonic-resonance/continuations/higher-digamma-powers/`, `reports/stieltjes-harmonic-resonance/continuations/canonical-digamma-fifth/`, and `reports/stieltjes-correlation-calculus/continuations/mixed-resolvents-harmonic-powers/`. Their all-power and fifth-moment cores are reconciled in Chapter 8; other Gamma, resolvent, collision and ordered branches retain separate audit scopes.

[132] NIST Digital Library of Mathematical Functions, Sections 5.5 (digamma recurrence and reflection), 5.7 (Taylor and reciprocal-Gamma series), and 5.11 (sectorial asymptotics). In particular formulas 5.5.2, 5.5.4, 5.7.4 and 5.11.2. https://dlmf.nist.gov/5. Inspected 11 October 2026 UTC.

[133] K. Ihara, M. Kaneko and D. Zagier, Derivation and double shuffle relations for multiple zeta values, *Compositio Mathematica* **142** (2006), 307–338. https://doi.org/10.1112/S0010437X0500182X.

[134] P. T. Young, Series of height one multiple zeta functions, *Integers* **24** (2024), A43, published 20 May 2024. https://math.colgate.edu/~integers/y43/y43.pdf. https://doi.org/10.5281/zenodo.11221606. The coefficient-description check in Appendix A concerns the inspected publisher PDF, printed page 12; its ordinary Stieltjes formula is retained.

[135] M. Can, L. Kargın, M. Cenkci and A. Dil, *On certain harmonic zeta functions*, arXiv:2403.07123v1, 11 March 2024, printed page 1. https://arxiv.org/pdf/2403.07123v1. The initial-index correction in Appendix A is scoped to that version's introductory Stieltjes-limit display. Inspected 11 October 2026 UTC.

[136] ProveIt research continuations, *Bernoulli Determinants, Cubic Tornheim Jets, and Canonical Digamma Integrals*, *Differential Polygamma and Lerch Identities*, and *Exact Polylogarithm Parity, Harmonic-Tail Reconstruction, and Higher Polygamma Moments*, 11 October 2026. Original packages preserved under `reports/stieltjes-harmonic-resonance/continuations/bernoulli-tornheim-digamma/`, `reports/stieltjes-harmonic-resonance/continuations/differential-polygamma-lerch-jets/`, and `reports/stieltjes-harmonic-resonance/continuations/polygamma-tail-identities/`. Their common differential-polynomial analytic core is reconciled in Chapter 8; their remaining coordinate, rank and Lerch claims retain separate audit scopes.

[137] ProveIt research continuation, *Beta Mixtures Across Integer Twists: Finite spectral corrections, harmonic logarithmic moments, and Stieltjes–polylogarithm primitives*, 10 October 2026. Original article, PDF and verification evidence preserved under `reports/stieltjes-correlation-calculus/continuations/beta-crossing/`. The affine crossing, integer-total mixture and balanced-jet cores are reconciled in Chapter 8; the later logarithmic Mellin and primitive families retain their separate integration scope.

[138] ProveIt research continuations, *Alternating Tails and Spectral Defects*, *Quartic Spectral Reconstruction*, *Bernoulli Tails and Spectral Faces*, *Bilinear Tails and Harmonic Determinants*, and *Tail Kernels and Boundary Products*, 11 October 2026. Original sources and exact evidence preserved on the harmonic-resonance spine and under `reports/tail-boundaries/`. The common centered coefficient model, retained determinant and effective eventual proof are reconciled in Chapter 8. Complete orders ten and twelve retain report attribution; the collective order-fourteen extension has its full exact certificate under `verification/prefix-order-fourteen/`.

[139] ProveIt research continuations, *Universal Gamma Normal Forms: Rational Kernels, Harmonic Sums, and Stieltjes Primitives*, and *A Universal Beta Formula for Gamma Normal Forms*, 11 October 2026. Original articles, PDFs and exact evidence preserved under `reports/stieltjes-harmonic-resonance/continuations/universal-gamma-coefficients/` and `reports/stieltjes-harmonic-resonance/continuations/universal-beta-normal-forms/`. Their common all-index coefficient formula and ordinary finite projections are reconciled in Chapter 8. The collective analytic-jet extension gives an actual locally convergent infinite normal generator.

[140] ProveIt research continuations, *Higher Identities*, *Cubic Gamma Normal Form*, and *Universal Gamma Coefficients*, 11–12 October 2026. Delivered articles and exact evidence preserved on the harmonic-resonance continuation spine. The compensator construction uses the classical Euler homotopy operator: D. Poole and W. Hereman, *The homotopy operator method for symbolic integration by parts and inversion of divergences with applications*, https://arxiv.org/abs/0908.0399; W. Hereman, B. Deconinck and L. D. Poole, *Continuous and Discrete Homotopy Operators: A Theoretical Approach made Concrete*, https://arxiv.org/abs/nlin/0510064. The hierarchy specialization, analytic primitive domain and independent normal-generator proof are reconciled in Chapter 8. The collective continuation proves every primitive coefficient and full support/sign, with convergence on the full input disk; fixed-gap analytic completion remains a separate pending proof scope.

[141] ProveIt research continuation, *A Finite Polylogarithm Alphabet for Rational Hankel Sums*, 11 October 2026. Original article, compiler and recorded evidence preserved under `reports/stieltjes-harmonic-resonance/continuations/rational-hankel-polylogarithms/`. The confluent weight budget, anchored two-letter compiler and quadratic denominator evaluation are reconciled in Chapter 8. The classical tools retain their attribution: M. E. Hoffman, *Quasi-shuffle products*, J. Algebraic Combin. **11** (2000), 49–68, https://arxiv.org/abs/math/9907173; E. Panzer, *Algorithms for the symbolic integration of hyperlogarithms with applications to Feynman integrals*, Comput. Phys. Commun. **188** (2015), 148–166, https://arxiv.org/abs/1403.3385.

[142] ProveIt research continuations, *Resonant Polylogarithmic Jets and Convergent Inverse Transseries* and *Polylogarithm Order Jets and Resonant Inversion*, 11–12 October 2026. Original sources and evidence are preserved under `reports/uniform-transition-continuation/continuations/`, in `normalized-resonant-jets/` and `order-jets-resonant-inversion/`. The

fixed-sector theorem extends R1 of the preserved `resonance-correlations-nonseparable-jets/` report. Reciprocal-Gamma normalization is credited to both overlapping continuations. Section 1.6 develops the exact order-lowering and arbitrary repeated-primitive identities from that normalization. Inverse geometry and Borel reconstruction retain separate review scopes.

[143] ProveIt research continuations, *Resonant Polylogarithm Transseries* and *Boundary Completion for Quadratic Polylogarithm Feedback*, 11–12 October 2026. Original articles, PDFs and evidence are preserved under `Analysis/Transseries/docs/series-and-transseries/`, in `Resonant_Polylogarithm_Transseries/` and `Quadratic_Polylogarithm_Feedback_Boundary_Completion/`. The first report establishes the interior fold and poses the boundary question; the second proves its completion. Section 1.7 reconciles that classification and develops finite Bessel-polynomial coefficient and order-spectrum identities. Nonmarginal critical charts and crossover refinements retain separate integration scopes.

[144] NIST Digital Library of Mathematical Functions, §18.34, *Bessel Polynomials*, especially (18.34.1)–(18.34.5), https://dlmf.nist.gov/18.34. The convention here is $y_n(x) = y_n(x; 2)$.

[145] P. Flajolet and A. Odlyzko, *Singularity analysis of generating functions*, SIAM J. Discrete Math. **3** (1990), 216–240.

[146] NIST Digital Library of Mathematical Functions, §4.13, *Lambert W-function*, https://dlmf.nist.gov/4.13.

[147] A. G. Pakes, *Affine relation between an infinitely divisible distribution function and its Lévy measure*, J. Math. Anal. Appl. **499** (2021), article 125008, https://doi.org/10.1016/j.jmaa.2021.125008. Related lower-branch Laplace-transform literature; the normalization and identities used in Section 1.8 are proved there.

[148] ProveIt research continuation, *Mixed Hurwitz-Tail Moments, Zeta-Anchor Reductions, and Stieltjes Transport*, 11 October 2026, preserved article and verification evidence in `reports/stieltjes-harmonic-resonance/continuations/mixed-tail-anchor-reductions/`. Its third research question asks for rational polynomial combinations for $(2, 2, 3)$ and $(2, 3, 4)$. Section 8.48 develops the first case within a precisely specified polynomial certificate class.

[149] ProveIt research continuation, *Double Resonance and Nested Stokes Geometry in Polylogarithmic Lambert Inversion*, 12 October 2026. Preserved source and evidence in `Analysis/Transseries/docs/series-and-transseries/Resonant_Lambert_Inverse_Transseries/`. The finite inverse formula and the conjectured hyperbolic amplitudes motivate Section 1.9.

[150] ProveIt research continuation, *Summation and Singularities under Polylogarithmic Reversion*, 12 October 2026, in the preceding report's `continuations/summation-singularities/` directory. Its Lambert section proves the leading amplitudes and first correction; its all-order question is resolved by Theorem 1.26. Original source evidence and fresh isolated verification are preserved.

[151] ProveIt research continuation, *Sharp Late Coefficients and Borel Continuation of Resonant Lambert Inverses*, 12 October 2026, in the resonant Lambert inverse report's `continuations/late-coefficients/` directory. Original source evidence and a complete isolated replay are preserved. Its inverse-index all-order question is resolved by Theorem 1.26; its nonlinear continuation argument is developed in Theorem 1.28.

[152] D. Sauzin, *Nonlinear analysis with resurgent functions*, Annales scientifiques de l'École Normale Supérieure (4) **48** (2015), 667–702. arXiv:1212.4477. Theorems 3 and 5 establish analytic substitution and compositional inversion for an additive closed discrete singular set; Theorems 3′ and 5′ give the corresponding summability statements.

[153] ProveIt research continuation, *Order-Jet Filters for Polylogarithmic Transseries: Sharp cancellation costs, Stokes-neutral curves, and a Stieltjes–digamma inverse*, 12 October 2026, in the resonant polylogarithm report's `continuations/order-jet-filters/` directory. Original source evidence and two isolated replays are preserved. Theorem 1.41 answers its fixed-filter neutral inverse-transport question.

[154] NIST Digital Library of Mathematical Functions, Gamma functional relations and Gamma series expansions.

[155] NIST Digital Library of Mathematical Functions, Gamma and digamma integral representations and sectorial asymptotic expansions.

[156] M. Karatarakis and F. Wiedijk, *A formalization of the Gelfond–Schneider theorem*, arXiv:2603.24823v1 (2026), Theorem 3.1. Used for the classical transcendence theorem with a chosen complex logarithm; this citation does not formalize the present filters.

[157] ProveIt research report, *Nonlinear Stokes Transport and Logarithmic Inversion*, in the Transseries collection at `Analysis/Transseries/docs/series-and-transseries/Nonlinear_Stokes_Transport_Logarithmic_Inversion/`. The order-zero averaging defect specializes its nonlinear inverse mechanism.

[158] ProveIt research continuation, *Uniform Coefficient Transseries across the Polylogarithmic Order-Three Resonance*, 12 October 2026, in the quadratic feedback boundary report's `continuations/joint-coefficient-transseries/` directory. The original article, PDF and evidence, together with an isolated replay, are preserved. The matched-derivative comparison in Theorem 1.63 develops its third research question.

[159] ProveIt research collection. *Resonant Order Jets and Convergent Inverse Transseries: for Polylogarithms and Logarithmically Weighted Tails*. 12 October 2026. Pinned article source. In particular, the noninteger inverse charts and research question 9, "Global geometry beyond integer orders."

[160] NIST Digital Library of Mathematical Functions. *Chapter 12, Parabolic Cylinder Functions*: §12.2, definitions and initial values; §12.5, integral representations, especially formula 12.5.4; §12.9, large-variable asymptotic expansions. Accessed 12 October 2026.

[161] H. Lam, J. Blanchet, D. Burch, and M. Z. Bazant. Corrections to the Central Limit Theorem for Heavy-tailed Probability Densities. *Journal of Theoretical Probability* **24**(4) (2011), 895–927. doi:10.1007/s10959-011-0379-y. Author-hosted final text.

[162] Q. Berger, M. Birkner, and L. Yuan. Collective vs. individual behaviour for sums of i.i.d. random variables: appearance of the one-big-jump phenomenon. *Annales de la Faculté des Sciences de Toulouse*, Series 6, **33**(5) (2024), 1413–1485; published online 24 April 2025. doi:10.5802/afst.1802.

[163] ProveIt research continuation. *Balanced Stieltjes Moments: Bernoulli Closure, Harmonic Filters, and Polylogarithmic Resonances*. 11 October 2026. Preserved article, Balanced scalar

kernels and their Mellin transform; Section 7 gives mixed logarithmic moments of filtered Stieltjes functions.

[164] ProveIt research continuation. *Gamma Determinants and Colored Stieltjes Moments: Closed Normalization Constants, Finite Gaussian Moment Laws, and Clustered Polylogarithm Identities*. 12 October 2026. Preserved colored-moment section. The colored scalar transform, Stieltjes dictionary, and first exceptional moment are established there; the further-research section asks about simultaneous tail zeros.

[165] NIST Digital Library of Mathematical Functions. §2.5, Mellin inversion and Parseval, Eq. 15.6.1, Euler's hypergeometric integral, and Table 18.3.1, Laguerre orthogonality. Accessed 12 October 2026.

[166] J. Brillhart. On the Euler and Bernoulli polynomials. *Journal für die reine und angewandte Mathematik* **234** (1969), 45–64. doi:10.1515/crll.1969.234.45. The multiple-root theorem is also stated explicitly in F. T. Howard, *Roots of the Euler polynomials*, Pacific Journal of Mathematics **64**(1) (1976), 181–191, p. 181, publisher text, and NIST DLMF, §24.12(iv).

[167] NIST Digital Library of Mathematical Functions. §25.15, Dirichlet L-functions: functional equation, Gauss sums, Euler products, and trivial zeros. Accessed 12 October 2026.